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ELECTRON AND HOLE SPINS IN QUANTUM DOTS

by

Joseph Albert Ferguson Pingenot

An Abstract

Of a thesis submitted in partial fulfillment of the requirements for the Doctor of
Philosophy degree in Physics in the Graduate College of
The University of Iowa

May 2009

Thesis Supervisor: Professor Michael E. Flatté

As the technology underlying modern electronics advances, it is unlikely that previous rates of power use and computational speed improvement can be maintained. Devices using the spin of an electron or hole, “spintronic” systems, can begin to address these problems, creating new devices which can be used as a continuation and augmentation of existing electronic systems. In addition, spintronic devices could make special use of coherent quantum states, making it feasible to address certain problems which are computationally intractable using classical electronic components. Unlike higher-dimensional nanostructures such as quantum wires and wells, quantum dots allow a single electron or hole to be confined to the dot. Through the spin-orbit effect, the electron and hole g-tensor can be influenced by quantum dot shape and applied electric fields, leading to the possibility of gating a single quantum dot and using a single electron or hole spin for quantum information storage or manipulation.

In this thesis, the spin of electrons and holes in isolated semiconductor quantum dots are investigated in the presence of electric and magnetic fields using realspace numerical 8-band strain-dependent $k \cdot p$ theory. The calculations of electron and hole g-tensors are then used to predict excitonic g-tensors as a function of electric field. These excitonic g-factors are then compared against existing experimental work, and show that in-plane excitonic g-factor dependence on electric field is dominated by the hole g-factor. The dependence of the electron and hole g-tensors on the applied electric field are then used to propose a class of novel quantum dot devices which manipulate the electron or hole spins in either a resonant or a non-resonant mode. Because of the highly parabolic dependence of some components of the hole g-tensor on the applied electric field, a shift in the Larmor frequency and an additional

resonance are predicted, with additional shifts and resonances occurring for higher-order dependencies. Spin manipulation times down to 3.9ns for electrons and 180ps for holes are reported using these methods.

Abstract Approved:

Thesis Supervisor

Title and Department

Date

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Joseph Albert Ferguson Pingenot

A thesis submitted in partial fulfillment of the requirements for the Doctor of
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May 2009

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Graduate College
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CERTIFICATE OF APPROVAL

PH.D. THESIS

This is to certify that the Ph. D. thesis of

Joseph Albert Ferguson Pingenot

has been approved by the Examining Committee for the thesis requirement for the
doctor of Philosophy degree in Physics at the May 2009 graduation.

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John Prineas

David Andersen

Dedicated to my lov{ely,ing} wife Shannon, who dared to come with me into the
Frozen North so that this work might be achieved.

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CHAPTER I INTRODUCTION

Electronic devices in use today have been designed solely around the charge property of the electron—despite the fact that an electron possesses both charge and spin (innate angular momentum). Meanwhile, integrated circuit technology continues to shrink devices to ever smaller sizes. Intel and AMD, leading computer chip fabrication companies, recently left the old standard of 65nm-wide transistors and transitioned to fabricating 45nm-wide transistors industrially starting at the end of 2007.[4][5] As if to underscore how rapidly chip technology progresses, in the two years since the author’s dissertation proposal was written (2007), Intel is now expected to be producing 32nm transistors industrially this year, with 22nm parts scheduled for 2011.[6][7] AMD expects to be producing 32nm parts in 2011.[8] At these length scales, quantum effects become increasingly important. Although smaller transistor widths provide greater power savings and greater transistor density, critical factors such as power consumption and radio signaling circuits fail to continue scaling as they have in the past (i.e. at larger length scales).[9] Indeed, the International Technology Roadmap for Semiconductors (ITRS) has stated in its 2005 and 2007 reports that “[i]t is [...] difficult for most people in the semiconductor industry to imagine how we could continue to afford the historic trends of increase in process equipment and factory costs for another 15 years!”[10] Because of this limitation of existing technologies, the ITRS and the computer industry in general have been looking toward new computing device technologies.[11] Many new devices

and ideas have been suggested, but quantum computing and especially spintronics are of particular interest for this dissertation.

I.1 Spintronics and Quantum Computing

In spintronics, the spin of the electron is used to store or manipulate information. Despite perhaps sounding exotic and futuristic, spintronic devices are already in mainstream use. The most common form of spintronic device in use today consists of two or three metal layers, and is the sensor used in today’s hard drives.[12]

The spin of an electron is related to its intrinsic magnetic moment, and is a two-state quantum system. Although a given measurement will find the electron either in the “up” or “down” state, the spin can exist in a coherent superposition of up and down states until measured. Precisely this coherent superposition, coupled with fact that it is a two-state system, makes electron spins so appealing for computational uses. Spintronics may thus provide an avenue for the enhancement of conventional electronics, or its quantum nature may be used in a new, *quantum* computational system.

In conventional computational systems, a number is represented by a sequence of bits (i.e. is converted to a binary number). These bits may be represented physically in a number of ways, most notably by having one voltage represent a 1 and another voltage represent 0, or else by having the orientation of magnetic domains in a magnetic medium represent 1 or 0 as is the case in hard drives. Because the spin of an electron is a two-state system, it is a natural choice for representing binary information.

Modern conventional computers (also called classical computers to contrast them with quantum computational systems) find some calculations very difficult to perform quickly. One notable theoretical application of quantum computation is Shor's algorithm, which can factor the product of two large primes more efficiently than is possible with a classical computer.[13] Of particular note is the problem of factorizing the product of two very large prime numbers, which forms the basis for public key encryption. This makes quantum computing so appealing, especially to government agencies. The source of the calculation speed-up of quantum computing is the strange ability of a bit to not just be in the 0 or 1 state, but in a *both* states simultaneously—and to be coherently manipulated in a quantum algorithm. This is the foundation of quantum computing—using the natural ability of quantum systems to be in a coherent combination of multiple states simultaneously.[14][15]

In 1998, Loss and DiVincenzo suggested a design for a quantum computer using electrons bound within lithographic quantum dots (quantum dots formed in a 2-degree electron gas by applying electric potentials in order to restrict electron movement within the gas).[16] In this proposal, the strength of interaction between two electrons, each in its own quantum dot, is controlled with a voltage that changes the barrier between the dots, thus controlling the electron overlap. That same year, Kane proposed a design for a quantum computer using the nuclear spins of P atoms embedded in Si qubits.[17] In his proposal, gates positioned above and between the atoms controlled the interaction of electron spins with a global AC magnetic field, and also controlled the interactions between nuclear spins (via electrons).

Several other quantum computing architectures have been devised, most notably those based on superconductors and Josephson junctions[18][19] and trapped ions[20] In 2007, a company named D-Wave claimed to have developed and demonstrated the “world’s first commercially viable quantum computer.”[21][22] Although it remains to be seen if the D-Wave device fits the definition of a quantum computer, it points toward ever-increasing commercial interest in the field of quantum computation.

Spintronic components can also work within a classical logic gate alongside conventional electronics. Datta and Das in 1989 proposed a transistor which uses spin-polarized electrons injected from a ferromagnetic metal contact, using the presence (or lack of) electric field from the gate to precess the spins.[23] The electrons are then either permitted to flow or are blocked depending on whether the spins are aligned or anti-aligned with the ferromagnet on the other side of the device. Since then, new designs have been proposed which remove the necessity of having magnetic materials in the device.[24]

I.2 Spintronic Material Systems

Because of their properties, semiconductors are uniquely positioned to aid in the control and storage of the electron, and thereby also its spin. For example, by adding a relatively small amount of a material (dopant) to a semiconductor, one can change its conductive properties greatly, the resulting effect depending upon the dopant material.[25] Additionally, a wide variety nanostructures can be constructed from layers of different—or just differently-doped—materials, creating a range of de-

vices to perform many different tasks. Finally, there is a large body of technological knowledge of semiconductor device design, construction, and behavior—both practical as well as theoretical—allowing prediction of material and device properties.

The work which will be discussed in this proposal relates to quantum dots, particularly InAs dots embedded in GaAs. Quantum dots are of particular interest for spintronic applications, since they have a discrete, atomic-like electronic structure and can contain a very small number of electrons due to their low-dimensionality. This makes them capable of containing a controlled number of electrons, even down to just a single electron.[26]

There are four main kinds of quantum dots. The first is simply a special kind of defect in a quantum well where the walls of the quantum well are pitted and thus there is a small area where an electron can become trapped.[27] There are also quantum dots formed by a 2-dimensional electron gas (2DEG) (narrow, doped quantum well) with gates deposited on top.[26] Electric potentials are applied to the gates and extend downward into the 2DEG to create confinement within the 2DEG by creating an electrical barrier to electron movement. Quantum dots can also be formed as small spheres and embedded in another material or suspended in a liquid[28][29]. Most importantly for this paper, quantum dots can be formed by molecular beam epitaxy (MBE) in the Stranski-Krastonow growth mode. These dots grow as islands of material when deposited thinly on a substrate, forming pyramids or lens-like structures as they grow.[30] Because the structures arise without human intervention due to innate processes, these are known as “self-assembled” quantum

dots. It is the behavior of electrons inside of these self-assembled dots that will be the primary subject of this proposal.

I.3 Electron and Hole Spins in Quantum Dots

Because a quantum bit (“qubit”) can be represented by a single spin in a spintronic quantum computer, one must first understand what influences the electron and hole spins in quantum dots. The general outline of one method used to calculate electron properties in semiconductors is discussed in section I.4 in order to understand how these influences and effects are calculated. The method used in this dissertation to calculate electron and hole energies and wavefunctions in quantum dots, $k \cdot p$ theory, will be derived, along with some examples. Section I.5 introduces basic methods used for this dissertation to calculate strain in semiconductors, as it is present in and around quantum dots and can have a non-negligible effect on the quantum dot states. Section I.6 discusses how the electron spin fits into the picture and how it interacts with magnetic fields. Additional details about how the calculations are performed are explained in chapter II. Single electron spins in InAs/GaAs and $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dots are dealt with in chapter III. Hole spins in InAs/GaAs and $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dots are discussed in IV. The results of chapters III and IV are used to discuss excitonic g-factors in chapter V. Finally, the electron and hole g-tensors in chapters III and IV are used to provide information on using resonant and nonresonant means to coherently manipulate the carrier spins. Chapter VI discusses using the electric field dependence of the electron and hole g-tensors for g-tensor modulation resonance. Chapter VII discusses

nonresonant spin manipulation techniques.

I.4 Electrons in Quantum Dots

To investigate the electronic behavior in quantum dots, the electronic behavior in bulk materials must first be understood. Although there are several methods of calculating the band structure of bulk semiconductors, this work will concentrate on using the $k \cdot p$ approximation. Other methods have been used, most commonly the tight-binding and pseudopotential models, which are discussed in chapter 2 of *Fundamentals of Semiconductors*[25] as well as in other places in the literature.[31][32] In addition, a boundary-element method[33] and a Green's function method[34] have been proposed and used to a much more limited extent.

Tight-binding methods have been used to calculate quantum dot properties, e.g. in CdSe nanocrystals[35], Si/Ge quantum dots[36], and self-assembled In-GaAs/GaAs quantum dots[37, 38]. The effective bond orbital model used by the references [37] and [38] was compared to effective mass and 8-band $k \cdot p$ in a recent paper by Sheng et al.[39] for calculating multiexciton complexes in a 2.3nm-high circular-footprint dot with footprint radius of 25.4nm self-assembled In_{0.5}Ga_{0.5}As/GaAs /GaAs quantum dots. Although their effective bond orbital model (EBOM) calculations and $k \cdot p$ calculations were similar near the Γ point, they diverged further out in k space. Qualitative differences were found when calculating the energy structure of this small dot using $k \cdot p$ and the EBOM tight binding model. The disagreement was attributed to the very short dots, as taller dots did not show significant disagreement. However, the qualitative differences in multiexciton spec-

tra were in good agreement, although the energies found using the 8-band $k \cdot p$ approximation were slightly higher than was found for the tight binding model. Because of the atomistic nature of tight-binding methods, tight-binding calculations require much more computational power than similar $k \cdot p$ calculations. Due to the large number of dots which needed to be calculated and due to the tall dots being calculated, tight-binding methods were not used in this dissertation.

The pseudopotential method has been employed to some extent in quantum dot calculations. Reference [40] discusses using pseudopotential methods in “large” dots of up to 700 atoms, in contrast to the work of Sheng et al.[39, 37, 38] which used between 50,000 and 500,000 atoms in their calculations. As the number of atoms required for dots of the size to be investigated were significantly larger than the 500,000 used in the EBOM calculations, pseudopotentials were not able to be used for the calculations in this dissertation. In addition, the realspace quantum dot grid is more readily usable in $k \cdot p$ than pseudopotential methods, as the pseudopotential calculations are done in momentum space.

$k \cdot p$ calculations, however, have been used to a very wide extent in semiconductor nanostructures, and are well understood. Effective mass[41, 42], Luttinger (4 valence bands)[43, 44], 6-band (4 valence + 2 conduction) [45, 46], 8-band (6 valence + 2 conduction) [47, 48, 49, 50, 51, 52, 53, 54, 55, 56] and even 10 bands (6 valence + 2 conduction + 2 “N”)[57, 58] have been used in the literature. Because the number of bands used is adjustable, the computation speed can be adjusted, as increasing the number of bands increases the time required to perform the calculation. A compar-

ison by Pryor[59] showed that 8 bands is necessary in dot calculations. Therefore, the calculations performed in this dissertation used the 8-band $k \cdot p$ model

1.4.1 Bulk Electronic States

As shown in figure 1, a semiconductor’s energy structure consists of ranges of allowed energies (“energy bands”) separated by a forbidden energy region, called the band gap (E_g). Additionally, the Fermi energy (E_F) (the highest occupied electron energy level at a temperature of 0K) of the material falls within the band gap at zero temperature. The electron states are then split into two main types: a mostly-empty set of states above the band gap and a mostly-filled set of states below the band gap. The higher-energy, mostly-empty states are the conduction band; the lower-energy, mostly-filled electronic states are the valence band. It is sometimes easier to view the mostly-filled electron states as being mostly-empty hole states, with higher-energy electron states corresponding to lower-energy hole states. By investigating the valence electron states, then, one is also investigating the hole states.

In figure 1 is an illustration of a four-band model of a semiconductor’s energy structure. The valence-band (red and blue) maximum is directly below the conduction-band (black) minimum and therefore figure 1 is an illustration of a “direct-gap” semiconductor. The vertical axis is energy, and plotted as a function of crystal momentum magnitude k along some direction. (Crystal momentum will be discussed in greater detail momentarily.) $\mathbf{k} = 0$ is at the center of the diagram, and left or right from it may be different $\mathbf{k}$ directions. The bands are roughly

parabolic near “zone center” ($k = 0$). The bands are not generally parabolic in k further out from zone center. The band gap is noticeable between the lower states (valence band, in red and blue) and upper states (conduction band, in black), with a prominent spin-orbit splitting (lowest valence state). In figure 1, the Fermi energy (E_F) is denoted by the dashed line. Note that it falls within the band gap, so the states are not occupied past the upper-most-energy valence state. As the temperature increases, however, the electrons in the valence band may begin to jump up into the conduction band as thermal excitations promote electrons into higher energy levels.

The lowest conduction band state is s-orbital-like—that is, it transforms like a scalar—and is usually referred to as “S”. There are 3 valence states, each with the symmetry behavior of the p-orbitals, which are usually represented by the letters “X”, “Y”, and “Z”, as they transform like a vector. There are additional states in both the valence band and the conduction band which are not shown in figure 1. However, the further away from the band gap the energy is, the less the states affect the semiconductor behavior. There are two sets of states that are most commonly used outside of the s-like conduction and p-like valence bands, one set of which is depicted in figure 2. Above the lowest conduction s-like state, there are 3 conduction states with p-like character; below the three p-like valence band states lies a valence-band s-like state (not shown).

Because the electron possesses an intrinsic angular momentum (spin- $\frac{1}{2}$), each of these states has a twofold spin degeneracy when no magnetic field is applied. The

six valence electron states (X, Y, and Z with both up and down spins) are degenerate at zone-center. However, if one includes spin-orbit coupling, one of the three states is lowered in energy and becomes the “split-off” or “spin-orbit” valence state. Note, however, that including spin-orbit coupling mixes the orbital and spin degrees of freedom, and thus one can no longer simply speak of spins, but must rather speak of “pseudospins”. Because s-like states have no orbital angular momentum, however, they remain true spin- $\frac{1}{2}$ states. Pseudospins will be discussed in more detail later, in section I.6.

Introduction to $k \cdot p$ Calculations In the $k \cdot p$ approximation, the electron wavefunction is considered a function of both position (x) and crystal momentum (k). Due to the periodicity of the lattice, the electron wavefunction is composed of a plane wave multiplied by a periodic function on the lattice. This function is the Bloch function, and has the property

$$u(\vec{k}, \vec{x} + \vec{X}) = u(\vec{k}, \vec{x}) \quad (1)$$

where $\vec{X}$ is a member of the set of all lattice vectors. It is an eigenfunction of the Hamiltonian

$$H_{electron} = H = \frac{p^2}{2m} + V(\vec{x}) \quad (2)$$

where $V(\vec{x})$ is the potential from the ions within the crystal (i.e. the potential at the lattice sites, or the “crystal potential”). The Bloch functions at zone-center ($k = 0$),

when multiplied by $e^{i\vec{k}'\cdot\vec{x}}$, form a complete basis at wave vector $\vec{k}'$. [60] One therefore needs only use the zone-center Bloch functions of the material under investigation, multiplied by a plane wave envelope function to construct the wavefunction. One then sums over all zone-center Bloch functions:

$$\Psi(\vec{k}, \vec{x}) = \sum_n e^{i\vec{k}\cdot\vec{x}} u_n(\vec{0}, \vec{x}) = \sum_n e^{i\vec{k}\cdot\vec{x}} u_n(\vec{x}) \quad (3)$$

where $u(\vec{x})$ is the zone-center Bloch function. Applying the Hamiltonian (equation 2) to this new Ψ , $H_{electron}$ becomes

$$u_l(\vec{x}) E u_n(\vec{x}) = \frac{\hbar^2 k^2}{2m} u_l(\vec{x}) u_n(\vec{x}) + \frac{\hbar \vec{k} \cdot (u_l(\vec{x}) \vec{p} u_n(\vec{x}))}{m} + u_l(\vec{x}) \frac{p^2}{2m} u_n(\vec{x}) + V(\vec{x}) u_l(\vec{x}) u_n(\vec{x}) \quad (4)$$

where $\vec{p}$ is the momentum operator. This result may be used in one of two ways. The terms introduced from the plane wave (the terms involving $\vec{k}$) may be used to perturb the $k = 0$ (zone-center) Hamiltonian (which is the term in square brackets in equation 4). Alternately, since the Bloch functions are a complete orthonormal basis over the unit cell, one may simply choose to integrate both sides of equation 4 over the unit cell. Both of these methods will be demonstrated shortly.

The Bloch functions form an infinite basis. Therefore, to make the calculations tractable, one must truncate the basis to a subset of the Bloch functions. Although one may technically use any number of bands one wishes, there is a trade-off between speed and accuracy. Increasing number of bands results in a more accurate calculation since it is getting closer to the full set of Bloch functions. However, it

also takes increasingly longer to perform the calculation since the time required to diagonalize a matrix of size N^2 (i.e. with N bands) is proportional to N^3 . [61] 24 bands [62] or more have been used in research, although calculations involve two, four, eight, or 14 bands are more common. An example of the band structure of a seven-band $k \cdot p$ model is in figure 2. As a middle ground between speed and accuracy, one may choose to include the influence of the more remote bands to improve the accuracy of the calculation without having to include them explicitly. One then must use Löwdin perturbation theory to include the effects from these extra bands. [63]

$k \cdot p$ Example: Effective Mass Approximation As an example of using $k \cdot p$ in bulk, the effective mass approximation will be derived. The effective mass approximation can be used to predict electron behavior within a direct-gap semiconductor close to zone center, where the states are approximately parabolic (this example is based on Yu and Cardona's *Fundamentals of Semiconductors* [25]). In the effective mass approximation, the energy of an electron in band n is just

$$E_n(\vec{k}) = E_{n,0} + \frac{\hbar^2 k^2}{2m_n^*} \quad (5)$$

with m^* determined by the second-order perturbation expansion of the Hamiltonian, and $E_{n,0}$ is the zone-center energy of band n . One must, then, determine what the effective mass of the electron is for band n . To do this, one assumes that the k -terms of the basic $k \cdot p$ Hamiltonian (equation 4) are small enough to use as a perturbation,

calling the k -terms $H_{\vec{k}\cdot\vec{p}}$

$$H_{\vec{k}\cdot\vec{p}} = \frac{\hbar^2 k^2}{2m} u_l u_n + \frac{\hbar \vec{k} \cdot (u_l \vec{p} u_n)}{m} \quad (6)$$

(note that the position-dependency of the Bloch functions has been dropped for simplicity of notation). Since the Bloch functions are the 0th-order eigenstates of the perturbation, the energy of the n^{th} band is, to second order,

$$E_n(\vec{k}) = E_{n,0} + \langle u_n | (\vec{x}) | H_{\vec{k}\cdot\vec{p}} | u_n \rangle + \langle u_n | H_{\vec{k}\cdot\vec{p}} \left(\sum_{n' \neq n} \frac{|u_{n'} \rangle \langle u_{n'}| H_{\vec{k}\cdot\vec{p}} | u_n \rangle}{E_{n,0} - E_{n',0}} \right) \quad (7)$$

where n, n' are zone-center band index numbers up to N . Rewriting this in a more physically transparent manner and setting it equal to E_n from equation 5,

$$E_{n,0} + \frac{\hbar^2 k^2}{2m} + \frac{\hbar^2}{m^2} \sum_{n' \neq n} \frac{|\vec{k} \cdot \langle u_{n'} | \vec{p} | u_n \rangle|^2}{E_{n,0} - E_{n',0}} = E_{n,0} + \frac{\hbar^2 k^2}{2m_n^*} \quad (8)$$

Terms linear in crystal momentum magnitude k have been neglected, as $E_{n,0}$ is assumed to be an extremum ($k = 0$), and a nonzero linear k term would shift the extremum away from $k = 0$. Terms of order higher than k^2 have also been neglected, as they are assumed to be sufficiently small. Thus, the effective mass is

$$\frac{1}{m_n^*} = \frac{1}{m} + \frac{2}{m^2 k^2} \sum_{n' \neq n} \frac{|\vec{k} \cdot \langle u_{n'} | \vec{p} | u_n \rangle|^2}{E_{n,0} - E_{n',0}} \quad (9)$$

Each band has its own effective mass. Because of their effective masses, the

upper two valence states are known as the heavy hole and light hole. It may also be noted from equation 9 that the higher the difference in energy, the smaller the contribution to m^* . One may conclude, then, that energetically remote bands have less influence on the effective masses of the bands close to the energy gap (the ones that are generally of primary interest).

General Bulk $k \cdot p$ Calculations Although the simple plane wave-plus-Bloch expansion in equation 3 suffices for finding the bulk band structure of *some* systems, band energy extrema need not lie at zone center in general. As a extreme example, Si has well-known conduction band minima (“valleys”) far away from zone center.

1.4.2 Electronic States in Quantum Dots

Although the plane wave-and-Bloch expansion of the wavefunction works well in bulk systems, nanostructures present new difficulties, owing to the different band structures of the materials within the nanostructure. For a quantum dot, the uppermost valence band energy (lowermost conduction band energy) of the dot material falls above the uppermost valence band energy (below the lowermost conduction band energy of the barrier material), leading to confinement of the electron envelope function. Other band configurations are possible. The aforementioned configuration is Type I, but other, semi-confined configurations are possible and are called Type IIA and IIB.[25] The three possible band edge configurations are illustrated in figure 3 Only the fully-confined (Type I) configuration will be discussed in this paper, as it provides confinement for both electron and hole states.

The difference in energy levels between the dot and surrounding “barrier” material provides the electron containment in the quantum dot. Like the difference in energy levels in the quantum well, energy levels below (above) the barrier material’s conduction (valence) energy band edge but above (below) the dot material’s conduction (valence) band edge become discretized. Unlike the quantum well system, however, the quantum dot provides containment in all three directions, not just one, yielding discrete, atom-like states. [25]

Within restricted-dimensionality nanostructures, the unrestricted directions may be expanded in a plane-wave representation as before. However, in the constrained direction, the wavefunction is no longer a simple plane wave. It is better represented as a function of position

$$\Psi(\vec{x}) = \sum_n F_n(\vec{x}) u_n(\vec{x}) \quad (10)$$

where $F_n(\vec{x})$ is the envelope function for band n in real space. For a semi-confined structure, such as a quantum wire along the z -axis, $F_n(\vec{x})$ is split into two pieces: a real space envelope function in the x and y directions, and a plane wave along the z direction:

$$\Psi(\vec{x}) = \sum_n F_{n,xy}(\vec{x}) e^{ikz} u_n(\vec{x}) \quad (11)$$

One may choose to solve the envelope functions analytically or numerically. To solve the envelope functions numerically, one changes the derivatives in the Hamiltonian

into finite differences where[64]

$$\frac{d\psi}{dx} \approx \frac{\psi(x+d) - \psi(x)}{d} \quad (12)$$

For a quantum dot, which is confined in all directions, no plane wave component is valid in $F_n(\vec{x})$. Rather, the Hamiltonian becomes a system of N coupled equations, one for each Bloch function included:

$$\sum_n^N E F_n(\vec{x}) u_n(\vec{x}) = \sum_n^N \left(\frac{p^2}{2m} + V \right) F_n(\vec{x}) u_n(\vec{x}) \quad (13)$$

$$= \sum_n -\frac{\hbar^2}{2m} (\nabla^2 F_n(\vec{x})) u_n(\vec{x}) - \frac{i\hbar}{m} (\vec{\nabla} F_n(\vec{x})) \cdot (\vec{p} u_n(\vec{x})) + F_n(\vec{x}) \left(\frac{p^2}{2m} + V(\vec{x}) \right) u_n(\vec{x}) \quad (14)$$

which becomes

$$\begin{aligned} = \sum_n \int_{unitcell} & -\frac{\hbar^2}{2m} (\nabla^2 F_n(\vec{x})) u_{n'}(\vec{x}) u_n(\vec{x}) - \frac{i\hbar}{m} (\vec{\nabla} F_n(\vec{x})) \cdot (u_{n'}(\vec{x}) \vec{p} u_n(\vec{x})) \\ & + F_n(\vec{x}) u_{n'}(\vec{x}) \left(\frac{p^2}{2m} + V(\vec{x}) \right) u_n(\vec{x}) dx^3 \end{aligned} \quad (15)$$

after multiplying both sides of the equation by $u_n(\vec{x})$ and integrating over the unit cell. Aside from the integration, this is the Hamiltonian for a plane wave, equation 4, if $F_n(\vec{x})$ is $e^{i\vec{k} \cdot \vec{x}}$. Instead of perturbatively expanding H_0 in with $H_{\vec{k}, \vec{p}}$, the orthonormality of the zone-center Bloch functions will be used and the envelope

functions $F_n(\vec{x})$ will be assumed to vary slowly over the unit cell (i.e. they and their derivatives are approximately constant), such that:

$$\int_{unitcell} u_{n'}(\vec{x}) F_n(\vec{x}) u_n(\vec{x}) d^3x \approx F_n(\vec{x}) \int_{unitcell} u_{n'}(\vec{x}) u_n(\vec{x}) d^3x = F_n(\vec{x}) \delta_{n',n} \quad (16)$$

$$EF_n(\vec{x}) = \sum_n -\frac{\hbar^2}{2m} (\nabla^2 F_n(\vec{x})) \delta_{n',n} - \frac{i\hbar}{m} (\vec{\nabla} F_n(\vec{x})) \cdot (\vec{p}_{n',n}) + F_n(\vec{x}) E_{n,0} \delta_{n',n} \quad (17)$$

where $\vec{p}_{n',n}$ is the momentum matrix element between zone-center Bloch states $u_{n'}(\vec{x})$ and $u_n(\vec{x})$. Thus, This is a system of N equations, forming an $N \times N$ matrix.

1.4.3 Spin-Orbit Interaction in $k \cdot p$ Calculations

The spin-orbit interaction is a relativistic effect on the electron spins due to the atomic potentials interacting with “orbiting” electrons[65]

$$H_{SO} = \frac{\hbar}{4m^2c^2} [\vec{\sigma} \times \vec{\nabla} V(\vec{x})] \cdot \vec{p} \quad (18)$$

Because the envelope functions only vary slowly on the scales of the individual atomic potentials, as well as because of ion screening effects, it is physically sensible to assume for now that the spin-orbit interaction only affects the zone-center Bloch functions.

The spin-orbit interaction, then, only interacts with the envelope functions indirectly for now, through the zone-center Bloch functions which form its basis. The effect of the spin-orbit interaction on the $u_{n,0}(\vec{x})$ is twofold. First, it causes the spin and orbital angular momentum parts of the p-like Bloch functions to mix and form

four degenerate spin- $\frac{3}{2}$ states and two degenerate spin- $\frac{1}{2}$ states. Second, it causes the spin- $\frac{3}{2}$ states to be higher in energy than the spin- $\frac{1}{2}$ states whereas before this, they were all degenerate at the same energy.

The impact on calculations of the envelope functions $F_{n,\vec{k}}(\vec{x})$ depends on the choice of basis states. One may choose the simpler $|S \uparrow\rangle |X \uparrow\rangle |Y \uparrow\rangle |Z \uparrow\rangle |S \downarrow\rangle |X \downarrow\rangle |Y \downarrow\rangle |Z \downarrow\rangle$ basis (for an 8-band model) at the expense of having to explicitly include the spin-orbit interaction in the end Hamiltonian.[66] Alternately, one may choose a more complicated set of basis states which include the spin-orbit interaction already and need only account for the different zone-center Bloch function energies, as did Johnson et al.[67]

The spin-orbit interaction provides the primary means of coupling the electron spin component to the spatial component of its wavefunction. This is of fundamental importance, as the spin and spatial portions would otherwise remain entirely separate. The spin-orbit interaction, then, provides an avenue for the spin to be influenced by non-spin environmental factors, such as the electric field. As shall be seen in the next section, the spin-orbit interaction also has additional, interesting effects that are not immediately apparent. However, to understand these effects, spin and angular momentum interactions with a magnetic field must first be discussed.

I.5 Strain in Semiconductor Systems

In the absence of an outside force, the unstrained crystal consists of atoms at specific points in space. Examining one of these atoms, one can then visualize it

being pushed out of its unstrained position. The displacement, $(\vec{u})$, is defined as

$$\vec{u} = \vec{x}' - \vec{x} \quad (19)$$

where $\vec{x}'$ is the displaced position of the atom, and $\vec{x}$ is its unstrained position.

Reversing this equation and finding the strained position of the atom, one has

$$\vec{x}' = \vec{x} + \vec{u} \quad (20)$$

This gives the final position of the atom based on its initial position and its displacement from that position. Note that a stress (external set of forces pushing on the crystal) can displace an atom in *any* direction.

More generally, one can look at strain as deforming the lattice coordinate system.

A length dl in the presence of strain becomes

$$dl'^2 = dx_i'^2 = (dx_i + du_i)^2 \quad (21)$$

(using implicit summation of dx_i). Now, one may define du thus:

$$du_i = \sum_j \frac{\partial u_i}{\partial x_j} dx_j \quad (22)$$

(again using implicit summation over j). The length becomes

$$dl'^2 = \sum_i dx_i^2 + 2 \sum_j \sum_i \frac{\partial u_i}{\partial x_j} dx_j dx_i + \sum_k \sum_j \sum_i \frac{\partial u_i}{\partial x_j} dx_j \frac{\partial u_i}{\partial x_k} dx_k \quad (23)$$

Now, one may assume that the deformation is small, such that the second-order term in $\frac{\partial u_i}{\partial x_j}$ disappears. The equation then becomes

$$dl'^2 = dl^2 + 2 \sum_i \sum_j \frac{\partial u_i}{\partial x_j} dx_j dx_i \quad (24)$$

Since this is a sum over both i and j , one can rearrange the summation as needed.

Noting that equation 24 effectively is adding two sets of identical sums, and that there are in each set of sums one of each possible combination of i and j , equation 24 becomes

$$dl'^2 = dl^2 + \sum_i \sum_j \frac{\partial u_i}{\partial x_j} dx_j dx_i + \sum_i \sum_j \frac{\partial u_i}{\partial x_j} dx_j dx_i \quad (25)$$

$$= dl^2 + \sum_i \sum_j \frac{\partial u_i}{\partial x_j} dx_j dx_i + \sum_i \sum_j \frac{\partial u_j}{\partial x_i} dx_i dx_j \quad (26)$$

$$= dl^2 + \sum_i \sum_j \left(\frac{\partial u_i}{\partial x_j} dx_j dx_i + \frac{\partial u_j}{\partial x_i} dx_i dx_j \right) \quad (27)$$

$$= dl^2 + \sum_i \sum_j \left(\frac{\partial u_i}{\partial x_j} dx_j dx_i + \frac{\partial u_j}{\partial x_i} dx_j dx_i \right) \quad (28)$$

$$= dl^2 + \sum_i \sum_j \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) dx_j dx_i \quad (29)$$

$$= \sum_i dx_i dx_i + \sum_i \sum_j \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) dx_j dx_i \quad (30)$$

Therefore, the strain tensor is defined as:[68][25]

$$e_{ij} = e_{i,j} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (31)$$

which yields a length of

$$dl'^2 = \sum_j \sum_i (\delta_{ij} + 2e_{ij}) dx_i dx_j \quad (32)$$

Thus, one can look at the strain tensor as a fractional change in each direction relative to another (or the same) direction. To change a lattice vector, for instance, using the matrix equation

$$a'_i = \sum_j (\delta_{ij} + e_{ij}) a_j \quad (33)$$

suffices, where δ_{ij} is the Kronecker delta.

With this information, then, one is in a position to start examining strain effects in semiconductors. The effects of strain in bulk in general will be discussed first, and then how to calculate them for an 8-band model. Finally, strain in quantum dots will be discussed.

1.5.1 Calculating strain distribution

To calculate strain effects, one must first be able to calculate the strain distribution in a nanostructure. Two methods have commonly been used to calculate strains in nanostructures: the valence force field (VFF) approach, and the continuum elasticity method.

The continuum elasticity approach simply assumes a solid material and deforms it as needed. For instance, a realspace grid might be created with the GaAs barrier material lattice constant and then include a slab of InAs inside of this barrier to form a well. The cells of InAs are allowed to expand or contract and push on the neighboring cells of material until equilibrium is reached. This model will be described in more detail in the upcoming section. The total energy is described as it is in bulk systems, assuming bulk-like individual cells on the realspace grid.[68]

The valence force field approach is based upon a paper by Keating, which was further adapted by Kane to remove second- and higher-order anharmonic effects in bond stretching and bending.[69, 70] It is called in [71] the “atomic elasticity” model, to reflect the bond-oriented nature of the model. The total energy of the system is based upon the stretching and bending of the bonds between atoms, and may be expanded to higher-order terms, e.g. second-nearest-neighbor bond stretching.

The VFF and CE models have been compared in work by Pryor et al.[71] and Stier et al.[72]. The conclusions were similar. The adjusted VFF method and CE methods differed mostly at interfaces, the apex of the pyramidal dots being investigated, and along the in-plane directions. Because the dots investigated in this dissertation are all lens-shaped, the differences they found for the pointed apex of their pyramidal dots is not likely to cause trouble. Because of the more atomic nature of the VFF calculations, they are better able to reproduce the tetrahedral symmetry of the underlying crystal, leading to a very minor difference (less than 1 percent) in the in-plane ($[110]$ vs $[1\bar{1}0]$) strains. The most important difference for

this research is the difference at the interfaces between the dot and barrier materials. However, the differences are fairly small and well-localized. Therefore, continuum elasticity was used to determine strain distributions due to its speed.

Determining strain from the continuum elasticity model Once mapped onto a realspace grid, the materials must then be strained. This is done by numerically reducing the strain energy[68, 73]

$$E = \int \frac{1}{2} \lambda_{\alpha\beta\gamma\delta} \sigma_{\alpha\beta} \sigma_{\gamma\delta} d^3r \quad (34)$$

which reduces to

$$E = \int \frac{1}{2} C_{xxxx} (\sigma_{xx}^2 + \sigma_{yy}^2 + \sigma_{zz}^2) + C_{xxyy} (\sigma_{xx}\sigma_{yy} + \sigma_{xx}\sigma_{zz} + \sigma_{yy}\sigma_{zz}) \quad (35)$$

$$+ 2C_{xyxy} (\sigma_{xy}^2 + \sigma_{xz}^2 + \sigma_{yz}^2) \quad (36)$$

($C_{\alpha\beta\gamma\delta}$ is a material parameter for our purposes) due to the symmetry of a cubic lattice by straining the grid sites.[74, 71, 59]

If the realspace grid was chosen to coincide with the cubic lattice spacing of the barrier material, the barrier material is unstrained, as expected. However, in any quantum dot regardless of size, straining the coordinate system (which is set to the barrier material) will introduce a strain which is unphysical and due purely to the strained coordinate system. This is solved in a 1997 paper by Pryor et al.[74] by

adding a term to the energy

$$F = E - \alpha (\sigma_{xx} + \sigma_{yy} + \sigma_{zz}) \quad (37)$$

$$\alpha = (C_{xxxx} + 2C_{xxyy}) \frac{a_D - a_B}{a_B} \quad (38)$$

where a_D is the dot material lattice constant and a_B is the barrier material lattice constant.

Once the strain is known throughout the system, one may proceed to calculate the electron states within the structure using $\vec{k} \cdot \vec{p}$.

1.5.2 Calculating with strain in the $k \cdot p$ method

In bulk, strain lowers the symmetry of the crystal, thereby causing some degeneracies to be lifted. For instance, the conduction band of Si has a valley (minimum in the dispersion relation) at a certain point along the x, y, and z directions (both at positive and negative points). With the introduction of strain, however, the energy of the valleys along the axis of the strain are raised while the energies of the valleys perpendicular to it are lowered.[75]

There have been many papers calculating the Hamiltonian of an electron in a strained semiconductor.[76, 77] The discussion in this section, which follows the work done by Pryor in his dotcode, follows work by Bahder.[1] As will be shown, it recovers the usual hydrostatic deformation potential interactions a_i and the shear interactions b and d . [1, 78]. Although the a_i , b , and d interactions are sufficient to describe some of the interactions in band structure calculations, additional terms

arise when considering nanostructures. These terms are discussed following Bahder's derivation.

When a crystal is strained, the atoms move from their unstrained positions, as discussed previously. From equation 33 and the Hamiltonian (equation 13 on page 17) with spin-orbit piece (equation 18, page 18), one has a starting point for calculating the effect of strain on electrons.

Following Bahder (who in turn follows Pikus and Bir[76][77]), one starts with a strained crystal and examines its symmetries. In Cartesian coordinates, the symmetries it possessed while unstrained are broken, as the basis vectors are now transformed as described by equation 33. However, if the coordinate system is transformed away from the Cartesian one to correspond to the *strained* coordinate system, one finds that the symmetries have been regained. One must then transform back to regular Cartesian coordinates to get the final results of the calculations in the laboratory coordinates. The transformation to get back to the original coordinate system needed is:

$$x_i = \sum_j (\delta_{ij} + e'_{ij}) x'_j \quad (39)$$

maintaining the convention that $\vec{x}$ is a vector in the laboratory coordinate system, and $\vec{x}'$ is a vector in the strained coordinate system. This is a matrix equation—the

inverse of the original transform in equation 33. However,

$$e'_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x'_j} + \frac{\partial u_j}{\partial x'_i} \right) \quad (40)$$

$$= e_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial (x_j + \sum_k e_{jk} x_k)} + \frac{\partial u_j}{\partial (x_i + \sum_k e_{ik} x_k)} \right) \quad (41)$$

$$\approx \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (42)$$

$$= e_{ij} \quad (43)$$

Thus, this becomes

$$x_i = \sum_j (\delta_{ij} + e_{ij}) x'_j \quad (44)$$

because only first-order terms in strain are kept.

One must now transform the wavefunction and Hamiltonian from the strained coordinate system to the original (laboratory) coordinate system by applying this transform to the wavefunction and Hamiltonian. As mentioned before, it is assumed that the deformations are small, and the Hamiltonian is expanded perturbatively around the strained coordinate system because it is only in the *strained* coordinate system that the crystal symmetries are present. Although one could simply expand perturbatively around the original coordinate system, the potential would not contain the symmetries of the original crystal. Note that both the position vector $\vec{x} \rightarrow \vec{x}'$ and momentum $\vec{p} \rightarrow \vec{p}' = \frac{\hbar}{i} \vec{\nabla}_{\vec{x}'}$ are with respect to the *strained* coordinate

system. The Hamiltonian, then, consists of two parts:

$$H_S = \frac{p'^2}{2m} + V(\vec{x}') \quad (45)$$

$$H_{SO} = \frac{\hbar^2}{4m^2c^2} \left[\vec{\sigma} \times \vec{\nabla}_{\vec{x}'} V(\vec{x}') \right] \cdot \vec{p}' \quad (46)$$

and will be expanded perturbatively back into the laboratory (original) coordinate system, expanding $V(\vec{x}')$ to first order in strain. $\vec{p}'$ will end up acting on $\vec{x}'$, so it will remain. One applies the wavefunction, composed of Bloch functions and periodic envelope functions in the laboratory coordinate system:

$$\Psi_{\vec{k}}(\vec{x}) = \sum_n e^{i\vec{k} \cdot \vec{x}} u_{n,\vec{0}}(\vec{x}) \quad (47)$$

Assuming that

$$u_{n,\vec{0}}(\vec{x}) \approx u_{n,\vec{0}}(\vec{x}') \quad (48)$$

one then has

$$\Psi_{\vec{k}}(\vec{x}') = \sum_n e^{i\vec{k} \cdot \vec{x}'} e^{i\vec{k} \cdot \hat{e} \vec{x}'} u_{n,\vec{0}}(\vec{x}') \quad (49)$$

Because of this, $\vec{p}'$ operating on $\Psi_{\vec{k}}(\vec{x}')$ will be

$$\vec{p}' \Psi_{\vec{k}}(\vec{x}') = \hbar(\vec{k} + \vec{k} \hat{e}) \Psi_{\vec{k}}(\vec{x}') + \sum_n e^{i\vec{k} \cdot (\hat{e} + \vec{e}) \vec{x}'} \vec{p}' u_{n,\vec{0}}(\vec{x}') \quad (50)$$

Now, expanding V to first order in e_{ij} gives:

$$V(\vec{x}) \approx V(\vec{x}') + \sum_{i,j} \frac{\partial}{\partial e_{ij}} [V(\vec{x}')]_{e_{ij}=0} e_{ij} \quad (51)$$

and can now put the whole Hamiltonian together. Equation 45, in the presence of strain becomes (remembering how $\vec{p}'$ operates on the wave function):

$$H_S = \frac{\hbar^2}{2m} (\vec{k} + \vec{k}\hat{e})^2 \Psi_{\vec{k}}(\vec{x}') \quad (52)$$

$$+ \frac{\hbar}{m} \sum_n e^{i\vec{k} \cdot (\hat{\delta} + \hat{e})\vec{x}'} (\vec{k} + \vec{k}\hat{e}) \cdot \vec{p}'(u_{n,\vec{0}}(\vec{x}')) \quad (53)$$

$$+ \frac{1}{2m} \sum_n e^{i\vec{k} \cdot (\hat{\delta} + \hat{e})\vec{x}'} \vec{p}'^2(u_{n,\vec{0}}(\vec{x}')) \quad (54)$$

$$+ V(\vec{x}') \Psi_{\vec{k}}(\vec{x}') \quad (55)$$

$$+ \sum_{i,j} \frac{\partial}{\partial e_{ij}} (V(\vec{x}'))_{e_{ij}=0} e_{ij} \Psi_{\vec{k}}(\vec{x}') \quad (56)$$

Which then becomes, after expanding squares, collecting $\hat{e}$ terms, and keeping only

linear or lower $\hat{e}$ terms:

$$\begin{aligned}
H_S = & \frac{\hbar^2}{2m} k^2 \Psi_{\vec{k}}(\vec{x}') \\
& + \frac{\hbar^2}{m} \vec{k} \cdot (\vec{k} \hat{e}) \Psi_{\vec{k}}(\vec{x}') \\
& + \frac{\hbar}{m} \sum_n e^{i\vec{k} \cdot (\hat{\delta} + \hat{e}) \vec{x}'} \vec{k} \cdot \vec{p}(u_{n,\vec{0}}(\vec{x}')) \\
& + \frac{\hbar}{m} \sum_n e^{i\vec{k} \cdot (\hat{\delta} + \hat{e}) \vec{x}'} (\vec{k} \hat{e}) \cdot \vec{p}(u_{n,\vec{0}}(\vec{x}')) \\
& + \frac{1}{2m} \sum_n e^{i\vec{k} \cdot (\hat{\delta} + \hat{e}) \vec{x}'} \vec{p}^2(u_{n,\vec{0}}(\vec{x}')) \\
& + V(\vec{x}') \Psi_{\vec{k}}(\vec{x}') \\
& + \sum_{i,j} V_{ij}(\vec{x}') e_{ij}
\end{aligned}$$

where V_{ij} is part 56 above. Similarly, the spin-orbit portion of the Hamiltonian (equation 46) becomes, to first order in strain:

$$\begin{aligned}
H_{SO} = & \frac{\hbar^2}{4m^2 c^2} \left[\vec{\sigma} \times \vec{\nabla}_{\vec{x}'} V(\vec{x}') \right] \cdot \vec{k} \\
& + \frac{\hbar^2}{4m^2 c^2} \left[\vec{\sigma} \times \vec{\nabla}_{\vec{x}'} V(\vec{x}') \right] \cdot \vec{k} \hat{e} \\
& + \frac{\hbar^2}{4m^2 c^2} \sum_{i,j} \left[\vec{\sigma} \times \vec{\nabla}_{\vec{x}'} V_{ij}(\vec{x}') e_{ij} \right] \cdot \vec{k} \\
& + \frac{\hbar^2}{4m^2 c^2} \sum_{i,j} \left[\vec{\sigma} \times \vec{\nabla}_{\vec{x}'} V_{ij}(\vec{x}') e_{ij} \right] \cdot \left[\sum_n e^{i\vec{k} \cdot (\hat{\delta} + \hat{e}) \vec{x}'} \vec{p}(u_{n,\vec{0}}(\vec{x}')) \right] \\
& + \frac{\hbar^2}{4m^2 c^2} \left[\vec{\sigma} \times \vec{\nabla}_{\vec{x}'} V(\vec{x}') \right] \cdot \left[\sum_n e^{i\vec{k} \cdot (\hat{\delta} + \hat{e}) \vec{x}'} \vec{p}(u_{n,\vec{0}}(\vec{x}')) \right]
\end{aligned}$$

where $\vec{p} \Psi_{\vec{k}}(\vec{x}')$ has been expanded out. Although Bahder continues on using Löwdin

perturbation theory to make his 8-band model, this suffices, as the core of the model has been re-derived. When extending the model to 14 bands, one may use the direct approach outlined in section I.4.2.

The spatial strain interactions are, in his basis, [1] with the following definitions:

	$\Gamma_{6,-\frac{1}{2}}$	$\Gamma_{6,+\frac{1}{2}}$	$\Gamma_{8,-\frac{3}{2}}$	$\Gamma_{8,-\frac{1}{2}}$	$\Gamma_{8,+\frac{1}{2}}$
$\Gamma_{6,-\frac{1}{2}}$	$a'e$	0	$(t^* - v^*)$	0	$-\sqrt{3}(t + v)$
$\Gamma_{6,+\frac{1}{2}}$	0	$a'e$	$\sqrt{2}(w + u)$	$-\sqrt{3}(t^* - v^*)$	0
$\Gamma_{8,-\frac{3}{2}}$	$(t - v)$	$\sqrt{2}(w^* + u^*)$	$-p + q$	$-s^*$	r
$\Gamma_{8,-\frac{1}{2}}$	0	$-\sqrt{3}(t - v)$	$-s$	$-p - q$	0
$\Gamma_{8,+\frac{1}{2}}$	$-\sqrt{3}(t^* - v^*)$	0	r^*	0	$-p - q$
$\Gamma_{8,+\frac{3}{2}}$	$\sqrt{2}(w^* + u^*)$	$(t^* + v^*)$	0	r^*	s
$\Gamma_{7,-\frac{1}{2}}$	$(w^* + u^*)$	$-\sqrt{2}(t^* + v^*)$	$\sqrt{\frac{3}{2}}s^*$	$-\sqrt{2}r^*$	$\frac{1}{\sqrt{2}}s$
$\Gamma_{7,+\frac{1}{2}}$	$\sqrt{2}(t - v)$	$(w - u)$	$-\sqrt{2}q$	$\frac{1}{\sqrt{2}}s^*$	$\sqrt{2}r$

Table 1: Spatial portion of the Hamiltonian strain term from Bahder[1]; first 6 columns

	$\Gamma_{8,+\frac{3}{2}}$	$\Gamma_{7,-\frac{1}{2}}$	$\Gamma_{7,+\frac{1}{2}}$
$\Gamma_{6,-\frac{1}{2}}$	$\sqrt{2}(2 + u)$	$(w + u)$	$\sqrt{2}(t^* - v^*)$
$\Gamma_{6,+\frac{1}{2}}$	$(t + v)$	$-\sqrt{2}(t + v)$	$(w^* - u^*)$
$\Gamma_{8,-\frac{3}{2}}$	0	$\sqrt{\frac{3}{2}}s$	$-\sqrt{2}q$
$\Gamma_{8,-\frac{1}{2}}$	r	$-\sqrt{2}r$	$\frac{1}{\sqrt{2}}s$
$\Gamma_{8,+\frac{1}{2}}$	s^*	$\frac{1}{\sqrt{2}}s^*$	$\sqrt{2}r^*$
$\Gamma_{8,+\frac{3}{2}}$	$-p + q$	$\sqrt{2}q$	$\sqrt{\frac{3}{2}}s^*$
$\Gamma_{7,-\frac{1}{2}}$	$\sqrt{2}q$	$-ae$	0
$\Gamma_{7,+\frac{1}{2}}$	$\sqrt{\frac{3}{2}}s$	0	$-ae$

Table 2: Spatial portion of the Hamiltonian strain term from Bahder[1]; last two columns

$$e = e_{xx} + e_{yy} + e_{zz} \quad (57)$$

$$t = \frac{1}{\sqrt{6}}b'(e_{xz} + \imath e_{yz}) \quad (58)$$

$$v = \frac{1}{\sqrt{6}}P_0 \sum_j (e_{xj} - \imath e_{yj}) k_j \quad (59)$$

$$w = \frac{\imath}{\sqrt{3}}b'e_{xy} \quad (60)$$

$$u = \frac{1}{\sqrt{3}}P_0 \sum_j e_{zj}k_j \quad (61)$$

$$s = -d(e_{xz} - \imath e_{yz}) \quad (62)$$

$$r = \frac{\sqrt{3}}{2}b(e_{xx} - e_{yy}) - \imath de_{xy} \quad (63)$$

$$p = ae \quad (64)$$

$$q = b \left[e_{zz} - \frac{1}{2}(e_{xx} + e_{yy}) \right] \quad (65)$$

Additional terms derived from the strain-dependent spin-orbit terms are derived, but will not be reproduced here for brevity. They are not used in existing code at Iowa.

The a , b , and d terms are the Pikus-Bir deformation potentials, and are related to terms derived in Bahder's paper by the relations

$$a = \frac{1}{3}(l + 2m) \quad (66)$$

$$b = \frac{1}{3}(l - m) \quad (67)$$

$$d = \frac{1}{\sqrt{3}}n \quad (68)$$

where l , m , and n are terms derived in Bahder's work. The a' and b' terms in 1 are also derived quantities. Because of the equivalence between a , b , and d terms and the Bahder terms l , m , and n , either the derived terms or empirically-determined Bir-Pikus terms may be used.

In Craig Pryor's dotcode, the spin-orbit strain interaction is discarded, as are the w and t terms.[59] If improved accuracy is desired, these small terms may be included, at the expense of additional computation time.

1.5.3 Strain in Quantum Dots

Although materials possessing the same symmetries will have *similar* lattice vectors, they will generally not be exactly the same. Therefore, when a material is grown on top of another material (i.e. grown on a substrate), there is some strain associated with the fact that the inter-atomic distances at and near the interface between the two materials are not what the bulk distances would be.

When the strain gets too large, defects arise in the material. For instance, cracks may form in the newly-deposited material.[79] As one would expect, the further away from the interface, the less the deviation from bulk inter-atomic distances, and hence the less the strain.

One then expects strain to exist around the quantum dot interfaces, decreasing as one moves away from the interfaces. An example of a calculated quantum dot strain along the $[001]$ axis is provided in figure 4.

I.6 Magnetic Field Effects in Quantum Dots

I.6.1 Zeeman Effect

The Zeeman effect is linear with magnetic field, and takes the form

$$H_{Zeeman} = -\mu \vec{J} \cdot \vec{B} = -\frac{g\mu_B}{\hbar} \vec{J} \cdot \vec{B} \quad (69)$$

Where J is the spin operator for the object under consideration (here, either the individual band pseudospin if appropriate or else the global pseudospin), μ_B is the Bohr magneton, and $\vec{B}$ is the magnetic field. The Zeeman Effect is, then, the magnetic moment of the electron interacting with the surrounding magnetic field. The magnetic moment of the electron is dependent upon the material in which it is located as well as the band state being investigated, and is expressed as a constant (g) multiplying a fundamental magnetic moment, the Bohr magneton. This constant is known as the *g-factor*, and is approximately 2 for free electrons. This is notably not true for electrons within bulk semiconductors, nor is it true for an electron orbiting an atom. To find g in a material, one must calculate the magnetic field dependence of the electron states and find the global pseudospin orientation (to find the sign of the g-factor) and the spin splitting (to find the magnitude of the g-factor).

To understand the Zeeman effect, then, one must understand the underlying spins. As has been discussed above, each bulk band state is degenerate. Both the lowest conduction band state and the split-off state (blue curve in figure 1) act like

spin- $\frac{1}{2}$ objects and are therefore twofold degenerate. The heavy-hole and light-hole states, on the other hand, both have a pseudospin of $\frac{3}{2}$ and are fourfold degenerate at zone center. However, they are not degenerate with each other at arbitrary $\vec{k}$ as can be seen in figures 1 and 2, and thus have only two degenerate states each (magnetic numbers $\pm\frac{3}{2}$ and $\pm\frac{1}{2}$ for heavy hole and light hole states, respectively).

This picture is complicated by bulk state mixing within a bound state. As discussed in the derivation of quantum dot $k \cdot p$ theory, an energy eigenstate within a quantum dot is actually composed of one envelope function in real space for each band being considered, and thus the overall state is composed of a *mix* of Bloch states. Therefore, one cannot in general consider a single “spin” but rather a global pseudospin composed of the spin or pseudospin of each band state being considered. Instead of the simple spin operator for an electron:

$$\vec{J}_{\frac{1}{2}} = \hbar \frac{1}{2} \vec{\sigma} \quad (70)$$

where $\vec{\sigma}$ is the familiar Pauli matrix[80], or alternately the $\vec{J}_{\frac{3}{2}}$ for the spin- $\frac{3}{2}$ object:

$$\vec{J}_{\frac{3}{2}}^T = \hbar \frac{1}{2} \left(\begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix} \begin{pmatrix} 0 & \sqrt{3} & 0 & 0 \\ \sqrt{3} & 0 & 2 & 0 \\ 0 & 2 & 0 & \sqrt{3} \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 0 & -i\sqrt{3} & 0 & 0 \\ i\sqrt{3} & 0 & -i2 & 0 \\ 0 & i2 & 0 & -i\sqrt{3} \\ 0 & 0 & i\sqrt{3} & 0 \end{pmatrix} \right) \quad (71)$$

one in actuality has the global pseudospin operator:

$$\vec{J}_{global} = \vec{J}_g = \begin{pmatrix} \vec{J}_{c\frac{1}{2}} & 0 & 0 \\ 0 & \vec{J}_{HH,LH\frac{3}{2}} & 0 \\ 0 & 0 & \vec{J}_{SO\frac{1}{2}} \end{pmatrix} \quad (72)$$

in the $|j_{conduction} = \frac{1}{2} > |j_{HH,LH} = \frac{3}{2} > |j_{SO} = \frac{1}{2} >$ basis. (note that a somewhat unconventional notation for has been used for simplicity; this is a vector of matrices composed of $\vec{J}_{\frac{1}{2}}$ and $\vec{J}_{\frac{3}{2}}$ as described by equation 72). For electrons contained entirely within one band state, the global pseudospin is not an issue. One need merely use the spin of the appropriate band for Zeeman splitting calculations. For a mixed-band-state wavefunction, however, *only* the global pseudospin will give an accurate assessment of the Zeeman splitting, and the full pseudospin operator must be used.

To find the total Zeeman splitting, one must use the g-factors of the constituent parts of the angular momenta:

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \quad (73)$$

Thus, the Zeeman splitting of the global pseudospin ignoring any crystalline features

is expected to be related to

$$\begin{pmatrix} 2\vec{J}_{c\frac{1}{2}} & 0 & 0 \\ 0 & \frac{4}{3}\vec{J}_{HH,LH\frac{3}{2}} & 0 \\ 0 & 0 & \frac{2}{3}\vec{J}_{SO\frac{1}{2}} \end{pmatrix} \cdot \vec{B} \quad (74)$$

I.6.2 Calculating Orbital Magnetic Field Effects

In this subsection, the spin and orbit components will be treated separately. The magnetic field enters the Hamiltonian by the canonical momentum, following the method in Sakurai's quantum mechanics book[80]:

$$\vec{p} \rightarrow \vec{p} + q_e \vec{A} \quad (75)$$

where $\vec{A}$ is the vector potential of the magnetic field. Without spin-orbit interaction, the magnetic interaction with an electron in the semiconductor is straightforward.

One may select a gauge such that

$$\vec{A} = -\frac{1}{2} \begin{pmatrix} B_z y - B_y z \\ B_x z - B_z x \\ B_y x - B_x y \end{pmatrix} \quad (76)$$

where B_i is the i component of $\vec{B}$, the magnetic field.

The basic Hamiltonian is now:

$$\frac{1}{2m} \left(\frac{\hbar}{i} \vec{\nabla} + q_e \vec{A}(\vec{x}) \right)^2 + V(\vec{x}) \quad (77)$$

From equation 76, one sees that $\vec{\nabla} \cdot \vec{A} = 0$, and hence $\vec{p}$ and $\vec{A}$ commute. Thus, one has

$$H = -\frac{\hbar^2}{2m} \vec{\nabla}^2 + \frac{q_e}{m} \vec{p} \cdot \vec{A}(\vec{x}) + \frac{e^2}{2m} A^2(\vec{x}) + V(\vec{x}) \quad (78)$$

Now, examining the $\vec{p} \cdot \vec{A}(\vec{x})$ term more closely, one can see that

$$\vec{p} \cdot \vec{A} = -\frac{1}{2} \vec{p} \cdot \begin{pmatrix} B_z y - B_y z \\ B_x z - B_z x \\ B_y x - B_x y \end{pmatrix} = -\frac{1}{2} (p_x B_z y - p_x B_y z + p_y B_x z - p_y B_z x + p_z B_y x - p_z B_x y) \quad (79)$$

Gathering the B_x , B_y , and B_z terms, one gets

$$\vec{p} \cdot \vec{A} = -\frac{1}{2} [B_x (p_y z - p_z y) + B_y (p_z x - p_x z) + B_z (p_x y - p_y x)] \quad (80)$$

This may look familiar, because

$$\vec{L} = \vec{r} \times \vec{p} = \begin{pmatrix} yp_z - zp_y \\ zp_x - xp_z \\ xp_y - yp_x \end{pmatrix} \quad (81)$$

Thus, the second term in equation 78 is just

$$-\frac{e}{2cm}\vec{L} \cdot \vec{B} \quad (82)$$

and thus the Hamiltonian becomes, including electron spin:

$$H = -\frac{\hbar^2}{2m}\vec{\nabla}^2 - \frac{e}{2cm}\vec{L} \cdot \vec{B} + \frac{e^2}{2mc^2}A^2(\vec{x}) + V(\vec{x}) + 2\mu_B\vec{\sigma} \cdot \vec{B} \quad (83)$$

which now accounts for magnetic field effects on the orbital angular momentum as well as the electron spin. Using this, one may now proceed as in the first subsection to create a $k \cdot p$ Hamiltonian for the system either perturbatively or directly to include crystal effects.

1.6.3 Spin-Orbit Interaction Revisited

With the introduction of the spin-orbit interaction, one must choose the method one will use. As mentioned in the previous subsection, there are two bases to choose from when the spin-orbit interaction is non-negligible. The first choice of basis is keeping the orbit and spin portions of the basis separate, at the cost of having to directly introduce spin-orbit coupling in the Hamiltonian used. This method is simple as long as one remembers to add on the spin-orbit portion of the Hamiltonian. However, some of the more interesting effects of the spin-orbit interaction are not as apparent if one uses this method.

The second option is to choose a basis where the spin-orbit interaction is built in. The valence (and p-like conduction for the 7-band model) states then have two

separate parts: four degenerate (at zone-center) spin- $\frac{3}{2}$ states, and two degenerate (at zone-center) spin- $\frac{1}{2}$ states, at a different energy. One may follow symmetry arguments similar to those of Luttinger[81] to determine the magnetic field interaction at zone center, and then use $k \cdot p$ theory to get the behavior for some k based upon the zone-center energy splittings. In 1956, Luttinger used this method to analyze the top valence states and found that the symmetries of the Hamiltonian and its constituents require the introduction of two new parameters, κ and q :

$$H_{SO-Magnetic} = \frac{e}{c} \kappa \vec{J} \cdot \vec{B} + \frac{e}{c} q (J_x^3 B_x + J_y^3 B_y + J_z^3 B_z) \quad (84)$$

Luttinger's work was subsequently used by Roth et al. in 1959 to determine the g-factor for the conduction-band electron:[82]

$$g = 1 - \left(\frac{m}{m^*} - 1 \right) \frac{\Delta}{3E_g + 2\Delta} \quad (85)$$

The Roth formula, as it has come to be known, has been used in bulk semiconductors with considerable success. In recent years, however, it has failed to predict proper g-factors found in nanostructures.

Roughly 44 years after Luttinger's paper, Winkler et al. used Luttinger's work to explain the anisotropic g-factors of the 3 p-like valence states for a quantum wells. Specifically, the g-factors for magnetic fields applied along the in-plane directions are be much smaller than the g-factor of the out-of-plane direction.[44] This effect has been seen in quantum wells, dots, and wires, both in experiment [83][84][85][86]

as well as in theoretical work.[36][87]

From the one-dimensional confinement of the quantum well and the resulting two different g-factors, it is logical to ask if a quantum dot—where the electron is confined in all three directions—might have a different g-factor along its three principal axes.

I.6.4 Envelope and Bloch Angular Momenta

Starting with the definition of angular momentum and the wavefunction

$$\psi_i(\vec{r}) = \sum_a \eta_{ia}(\vec{r}) u_a(\vec{r}) \quad (86)$$

The Zeeman term acting on states i and j then becomes

$$\begin{aligned} H_{Z,ij} &= \mu_B g_{ij} \vec{B} \cdot \sum_{a,b} \left(\eta_{ib}^\dagger u_b \vec{L} \eta_{ja} u_a \right) \\ &= \mu_B g_{ij} \vec{B} \cdot \sum_{a,b} \int_{\mathbb{R}} \int_{\Omega} \left[\eta_{ib}^\dagger u_b^\dagger \left(\vec{L} \eta_{ja} \right) u_a \right] + \eta_{ib}^\dagger u_b^\dagger \eta_{ja} \left(\vec{L} u_a \right) \\ &= \mu_B g_{ij} \vec{B} \cdot \sum_{a,b} \int_{\mathbb{R}} \int_{\Omega} \left[\left(\eta_{ib}^\dagger \vec{L} \eta_{ja} \right) u_b^\dagger u_a + \eta_{ib}^\dagger \eta_{ja} \left(u_b^\dagger \vec{L} u_a \right) \right] \\ &= \mu_B g_{ij} \vec{B} \cdot \sum_{a,b} \left[\int_{\mathbb{R}} \eta_{ib}^\dagger \vec{L} \eta_{ja} \int_{\Omega} u_b^\dagger u_a + \int_{\mathbb{R}} \eta_{ib}^\dagger \eta_{ja} \int_{\Omega} u_b^\dagger \vec{L} u_a \right] \\ &= \mu_B g_{ij} \vec{B} \cdot \sum_{a,b} \left[\int_{\mathbb{R}} \eta_{ib}^\dagger \vec{L} \eta_{ja} \delta_{ab} + \int_{\mathbb{R}} \eta_{ib}^\dagger \eta_{ja} \int_{\Omega} u_b^\dagger \vec{L} u_a \right] \end{aligned}$$

Assuming $\vec{B} = B \hat{z}$, this reduces to

$$H_{Z,ij} = \mu_B g_{ij} B \sum_{a,b} \left[\int_{\mathbb{R}} \eta_{ib}^\dagger L_z \eta_{ja} \delta_{ab} + \int_{\mathbb{R}} \eta_{ib}^\dagger \eta_{ja} \int_{\Omega} u_b^\dagger L_z u_a \right] \quad (87)$$

Or, since u_a is an eigenvalue of L_z ,

$$\begin{aligned} H_{Z,ij} &= \mu_B g_{ij} B \sum_{a,b} \left[\int_{\mathbb{R}} \eta_{ib}^\dagger L_z \eta_{ja} \delta_{ab} + \int_{\mathbb{R}} \eta_{ib}^\dagger \eta_{ja} L_{za} \delta_{ab} \right] \\ &= \mu_B g_{ij} B \sum_a \left[\int_{\mathbb{R}} \eta_{ia}^\dagger L_z \eta_{ja} + \int_{\mathbb{R}} \eta_{ia}^\dagger \eta_{ja} L_{za} \right] \end{aligned}$$

Assuming no or negligible band mixing in the η_{ia} , this gives a result of

$$\begin{aligned} H_{Z,ij} &= \mu_B g_{ij} B \left[\int_{\mathbb{R}} \eta_i^\dagger L_z \eta_j + \int_{\mathbb{R}} \eta_i^\dagger \eta_j L_{za} \right] \\ H_{Z,ij} &= \mu_B g_{ij} B \left[\int_{\mathbb{R}} \eta_i^\dagger L_z \eta_j + \delta_{ij} L_{za} \right] \end{aligned}$$

In quantum dots, as the size of the dot is reduced, the eigenstates get further and further apart in energy. This leads to the first term going to zero leaving only the latter term,

$$H_{Z,ij} = \mu_B g_{ij} \delta_{ij} B L_{za} \quad (88)$$

where the a reflects the fact that the solution ψ_i consists entirely of one band. For conduction band states, this reduces to $g_{ij} = 2$; for heavy and light hole, $g_{ij} = \frac{4}{3}$, and for $g_{ij} = \frac{2}{3}$. This effect is referred to as “angular momentum quenching”. [88]

I.7 g-Tensors in Quantum Dots and Spin Precession

As has been previously shown, the g-factor of an electron in a bulk semiconductor can vary depending upon its band structure. In a nanostructure, confinement plays a central role alongside bulk band structure. The confinement causes band mixing within the electron wave function, which causes g-factors to change from their bulk

values. The g-factor must therefore be considered a tensor, as different directions may be more or less confined or restricted than others. For example, a quantum well has confinement along the growth direction, but confined electrons are free to move within the in-plane direction. Thus, one would readily expect the g-factor out of plane to differ from the g-factor in-plane, which is what has been found both for s-like conduction as well as for p-like valence states.[44][86][84]

In a system with cubic or higher symmetry, the g-tensor is simply a scalar multiplied by the unit matrix. In systems with lower symmetry, however, the g-tensor is a diagonal tensor if the magnetic field is applied along the 3 high-symmetry axes. For example, in a quantum dot with major axes along $[001]$, $[110]$, and $[1\bar{1}0]$, we expect a g-tensor of

$$\mathbf{g} = \begin{pmatrix} g_{[001]} & 0 & 0 \\ 0 & g_{[110]} & 0 \\ 0 & 0 & g_{[1\bar{1}0]} \end{pmatrix} \quad (89)$$

1.7.1 The g-Tensor and the Spin Precession Axis

Using the g-tensor instead of a scalar g-factor, the Zeeman splitting becomes

$$H_Z = -\frac{\mu_B}{\hbar} \vec{S} \cdot \tilde{\mathbf{g}} \cdot \vec{B} = -\frac{\mu_B}{\hbar} \vec{S} \cdot \begin{pmatrix} g_{[001]} & 0 & 0 \\ 0 & g_{[110]} & 0 \\ 0 & 0 & g_{[1\bar{1}0]} \end{pmatrix} \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} \quad (90)$$

The product of $\tilde{\mathbf{g}}$ and $\vec{B}$ is defined to be $\vec{\Omega}$, and regain a simple-yet insightful-

expression for the Zeeman splitting:

$$\vec{\Omega} = \begin{pmatrix} g_{[001]} & 0 & 0 \\ 0 & g_{[110]} & 0 \\ 0 & 0 & g_{[1\bar{1}0]} \end{pmatrix} \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} \quad (91)$$

$$H_Z = -\frac{\mu_B}{\hbar} \vec{S} \cdot \vec{\Omega} \quad (92)$$

From equation 91, the spin precession axis can be thought of as depending upon the g-tensor and magnetic field—both magnitude *and direction*. To make the equation more explicit,

$$\vec{\Omega} = \begin{pmatrix} g_{[001]}B_{[001]} \\ g_{[110]}B_{[110]} \\ g_{[1\bar{1}0]}B_{[1\bar{1}0]} \end{pmatrix} \quad (93)$$

Clearly, the spin precession axis *may point in a direction different from the magnetic field* as the g-tensor changes. In a crystal with cubic or greater symmetry, the g-tensor is a scalar multiplied by the unit matrix, and the spin precession axis must point parallel or anti-parallel to the magnetic field. When the cubic symmetry is broken, such as in a hemispherical quantum dot such as those examined above, however, the g-tensor may cause the spin precession axis to point in a direction different from that of the magnetic field. If the g-tensor is sensitive to electric field, the electric field may change the position and magnitude of the spin precession axis, permitting both resonant and non-resonant coherent spin manipulation.

I.7.2 Electron Spin Resonance

In electron spin resonance (ESR), a static spin precession axis is used to cause the spin states to split. This axis will hereafter be referred to as Ω_0 , and the “z” axis for the purposes of spin manipulation will be set to be parallel to it. In addition, a small spin precession axis component is given that causes the spin precession axis to rotate at a frequency ω . Diagrammatically, the total spin precession axis looks like the illustration in figure 6.

The overall spin precession vector is, then

$$\vec{\Omega}(t) = \Omega_0 \hat{z} + \Omega_t (\cos(\omega t) \hat{x} + \sin(\omega t) \hat{y}) \quad (94)$$

Because it is difficult to create a rotating spin precession vector in a microwave cavity in conventional ESR, an alternate approach is used (Figure 7). An oscillating transverse spin precession axis component is applied instead of a rotating one. The two are roughly equivalent:

$$\vec{\Omega}_t(t) = \Omega_t \cos(\omega t) \hat{x} \quad (95)$$

$$= \frac{\Omega_t}{2} (\cos(\omega t) \hat{x} + \sin(\omega t) \hat{y} + \cos(\omega t) \hat{x} - \sin(\omega t) \hat{y}) \quad (96)$$

$$= \frac{\Omega_t}{2} (\cos(\omega t) \hat{x} + \sin(\omega t) \hat{y} + \cos(-\omega t) \hat{x} + \sin(-\omega t) \hat{y}) \quad (97)$$

$$= \frac{\Omega_t}{2} (\cos(\omega t) \hat{x} + \sin(\omega t) \hat{y}) + \frac{\Omega_t}{2} (\cos(-\omega t) \hat{x} + \sin(-\omega t) \hat{y}) \quad (98)$$

That is, applying an oscillating transverse spin precession axis component is equiv-

alent to applying a half-amplitude rotating and counter-rotating transverse spin precession axis components. To the first-order approximation, this counter-rotating field produces a small offset to the resonant frequency,[89] which will be neglected for the rest of this dissertation.

Conventional ESR In conventional ESR, the g-tensor is a scalar. Therefore, a static magnetic field is applied to a sample while a high-frequency (GHz) oscillating field is applied perpendicular to it. The static field causes electron spins to polarize with the field, while the small transverse field oscillating at the Larmor frequency causes the electrons to oscillate between spin-up and spin-down states. An oscillating magnetic field pulse which spins the electrons from up to down (for example) is called a “ π pulse”, and a magnetic field pulse which spins the electrons from up to down and back up again is called a “ 2π pulse”.

Spin precession Starting with equation 94, it may be seen that there are two spin precession axis components: the static magnetic field and the transverse oscillating magnetic field. Clearly, the spin will precess about the static magnetic field at a frequency ω_0 , the Larmor frequency:

$$e^{-i\frac{H_z}{\hbar^2}t} |\psi\rangle = e^{i\frac{g\mu_B B_0 t}{\hbar^2} \frac{\hbar}{2} \sigma_z} (\alpha |\uparrow\rangle + \beta |\downarrow\rangle) \quad (99)$$

$$= \alpha |\uparrow\rangle e^{i\frac{g\mu_B B_0}{2\hbar}t} + \beta |\downarrow\rangle e^{-i\frac{g\mu_B B_0}{2\hbar}t} \quad (100)$$

Applying the original state on the left to find the probability amplitude of returning to the original state,

$$(\langle \uparrow | \alpha^\dagger + \langle \downarrow | \beta^\dagger) \left(\alpha |\uparrow\rangle e^{i \frac{g\mu_B B_0}{2\hbar} t} + \beta |\downarrow\rangle e^{-i \frac{g\mu_B B_0}{2\hbar} t} \right) \quad (101)$$

$$= |\alpha|^2 e^{i \frac{g\mu_B B_0}{2\hbar} t} + |\beta|^2 e^{-i \frac{g\mu_B B_0}{2\hbar} t} \quad (102)$$

remembering the normalization condition $|\alpha|^2 + |\beta|^2 = 1$. This reduces to a cosine if $|\alpha| = |\beta|$ (e.g. x or y state), and a simple phase factor if fully in the $|\uparrow\rangle$ or $|\downarrow\rangle$ state. Also, it is clear that the time required to return to the original state is $\frac{2\hbar}{g\mu_B B_0}$, which is double the Larmor time of The expectation value of the spin operator is then

$$\tilde{\mathbf{S}} = \frac{\hbar}{2} \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right)^T \quad (103)$$

in the arbitrary spinor state

$$|\chi\rangle = \begin{pmatrix} |\alpha| \\ |\beta| e^{i\delta} \end{pmatrix} e^{i\phi} \quad (104)$$

where $|\beta|$ is retained for clarity, although it's fully determined by $|\alpha|$ due to normalization, as outlined above. $e^{i\phi}$ is an overall phase of the state. The expectation

value of the spin vector is

$$\langle \chi | \tilde{\mathbf{S}} | \chi \rangle = \begin{pmatrix} |\alpha| & |\beta|e^{-i\delta} \end{pmatrix} \frac{\hbar}{2} \left(\begin{pmatrix} |\beta|e^{i\delta} \\ |\alpha| \end{pmatrix} \begin{pmatrix} -i|\beta|e^{i\delta} \\ i|\alpha| \end{pmatrix} \begin{pmatrix} |\alpha| \\ -|\beta|e^{i\delta} \end{pmatrix} \right)^T \quad (105)$$

$$= \frac{\hbar}{2} \begin{pmatrix} |\alpha||\beta| \cos(\delta) & |\alpha||\beta| \sin(\delta) & |\alpha|^2 - |\beta|^2 \end{pmatrix}^T \quad (106)$$

Now, using the state $|\psi\rangle$ from equation 99 which only introduces phase factors of $e^{\pm i\frac{1}{2}\omega_0 t}$ to the components of the left and right states, the spin vector at time t is

$$\langle \psi(t) | \tilde{\mathbf{S}} | \psi(t) \rangle = \frac{\hbar}{2} \begin{pmatrix} |\alpha||\beta| \cos(\omega_0 t + \delta) & |\alpha||\beta| \sin(\omega_0 t + \delta) & |\alpha|^2 - |\beta|^2 \end{pmatrix}^T \quad (107)$$

Thus the spin vector returns to its original position in the Larmor time $\frac{\hbar}{g\mu_B B_0}$, although the state itself requires double the time to return to the original state.

1.7.3 Rabi Oscillation

A transverse, rotating magnetic field is now added to the static magnetic field (figure 6). The Hamiltonian of this system is[89]

$$H = H_0 + H_t = -\frac{g\mu_B B_0}{\hbar} \mathbf{S}_z - \frac{g\mu_B B_t}{\hbar} \cos(\omega t) \mathbf{S}_x + -\frac{g\mu_B B_t}{\hbar} \sin(\omega t) \mathbf{S}_y \quad (108)$$

$$= H_0 + H_t = -\frac{g\mu_B B_0}{\hbar} \mathbf{S}_z - \frac{g\mu_B B_t}{2\hbar} (e^{i\omega t} + e^{-i\omega t}) \mathbf{S}_x - \frac{g\mu_B B_t}{2\hbar} (-i) (e^{i\omega t} - e^{-i\omega t}) \mathbf{S}_y \quad (109)$$

$$= H_0 + H_t = H_0 - \frac{g\mu_B B_0}{2\hbar} (e^{i\omega t} (\mathbf{S}_x - i\mathbf{S}_y) + e^{-i\omega t} (\mathbf{S}_x + i\mathbf{S}_y)) \quad (110)$$

$$= H_0 + H_t = H_0 - \frac{g\mu_B B_0}{2\hbar} (e^{i\omega t} \mathbf{S}_- + e^{-i\omega t} \mathbf{S}_+) \quad (111)$$

where B_t is the transverse rotating magnetic field strength. Transforming our Hamiltonian into a frame rotating at angular frequency ω , the transformation operator is

$$\mathbf{U} = e^{-i\frac{1}{\hbar}\mathbf{S}_z\omega t} \quad (112)$$

$$= \mathbb{1} \cos\left(\frac{\omega}{2}t\right) - i\sigma_z \sin\left(\frac{\omega}{2}t\right) \quad (113)$$

(negative due to rotating the coordinate system, equivalent to rotating the objects positively) which, when applied to the Hamiltonian on the left and right, gives

$$H' = \mathbf{U}^\dagger H \mathbf{U} \quad (114)$$

$$= H_0 - \frac{g\mu_B B_t}{\hbar} (e^{i\omega t} \mathbf{U}^\dagger \mathbf{S}_- \mathbf{U} + e^{-i\omega t} \mathbf{U}^\dagger \mathbf{S}_+ \mathbf{U}) \quad (115)$$

The first part, involving S_- , becomes

$$\mathbf{U}^\dagger \mathbf{S}_- \mathbf{U} = \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) + i\sigma_z \sin\left(\frac{\omega}{2}t\right) \right) (\mathbf{S}_x - i\mathbf{S}_y) \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) - i\sigma_z \sin\left(\frac{\omega}{2}t\right) \right)$$

Examining $\mathbf{S}_x$,

$$\begin{aligned}
\mathbf{U}^\dagger \mathbf{S}_x \mathbf{U} &= \frac{\hbar}{2} \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) + \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \sigma_x \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) - \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) + \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \left(\mathbb{1} \sigma_x \cos\left(\frac{\omega}{2}t\right) - \imath \sigma_x \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) + \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \left(\sigma_x \cos\left(\frac{\omega}{2}t\right) - \sigma_y \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) \sigma_x \cos\left(\frac{\omega}{2}t\right) - \mathbb{1} \cos\left(\frac{\omega}{2}t\right) \sigma_y \sin\left(\frac{\omega}{2}t\right) \right) \\
&\quad + \frac{\hbar}{2} \left(\imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \sigma_x \cos\left(\frac{\omega}{2}t\right) - \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \sigma_y \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\cos\left(\frac{\omega}{2}t\right) \sigma_x \cos\left(\frac{\omega}{2}t\right) - \cos\left(\frac{\omega}{2}t\right) \sigma_y \sin\left(\frac{\omega}{2}t\right) - \sigma_y \sin\left(\frac{\omega}{2}t\right) \cos\left(\frac{\omega}{2}t\right) \right) \\
&\quad - \frac{\hbar}{2} \left(\sigma_x \sin\left(\frac{\omega}{2}t\right) \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\sigma_x \left(\cos^2\left(\frac{\omega}{2}t\right) - \sin^2\left(\frac{\omega}{2}t\right) \right) - \sigma_y 2 \cos\left(\frac{\omega}{2}t\right) \sin\left(\frac{\omega}{2}t\right) \right)
\end{aligned}$$

And for $\mathbf{S}_y$

$$\begin{aligned}
\mathbf{U}^\dagger \mathbf{S}_y \mathbf{U} &= \frac{\hbar}{2} \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) + \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \sigma_y \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) - \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) + \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \left(\sigma_y \cos\left(\frac{\omega}{2}t\right) - \imath \sigma_y \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) + \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \right) \left(\sigma_y \cos\left(\frac{\omega}{2}t\right) + \sigma_x \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\mathbb{1} \cos\left(\frac{\omega}{2}t\right) \sigma_y \cos\left(\frac{\omega}{2}t\right) + \mathbb{1} \cos\left(\frac{\omega}{2}t\right) \sigma_x \sin\left(\frac{\omega}{2}t\right) \right) \\
&\quad + \frac{\hbar}{2} \left(\imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \sigma_y \cos\left(\frac{\omega}{2}t\right) + \imath \sigma_z \sin\left(\frac{\omega}{2}t\right) \sigma_x \sin\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\sigma_y \cos^2\left(\frac{\omega}{2}t\right) + \sigma_x \cos\left(\frac{\omega}{2}t\right) \sin\left(\frac{\omega}{2}t\right) + \sigma_x \sin\left(\frac{\omega}{2}t\right) \cos\left(\frac{\omega}{2}t\right) \right) \\
&\quad - \frac{\hbar}{2} \left(\sigma_y \sin^2\left(\frac{\omega}{2}t\right) \right) \\
&= \frac{\hbar}{2} \left(\sigma_y \left(\cos^2\left(\frac{\omega}{2}t\right) - \sin^2\left(\frac{\omega}{2}t\right) \right) + \sigma_x 2 \cos\left(\frac{\omega}{2}t\right) \sin\left(\frac{\omega}{2}t\right) \right)
\end{aligned}$$

Putting these back together into $\mathbf{S}_-$ results in

$$\begin{aligned}
\mathbf{U}^\dagger \mathbf{S}_- \mathbf{U} &= \frac{1}{2} (\mathbf{U}^\dagger \mathbf{S}_- \mathbf{U} - \imath \mathbf{U}^\dagger \mathbf{S}_- \mathbf{U}) \\
&= \frac{1}{2} (\mathbf{U}^\dagger \mathbf{S}_- \mathbf{U} - \imath \mathbf{U}^\dagger \mathbf{S}_- \mathbf{U}) \\
&= \frac{\hbar}{4} \left[\sigma_x \left(\cos^2\left(\frac{\omega}{2}t\right) - \sin^2\left(\frac{\omega}{2}t\right) \right) - \sigma_y 2 \cos\left(\frac{\omega}{2}t\right) \sin\left(\frac{\omega}{2}t\right) \right] \\
&\quad - \frac{\hbar}{4} \left[\imath \sigma_y \left(\cos^2\left(\frac{\omega}{2}t\right) - \sin^2\left(\frac{\omega}{2}t\right) \right) - \imath \sigma_x 2 \cos\left(\frac{\omega}{2}t\right) \sin\left(\frac{\omega}{2}t\right) \right] \\
&= \frac{\hbar}{2} \left[\sigma_- \left(\cos^2\left(\frac{\omega}{2}t\right) - \sin^2\left(\frac{\omega}{2}t\right) \right) - \imath 2 \cos\left(\frac{\omega}{2}t\right) \sin\left(\frac{\omega}{2}t\right) \sigma_- \right] \\
&= \frac{\hbar \sigma_-}{2} \left[\cos^2\left(\frac{\omega}{2}t\right) - \sin^2\left(\frac{\omega}{2}t\right) - \imath 2 \cos\left(\frac{\omega}{2}t\right) \sin\left(\frac{\omega}{2}t\right) \right] \\
&= \frac{\hbar \sigma_-}{2} \left[\cos\left(\frac{\omega}{2}t\right) - \imath \sin^2\left(\frac{\omega}{2}t\right) \right]^2 \\
&= \frac{\hbar \sigma_-}{2} \left[e^{-\imath \frac{\omega}{2}t} \right]^2
\end{aligned}$$

$$\mathbf{U}^\dagger \mathbf{S}_y \mathbf{U} = \mathbf{S}_- e^{-\imath \omega t} \quad (116)$$

Similarly, for S_+ ,

$$\mathbf{U}^\dagger \mathbf{S}_+ \mathbf{U} = e^{\imath \omega t} \mathbf{S}_+ \quad (117)$$

So, using the transformed Hamiltonian in the *transformed* time-dependent

Schrödinger equation,

$$U^\dagger H U |\chi\rangle = -i\hbar U^\dagger \frac{\partial}{\partial t} U |\chi\rangle \quad (118)$$

$$\left[H_0 - \frac{g\mu_B B_t}{\hbar} (e^{i\omega t} \mathbf{U}^\dagger \mathbf{S}_- \mathbf{U} + e^{-i\omega t} \mathbf{U}^\dagger \mathbf{S}_+ \mathbf{U}) \right] |\chi\rangle = i\hbar \left(U^\dagger \frac{\partial}{\partial t} U \right) |\chi\rangle + i\hbar \frac{\partial}{\partial t} |\chi\rangle \quad (119)$$

$$\left[H_0 - \frac{g\mu_B B_t}{\hbar} (e^{i\omega t} \mathbf{S}_- e^{-i\omega t} + e^{-i\omega t} \mathbf{S}_+ e^{i\omega t}) \right] |\chi\rangle = i\hbar (-i)\omega |\chi\rangle + i\hbar \frac{\partial}{\partial t} |\chi\rangle \quad (120)$$

$$\left[-\frac{g\mu_B B_0}{\hbar} \mathbf{S}_z - \frac{g\mu_B B_t}{\hbar} (\mathbf{S}_- + \mathbf{S}_+) - \mathbf{S}_z \omega \right] |\chi\rangle = i\hbar \frac{\partial}{\partial t} |\chi\rangle \quad (121)$$

$$[\omega_0 \mathbf{S}_z + \omega_t (\mathbf{S}_- + \mathbf{S}_+) - \omega \mathbf{S}_z] |\chi\rangle = i\hbar \frac{\partial}{\partial t} |\chi\rangle \quad (122)$$

$$(\omega_0 - \omega) \mathbf{S}_z + \omega_t (\mathbf{S}_- + \mathbf{S}_+) |\chi\rangle = i\hbar \frac{\partial}{\partial t} |\chi\rangle \quad (123)$$

where $\omega_t = g\frac{\mu_B}{\hbar} B_t$ and $\omega_0 = g\frac{\mu_B}{\hbar} B_0$. Because $\mathbf{S}_- + \mathbf{S}_+ = \frac{1}{2} [\mathbf{S}_x - i\mathbf{S}_y + \mathbf{S}_x + i\mathbf{S}_y] = \mathbf{S}_x$

$$(\omega_0 - \omega) \mathbf{S}_z + \omega_t \mathbf{S}_x |\chi\rangle = i\hbar \frac{\partial}{\partial t} |\chi\rangle \quad (124)$$

$$(\omega_0 - \omega) \sigma_z + \omega_t \sigma_x |\chi\rangle = i2 \frac{\partial}{\partial t} |\chi\rangle \quad (125)$$

which integrates to (again, this remains within the rotating frame)

$$|\chi(t)\rangle = e^{-it\left(\frac{\omega_0 - \omega}{2} \sigma_z + \frac{\omega_t}{2} \sigma_x\right)} |\chi_0\rangle \quad (126)$$

which is more straightforwardly written as

$$|\chi(t)\rangle = e^{-it\vec{\Omega}\cdot\vec{\sigma}} |\chi_0\rangle \quad (127)$$

where

$$|\vec{\Omega}| = \sqrt{\left(\frac{\omega_0 - \omega}{2}\right)^2 + \left(\frac{\omega_t}{2}\right)^2} \quad (128)$$

$$\Omega_x = \frac{1}{\sqrt{1 + \left(\frac{\omega_0 - \omega}{\omega_t}\right)^2}} \quad (129)$$

$$\Omega_z = \frac{1}{\sqrt{\left(\frac{\omega_t}{\omega_0 - \omega}\right)^2 + 1}} \quad (130)$$

It is to be noted that the components of $\hat{\Omega}$ are just the sine and cosine of the angle between the spin precession axis components in the rotating frame. The frequency $|\vec{\Omega}|$ is the Rabi frequency, the (inverse of the) time required to resonantly flip a spin from up to down and back again. To make it clear, the probability of finding an initially spin-up state in a spin-up state at time t is checked. Remaining in the

rotating frame,

$$P(t) = |\langle \uparrow | \chi(t) \rangle| \quad (131)$$

$$= \left| \langle \uparrow | e^{-i t \vec{\Omega} \cdot \vec{\sigma}} | \uparrow \rangle \right| \quad (132)$$

$$= |\langle \uparrow | \cos(\Omega t) - i \Omega_x \sigma_x \sin(\Omega t) - i \Omega_z \sigma_z \sin(\Omega t) | \uparrow \rangle| \quad (133)$$

$$= |\cos(\Omega t) - i \Omega_z \sin(\Omega t)| \quad (134)$$

$$= \cos^2(\Omega t) + \frac{1}{1 + \left(\frac{\omega_t}{\omega_0 - \omega} \right)^2} \sin^2(\Omega t) \quad (135)$$

$$= \frac{\cos^2(\Omega t) + \sin^2(\Omega t) \left(\frac{\omega_t}{\omega_0 - \omega} \right)^2 + \cos^2(\Omega t)}{1 + \left(\frac{\omega_t}{\omega_0 - \omega} \right)^2} \quad (136)$$

$$= \frac{1 + \left(\frac{\omega_t}{\omega_0 - \omega} \right)^2 \cos^2(\Omega t)}{1 + \left(\frac{\omega_t}{\omega_0 - \omega} \right)^2} \quad (137)$$

$$= \frac{(\omega_0 - \omega)^2 + \omega_t^2 \cos^2(\Omega t)}{(\omega_0 - \omega)^2 + \omega_t^2} \quad (138)$$

$$(139)$$

At time $t = 0$, the probability of being in the spin-up state is 1, as expected. It is a minimum at $t = \frac{n\pi}{2\phi}$. Similarly, for the probability of finding a spin-up state starting

from a spin-down state:

$$P(t) = |\langle \uparrow | \cos(\Omega t) - i\Omega_x \sigma_x \sin(\Omega t) - i\Omega_z \sigma_z \sin(\Omega t) | \downarrow \rangle| \quad (140)$$

$$= |\langle \uparrow | \cos(\Omega t) - i\Omega_x \sigma_x \sin(\Omega t) - i\Omega_z \sigma_z \sin(\Omega t) | \downarrow \rangle| \quad (141)$$

$$= |\langle \uparrow | -i\Omega_x \sigma_x \sin(\Omega t) | \downarrow \rangle| \quad (142)$$

$$= |-i\Omega_x \sin(\Omega t)| \quad (143)$$

$$= \frac{1}{\left(\frac{\omega_0 - \omega}{\omega_t}\right)^2 + 1} \sin^2(\Omega t) \quad (144)$$

which, as noted above and in Abragam's book[89], is just the square of the sine of the angle formed between the transverse magnetic fields. This angle, then, sets the maximum and minimum probabilities of finding the system in an up or down state at time t .

Near resonance ($\omega = \omega_0$), the probability of flipping a spin is maximum, as the coefficients of the sine and cosine go to 1. Although finding a spin-up or spin-down state at a time t is

I.8 In(x)Ga(1-x)As Ternary Alloys

Although only InAs/GaAs quantum dots have been discussed thus far, other quantum dot systems can be considered. One such system is an $\text{In}_x\text{Ga}_{1-x}\text{As}/\text{GaAs}$ quantum dot, which presents some similarities and some differences when compared to InAs/GaAs quantum dots.

Atomically, an InAs lattice can have Ga substituting for the In atoms (or, equivalently, In for Ga in GaAs). This is a ternary alloy of InAs and GaAs, and its

properties are related to the two materials which make it up: InAs and GaAs.

Because $(1 - x)$ percent of the barrier material is mixed in to the dot material (assuming a homogeneous mixing of the materials), the well depth is decreased by that amount. That is, to first order,

$$E_{g,In(x)Ga(1-x)As} = E_{g,InAs} + (E_{g,GaAs} - E_{g,InAs})x \quad (145)$$

or alternately

$$E_{g,In(x)Ga(1-x)As} = (1 - x)E_{g,InAs} + xE_{g,GaAs} \quad (146)$$

The latter form is a first-order Bezier curve between $E_{g,InAs}$ and $E_{g,GaAs}$. In many ternary alloys, a second-order dependence upon concentration is found, the “bowing parameter.” The bowing parameter b is then used as

$$E_{g,In(x)Ga(1-x)As} = (1 - x)E_{g,InAs} + xE_{g,GaAs} + bx(1 - x) \quad (147)$$

This is similar but not identical to a second-order Bezier curve:

$$E_{g,In(x)Ga(1-x)As} = (1 - x)^2E_{g,InAs} + x^2E_{g,GaAs} + bx(1 - x) \quad (148)$$

Although the start and end points are identical, the intermediate path is not, and a non-trivial coordinate transformation is required to switch from the common bowed equation to a second-order Bezier curve. The bowing parameter of $In(x)Ga(1-x)As$ is 0.477.[90].

In addition to the bandgap, the lattice constant is similarly dependent upon the InAs and GaAs lattice constants, with a bowing parameter of 2.61[90]. Because of this, the strain, which is dependent upon the lattice constant:

$$\sigma_{xx} = \frac{a_A - a_B}{a_A} \quad (149)$$

is less than in a pure InAs/GaAs dot.

Thus one can expect two things to be immediately different in an InGaAs dot than in an InAs dot. First, the well should be shallower and the electrons and holes therefore less confined and with fewer bound states. Second, the strain in and around the dot should be less, leading again to even lower carrier confinement within the dot.

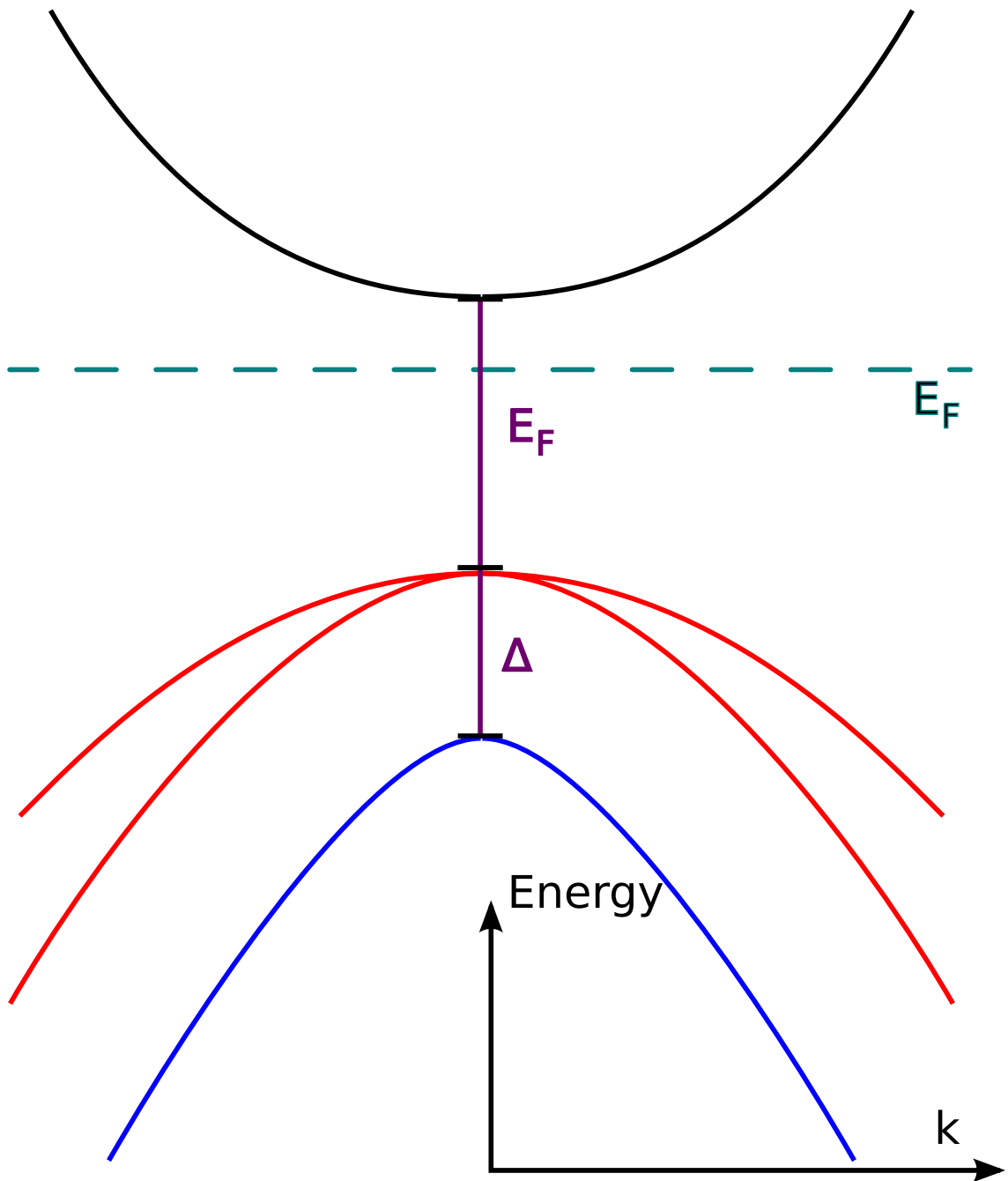


Figure 1: Sketch of a four-band model (Eight-band with pseudospin degeneracy). Vertical axis is energy; horizontal axis is crystal momentum $\vec{k}$ magnitude. The black line indicates the conduction band, the red lines indicate the heavy hole and light hole bands in the valence band, and blue indicates the split-off band. The purple lines and letters denote energy differences, namely the energy gap E_g and the spin-orbit splitting Δ .

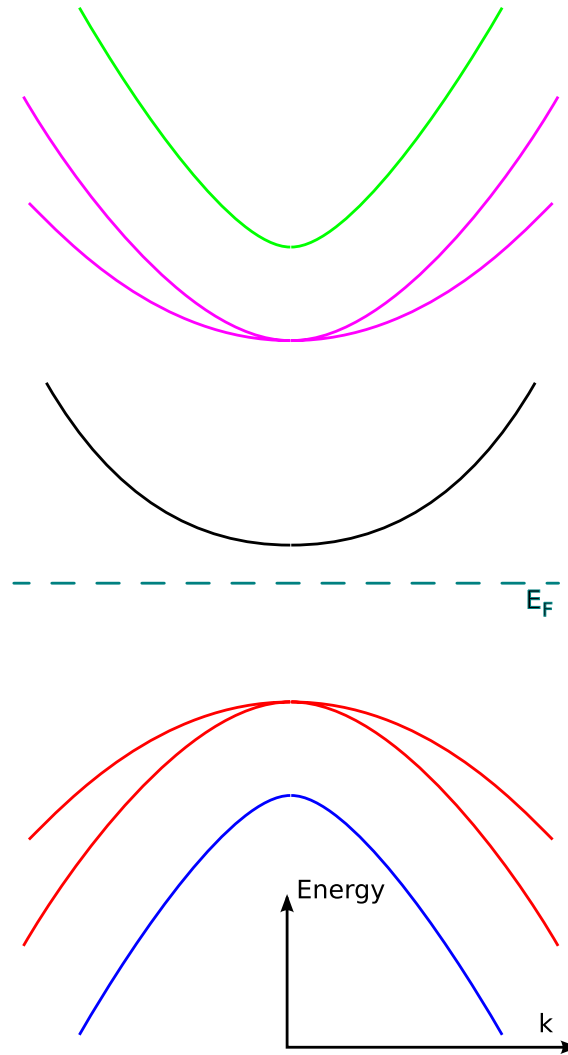


Figure 2: Seven-band model (14-band with pseudospin degeneracy). Vertical axis is energy; horizontal is crystal momentum $\vec{k}$ magnitude. Note that the new bands (green and pink) are added into the four-band model of figure 1.

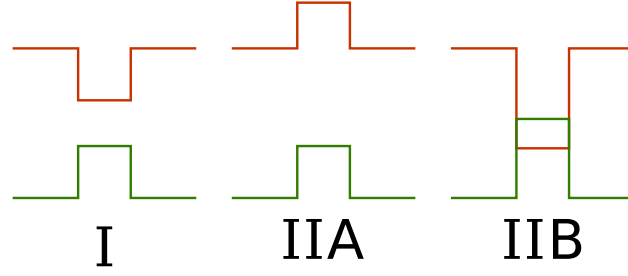


Figure 3: Possible band edge configurations for a quantum dot. Lines indicate the lowest conduction state (red) or highest valence state (green). The horizontal axis is position along a line running through the quantum dot

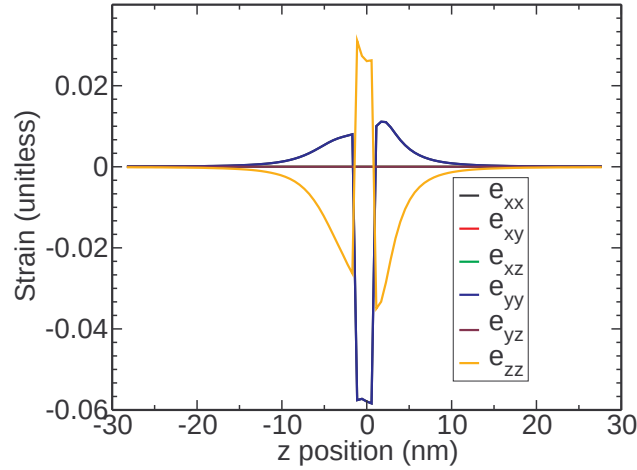


Figure 4: Strain tensor components in and around an InAs/GaAs quantum dot (circular footprint with nominal radius of 5.1nm and nominal height of 2.3nm) running along a line parallel to the [001] axis through the center of the dot. Horizontal axis is position along this line (nm). Vertical axis is strain tensor component. Black is e_{xx} , red is e_{xy} , green is e_{xz} , blue is e_{yy} , maroon is e_{yz} , and orange is e_{zz} . The dot itself is located in the center (centered on 0).

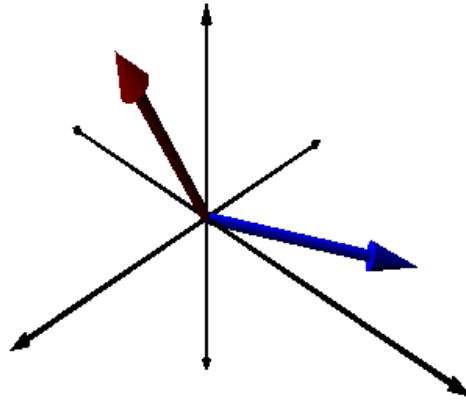


Figure 5: Illustration the spin precession axis (red) pointing in a different direction from the applied magnetic field (blue). Black arrows are the basis axes.

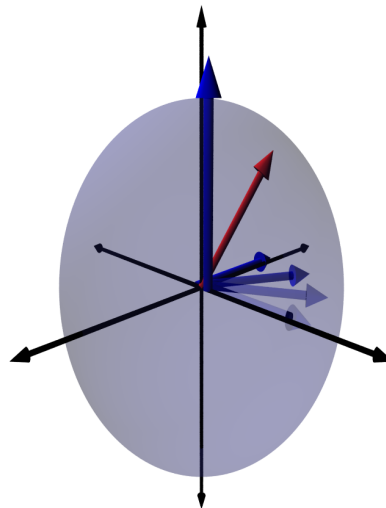


Figure 6: The stationary and rotating spin precession axis components (blue). The spin vector is red.

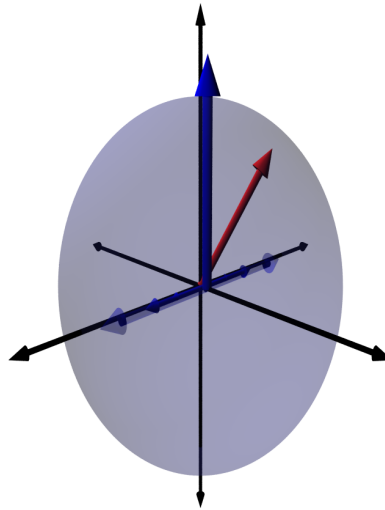


Figure 7: The stationary and oscillating spin precession axis components (blue). The spin vector is red.

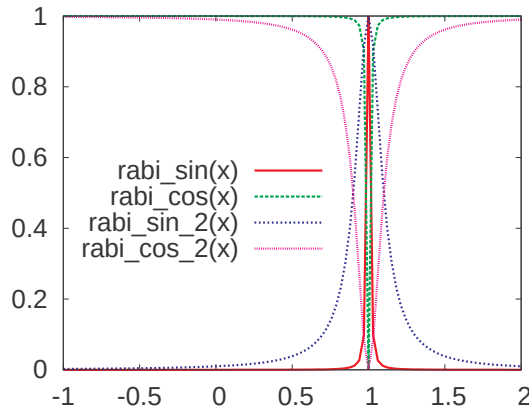


Figure 8: Maximum probability of finding the system in the up state when starting from the up (green) and down (red) state, as a function of frequency. ω_0 is set to 1, and ω_t is set to 0.01. Blue and purple curves are probability of finding the system in the up state when starting from up (purple) and down (blue) state, as a function of frequency, with $\omega_0 = 1$ and $\omega_t = 0.1$.

CHAPTER II

DETAILS OF THE CALCULATIONS PERFORMED

In order to address spin behavior in quantum dots, a suite of programs created by Craig Pryor was used. The suite is discussed in appendix A. This section addresses how the calculations were performed and the challenges that were presented.

II.1 Varying a parameter

When calculating the dependence on a certain parameter, e.g. quantum dot footprint radius, a number of dot structures had to be created and their properties calculated. Because of the similarity of the jobs, the work was done by perl script. Because a number of calculations were performed on a single dot structure, the structure was generated and then put into a library of structures. The structures were generated by a structure generator script, which created a number of structures which varied a parameter. An example of a structure generator script is in appendix F. The structure generator scripts are passed a job ID number, which they determine the structure to be generated.

Once the structures were generated, a second script was run, which started the calculations. As with the structure generator, the script was passed a job ID number, from which it determined the applied electric field, magnetic field strength and direction, and dot whose properties were to be calculated. An example calculation script is also in appendix F.

During the course of this dissertation work, both electric field and dot dimensions have been found to affect the energies, particularly the conduction electron states.

The hole (valence electron states) energies have been found to be much less sensitive to structure and electric field when determining this energy bracket. This is a qualitative assessment, however, and completely based upon the dots investigated during the course of this dissertation work. This information was important when creating a new data calculation script, as the energies must be bracketed for the entire range being calculated. A calculation was performed at the maximum and minimum of the parameter being varied to determine where the energy levels would lie. Most of the time a linear approximation was sufficient. Occasionally, due to how the energy levels varied as the parameter was changed, a middle point needed to be used to keep the lower or higher energy eigenstates out. If a few lower or higher eigenstates ended up in the calculation, those points could be recalculated.

After the dot properties were calculated, a script was called to gather and postprocess the data into one file. This script had the option of calling a sub-script (xdriver, since it is specified using the -x argument) to do additional postprocessing. Because the data was output in a structured fashion, this “gather-data-and-vars” script was able to determine what the parameters were, and output all of the information into a big table that could then be manually further postprocessed. The script used to gather and postprocess the data is in appendix F

II.2 Calculating the electron g-tensor

The electron g-tensor magnitude is calculated by taking the difference in spin-split energies:

$$|g_i| = \frac{\Delta E}{\mu_B B_i} \quad (150)$$

for a magnetic field applied along i , one of the principal axes of the quantum dot.

The sign of the g-factor in that direction, however, was more difficult to determine.

Because the Hamiltonian is

$$H_z |\uparrow\rangle = -\frac{\mu_B}{\hbar} g_z B_z S_z |\uparrow\rangle$$

$$H_z |\downarrow\rangle = -\frac{\mu_B}{\hbar} g_z B_z S_z |\downarrow\rangle$$

$$H_z |\uparrow\rangle = -\frac{\mu_B}{2} g_z B_z |\uparrow\rangle$$

$$H_z |\downarrow\rangle = \frac{\mu_B}{2} g_z B_z S_z |\downarrow\rangle$$

implying that the $|\downarrow\rangle$ state is higher energy when the g-factor is positive. However,

because of the negative charge of the electron, the actual Hamiltonian is ($\mu_B \rightarrow$

$-\mu_B$)

$$H_z |\uparrow\rangle = \frac{\mu_B}{2} g_z B_z |\uparrow\rangle \tag{151}$$

$$H_z |\downarrow\rangle = -\frac{\mu_B}{2} g_z B_z S_z |\downarrow\rangle \tag{152}$$

i.e. the $|\downarrow\rangle$ state is lower energy for a positive g-factor. Thus, after determining $|g_i|$,

the sign was found by checking the angular momentum state. If the $m = -\frac{1}{2}$ state

was the lower energy, the g-factor sign was positive. Otherwise, the $m = -\frac{1}{2}$ state

was the higher energy, and the g-factor sign was positive.

II.3 Calculating the hole g-tensor

Because of strain and breaking growth-direction symmetry for elliptical dots, the heavy hole ($|\frac{3}{2} \pm \frac{3}{2}\rangle$; upper red curve in figure 1) and light hole ($|\frac{3}{2} \pm \frac{1}{2}\rangle$; lower red curve in figure 1) states are split at zone-center (as opposed to the bulk case, where they are degenerate at zone center). This split enables one to view the two m -state sets as quasi spin- $\frac{1}{2}$ systems. Although the spin-splitting of the $m = \pm\frac{1}{2}$ (light hole) states is straightforward, the spin splitting of the $m = \pm\frac{3}{2}$ states may be defined in one of two ways. One may either consider the states to be simply $m = \pm\frac{1}{2}$ states with a very large g-factor or $m = \pm\frac{3}{2}$ states with a smaller g-factor. In this work, the latter definition was used. Therefore, the same methods as used for determining the electron g-factor were used for the hole g-factor, albeit with one modification, namely that the spin operator for the heavy hole states was defined as

$$\vec{S}_h = 3\vec{S}_e \quad (153)$$

due to the heavy-hole states being $m_l = \pm\frac{3}{2}$ states.

As with finding electron g-factors, the g-factor magnitude along a specific direction was found by applying the magnetic field pointing along that direction. After calculating the energy splitting, the hole g-factor magnitude along that direction was found by using the equation

$$|g| = \frac{\Delta E}{3\mu_B B} \quad (154)$$

where ΔE is the energy splitting and B is the magnetic field magnitude. The factor of 3 comes from the definition of the hole spin angular momentum operator, $\vec{S}_h$.

The two systems are equivalent, provided the factor of 3 is maintained to keep the energy splitting correct. The g-tensors in the rest of the work may be multiplied by 3 to get the g-factors for calculations using the other definition.

II.4 Ensuring speed and accuracy

A central trade-off in doing numerical calculations is between speed and accuracy. At one extreme, one may have a calculation which is very fast but whose approximations make the calculation useless. At the other extreme, one may have an incredibly accurate calculation, but the calculation takes such a long time that it is also useless—particularly when one includes the time required to perform all of the minor modifications which are needed to ensure an accurate result. Therefore, one must carefully reduce the scope of the problem but check to ensure accuracy.

Some of the options are limited by what has been implemented in the code. For instance, current dotcode can calculate with at most 8 bands. Additional bands improve the accuracy of the calculation—for instance, by going to 14 bands, one may include the Dresselhaus[91] spin-orbit interaction in non-centrosymmetric zincblende structures. However, 8 bands give a sufficiently accurate result for the extra computation time required compared to using fewer bands.

One of the central issues when doing a numerical calculation on a grid is the size of the “box” in which the calculation is performed and the density of grid points. An ideal calculation would involve a continuum of points in all space. That is, an

infinite-sized “box” with an infinitesimal inter-point distance. Because computers have only a finite amount of storage, neither an infinitesimal inter-point distance nor an infinite box size is possible. Therefore, one must find a distance between points that is not so large as to render the calculation inaccurate (e.g. by having the dot fit within a grid site). As the grid density increases, the discretization error (see above) decreases. It is generally sufficient to choose a good grid spacing and use it consistently. Unless otherwise stated, grid spacings in this dissertation were chosen to be the lattice constant of the barrier material, GaAs (0.56532nm). This choice was made because the dots are sufficiently large compared to the grid size to not be excessively rough, and because the strain in the barrier would be guaranteed to be roughly 0 far away from the dot.

The calculation “box” must also be chosen carefully. There are two ways of looking at the condition for being “sufficiently large”. In the physical picture, it is important to ensure that the wavefunction or strain being calculated is approximately zero on the boundary of the calculation box. More generally, one must check to make sure that the value being calculated is relatively constant when the box size is increased.

The latter definition of “sufficiently large” is the definition used throughout this dissertation. The calculations in question were performed for a variety of box sizes, and the parameter being calculated—the g-factor along a certain direction—was checked. Because it would be excessively time-dependent to check the box size for every calculation to be performed, the box size was checked for the minimum and

maximum values of the parameter being changed—e.g. dot height. It was then reasonable to assume—because of the equivalent second definition of “sufficiently large”—that the wavefunction would leak out of the dot and onto the barrier to a monotonically increasing extent as the dot dimension being checked was decreased. Such a check is plotted in figure 9.

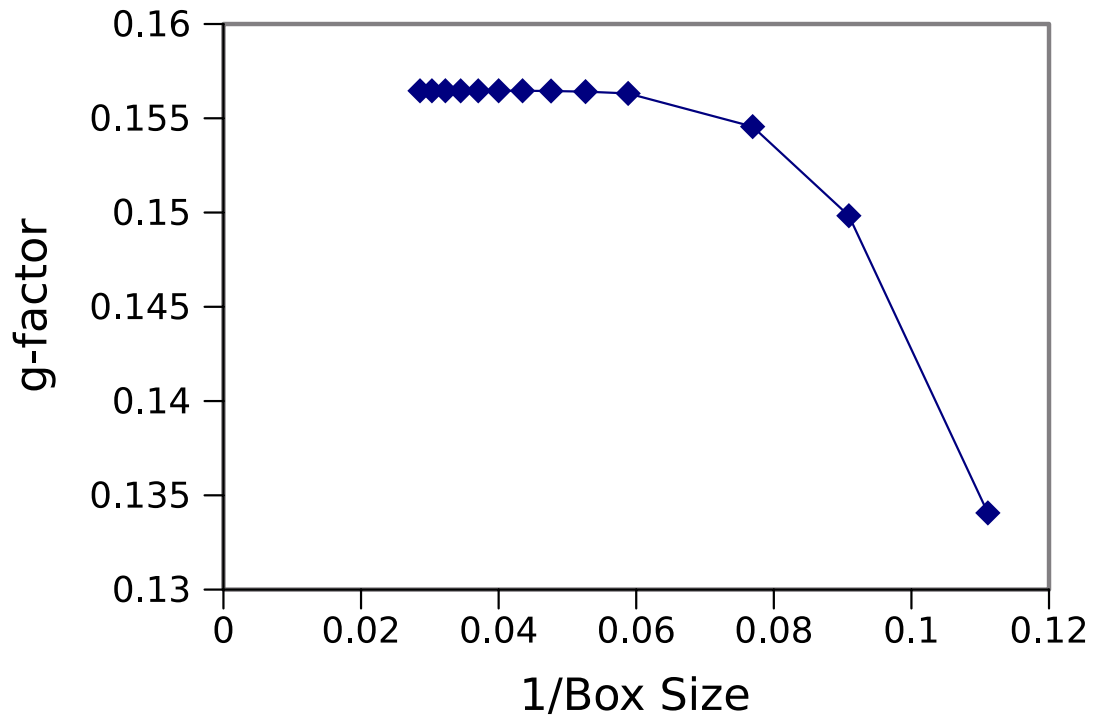


Figure 9: Plot of g-factor versus calculation box size (here, the length in grid sites of one edge of the cube).

II.5 Discretization errors and volume

When a real object is mapped onto a discrete grid, the discretization of the structure introduces error into the calculation. As an extreme example, one might attempt to specify a very small sphere and end up with a similar-sized cube of one grid site. More commonly, a smooth real object will become rough when dis-

cretized, due to the smooth portions being neither totally in nor totally outside of the applicable gridsite.

Therefore, if the size of a grid is fixed, a larger structure will more closely match the continuous geometry specified than will a small structure. Additionally, the discretized structure's dimensions may change due to the discretization, as the bounding surface may not lie precisely on a gridsite. It is useful, then, to find how much different the final structure is from the one originally intended.

For some of the subsequent calculations, the geometry file was created manually rather than by using `makeSphericalCap.pl` (see appendix A), in order to have greater control over the resulting structure and therefore match the structure being experimentally investigated. In the investigation, the discretization error was checked, resulting in the following tables. Both the continuous (first three columns) quantum dot axes and the discretized (second three columns) are listed. The last three columns are the final volume (in grid sites) after discretization, the discrepancy in volume between using the discretized dot axis lengths and the final discretized volume, and the discrepancy in volume between the continuous dot axis lengths and the final discretized volume.

In the following tables, columns are continuous lens-shaped quantum dot dimensions, error from discretization $((d_{continuous} - d_{discretized})/d_{discretized})$, actual volume of the final discretized quantum dot (in number of grid sites), error using half-ellipsoid volume and continuous dot dimensions $((V_{continuous} - V_{final})/V_{final})$, error using half-ellipsoid volume and discretized dot dimensions $((d_{discretized} - V_{final})/V_{final})$.

$d_{[001]}$	$d_{[110]}$	$d_{[1\bar{1}0]}$	$Err_{[001]}$	$Err_{[110]}$	$Err_{[1\bar{1}0]}$	V_{final}	$Err_{cont.}$	$Err_{disc.}$
6	10.1	15.1	0.0351	-0.010	0.021	10919	-0.028	-0.024
5.4	8.66	13.85	-0.045	0.032	0.019	7656	-0.19	-0.23
7.3	8.66	13.85	-0.0067	0.032	0.019	10329	-0.17	-0.59
5.7	9.21	14.81	0.0083	0.0017	0.029	8384	0.075	0.34
6.6	11.01	16.51	0.061	0.020	0.0074	13307	0.045	-0.042
5.4	10.01	16.01	0.061	0.0016	0.0013	9595	0.046	-0.018
5.7	10.01	16.01	0.0083	0.0016	0.0013	9880	0.072	0.06
5.4	9.01	13.31	0.061	-0.020	0.0090	7167	0.047	-0.0019

Table 3: Discretization errors for manually created dots.

$d_{[001]}$	$d_{[110]}$	$d_{[1\bar{1}0]}$	$Err_{[001]}$	$Err_{[110]}$	$Err_{[1\bar{1}0]}$	V_{final}	$Err_{cont.}$	$Err_{disc.}$
5.6	11.44	7.151	0	-0.013	-0.0062	4938	0.086	0.11
5.6	12.16	7.598	0	0.019	-0.050			
5.6	12.87	8.045	0	-0.024	0.0062	6108	0.11	0.13
5.6	13.59	8.492	0	-0.029	-0.034	6738	0.12	0.20
5.6	14.30	8.938	0	-0.033	0.016	7380	0.14	0.16
5.6	15.02	9.385	0	-0.037	-0.022	8100	0.14	0.21
5.6	15.73	9.832	0	-0.040	-0.54	8822	0.15	0.26
5.6	16.45	10.28	0	-0.043	-0.011	9584	0.16	0.22
5.6	17.16	10.73	0	-0.0016	-0.042	10380	0.16	0.21
5.6	17.88	11.17	0	0.073	-0.0018	11216	0.17	0.18
5.6	18.59	11.62	0	-0.010	-0.031	12018	0.18	0.23
5.6	19.31	12.07	0	-0.014	0.0062	12940	0.18	0.19

Table 4: Discretization errors for dots created with makeSphericalCap.pl.

CHAPTER III

A SINGLE ELECTRON SPIN IN AN ISOLATED QUANTUM DOT

The orbital angular momentum of the electron can become quenched in quantum dots due to the 3-dimensional confinement present within quantum dots becomes sufficiently high. The 3-dimensional confinement causes the electron energy levels to be completely discretized. This discretization is atom-like, and the Landau levels that form in the presence of a magnetic field in a continuum must instead work with the discrete energy levels of the quantum dot. Because the energy levels in the quantum dot are widely-spaced, the energy levels cannot readily be combined into angular momentum states, quenching the angular momentum. This g-factor modification has recently been studied and the inadequacy of the Roth formula for predicting g-factors in nanostructures has been further confirmed, particularly in light of angular momentum quenching.[92]

III.1 InAs/GaAs Quantum Dots

The g-tensor in lens-shaped self-assembled InAs/GaAs quantum dots (i.e. InAs quantum dots grown embedded in GaAs barrier material) has been investigated as part of this dissertation. In particular, the g-tensor's dependence on magnetic field, quantum dot footprint diameter, quantum dot footprint ellipticity, and quantum dot height has been examined. To find the g-tensor, a 1T magnetic field was applied along the $[001]$, $[110]$, and $[1\bar{1}0]$ directions, as these are the growth, elongation, and short axes of common dots, respectively. An 8-band $k \cdot p$ Hamiltonian to perform the calculations on a 3-dimensional grid, constructed as has been described previously

in this dissertation.

As shall be seen shortly, applying an electric field has the effect of changing the confinement of the electron within the dot. Since the force on the electron is in the direction opposite the electric field, one expects the electron to localize predominantly at the bottom of the quantum dot for a positive electric field applied along $[001]$, and at the top of the quantum dot for a negative electric field applied along $[001]$. This has the effect of not only confining the electronic wavefunction more than would be the case without the electric field, but due to the curvature of the top of the dot, it also confines the wavefunction to a lesser or greater extent laterally, as one can see from an illustration of a quantum dot being investigated (figure 10).

To get a basis for understanding how the g-tensor is modified in InAs/GaAs quantum dots, the g-tensor dependence on electric field (ranging from -150kV/cm to 150kV/cm) was examined in a round-footprint dot with a height of 6.2nm and a dot footprint radius of 6.2nm . Figures 11 and 12 show that the g-factor is parabolic and slightly off-center along the $[001]$ direction, but almost linear in the in-plane ($[110]$ and $[1\bar{1}0]$) directions. Tantalizingly, the g-factor along the $[110]$ direction changes sign as the electric field increases from 49 to 50kV/cm . This sign change is where the angular momentum states switch orderings as a function of electric field, and will prove useful for novel spintronic devices, discussed in chapter VII.

The electron g-tensor's dependence on electric field was examined while varying the quantum dot height. The results are in figures 13 (dot with geometric mean

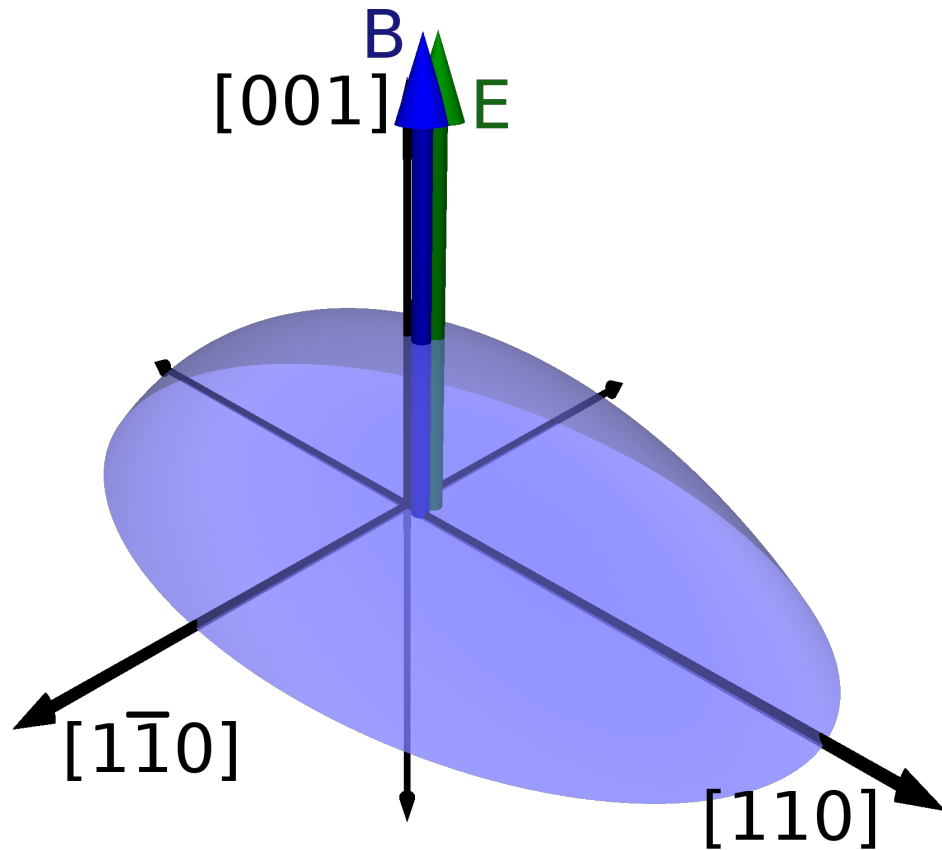


Figure 10: Illustration of a quantum dot, illustrating the curved top, flat bottom, and ellipticity (ratio of the $[110]$ to the $[1\bar{1}0]$ lengths). The electric field is applied along to the $[001]$ axis (although it may be negative). The magnetic field is applied along each of the principal axes, although it is applied along $[1\bar{1}0]$ in this figure.

footprint radius of 6.2nm) and 14 (geometric mean footprint radius of 5.6nm). Colors in the graphs indicate ellipticity ($e = d_{[110]}/d_{[1\bar{1}0]}$), with blue being round footprint dots, green being slightly elliptical ($e = 1.25$) and red being more elliptical ($e = 1.667$). In addition, the line style indicates electric field values; the dashed line is

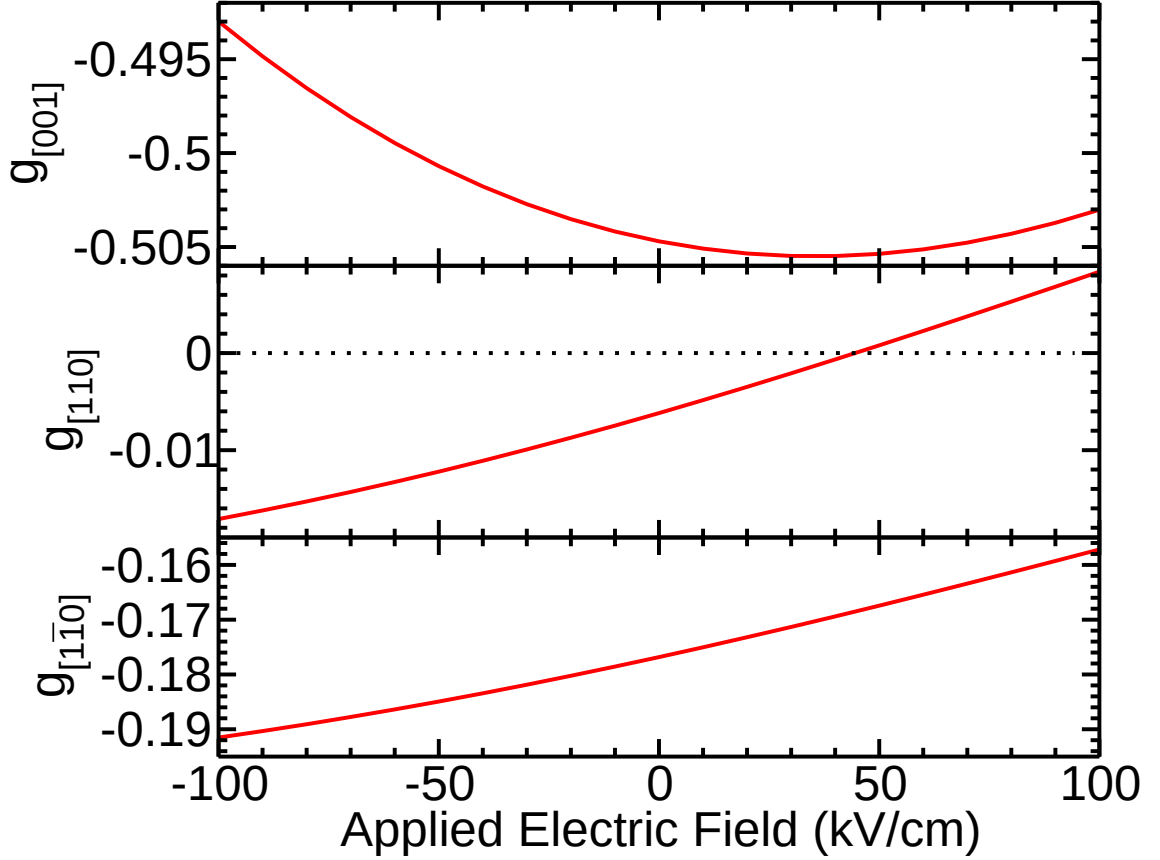


Figure 11: g-tensor (g-factor along $[001]$, $[110]$ $[1\bar{1}0]$ directions) as a function of electric field. The quantum dot has a height of 6.2nm, geometric mean radius of 5.6nm, and ellipticity $e = \frac{d_{[110]}}{[1\bar{1}0]} = \frac{5}{3}$. Note the sign change for $g_{[110]}$ as the electric field goes positive.

+100kV/cm, solid is no electric field, and dotted is -100kV/cm.

In figures 13 and 14, one can see the trends from figures 11 and 12 again, plotted against the energy gap E_g of the 0-electric field dot. E_g was taken as being the difference between the lowermost conduction state and the uppermost valence state energy (as opposed to choosing a different one of the spin-split doublets). Although this definition of E_g may not be used by all investigators, the differences are small (on the order of the spin splitting, and thus μeV). E_g is a function of confinement,

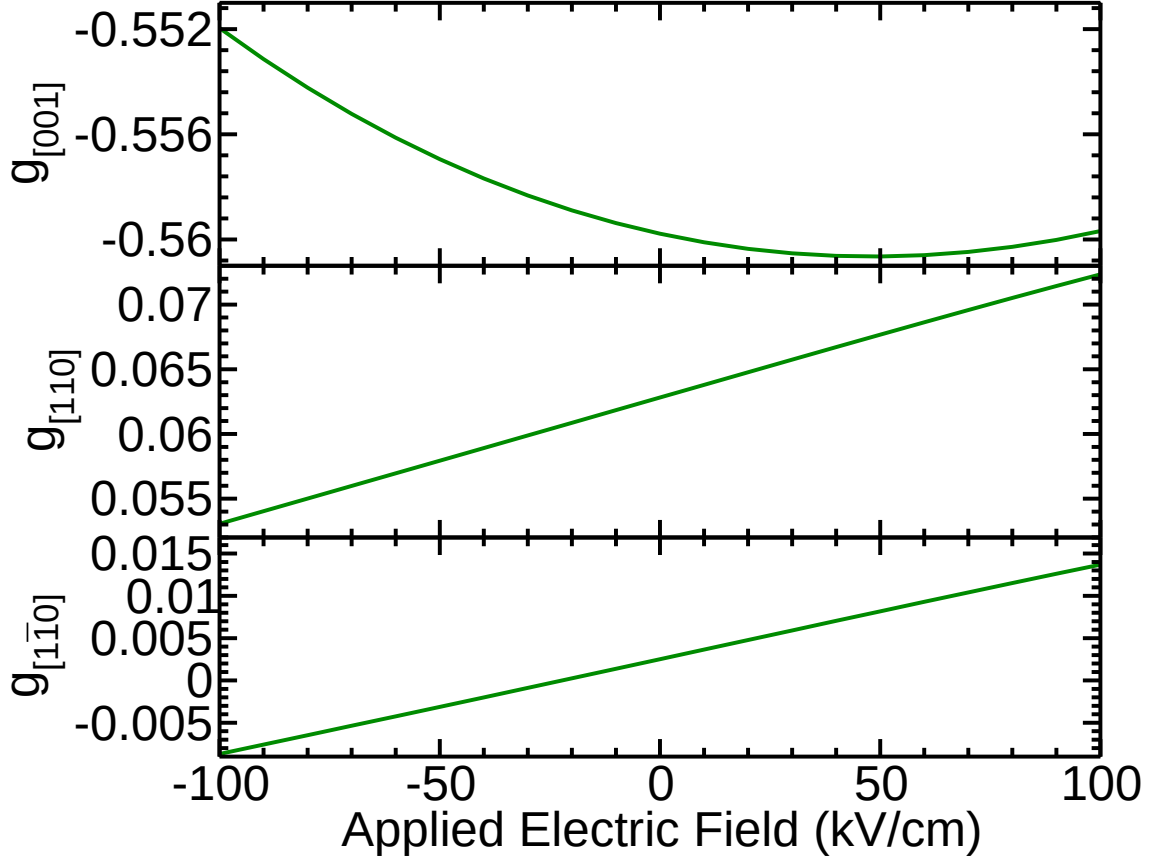


Figure 12: g-tensor (g-factor along $[001]$, $[110]$ $[1\bar{1}0]$ directions) as a function of electric field. The quantum dot has a height of 5.0nm, geometric mean radius of 6.2nm, and ellipticity $e = \frac{d_{[110]}}{[1\bar{1}0]} = \frac{5}{4}$. Note the sign change for $g_{[1\bar{1}0]}$ as the electric field goes negative.

as one might expect. As the electron becomes more and more confined (here, as the dot height decreases), E_g increases. Although E_g is dependent upon the electric field applied, the 0-field E_g was used for the E_g of all 3 ellipticities, for clarity.

As E_g increases (dot height decreases), the g-factors decrease. This is true of all g-factor components investigated, and continues beyond the points plotted here. Interestingly, increasing the ellipticity causes $g_{[001]}$ to decrease, but causes $g_{[1\bar{1}0]}$ to increase. Mixing the two, $g_{[110]}$ decreases with ellipticity at lower energy (taller dots)

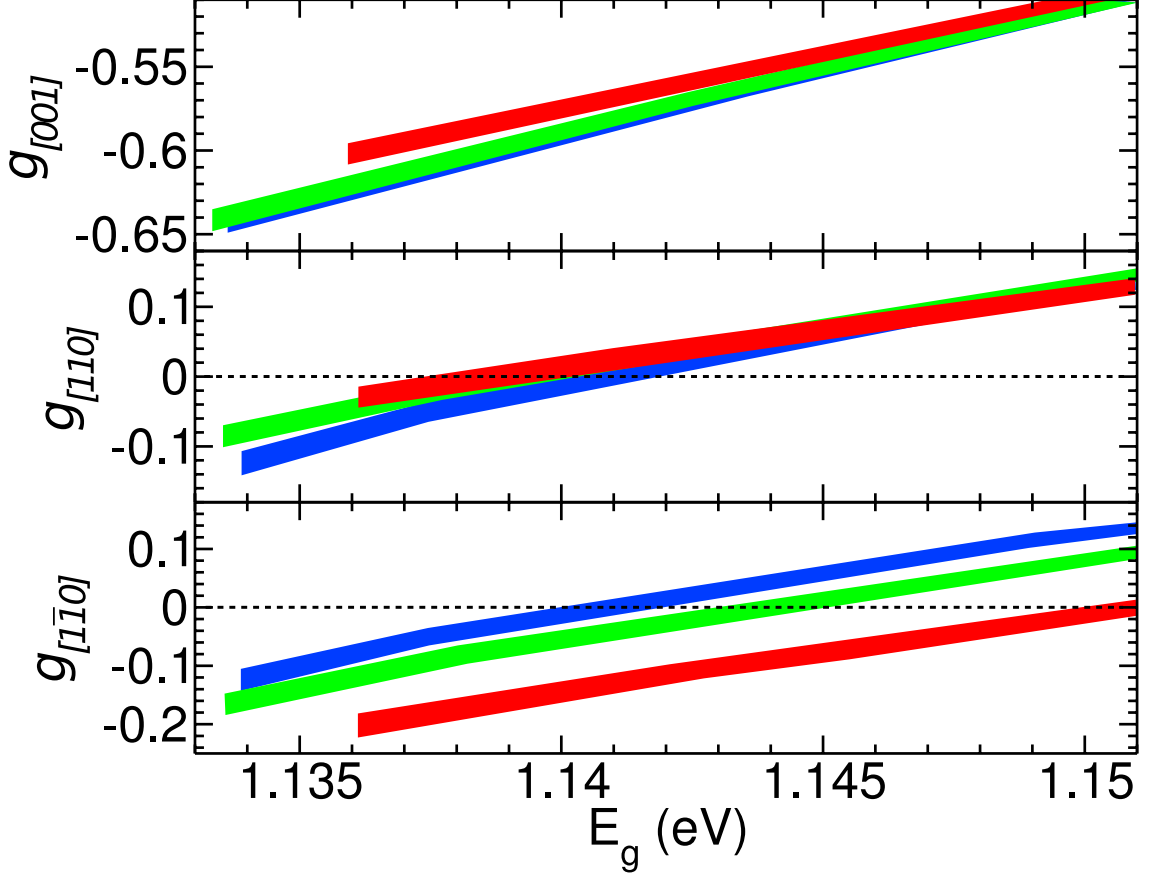


Figure 13: g-tensor (g-factor along $[001]$, $[110]$, $(1\bar{1}0)$ directions) as a function of electric field and quantum dot height. Color indicates ellipticity (blue $\rightarrow e = 0$, green $\rightarrow e = 1.25$, red $\rightarrow e = 1.667$). Application of electric field from -100kV/cm to $+100\text{kV/cm}$ changes the g-factor within the bands. Geometric mean dot footprint radius fixed at 6.2nm .

and then increases with ellipticity at higher energy (shorter dots). To understand these results, one must examine qualitatively the effect of size on the electronic wave function.

As quantum dot size decreases, the electron g-factor changes from approximately its bulk value in very large dots to the free electron g-factor of approximately 2. As the electric field increases, it causes the electron to be compressed into the apex or

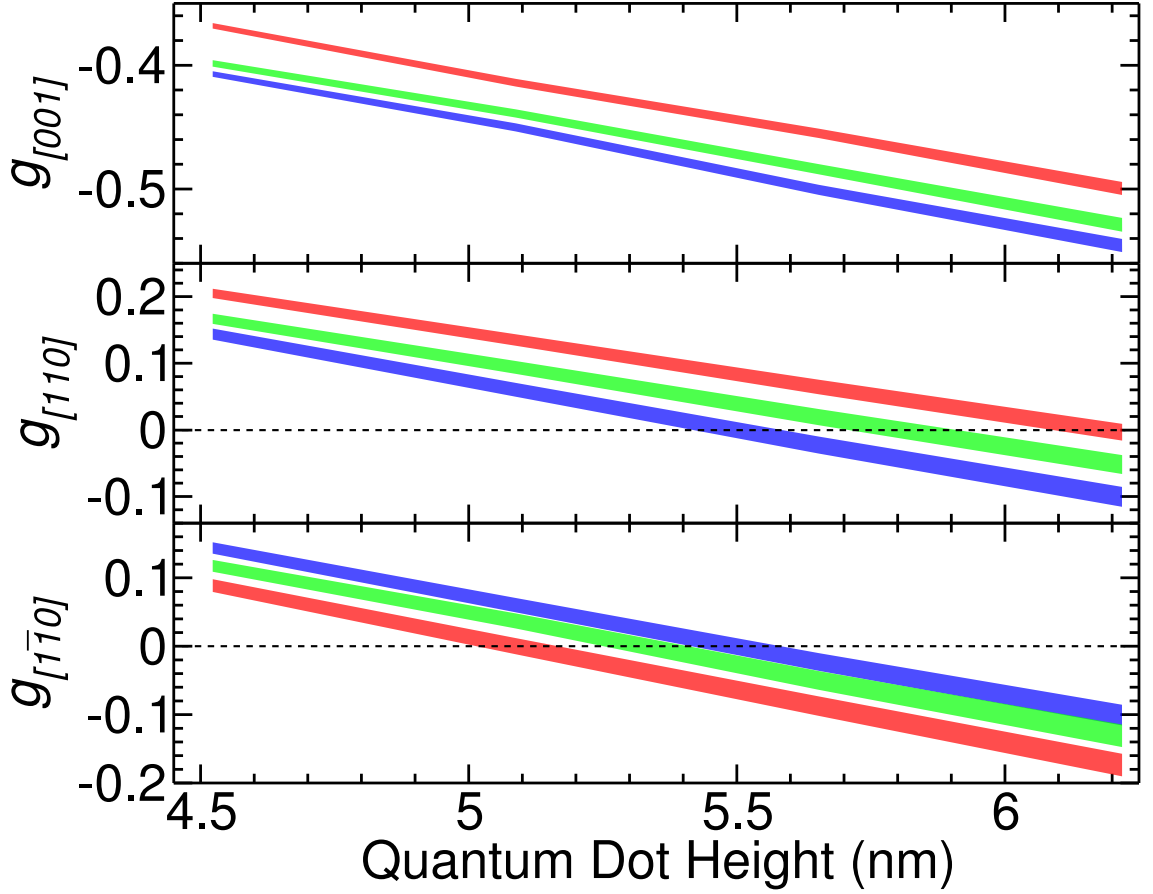


Figure 14: g-tensor (g-factor along $[001]$, $[110]$, $(1\bar{1}0)$ directions) as a function of electric field and quantum dot height. Color indicates ellipticity (blue $\rightarrow e = 0$, green $\rightarrow e = 1.25$, red $\rightarrow e = 1.667$). Application of electric field from -100kV/cm to $+100\text{kV/cm}$ changes the g-factor within the bands. Geometric mean dot footprint radius fixed at 5.6nm .

nadir of the dot, the electron confinement generally increases, shifting the g-factor.

It can be noted in figures 13 and 14 that both $g_{[110]}$ and $g_{[1\bar{1}0]}$ switch from positive to negative as the energy gap increases (height decreases). Indeed, in one of these dots—specifically, $g_{[1\bar{1}0]}$ in the dot with $e = 1.25$ and height of 5nm —changes from positive to negative as the electric field increases from -100kV/cm to 0 .

The g-tensor has also been investigated as a function of dot ellipticity, applied

electric field, and changing dot footprint radius. The results are summarized in figures 15 and 16.

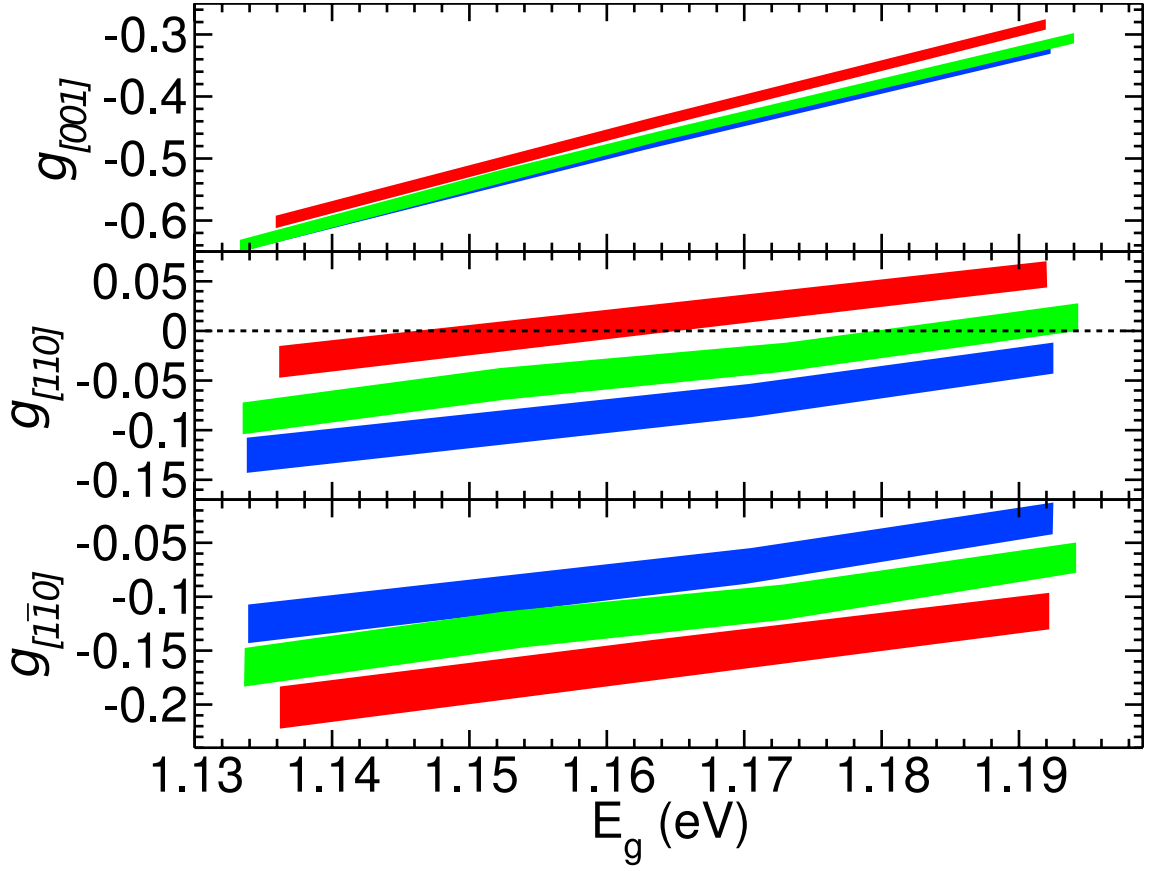


Figure 15: g-tensor (g-factor along $[001]$, $[110]$, $(1\bar{1}0)$ directions) as a function of electric field and quantum dot footprint radius. Color indicates ellipticity (blue $\rightarrow e = 0$, green $\rightarrow e = 1.25$, red $\rightarrow e = 1.667$). Application of electric field from -100kV/cm to +100kV/cm changes the g-factor within the bands. Quantum dot height fixed at 6.2nm.

Again one can see that increasing E_g (decreasing dot footprint radius) decreases the g-factor in all directions. Here, however, the ellipticity plays a greater role throughout the series, roughly doubling the g-factor in some cases from 0.11 to 0.19 for $e = 0$ and $e = 1.667$, respectively. Again, we see that for $g_{[001]}$, increasing ellipticity decreases the g-factor, although for $g_{[1\bar{1}0]}$, increasing ellipticity increases

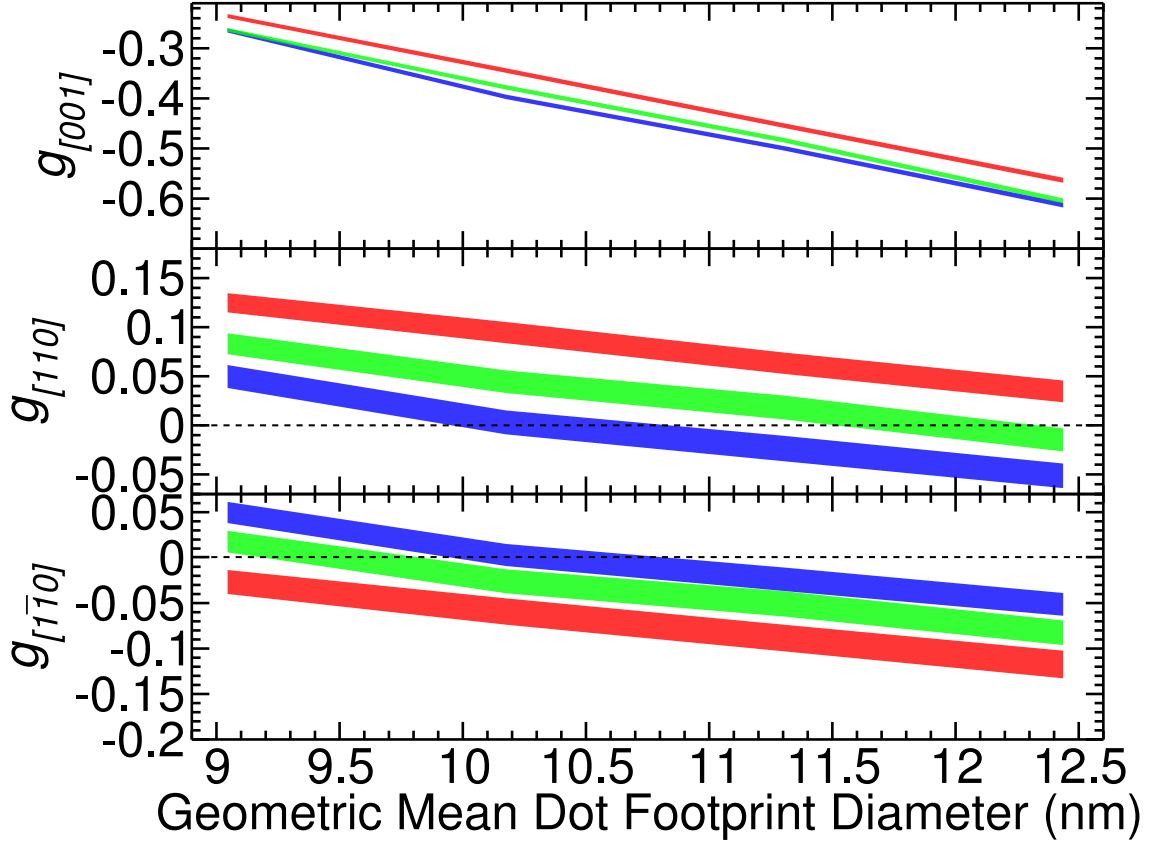


Figure 16: g-tensor (g-factor along $[001]$, $[110]$, $(1\bar{1}0)$ directions) as a function of electric field and quantum dot footprint radius. Color indicates ellipticity (blue $\rightarrow e = 0$, green $\rightarrow e = 1.25$, red $\rightarrow e = 1.667$). Application of electric field from -100kV/cm to $+100\text{kV/cm}$ changes the g-factor within the bands. Quantum dot height fixed at 5.6nm .

the g-factor. In this particular case, increasing ellipticity $g_{[110]}$ decreased the g-factor (note that this dot corresponds to the tallest height investigated, so this is in agreement with the previous findings). One may note how much larger the change in g-factor is as electric field is applied for the $[110]$ and $[1\bar{1}0]$ directions than for the $[001]$ direction. It is precisely this difference in g-factor changes which makes it possible for g-tensor modulation resonance (discussed in more detail in chapter VI) work, as an identical change in all directions would merely strengthen or weaken the

effective magnetic field, as one can see by reviewing equation 90.

Once again, a specific dot changes sign as a function of electric field. This dot has an ellipticity of 1.667, a height of roughly 6.2nm and a long dot radius of 5.6nm. In this case, however, the change comes from *increasing* the electric field from 0 to +100kV/cm.

III.2 $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ Quantum Dots

Calculations similar to that of the preceding section have been performed on $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dots. In $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ dots, however, the strain is greatly reduced from that in and around InAs/GaAs quantum dots. Comparing the strain in and around an InAs/GaAs dot (figure 4) to that in and around an $\text{InGaAs}/\text{GaAs}$ quantum dot (fig 17) shows that the strain components e_{zz} and e_{yy} in $\text{InGaAs}/\text{GaAs}$ dots is roughly half of their counterparts in InAs/GaAs . As the lattice constant in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ is half of the difference between the GaAs barrier and the pure InAs dot, this is not surprising. The band edge diagrams 18 and 19 also reflect this in that the potential depth in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ is roughly half that of InAs/GaAs .

Accordingly, one can expect fewer, less tightly bound conduction states in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ than in InAs/GaAs quantum dots. Calculations were performed on a number of $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dots. No electric field was initially applied. However, a magnetic field of 1T (table 5) and 7T (table 6) was applied.

As with InAs/GaAs quantum dots, the electron [001] g-factor depends sensitively on the dot confinement (i.e. E_g). [88] As a function of magnetic field, the g-factor

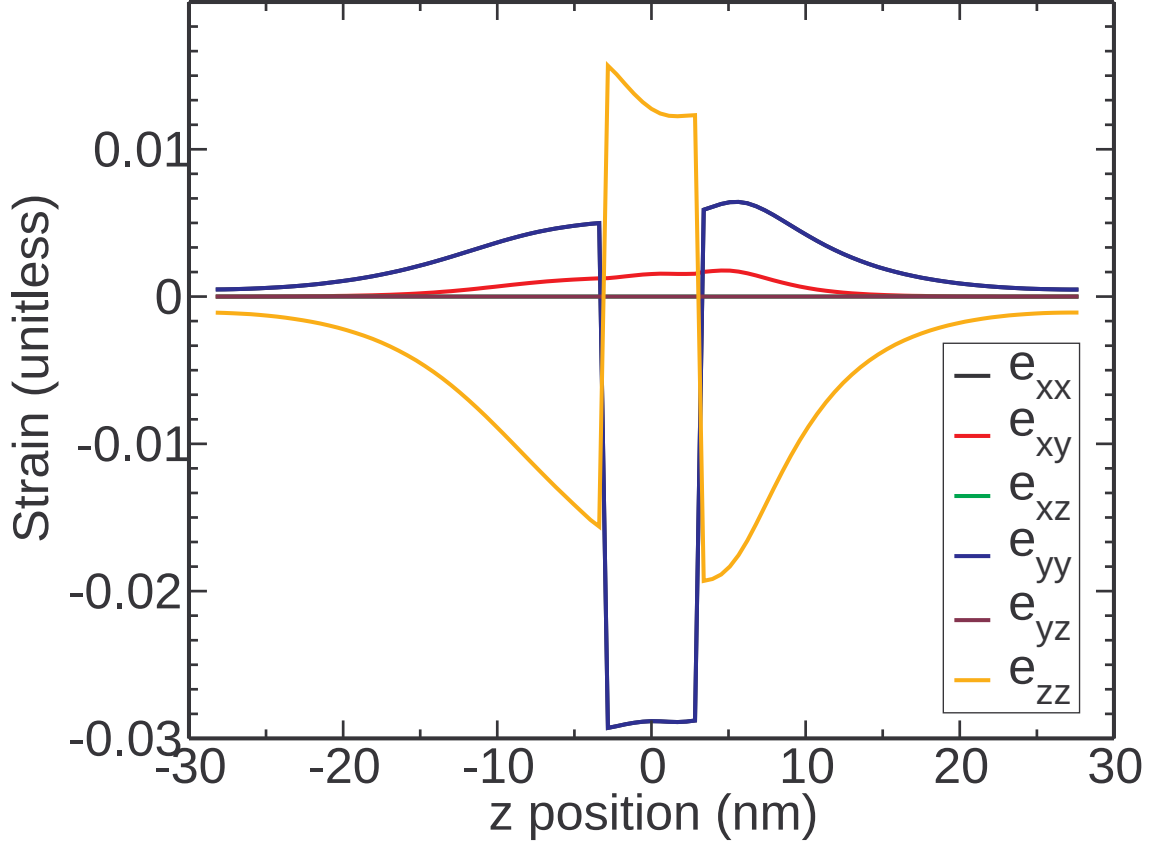


Figure 17: Strain tensor components in and around an $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dot ($d_{[110]} = 21.6\text{nm}$, $d_{[\bar{1}\bar{1}0]} = 32.8\text{nm}$, height of 6.2nm) running along a line parallel to the $[001]$ axis through the center of the dot. Horizontal axis is position along this line (nm). Vertical axis is strain tensor component. Black is e_{xx} , red is e_{xy} , green is e_{xz} , blue is e_{yy} , maroon is e_{yz} , and orange is e_{zz} . The dot itself is located in the center (centered on 0).

$E_g(\text{eV})$	$d_{[001]}$	$d_{[110]}$	$d_{[\bar{1}\bar{1}0]}$	e	$g_{el,[001]}$	$g_{el,[110]}$	$g_{el,[\bar{1}\bar{1}0]}$
1.271	5.65nm	16.8nm	27.2nm	1.62	-0.3482	-0.0168	0.0288
1.266	5.65nm	18.4nm	28.8nm	1.56	-0.374	-0.0189	0.026
1.266	7.35nm	16.8nm	27.2nm	1.62	-0.398	-0.112	-0.0478
1.256	5.09nm	20.0nm	32.0nm	1.60	-0.413	-0.0167	0.0254
1.256	5.65nm	20.0nm	32.0nm	1.60	-0.420	-0.0267	0.0167
1.252	6.22nm	20.0nm	29.6nm	1.48	-0.446	-0.063	-0.0196

Table 5: Summary of electron g-tensors at 1T without applied electric field; $e = \frac{d_{[\bar{1}\bar{1}0]}}{d_{[110]}}$.

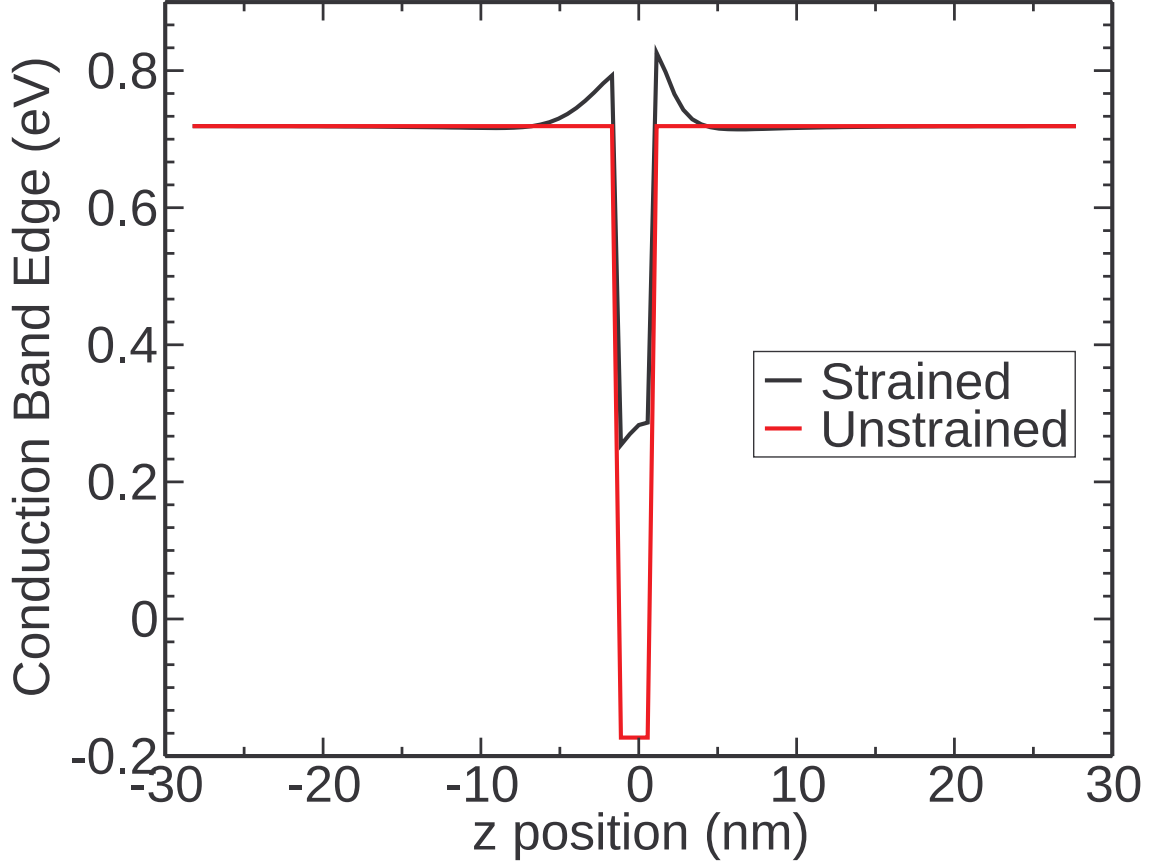


Figure 18: Strain influence on the conduction band edge in and around an InAs/GaAs quantum dot (circular footprint with nominal radius of 5.1nm and nominal height of 2.3nm) running along a line parallel to the [001] axis through the center of the dot. Horizontal axis is position along this line (nm). Vertical axis is the conduction band edge.

is dependent on the magnetic field to a very limited extent. The energy splitting is therefore almost entirely linear in magnetic field, in contrast to the case in hydrogenic donor systems.[48]

Two dots were investigated as a function of electric field. Because of the reduced confinement, the electron g-factors were only able to be reliably calculated over a range extending from -20kV/cm to $+20\text{kV/cm}$. Figure 20 is the calculated g-tensor of an $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ dot with a height of 5.1nm, $l_{[1\bar{1}0]} = 32.0\text{nm}$ and

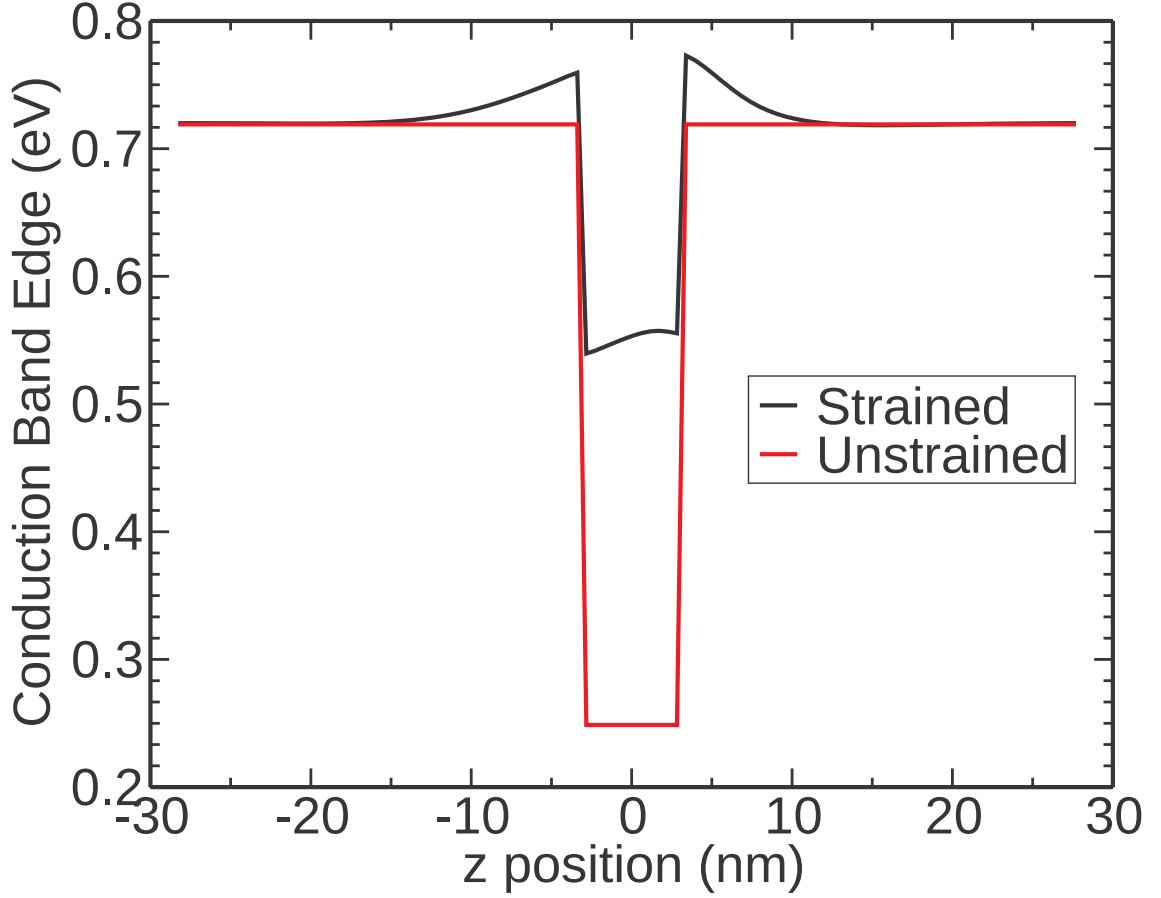


Figure 19: Strain influence on the conduction band edge in and around an $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dot ($d_{[110]} = 21.6\text{nm}$, $d_{[1\bar{1}0]} = 32.8\text{nm}$, height of 6.2nm) running along a line parallel to the $[001]$ axis through the center of the dot. Horizontal axis is position along this line (nm). Vertical axis is the conduction band edge.

$l_{[110]} = 20.0\text{nm}$ (figure 21 is the same dot at 7T). Figure 22 is an $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ dot with a height of 5.1nm , $l_{[1\bar{1}0]} = 32.0\text{nm}$ and $l_{[110]} = 20.0\text{nm}$.

A clear quadratic dependence of the electron g-tensor components is found for these dots, even in the in-plane directions. This is very different from the InAs/GaAs quantum dots seen earlier. As with the dots above, the magnetic field dependence of the g-tensors was small.

E_g (eV)	$d_{[001]}$	$d_{[110]}$	$d_{[1\bar{1}0]}$	e	$g_{el,[001]}$	$g_{el,[110]}$	$g_{el,[1\bar{1}0]}$
1.271	5.65nm	16.8nm	27.2nm	1.62	-0.344	-0.0126	0.0327
1.256	5.09nm	20.0nm	32.0nm	1.60	-0.408	-0.0132	0.0286
1.252	6.22nm	20.0nm	29.6nm	1.48	-0.441	-0.0599	-0.0168

Table 6: Summary of electron g-tensors at 7T without applied electric field; $e = \frac{d_{[1\bar{1}0]}}{d_{[110]}}$.

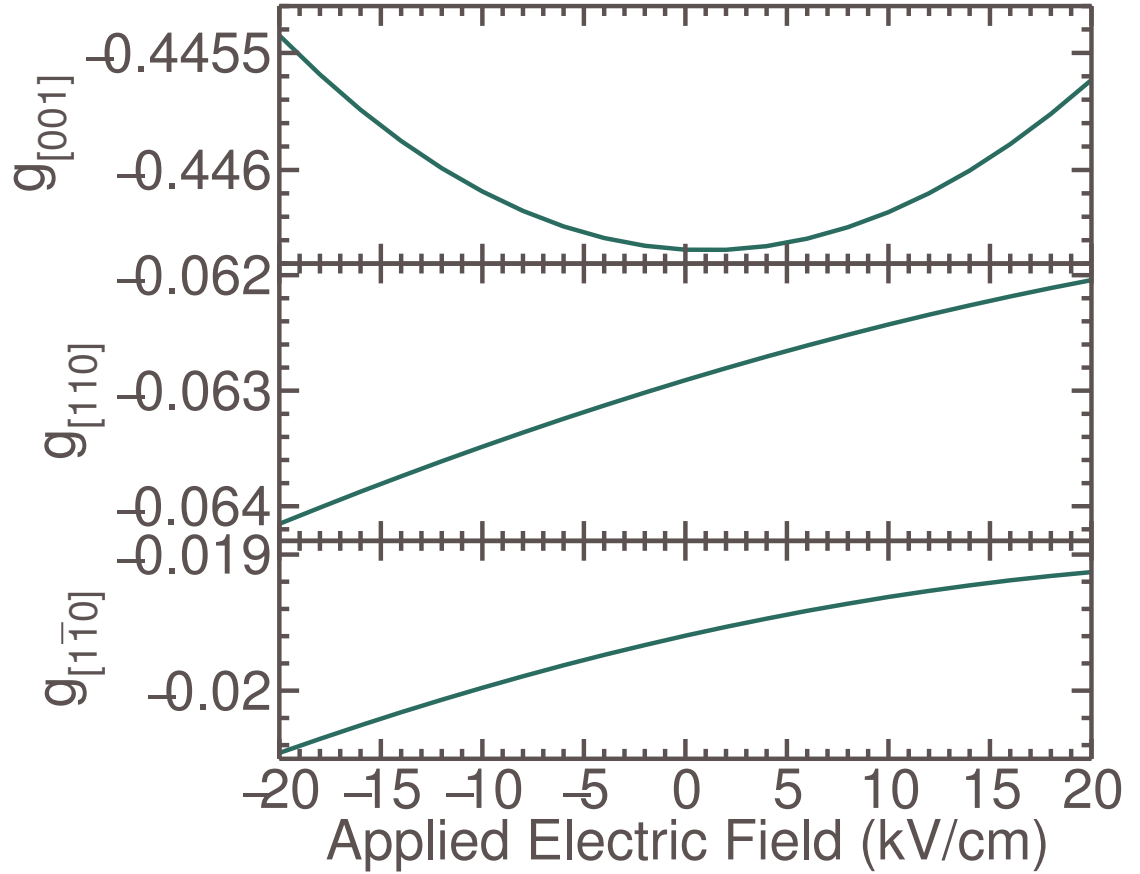


Figure 20: Electron g-tensor at 1T as a function of electric field in the range ± 20 kV/cm. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 6.2nm, $l_{[1\bar{1}0]} = 29.6$ nm, $l_{[110]} = 20$ nm, and energy gap $E_g = 1.252$ eV. Calculations performed at 0K.)

III.3 Comparison with previous research

This is the first conclusive demonstration of calculations of electric-field dependent g-tensors in self-assembled quantum dots. The sensitivities discussed in

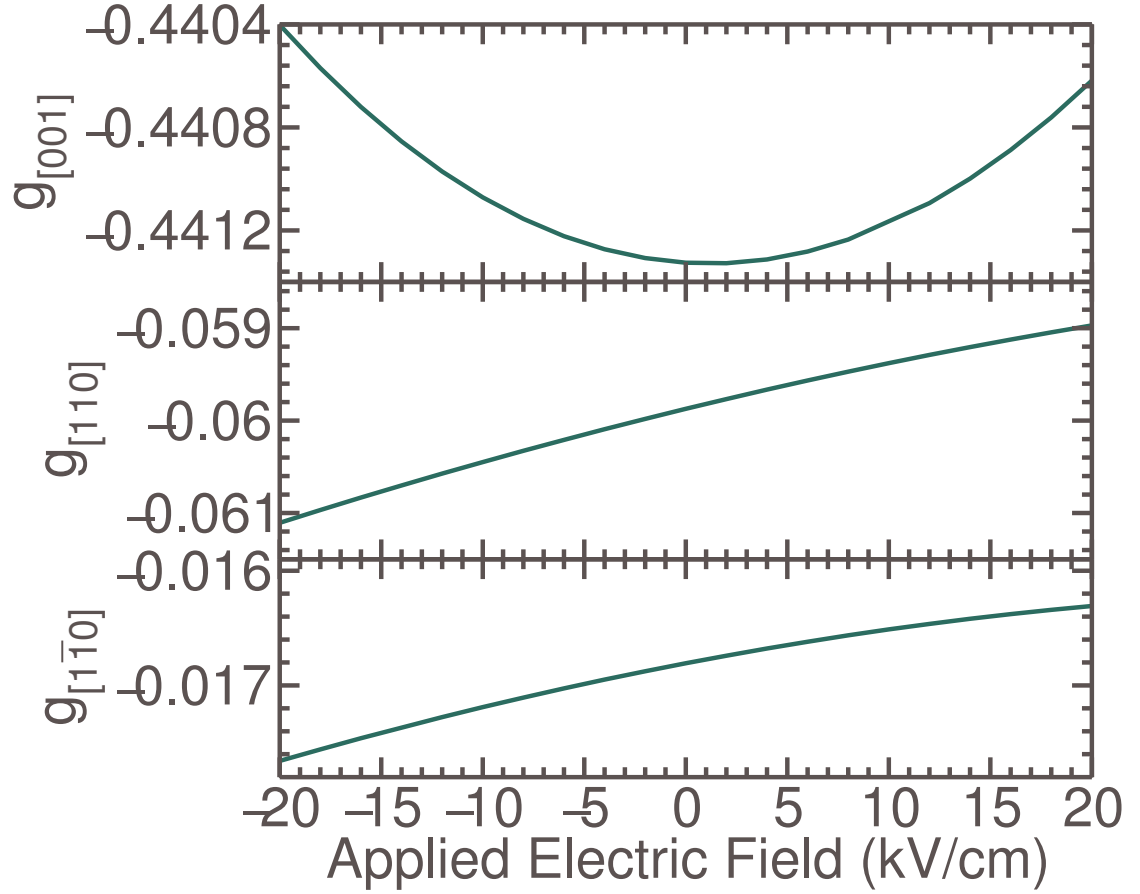


Figure 21: Electron g-tensor at 7T as a function of electric field in the range ± 20 kV/cm. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 6.2 nm, $l_{[1\bar{1}0]} = 29.6$ nm, $l_{[110]} = 20$ nm, and energy gap $E_g = 1.252$ eV. Calculations performed at 0 K.)

InAs/GaAs quantum dots follow the trends found in Pryor and Flatté.[92]. A recent paper has dealt with electric-field dependent g-tensors in hydrogenic donor systems.[48] Additionally, another recent paper calculates electron and hole g-factors in coupled truncated pyramidal InAs/GaAs quantum dots.[47], which will behave significantly differently from a single, isolated InAs/GaAs quantum dot. The 10-band calculations of $\text{InAs}_{1-x}\text{N}_x$ dots shows extremely large g-factors (-600)[57] and electrically tunable g-factors (including a 0 g-factor with electric field) in this

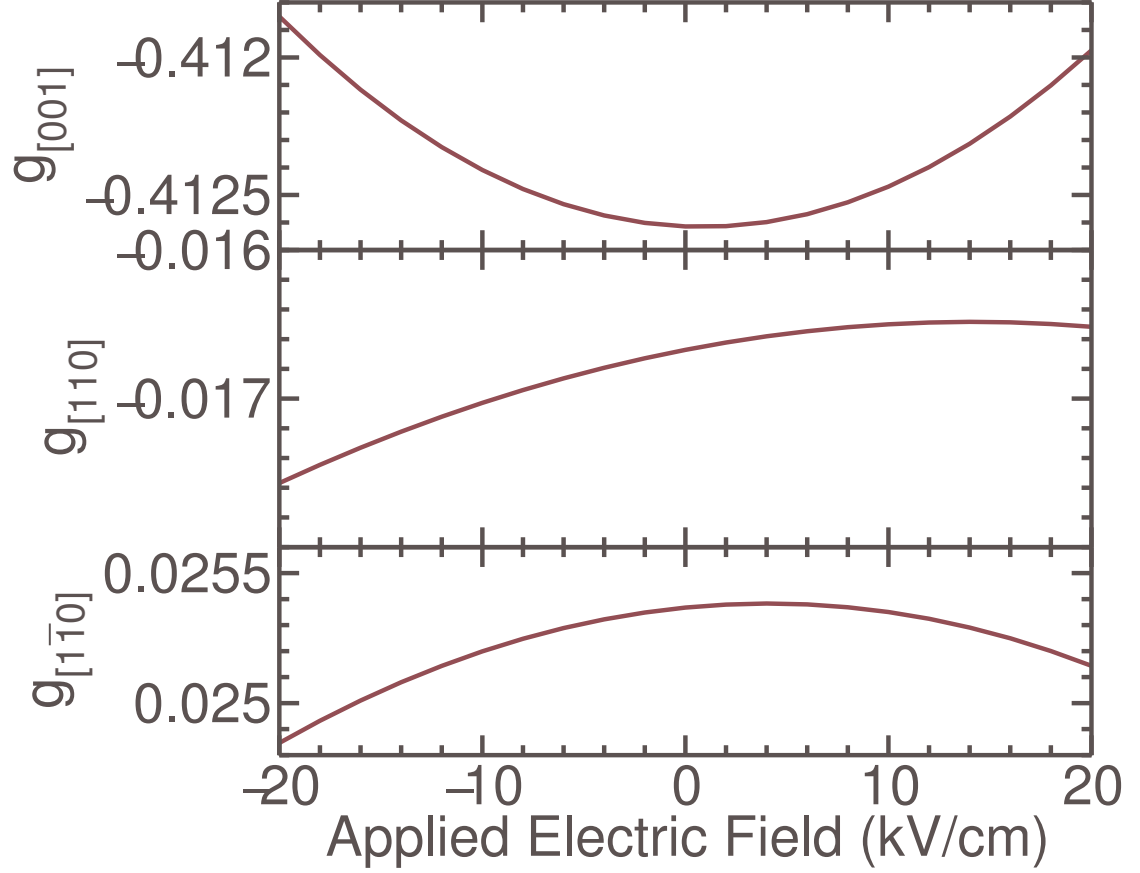


Figure 22: Electron g-tensor at 1T as a function of electric field in the range ± 20 kV/cm. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 5.1 nm, $l_{[1\bar{1}0]} = 32.0$ nm, $l_{[110]} = 20.0$ nm, and energy gap $E_g = 1.256$ eV. Calculations performed at 0 K.)

system[58]. However, their model did not include strain, and used infinitely high barrier.

A convergence toward 2 is found in Zhang et al.[51] when examining ellipsoidal InAs dots with infinitely high barrier and no strain, replicating the findings of [92]. Strong shape and size dependence of InAs nanocrystals was found in Prado et al.[93] for spherical and semispherical InAs quantum dots. Valín-Rodríguez et al.[42] also find a dependence of g on magnetic field strength and “tilt angle”, i.e. the angle of the magnetic field from the $[001]$ axis in the $[100]$ - $[001]$ plane in GaAs/AlGaAs

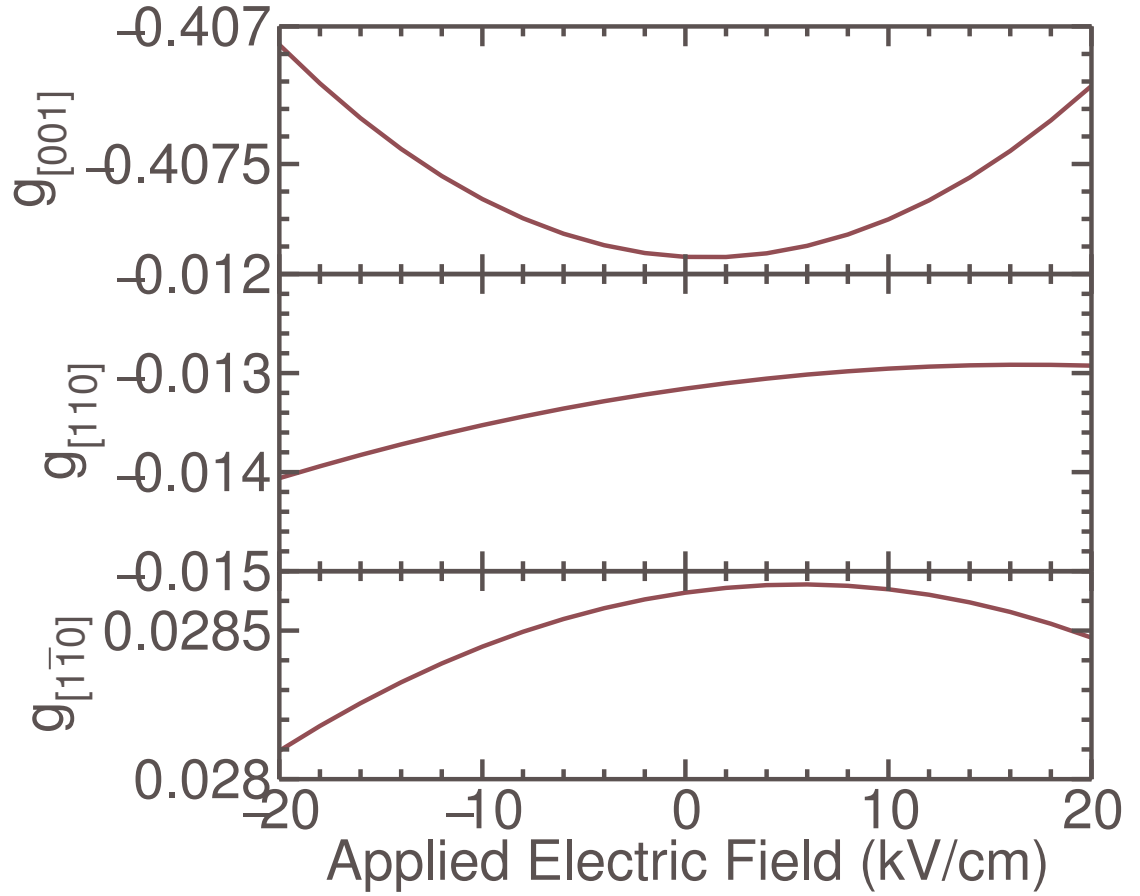


Figure 23: Electron g-tensor at 1T as a function of electric field in the range ± 20 kV/cm. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 5.1 nm, $l_{[1\bar{1}0]} = 32.0$ nm, $l_{[110]} = 20.0$ nm, and energy gap $E_g = 1.256$ eV. Calculations performed at 0 K.)

quantum dots. The angular dependence is not separated out into individual g-tensor components, however.

Different g-factors in InAs/GaAs quantum dots were seen by Medeiros-Ribeiro et al.[94] when a magnetic field was applied along the $[001]$, $[110]$, and $[1\bar{1}0]$ axes. A high angular dependence of the s-like state in self-assembled InAs/GaAs quantum dots, with higher states showed a less angular dependence, due to delocalization.[95] Electrical g-factor control of electron spins in self-assembled InGaAs/GaAs pyrami-

dal dots was seen by Nakaoka et al.[96] of about 8% for fields of up to 5kV/cm. This is larger than what was calculated in this chapter. However, the electric field was applied in plane instead of in the growth direction. An in-plane electric field could potentially lead to greater confinement, explaining the larger change in g-factor.

CHAPTER IV

A SINGLE HOLE SPIN IN AN ISOLATED QUANTUM DOT

IV.1 InAs/GaAs Quantum Dots

As with the first conduction states investigated in the previous section, the g-tensor was examined as a function of electric field. In this case, however, the g-tensor was calculated at three different height-width combinations. In figure 24, the g-tensor is plotted as a function of electric field (again being varied from -100 to +100kV/cm) for an electron in a 6.22nm high dot with a 6.22nm wide footprint (6.22nm long major axis for the elliptical dot). The g-factor in-plane ($g_{[110]}$, $g_{[1\bar{1}0]}$) is almost zero, but out of plane, g is much larger (around 2.8)—but *only* for the round-footprint dots. For elliptical-footprint dots, the in-plane g-factor, despite not being as large as the out-of-plane g-factor, is still an order of magnitude larger than that of the round-footprint dots. It can therefore be concluded that ellipticity has a large effect on the valence-state electron (and thus hole) g-tensor. It may also be interesting to note that the electric field effect on the g-tensor components is almost linear, except at sufficiently large field. Double-checking the wavefunctions has found no evidence of sufficient electron wave function leakage even at high electric field strength, and therefore expect that the evident non-linearity is physical. The non-linearity appears also to increase as ellipticity increases.

Figure 25 shows the same figure, but for a short dot (height of 3.40nm, major axis length of 6.22nm). The slopes are now much shallower, likely because of the decrease in available room in which the hole may be moved. The non-linearity is

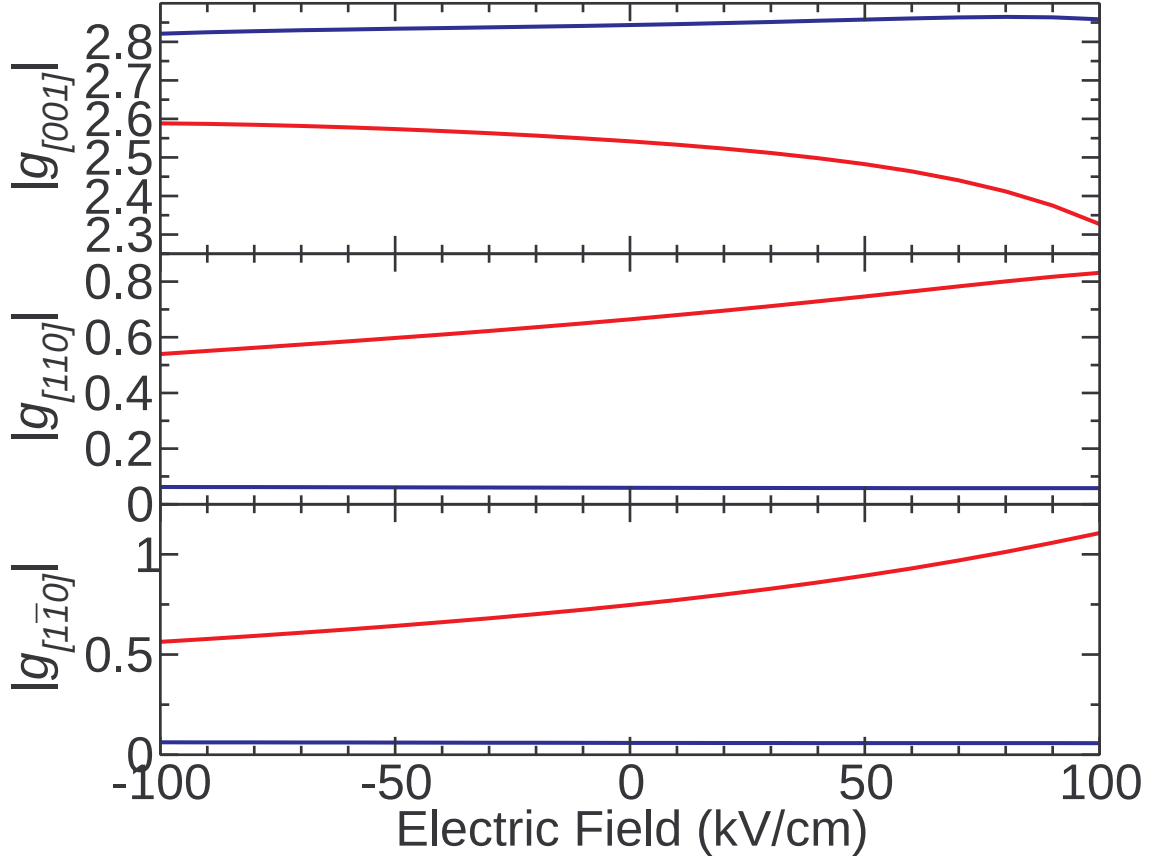


Figure 24: hole g-tensor component magnitude (g-factor magnitude along $[001]$, $[110]$, $[1\bar{1}0]$ directions) as a function of electric field for the 6.22nm high, 6.22nm wide footprint dot. Color indicates ellipticity (blue $\rightarrow e = 0$, red $\rightarrow e = 1.667$).

mostly gone although a small, almost parabolic curve can be seen in the out-of-plane g-factor ($g_{[001]}$).

Finally, in figure 26 is a narrow dot (height of 6.22nm and major axis length of 4.52nm). Here again one can see a pronounced non-linearity, as was the case in figure 24. The electric field influence is greater for this tall, narrow dot than for the tall, wide dot, likely owing to the increased electron penetration into the barrier material as discussed in the previous section on the conduction state g-tensor.

As with the conduction state g-tensor, the height dependency of the electric

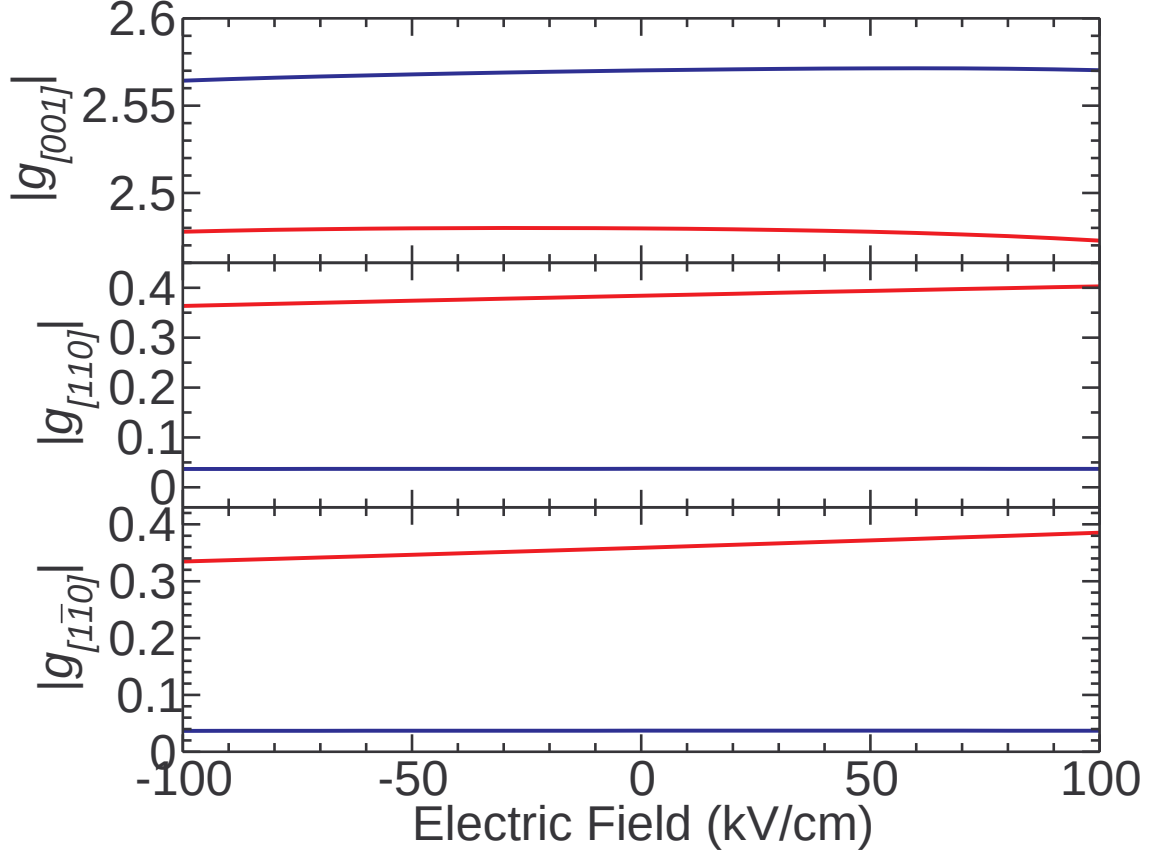


Figure 25: hole g-tensor component magnitude (g-factor magnitude along $[001]$, $[110]$, $[1\bar{1}0]$ directions) as a function of electric field for the 3.40nm high, 6.22nm wide footprint dot. Color indicates ellipticity (blue $\rightarrow e = 0$, red $\rightarrow e = 1.667$).

field effect on the valence state (hole) g-tensor has been investigated. The results of these calculations are plotted in figure 27. As with the previous section on the conduction band g-tensor, color indicates ellipticity (blue is a round footprint dot, green is $e=1.25$, and red is $e=1.667$) and line style indicates electric field strength (solid corresponds to no electric field applied, dashed is $+100\text{kV/cm}$, and dotted is -100kV/cm).

What one may notice is the large effect that ellipticity has on the valence g-factor. This is a much larger effect than what was seen with the conduction state

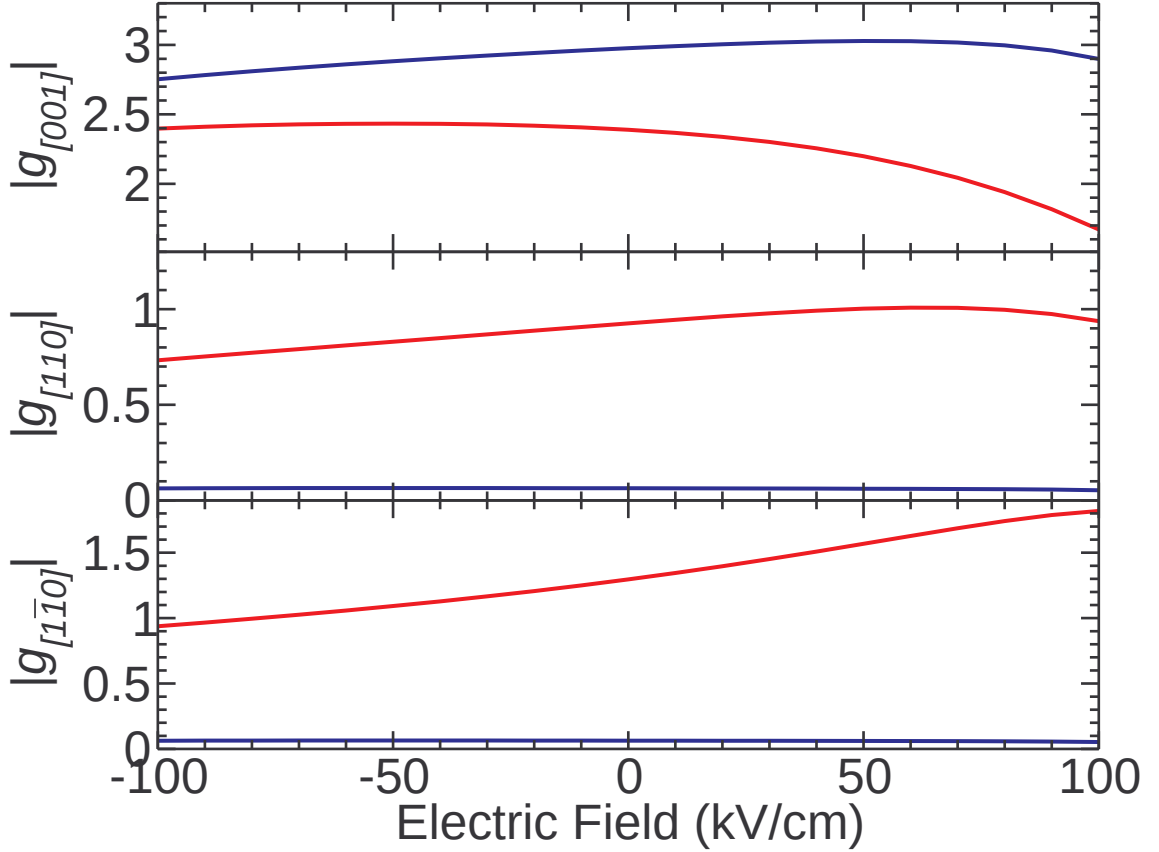


Figure 26: hole g-tensor component magnitude (g-factor magnitude along $[001]$, $[110]$, $[1\bar{1}0]$ directions) as a function of electric field for the 6.22nm high, 4.52nm wide footprint dot. Color indicates ellipticity (blue $\rightarrow e = 0$, red $\rightarrow e = 1.667$).

g-tensor. Although a noticeable change for $g_{[001]}$ occurs in the conduction band, the g-factor changes by about an *order of magnitude* when ellipticity goes from 1 to 1.667 (round to elliptical) in the valence band. Additionally, there is a very large anisotropy between the in-plane g-factors and the out-of-plane g-factor (even with the elliptical effects), which is in line with other groups' calculations and experimental results.[42][92][86][87][36]

As before, increasing the dot height gives the carrier more room to be pushed about by the electric field (and thereby allowing the electric field to change the

confinement of the electron more) and thus an applied electric field changes the g-factors for each direction more as the dot height is increased. The electric field effects are much less pronounced when the dot height is decreased.

For the vast majority of the dots and directions investigated, the g-factor is highest with a positive electric field applied, and lowest with a negative electric field applied. Interestingly, however, this trend is reversed along the $[001]$ direction, where a positive electric field causes the g-factor to be *lowest* and the negative electric field causes the g-factor to be *highest*. Again, there is an exception as both positive *and* negative electric fields increase $g_{[001]}$ when the dot footprint is round.

The dependency of the electric field effects on the dot footprint width (major axis length) for the tall (height of 6.22nm) dots has also been investigated, and the results are plotted in figure 28. As before, a large anisotropy may be noticed between the in-plane and out-of-plane g-factors for round dots, and the ellipticity effects also appear.

In contrast to the height dependency, the electric field has a larger effect on taller, narrower than on shorter, wider dots. This is also a much larger effect than was seen for the conduction electron states.

From these results, one can see that there is no point where a g-factor changes sign, despite not knowing the sign of the g-factors. Were this not the case, one would find an abrupt change in the height or footprint width trends where the magnitude decreases and then increases again as the sign changes. Further calculations were made in an attempt to find a sign change in a g-tensor component, but no sign

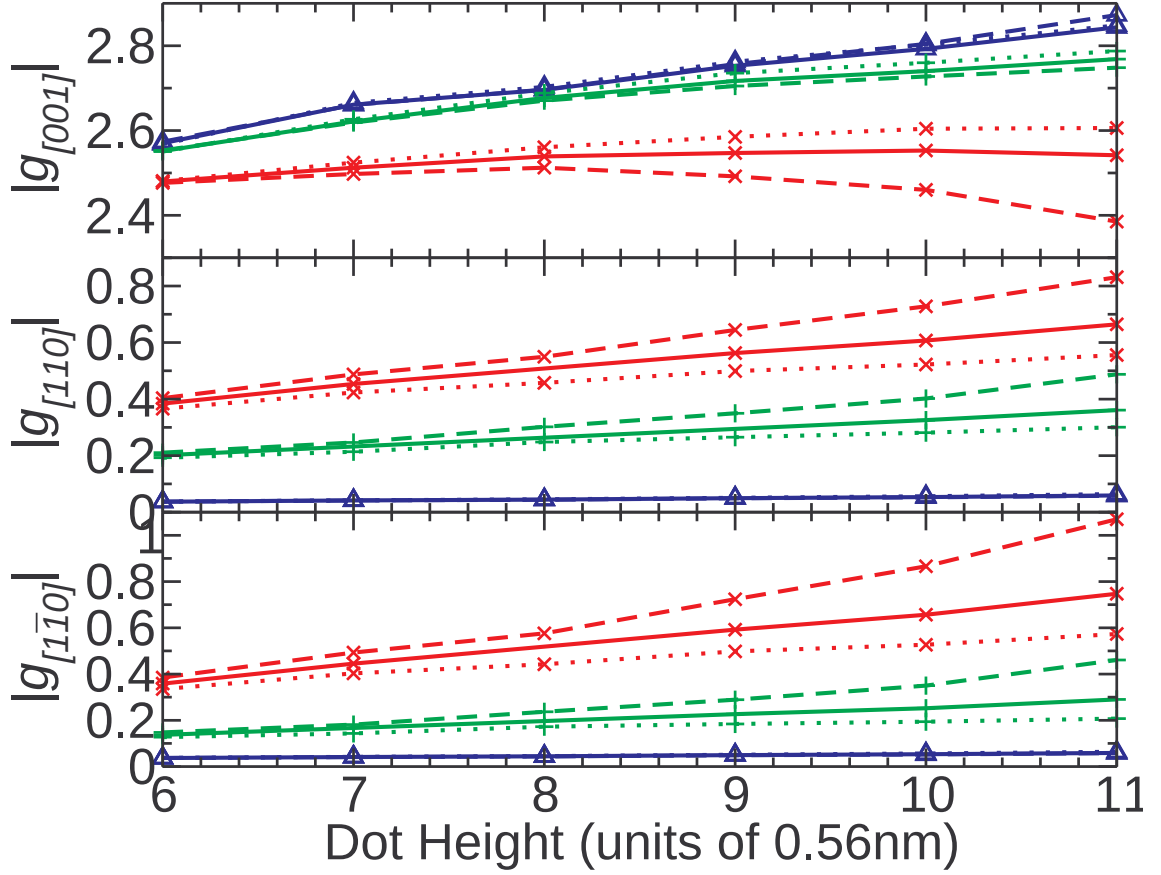


Figure 27: g -tensor component magnitude (hole g -factor magnitude along $[001]$, $[110]$, $[1\bar{1}0]$ directions) as a function of electric field and quantum dot height. Color indicates ellipticity (blue $\rightarrow e = 0$, green $\rightarrow e = 1.25$, red $\rightarrow e = 1.667$). Line style indicates applied electric field (dashed $\rightarrow E = +100 \text{ kV/cm}$, solid $\rightarrow E = 0 \text{ kV/cm}$, dotted $\rightarrow E = -100 \text{ kV/cm}$).

change was found.

Although no heavy hole states were found which were suited to non-resonant spin manipulation, the electric field effects are larger for valence states than for conduction states. This increased effect may be used either to decrease the electric field needed to achieve a certain resonant manipulation time, or alternately a faster time may be used. Additionally, larger electric fields may be used with holes than with electrons, due to their larger mass and therefore greater confinement.

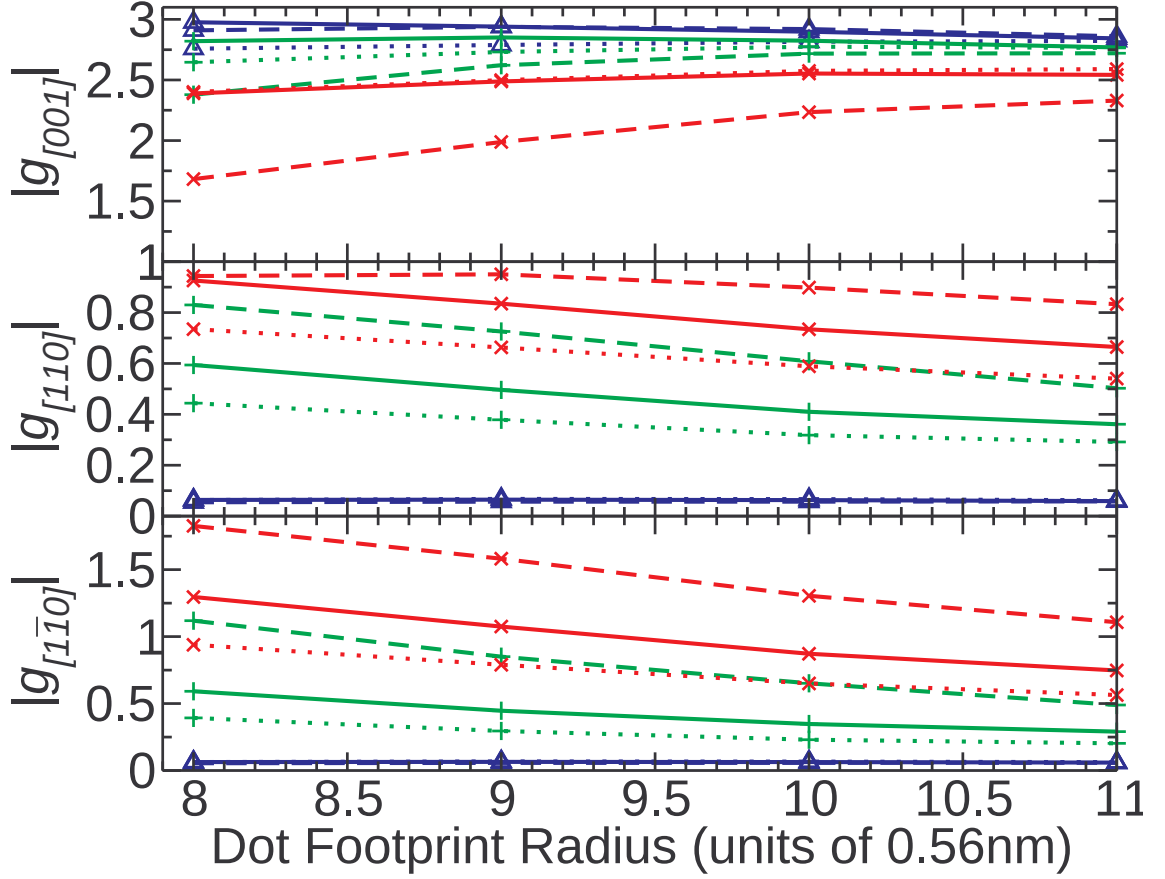


Figure 28: g-tensor magnitude (hole g-factor magnitude along $[001]$, $[110]$, $[1\bar{1}0]$ directions) as a function of electric field and quantum dot footprint radius. Color indicates ellipticity (blue $\rightarrow e = 0$, green $\rightarrow e = 1.25$, red $\rightarrow e = 1.667$). Line style indicates applied electric field (dashed $\rightarrow E = +100kV/cm$, solid $\rightarrow E = 0kV/cm$, dotted $\rightarrow E = -100kV/cm$).

IV.2 $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ Quantum Dots

As was discussed in the previous section, the strain in and around $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dots is reduced compared to that of InAs/GaAs quantum dots. The effects of strain on the valence band edge structure are different, however, from the conduction band effects. Figures 29 and 30 compare the strained and unstrained valence band structure of InAs/GaAs quantum dots and $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/$

GaAs quantum dots (respectively). Of particular note is the splitting of the heavy hole and light hole states in the presence of strain.

Perhaps interestingly, the well depths do not change to a large extent. The heavy hole confinement is additionally reduced due to the increased (lowered, in the hole picture) barrier potential in the immediate vicinity of the dot. The light hole confinement is increased, however, due to a lowering of the barrier potential (raising, in the hole picture). These features are significantly reduced in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ compared to InAs/GaAs . Combined with the small changes in well depth, this indicates that the InAs/GaAs dots investigated in the previous subsection are, when including strain effects, significantly larger than their $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ counterparts for the purposes of heavy hole spin confinement. Due to the increase in light hole confinement with increasing strain (similar to the situation in the conduction band (figures 18 and 19), it can be expected that the light holes will behave more like the electron states discussed in the previous chapter. Given these heavy hole strain effects, is perhaps unsurprising, then, that the GaAs heavy hole g-tensors in the previous subsection remain negative, indicating larger quantum dots whereas the g-tensors in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ in this section are almost entirely positive, despite being reasonably similar in size.

As with the conduction-band g-tensors in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$, valence-band g-tensors were calculated on the same dots, for the purposes of creating excitonic g-factors to compare with experiment (discussed in detail in the next chapter). These hole g-tensors are listed in table 7. The hole g-tensors also show a clear dependence

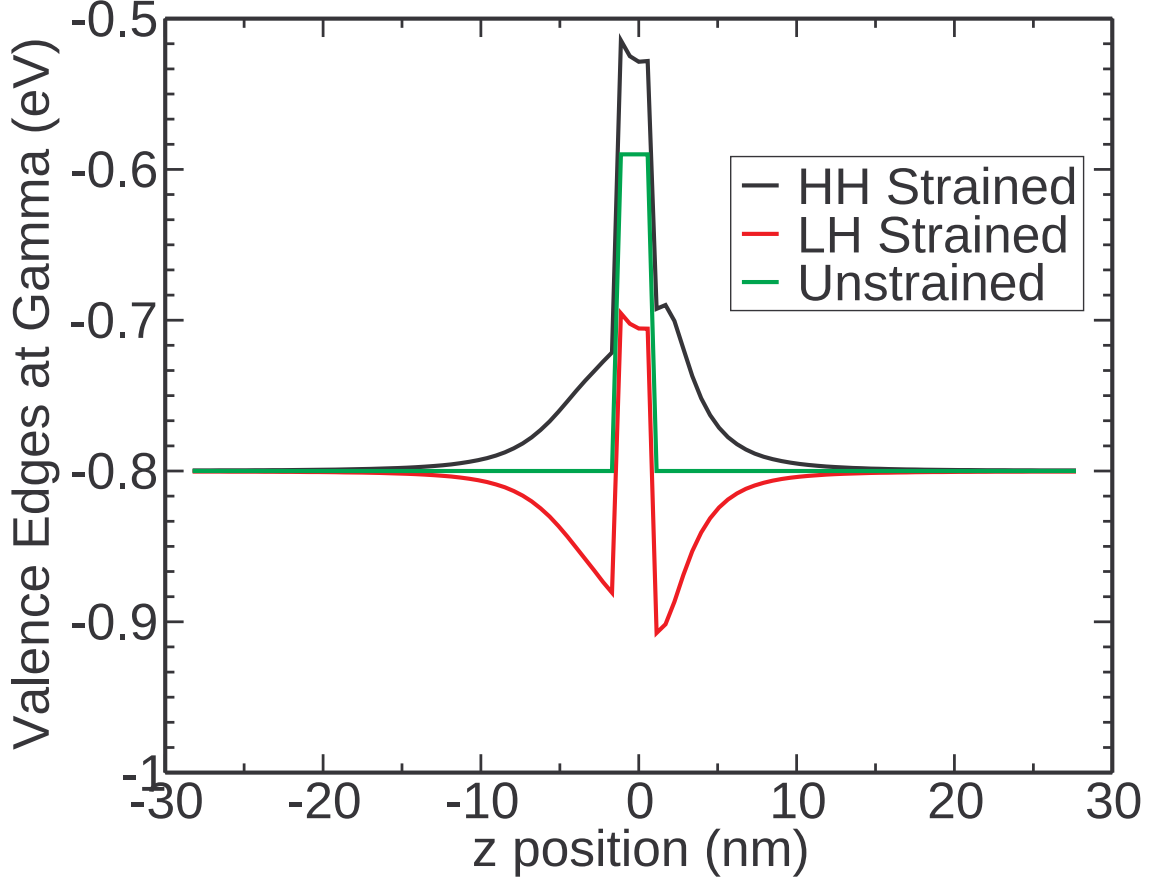


Figure 29: Strain influence on the valence band edges in and around an InAs/GaAs quantum dot (circular footprint with nominal radius of 5.1nm and nominal height of 2.3nm) running along a line parallel to the $[001]$ axis through the center of the dot. Horizontal axis is position along this line (nm). Vertical axis is the valence band edge.

upon confinement, as was the case for the electron g-tensors.

E_g (eV)	$d_{[001]}$	$d_{[110]}$	$d_{[1\bar{1}0]}$	e	$g_{hole,[001]}$	$g_{hole,[110]}$	$g_{hole,[1\bar{1}0]}$
1.271	5.65nm	16.8nm	27.2nm	1.62	0.497	0.208	0.279
1.266	7.35nm	16.8nm	27.2nm	1.62	0.401	0.258	0.346
1.266	5.65nm	18.4nm	28.8nm	1.56	0.384	0.193	0.261
1.256	5.65nm	20.0nm	32.0nm	1.60	0.171	0.170	0.233
1.256	5.09nm	20.0nm	32.0nm	1.60	0.185	0.167	0.228
1.252	6.22nm	20.0nm	29.6nm	1.48	0.117	0.156	0.225

Table 7: Summary of hole g-tensors at 1T without applied electric field. $e = \frac{d_{[1\bar{1}0]}}{d_{[110]}}$.

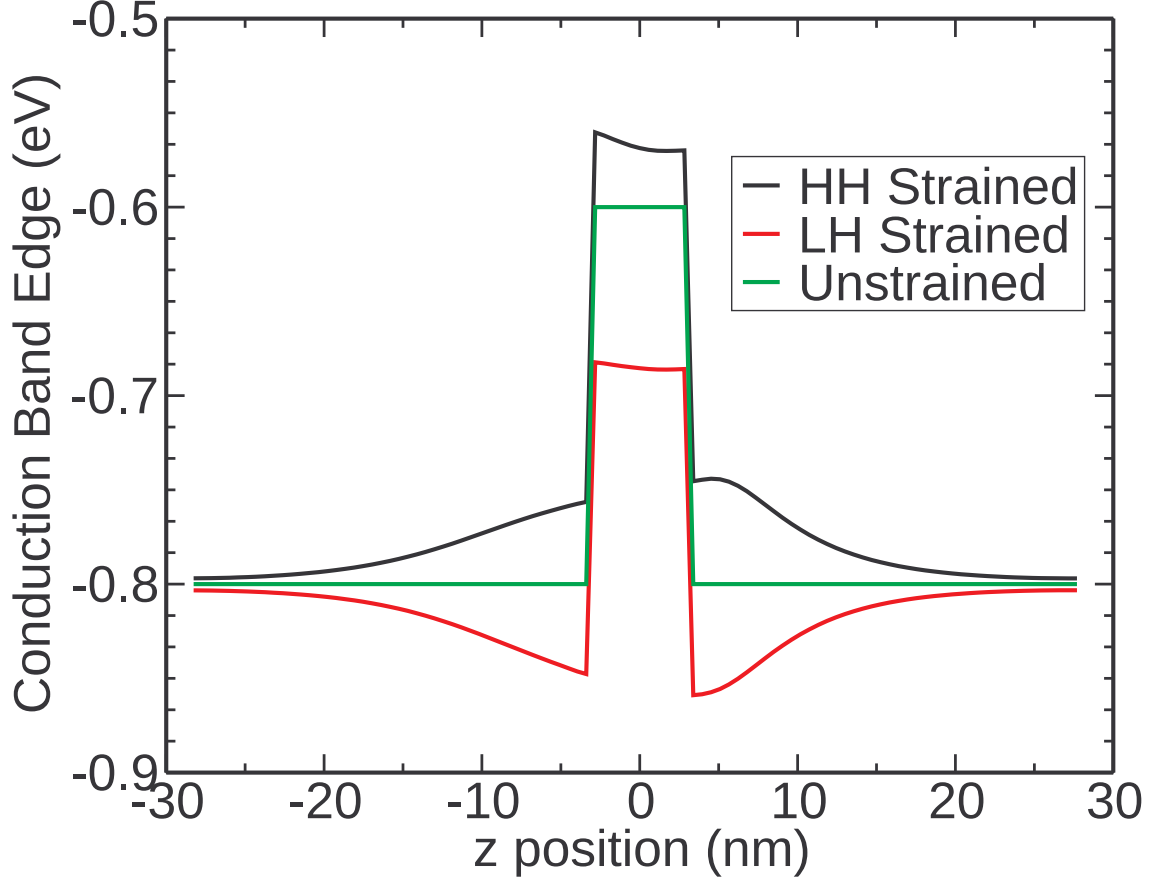


Figure 30: Strain influence on the valence band edges in and around an $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dot ($d_{[110]} = 21.6\text{nm}$, $d_{[1\bar{1}0]} = 32.8\text{nm}$, height of 6.2nm) running along a line parallel to the $[001]$ axis through the center of the dot. Horizontal axis is position along this line (nm). Vertical axis is the valence band band edge.

In addition to the data at 1T, calculations were performed on some of the dots at 7T (table 8). As with the electron g-tensors, the change going from 1T to 7T was small. In the $[001]$ direction, however, it shows a slightly larger dependence on magnetic field strength than the electrons.

As had been done for previous sections, the electric field dependence of the g-tensor was investigated. As was the case for electrons in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$, the electric field was initially limited to the $-20\text{kV}/\text{cm}$ to $+20\text{kV}/\text{cm}$ range. This was

E_g (eV)	$d_{[001]}$	$d_{[110]}$	$d_{[1\bar{1}0]}$	e	$g_{hole,[001]}$	$g_{hole,[110]}$	$g_{hole,[1\bar{1}0]}$
1.252	6.22nm	20.0nm	29.6nm	1.48	0.143	0.156	0.225
1.256	5.09nm	20.0nm	32.0nm	1.60	0.209	0.167	0.228
1.271	5.65nm	16.8nm	27.2nm	1.62	0.513	0.207	0.279

Table 8: Summary of hole g-tensors at 7T without applied electric field. $e = \frac{d_{[1\bar{1}0]}}{d_{[110]}}$.

done in order to ensure that the excitonic g-tensors would be accurate. These hole g-tensors are plotted in figures 31 and 32 for hole g-tensors in an $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ dot with a height of 6.2nm and lateral dimensions of $l_{[1\bar{1}0]} = 29.6\text{nm}$ and $l_{[110]} = 20\text{nm}$ at 1T and 7T, respectively.

The same calculation was performed on another $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dot, this time with a height of 6.2nm and lateral dimensions $l_{[1\bar{1}0]} = 29.6\text{nm}$ and $l_{[110]} = 20\text{nm}$ at 1T and 7T. These are plotted in 33 and 34 for 1T and 7T, respectively.

Unlike the electrons, which, due to their lower effective mass, are less confined within the dot, the hole g-factors were able to be accurately calculated up to $\pm 150\text{kV/cm}$. Therefore, the calculations were performed again, using this new range of electric fields. The results are plotted in figures 35 and 36 for the first dot at magnetic field strengths of 1T and 7T (respectively) and in figures 37 and 38 for magnetic field strengths of 1T and 7T, respectively. Because of limitations in the experiments with which the theoretical results were being calculated, the former dot only has g-factors along the $[001]$ direction.

In the process of calculating these g-tensors, a g-factor which changes sign as a function of electric field was found. The dot has a height of height of 6.2nm,

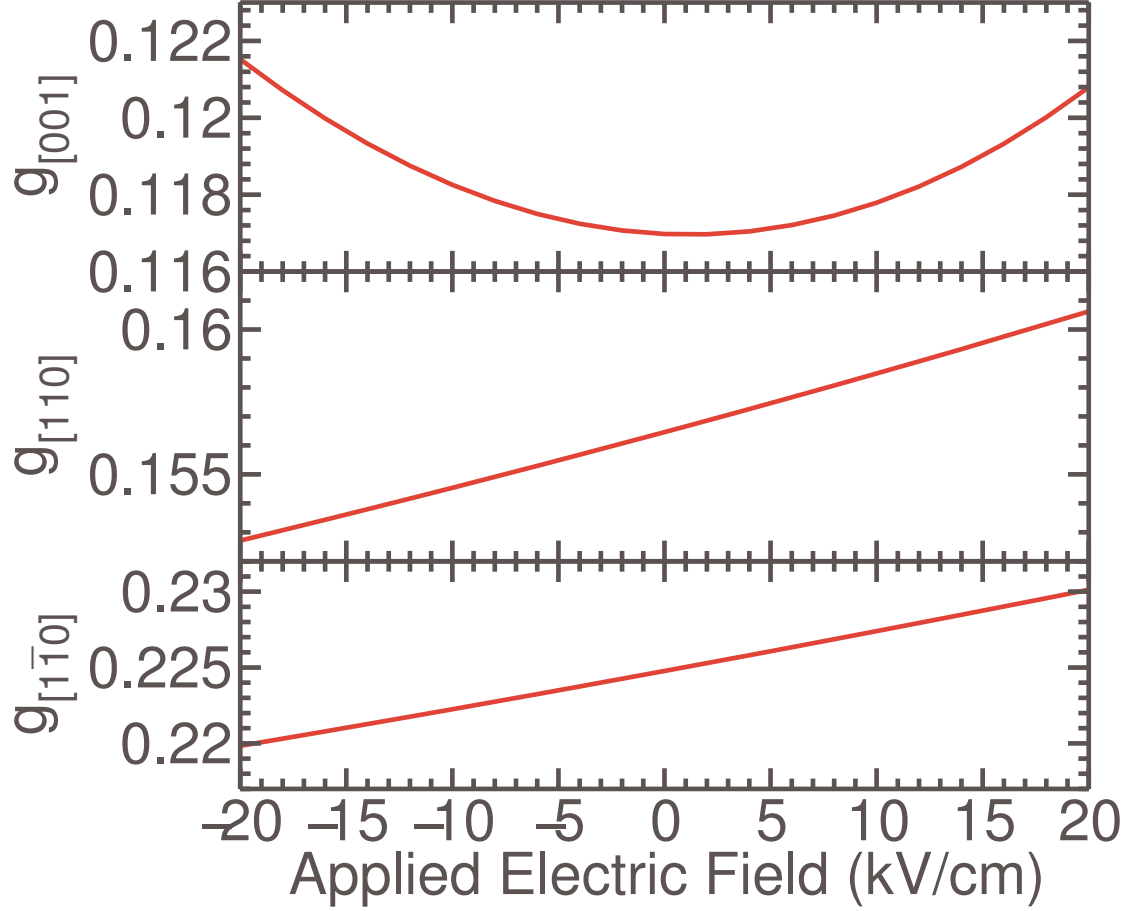


Figure 31: Hole g-tensor at 1T as a function of electric field in the range ± 20 kV/cm. For a magnetic field applied along the growth direction, the electron g-factor dominates the hole g-factor. The opposite is true, however, for a magnetic field applied in-plane (i.e. along $[110]$ or $[1\bar{1}0]$). (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 6.2 nm, $l_{[1\bar{1}0]} = 29.6$ nm, $l_{[110]} = 20$ nm, and energy gap $E_g = 1.252$ eV. Calculations performed at 0 K.)

footprint length along the $[110]$ direction of 21.6 nm, and footprint length along the $[1\bar{1}0]$ direction of 32.8 nm. This dot is slightly larger than some of the other $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ dots discussed to this point, and it makes sense that, as the confinement decreases, the g-tensor trends toward its negative bulk value.

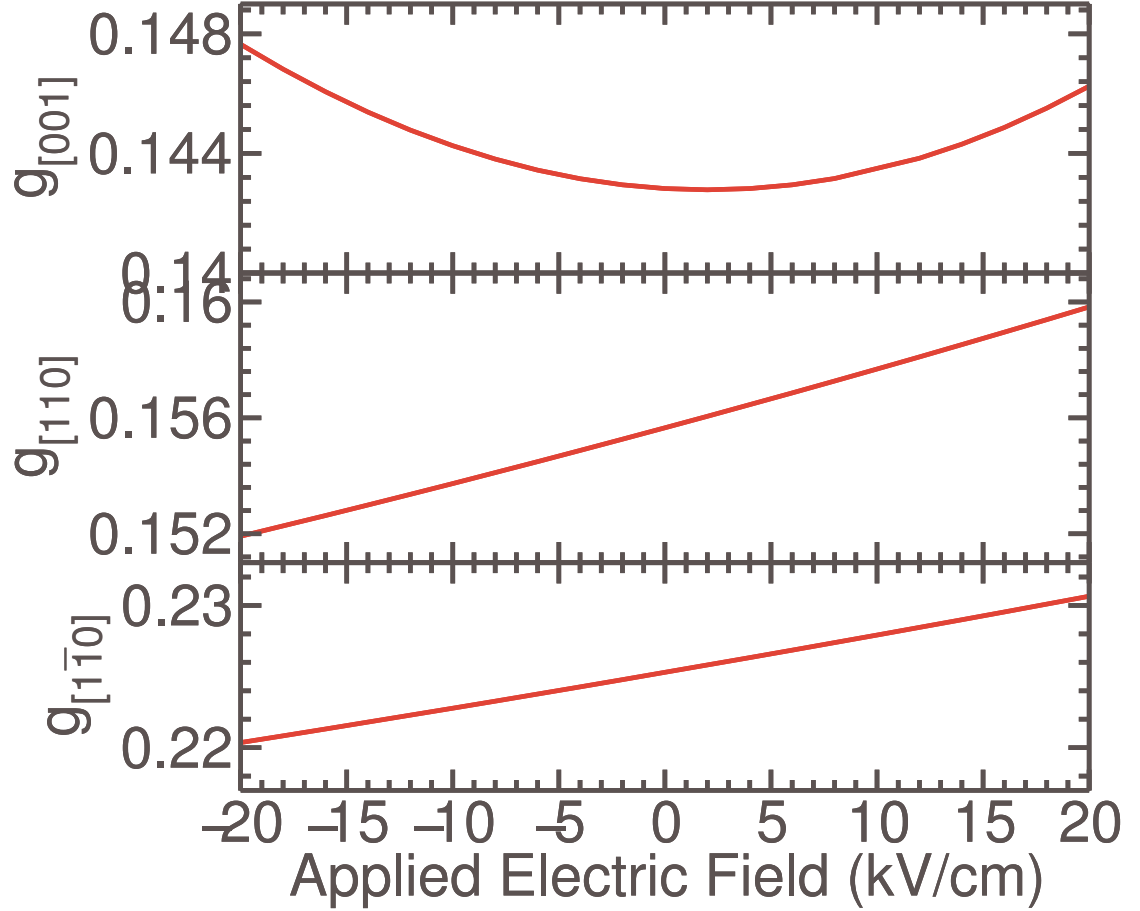


Figure 32: Hole g-tensor at 7T as a function of electric field in the range ± 20 kV/cm. For a magnetic field applied along the growth direction, the electron g-factor dominates the hole g-factor. The opposite is true, however, for a magnetic field applied in-plane (i.e. along $[110]$ or $[1\bar{1}0]$). (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 6.2 nm, $l_{[1\bar{1}0]} = 29.6$ nm, $l_{[110]} = 20$ nm, and energy gap $E_g = 1.252$ eV. Calculations performed at 0 K.)

IV.3 Comparison with previous research

Much less has been done with holes in quantum dots than with electrons. This is likely due to a perceived short lifetime, which is based upon the extremely short bulk spin lifetimes. In quantum dots, however, spin lifetimes are greatly increased.[97]

As with the electron calculations, the hole calculations here show distinct similarities with the hole g-factors calculated in Pryor and Flatté's paper.[92] In particular,

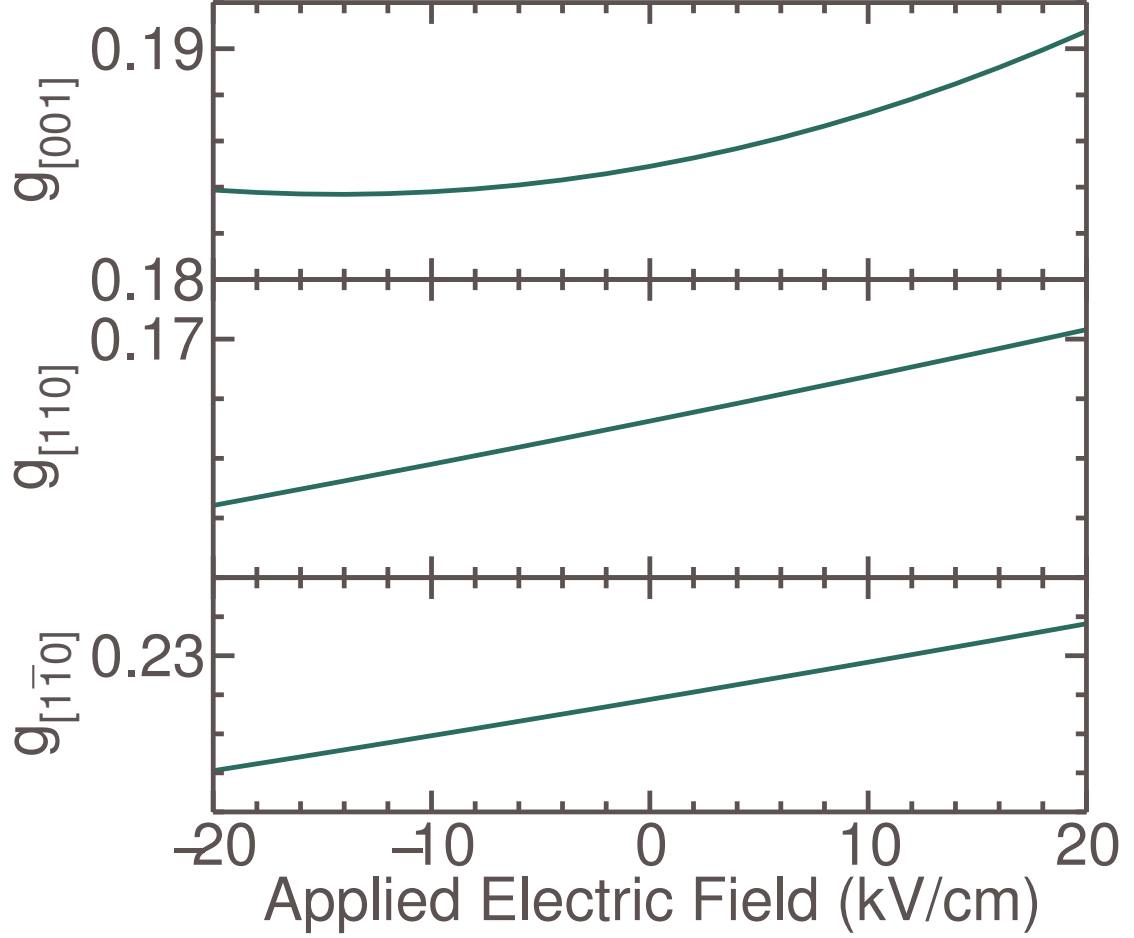


Figure 33: Hole g-tensor at 1T as a function of electric field in the range ± 20 kV/cm. For a magnetic field applied along the growth direction, the electron g-factor dominates the hole g-factor. The opposite is true, however, for a magnetic field applied in-plane (i.e. along $[110]$ or $[1\bar{1}0]$). (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 5.1 nm, $l_{[1\bar{1}0]} = 32.0$ nm, $l_{[110]} = 20.0$ nm, and energy gap $E_g = 1.256$ eV. Calculations performed at 0 K.)

the ellipticity dependence is quite remarkable both there and here. As they did with holes, Andlauer et al.[47] calculates the electric field g-factor dependence of holes in their double-dot truncated pyramidal structure. The g-factors they found behave quite differently, due to the double-dot structure.

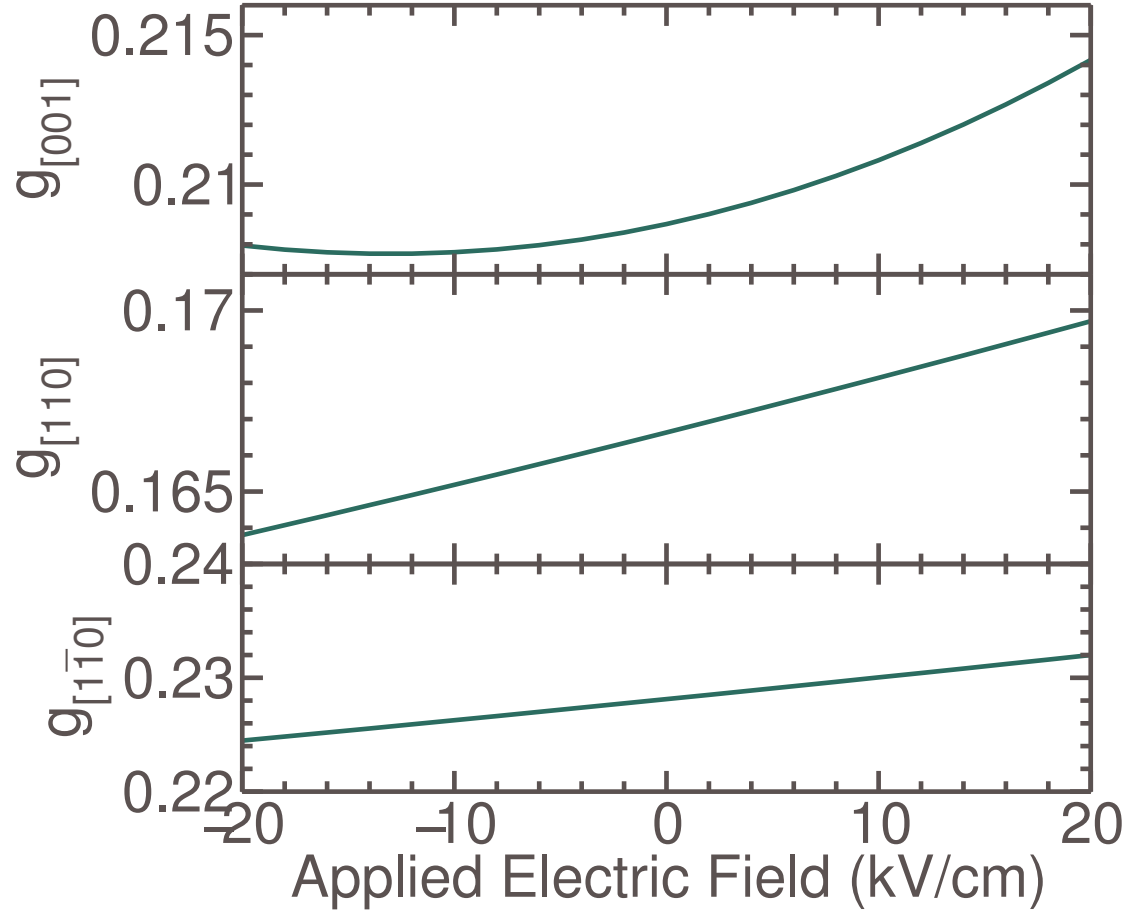


Figure 34: Hole g-tensor at 1T as a function of electric field in the range ± 20 kV/cm. For a magnetic field applied along the growth direction, the electron g-factor dominates the hole g-factor. The opposite is true, however, for a magnetic field applied in-plane (i.e. along $[110]$ or $[1\bar{1}0]$). (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 5.1 nm, $l_{[1\bar{1}0]} = 32.0$ nm, $l_{[110]} = 20.0$ nm, and energy gap $E_g = 1.256$ eV. Calculations performed at 0K.)

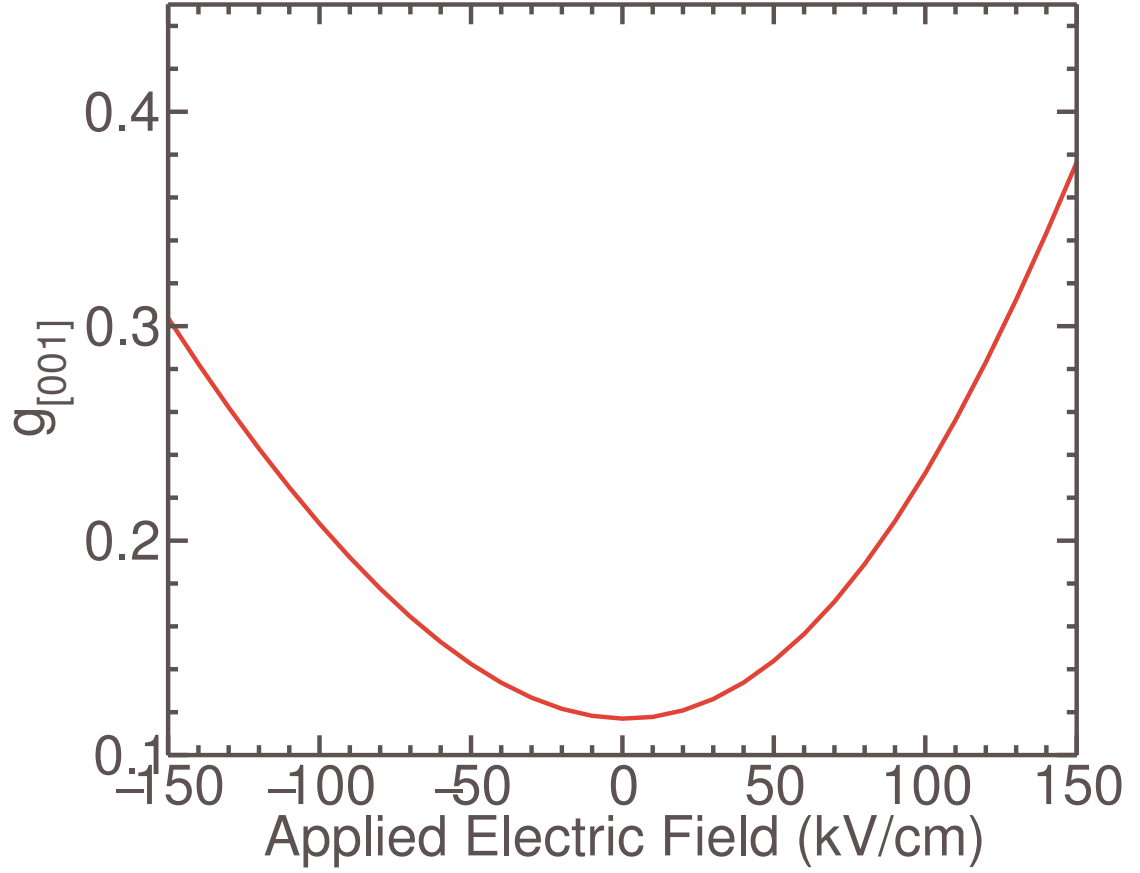


Figure 35: Hole g-tensor along the growth direction ($[001]$) as a function of electric field in the range ± 150 kV/cm, with an applied magnetic field strength of 1T. At high electric field magnitude, the g-factor more than doubles its 0-field value. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 6.2nm, $l_{[1\bar{1}0]} = 29.6$ nm, $l_{[110]} = 20$ nm, and energy gap $E_g = 1.252$ eV. Calculations performed at 0K.)

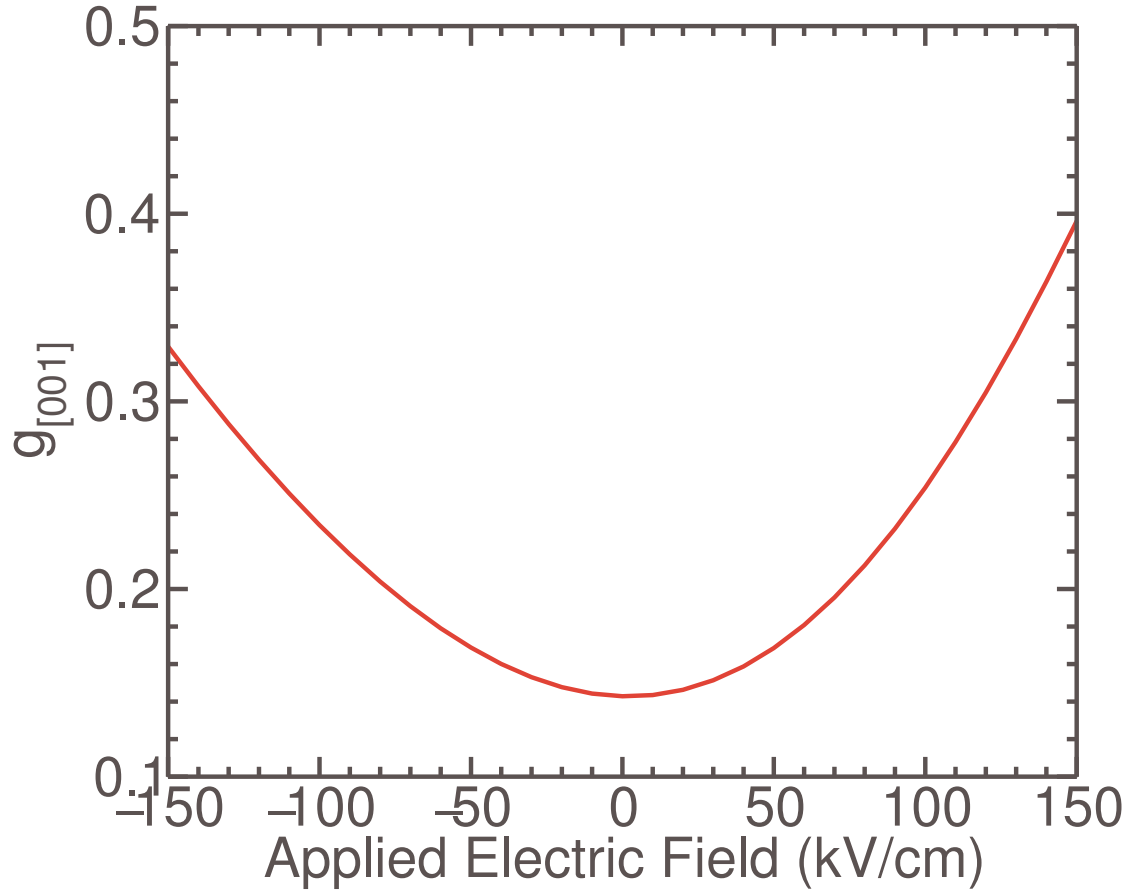


Figure 36: Hole g-tensor along the growth direction ([001]) as a function of electric field in the range ± 150 kV/cm, with an applied magnetic field strength of 7T. At high electric field magnitude, the g-factor more than doubles its 0-field value. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 6.2nm, $l_{[1\bar{1}0]} = 29.6$ nm, $l_{[110]} = 20$ nm, and energy gap $E_g = 1.252$ eV. Calculations performed at 0K.)

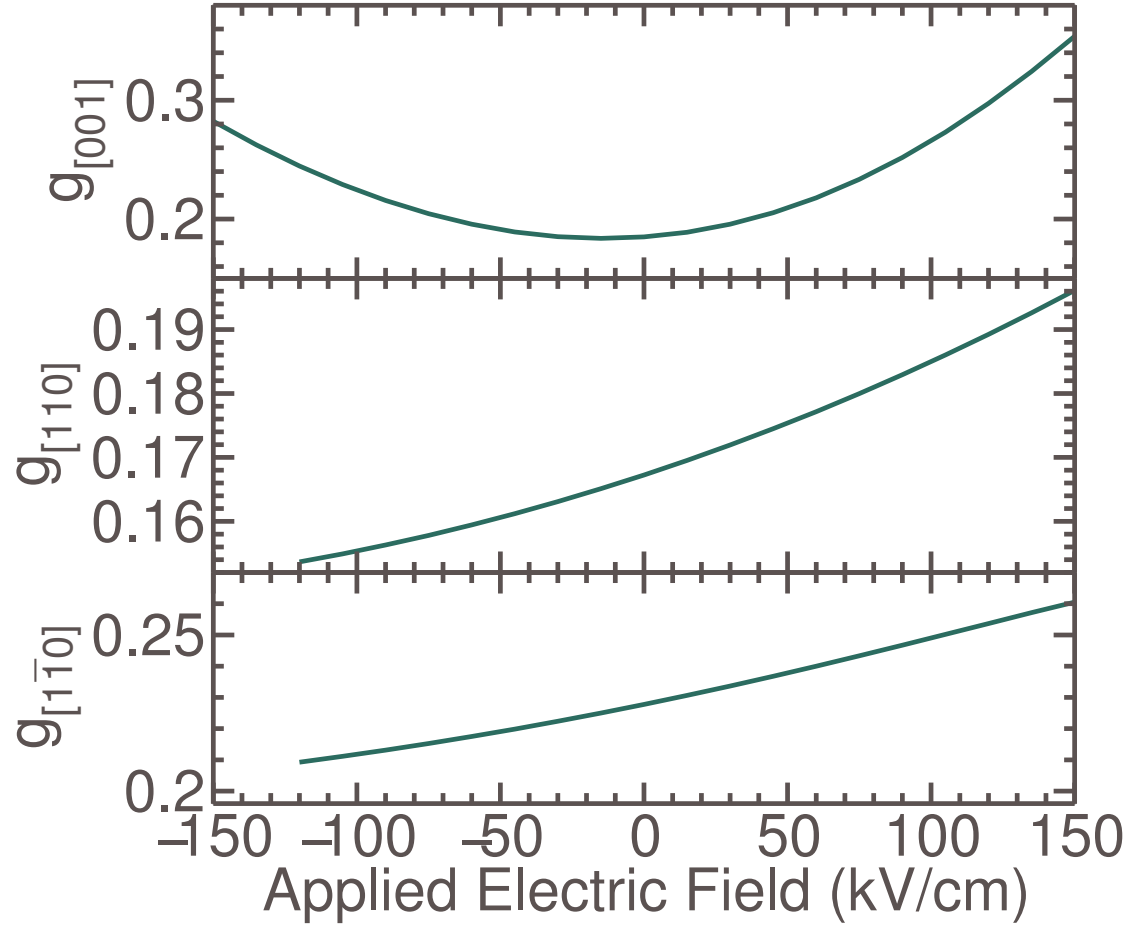


Figure 37: Hole g-tensor as a function of electric field in the range ± 150 kV/cm at a magnetic field strength of 1T. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 5.1nm, $l_{[1\bar{1}0]} = 32.0$ nm, $l_{[110]} = 20.0$ nm, and energy gap $E_g = 1.256$ eV. Calculations performed at 0K.)

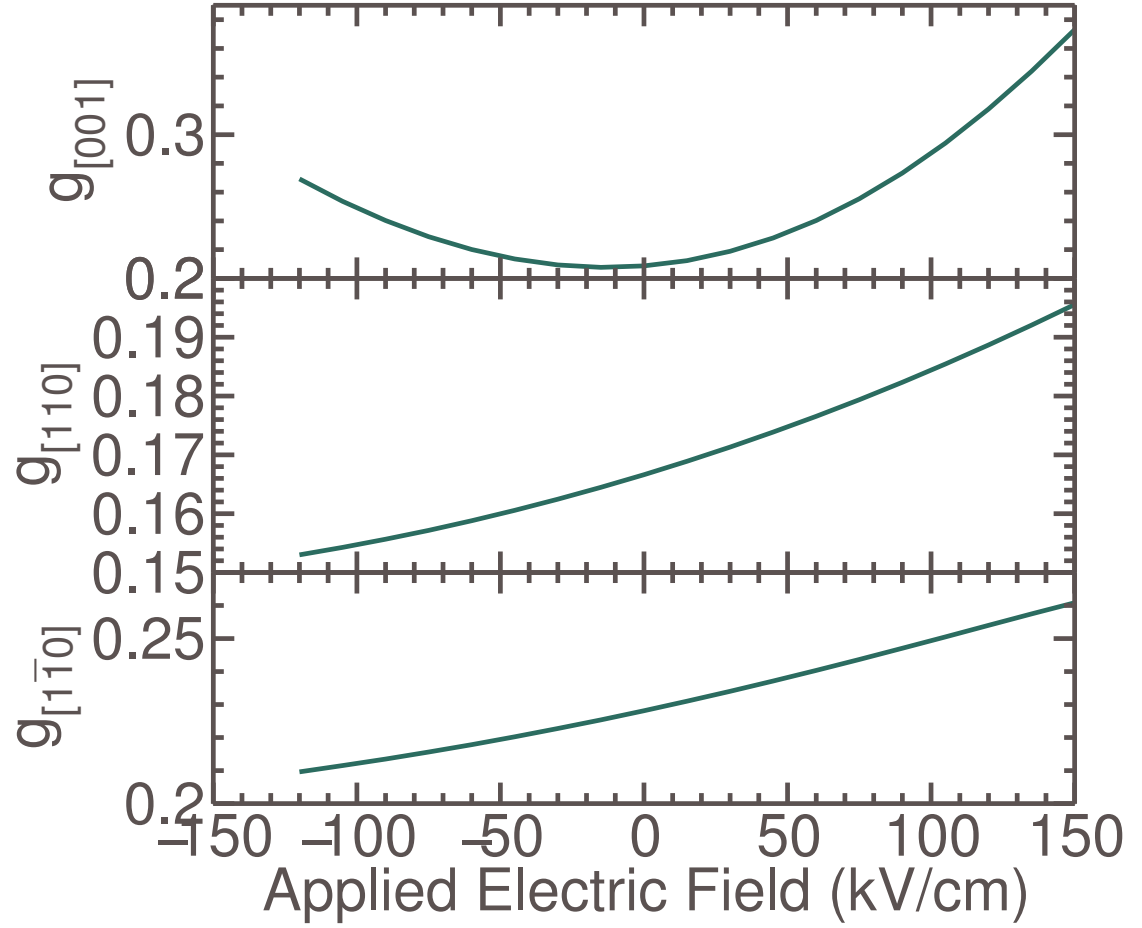


Figure 38: Hole g-tensors as a function of electric field in the range ± 150 kV/cm at a magnetic field strength of 7T. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 5.1nm, $l_{[1\bar{1}0]} = 32.0$ nm, $l_{[110]} = 20.0$ nm, and energy gap $E_g = 1.256$ eV. Calculations performed at 0K.)

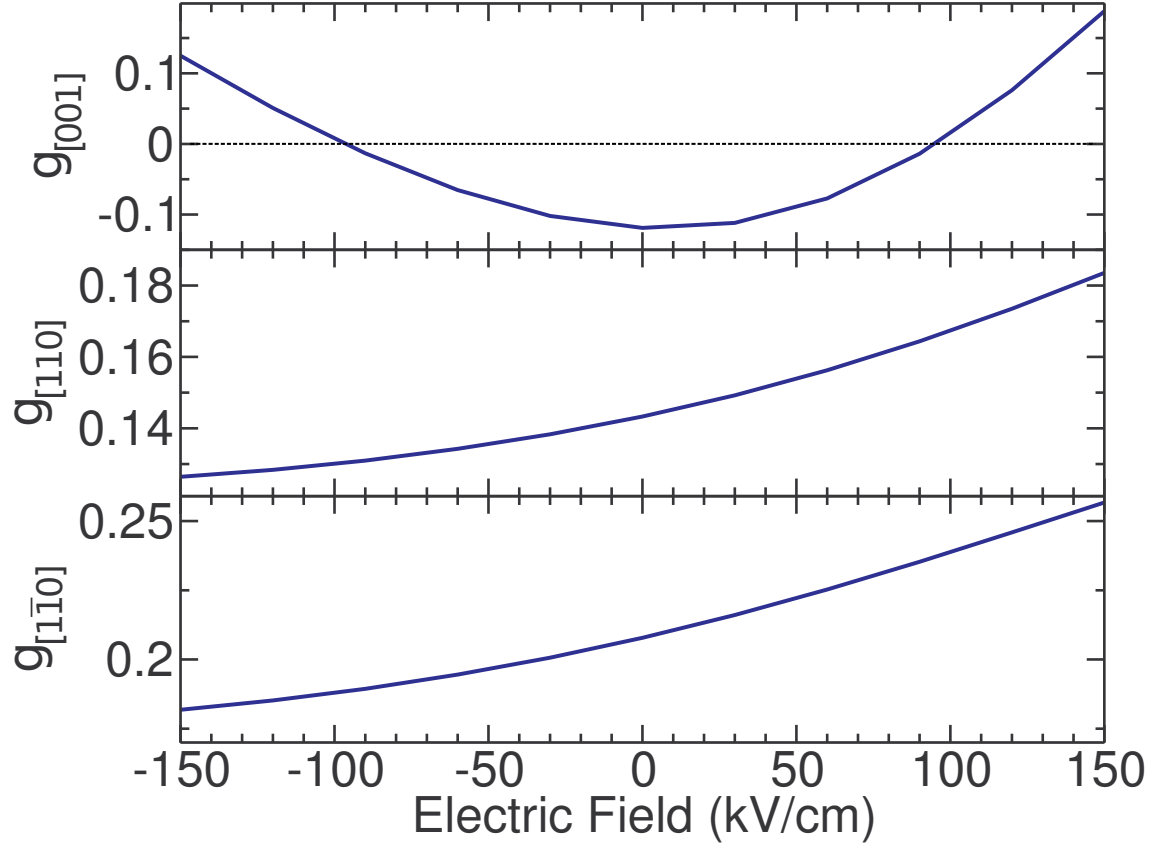


Figure 39: g-tensor of the uppermost hole state in an $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ dot as a function of electric field at 0K. Dot has a height of 6.2nm, footprint length along the $[110]$ direction of 21.6nm, and footprint length along the $[1\bar{1}0]$ direction of 32.8nm, giving it an ellipticity of 1.5. Of particular note is the sign change of the g-factor along the $[001]$ direction.

CHAPTER V

THE EXCITONIC G-TENSOR IN AN ISOLATED QUANTUM DOT

V.1 Experimental Work

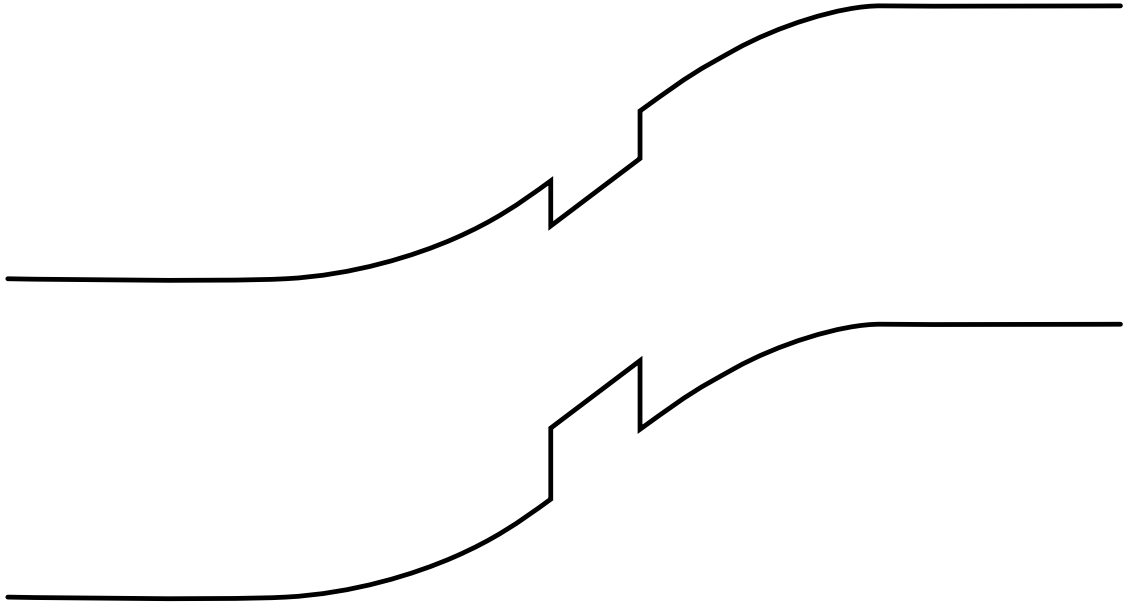


Figure 40: Illustration of a p-i-n diode. Positively and negatively-doped materials on the left and right (respectively) sandwich an intrinsic region in the middle. The intrinsic region contains a quantum dot. Because of the positively- and negatively-doped regions, an electric field forms across the intrinsic region (center), which interacts with carriers contained within the quantum dot.

Recently, the Koenraad group at the Eindhoven University of Technology has investigated the electric field dependence of excitonic g-factors in $\text{In}_x\text{Ga}_{1-x}\text{As}/\text{GaAs}$ quantum dots.[2][3][98] In their work, quantum dots were embedded in a diode with a p-i-n structure. That is, the diode consisted of a positively-doped region and

a negatively-doped region, which sandwiched an intrinsic region in the center (see figure 40).

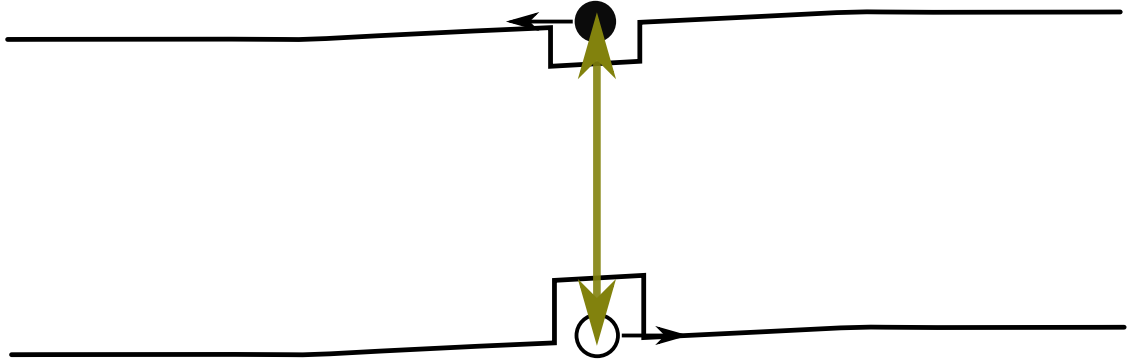


Figure 41: Illustration of a p-i-n diode with optical carrier injection. Positively and negatively-doped materials on the left and right (respectively) sandwich an intrinsic region in the middle. Laser excitations (yellow line) at the gap frequency $E_g/\hbar$ excite an electron (solid circle) from the heavy hole to the conduction state, leaving behind a hole (empty circle). The electron and hole move under the influence of the electric field (black arrows), but themselves counter the electric field while they remain in the intrinsic region. This may reduce or eliminate the electric field across the dot, depending on the number of carriers excited, which in turn depend on the power of the laser (i.e. the number of photons per unit time).

Their measurements were conducted at a temperature of 1.7K and with an a magnetic field of 7T applied along the growth direction ($[001]$). The p-i-n diode had a built-in electric field of 240kV/cm across the intrinsic region. Recombining electrons

and holes in excitonic pairs emitted circularly polarized light. The polarization is plotted in figure 42 and corresponds to the exciton polarization. A sign change in the circular polarization indicates a sign change in the exciton g-factor, as the exciton populates the lowest-energy state.

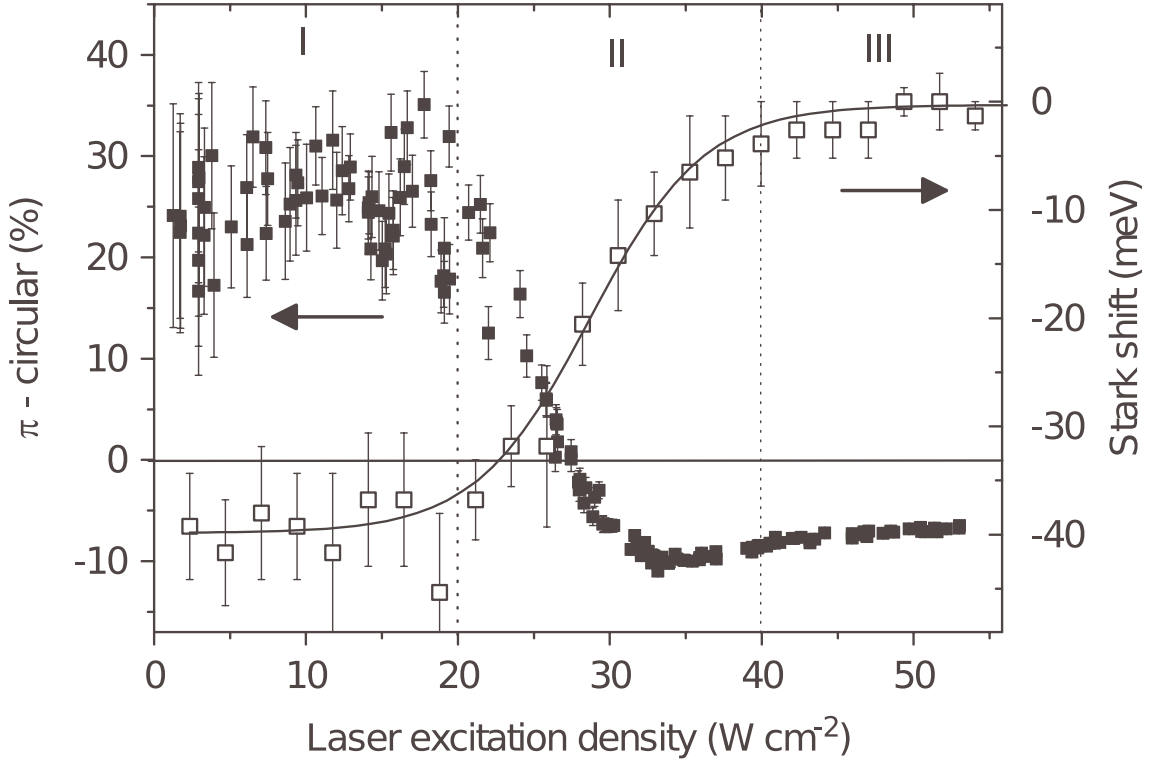


Figure 42: Circular polarization (solid squares) and Stark shift (empty squares) of excitons in an InGaAs/GaAs quantum dot as a function of applied laser power density. Of note is the circular polarization, which changes sign at approximately 25 W cm^{-2} . (Figure from G. W. W. Quax *et al.*, Physica E **40** 1832 (2008)[2])

Because the “excitonic g-factor” is defined as

$$g_X = \frac{E_{\sigma_+} - E_{\sigma_-}}{\mu_B B} \quad (155)$$

(where σ_i is the circular polarization of the light emitted by electron-hole recomb-

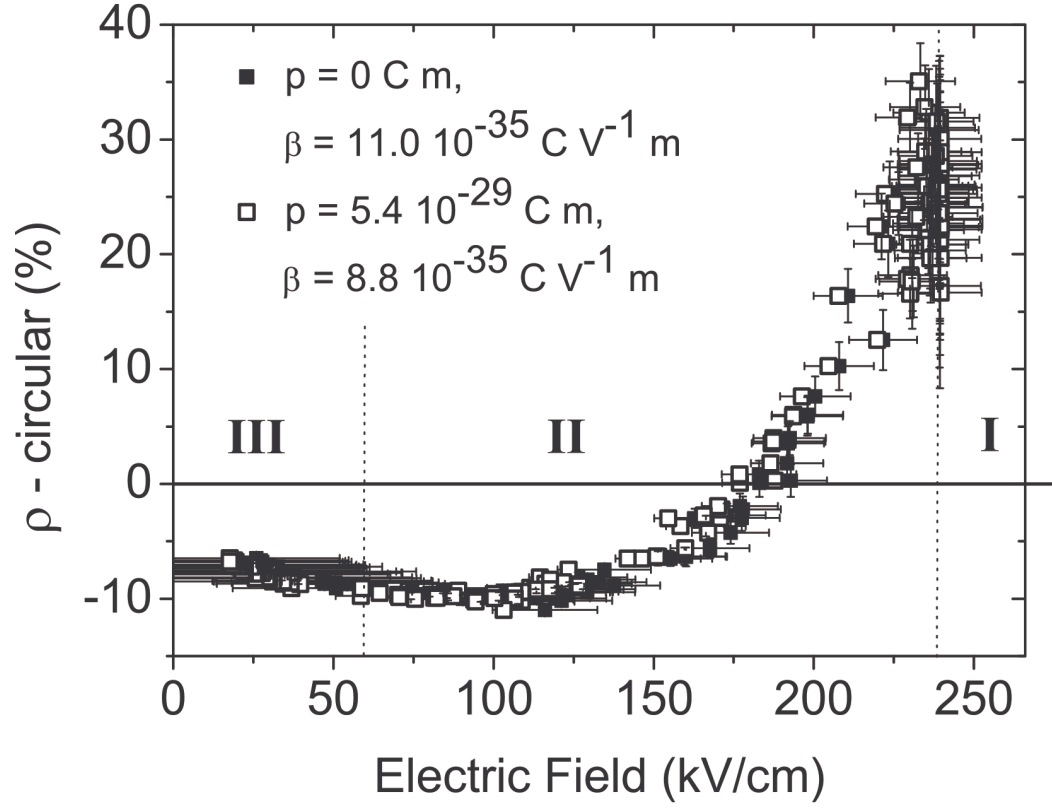


Figure 43: Circular polarization as a function of electric field, as determined from Stark shift information, the optical excitation density, and two different models (one with and the other without a built-in dipole). (Figure from the dissertation of G. W. W. Quax[3])

nation, and the electron spin splitting in the valence and conduction band is

$$E_{\sigma_+} = E_{c\downarrow} - E_{v\downarrow} \quad (156)$$

$$E_{\sigma_-} = E_{c\uparrow} - E_{v\uparrow} \quad (157)$$

(in the electron picture) the excitonic g-factor can be written as

$$\begin{aligned}
 g_X &= \frac{E_{c\downarrow} - E_{v\downarrow} - E_{c\uparrow} + E_{v\uparrow}}{\mu_B B} \\
 g_X &= \frac{E_{c\downarrow} - E_{c\uparrow}}{\mu_B B} + \frac{E_{v\uparrow} - E_{v\downarrow}}{\mu_B B} \\
 g_X &= -\frac{E_{c\uparrow} - E_{c\downarrow}}{\mu_B B} + \frac{E_{v\uparrow} - E_{v\downarrow}}{\mu_B B} \\
 g_X &= \frac{E_{v\uparrow} - E_{v\downarrow}}{\mu_B B} - \frac{E_{c\uparrow} - E_{c\downarrow}}{\mu_B B}
 \end{aligned}$$

$$g_X = g_{e,v} - g_{e,c} \quad (158)$$

again in the electron picture. The hole calculations, however, were performed in the hybrid picture, in which the sign of the energy is reversed. That is, the spin state parallel to the magnetic field which is the higher energy state (assuming a positive g-factor) in the electron picture becomes the lower energy state in the hole picture. However, the spin definitions also reverse, e.g. “up” becomes “down”. Because of this, a positive g-factor in the electron picture will remain a positive g-factor in the hole picture, and no further modification of the equation is needed with respect to that. Because the spin operator of the heavy hole state was defined as $\vec{S}_h = 3\vec{S}_e$, the equation must be modified slightly:

$$g_X = 3g_h - g_e \quad (159)$$

where g_h is the g-factor determined by the method in section II.3.

Now that the excitonic g-factor can be derived from the electron and hole g-factors, the issue then becomes finding a quantum dot which has an excitonic g-factor sign change, or which shows similar behavior, particularly at 0 electric field.

The Eindhoven group performed additional experiments on the dots under similar conditions, but with an in-plane (either along $[110]$ or $[1\bar{1}0]$; the direction was not recorded) magnetic field and using Kerr rotation instead of circular polarization of emitted light to measure spin polarization. The g-factors, attributed to valence electrons, were then plotted and appear in figure 44. From this experiment, they found that $|g_h(E = 0)| = 0.42 \pm 0.01$ and that the hole g-factor magnitude increases with increasing electric field (decreasing excitation power in figure 44).

V.2 Theoretical Methods

As in previous chapters, the dots were investigated theoretically using Craig Pryor's dotcode. Measurements on the dots were performed at Eindhoven and sent to the Flatté group at Iowa. Calculations with electric field were performed on the dots as described in previous chapters. Both electrons (chapter III) and holes (chapter IV) were included in the calculations, as excitonic g-factors were being detected by the Eindhoven group.

Because of the lower confinement in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ dots, electron g-factors were limited to $\pm 20\text{kV}/\text{cm}$. Hole g-tensors, due to the greater hole than electron confinement within the dot, were able to be calculated over a range of $\pm 150\text{kV}/\text{cm}$. Additionally, due to magnetic field orientation limitations in the experiments, some

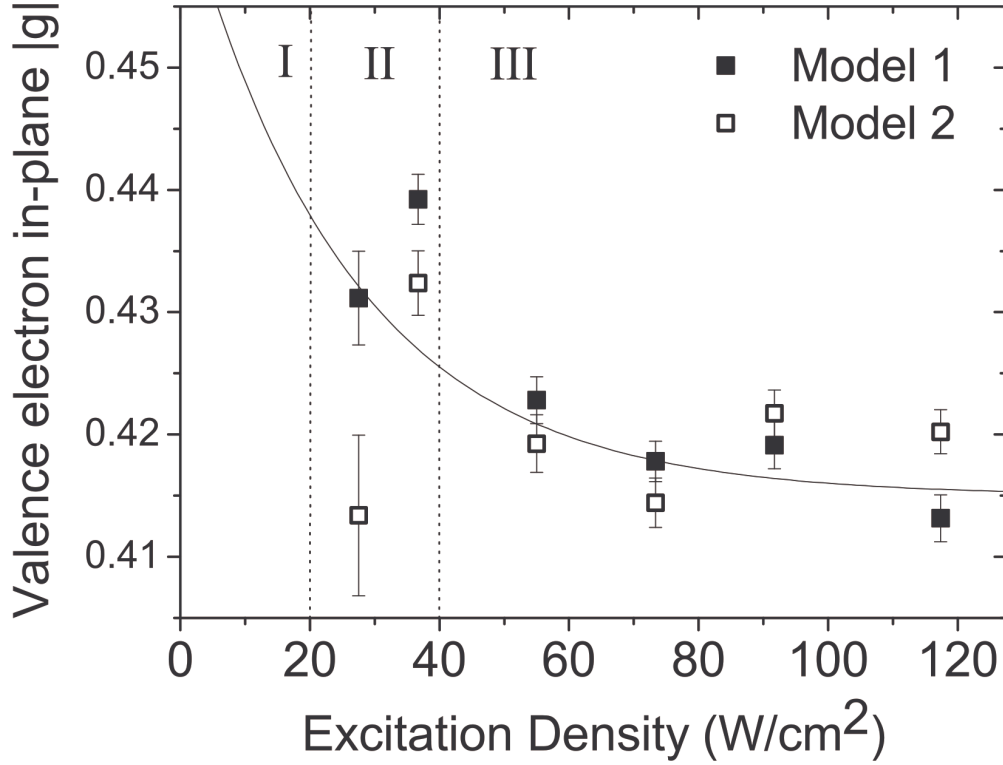


Figure 44: In-plane g-factor magnitude as a function of excitation density (that is, electric field, as described previously). High excitation densities imply low magnetic field; low densities imply high electric field. Two models were used to determine the g-factors (solid or empty squares), although that detail is outside the context of the current work. (Figure from the dissertation of G. W. W. Quax[3])

calculations involving only holes were performed only with the magnetic field oriented along the [001] axis.

V.3 Results

The Eindhoven group found the dots to be lens-shaped, with a height of 6.7 ± 1.5 nm, average base diameter of 25 ± 5 nm, average elongation of 1.6 nm, and quantum dot density of $4.5 \times 10^{10} \text{ cm}^{-2}$. From this, an initial calculation was performed on an isolated quantum dot with a height of 6 nm, length along the [110] axis of 20 nm,

and a length along the $[1\bar{1}0]$ axis of 30nm. Calculations were then performed for electric fields between -20kV cm^{-1} and 20kV cm^{-1} , and magnetic field strengths of 1T. The results are plotted in figures 20 and 31. The calculations were performed again at 7T, and are plotted in figures 21 and 32. Finally, because of their excellent confinement, only the hole states were calculated at 1T and 7T at electric fields between -150kV cm^{-1} and 150kV cm^{-1} and are plotted in figures 35 and 36 (respectively). The exciton g-tensors at 1T and 7T were then calculated from the electron and hole g-tensors using equation 159, and is plotted in figures 45 and 46 (respectively).

A trend is visible in figures 20-36. The electron g-factor dominates the hole g-factor in the excitonic g-factor (equation 159) when the magnetic field is applied along the growth ($[001]$) direction. The hole, however, dominates the electron g-factor when the magnetic field is applied along an in-plane direction ($[110]$ and $[1\bar{1}0]$). Additionally, at 1T, the electron shows much less dependence on electric field within the $\pm 20\text{kV/cm}$ range, changing by roughly .2 percent (change of approximately 0.001) along $[001]$ contrasted with the change in the hole g-factor of 4 percent (change of approximately 0.005). In the in-plane directions, the electron changes by 3 percent ($[110]$, change of 0.002) and 7 percent ($[1\bar{1}0]$, change of 0.0015). For the same axes, the hole changes by 4 percent (change of 0.0075) and 4.3 percent (change of 0.01), respectively. The variation of the excitonic g-tensor at 1T is therefore mostly driven by the holes.

Because holes are more sensitive to anisotropy[99], it is reasonable to expect

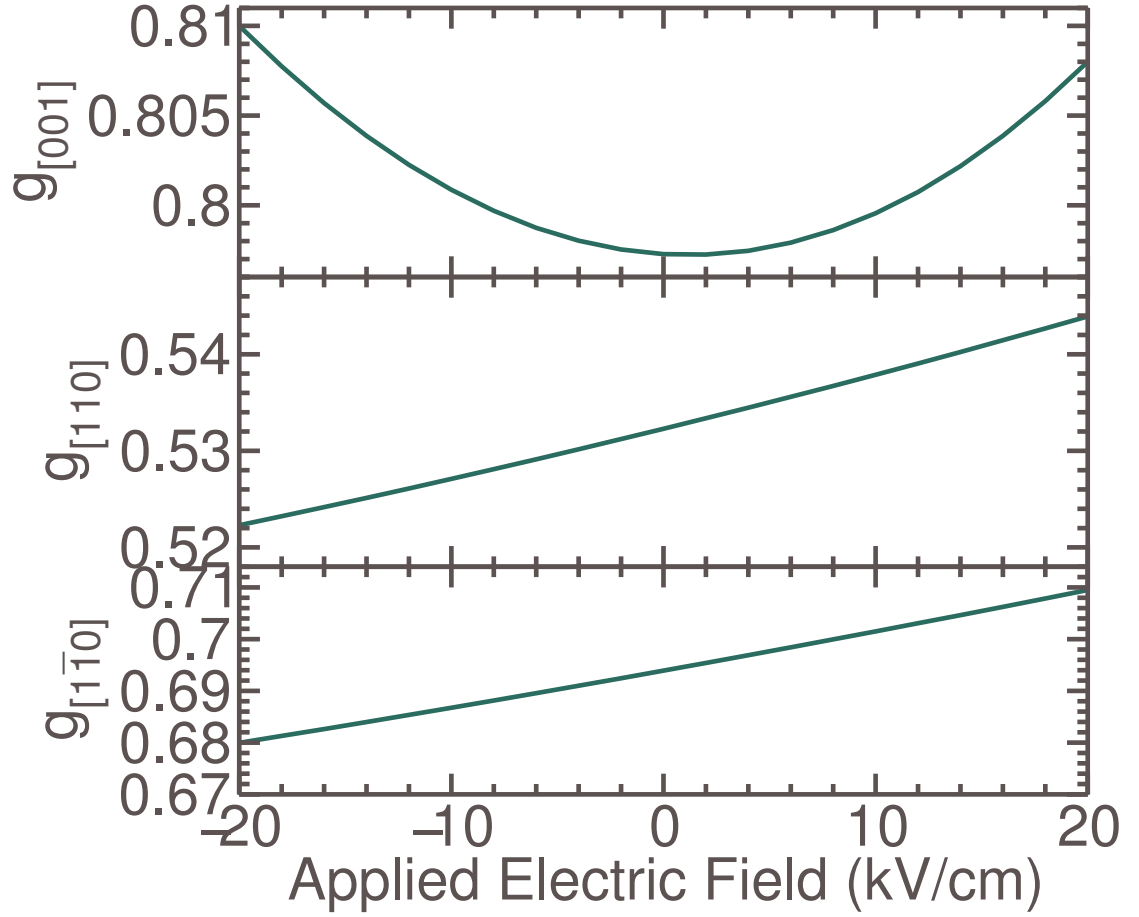


Figure 45: Excitonic g-tensor at 1T. No sign change is found within this range of electric field strengths. Electron (20) and hole (31) g-tensors at 7T as a function of electric field in the range ± 20 kV/cm. For a magnetic field applied along the growth direction, the electron g-factor dominates the hole g-factor. The opposite is true, however, for a magnetic field applied in-plane (i.e. along $[110]$ or $[1\bar{1}0]$). (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 6.2 nm, $l_{[1\bar{1}0]} = 29.6$ nm, $l_{[110]} = 20$ nm, and energy gap $E_g = 1.252$ eV. Calculations performed at 0K.)

the holes to be more sensitive to both higher electric fields and magnetic fields than electrons. At 7T, the electron g-factor along the growth direction ($[001]$) changed by 0.1 percent (change of 0.0006) compared to the holes' change of 4 percent (change of 0.006). In the in-plane directions, the electron g-factor changed by 3 percent ($[110]$, change of 0.002) and 8 percent ($[1\bar{1}0]$, change of 0.0015). For the same axes,

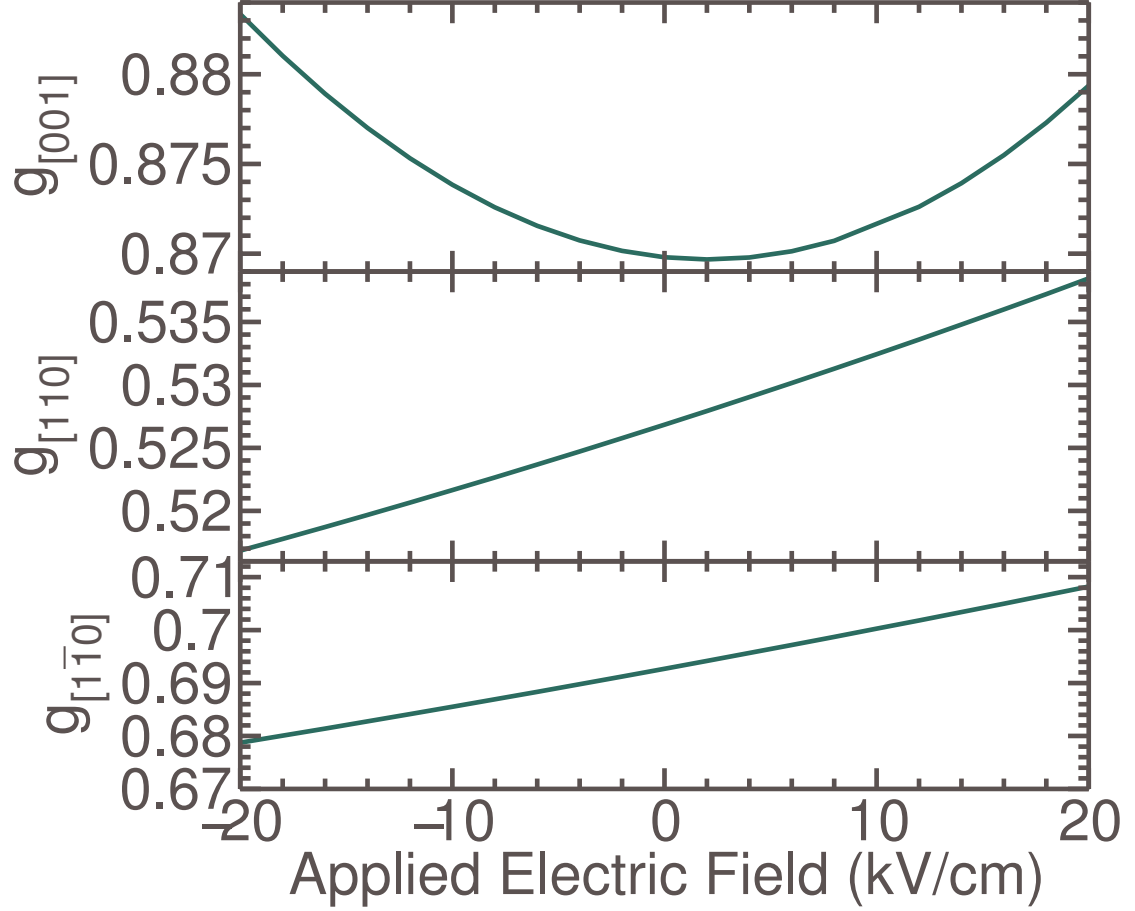


Figure 46: Excitonic g-tensor at 7T. No sign change is found within this range of electric field strengths. Electron (21) and hole (32) g-tensors at 7T as a function of electric field in the range ± 20 kV/cm. For a magnetic field applied along the growth direction, the electron g-factor dominates the hole g-factor. The opposite is true, however, for a magnetic field applied in-plane (i.e. along $[110]$ or $[1\bar{1}0]$). (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 6.2 nm, $l_{[1\bar{1}0]} = 29.6$ nm, $l_{[110]} = 20$ nm, and energy gap $E_g = 1.252$ eV. Calculations performed at 0 K.)

the holes changed by 5 percent (change of 0.008) and 4 percent (change of 0.01), respectively. The hole g-factors, then, are still driving the excitonic g-factor change.

The calculations were then compared against the values of E_g and the g-factor behavior that the Eindhoven group had obtained. Although agreement was good, it was determined that a closer fit would be possible if the dot were changed slightly.

A calculation of both a larger and smaller dot (1 gridsize in each dimension) were performed. It was then determined that a smaller and more elliptical dot would likely be the closest fit readily possible. The calculations performed on the original dot were then performed again on the smaller, more elliptical dot, and are plotted in figures 22, 33, 23, 34, 37, and 38. As with the original dot, the electron and hole g-tensors were used to calculate the exciton g-tensor as a function of electric field. These excitonic g-tensors at 1T and 7T are plotted in figures 47 and 48.

The new dot is shorter (5.1nm instead of 6.2nm) and more elliptical (ellipticity $\frac{d_{[1\bar{1}0]}}{d_{[110]}} = 1.6$ instead of 1.48). As seen in the previous chapters, increasing the ellipticity of a quantum dot has a larger effect on the hole g-factors than on the electron g-factors.

At 1T, the growth-direction ([001]) electron g-factor changed by approximately 1 percent (change of 0.0006) compared with the hole g-factor change of approximately 3 percent (change of 0.006). That is, the disparity between the electron and hole g-factor changes has approximately doubled from the original change of 0.0001 for electrons versus 0.0005 for holes. In the in-plane direction, the contrast is even more stark. The in-plane electron g-factor changes by 6 percent ([110], change of 0.001) and 2 percent ([1 $\bar{1}$ 0], change of 0.0006). The holes, on the other hand, changed by 5 percent (change of 0.008) and 4 percent (change of 0.01), respectively. The exciton g-factor is clearly driven almost exclusively by the holes.

At 7T, the picture is similar. Along the growth direction, the electrons change by 0.2 percent (change of 0.0001) compared to the change in the hole g-factor of 3

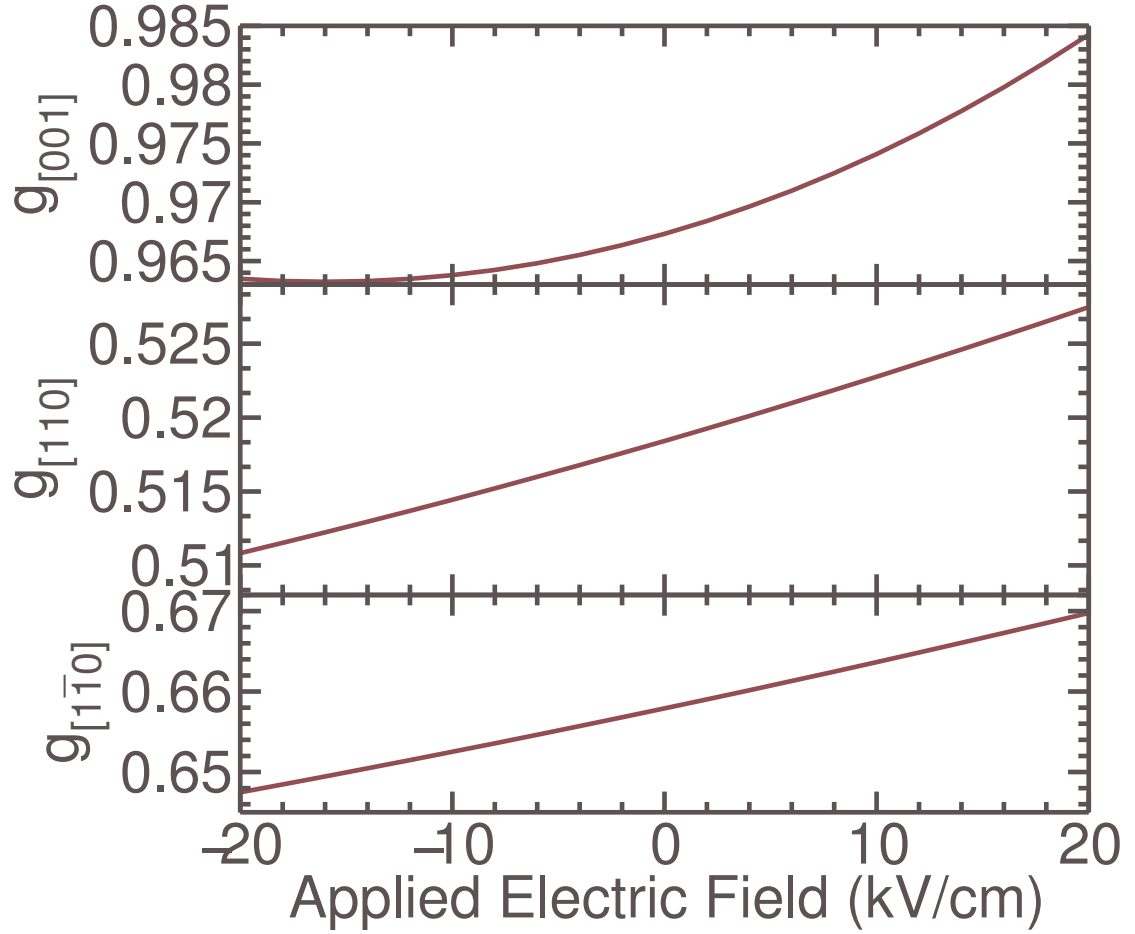


Figure 47: Excitonic g-tensor as a function of electric field in the range $\pm 20 \text{ kV/cm}$ at a magnetic field strength of 1T. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 5.1nm, $l_{[1\bar{1}0]} = 32.0 \text{ nm}$, $l_{[110]} = 20.0 \text{ nm}$, and energy gap $E_g = 1.256 \text{ eV}$. Calculations performed at 0K.)

percent (change of 0.006). Electron in-plane electron g-factor changes were approximately 7 percent ($[110]$, change of 0.001) and 2 percent (change of 0.0005). Hole in-plane g-factors were 4 percent (change of 0.006) and 3 percent (change of 0.008).

Unfortunately, the range over which the Iowa group was able to reliably calculate electron g-factors was quite limited—only 40 kV/cm . The trends from the electrons and holes provide information on how the electrons and holes may be expected to

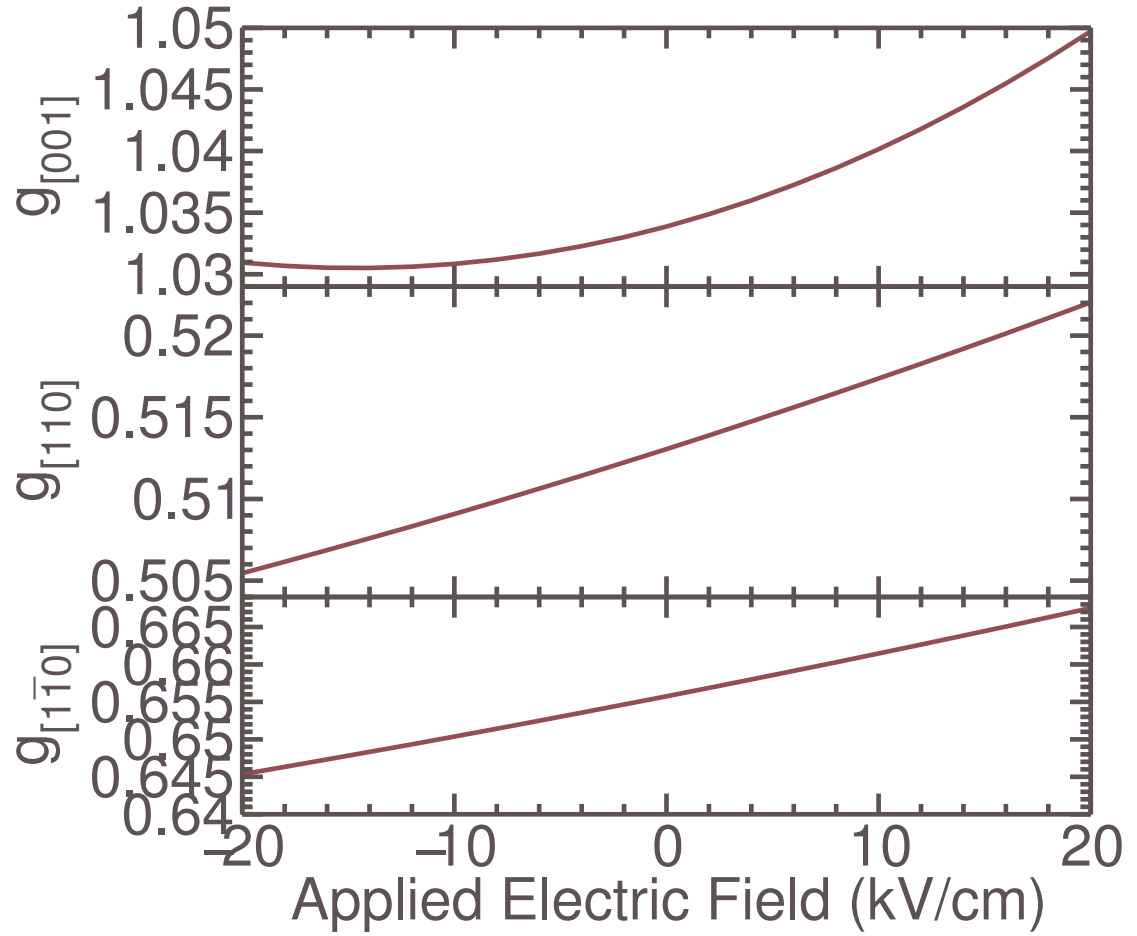


Figure 48: Excitonic g-tensor as a function of electric field in the range ± 20 kV/cm at a magnetic field strength of 7T. (Dot is $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}$ in GaAs with height of 5.1nm, $l_{[1\bar{1}0]} = 32.0$ nm, $l_{[110]} = 20.0$ nm, and energy gap $E_g = 1.256$ eV. Calculations performed at 0K.)

behave at higher electric fields. The ellipticity found by the Eindhoven group (1.6) corresponds to the “very elliptical” dots of the previous sections, and has a particularly large effect on the hole g-factors. Electron g-factors make up the majority of the excitonic growth-direction ($[001]$) g-factor, although the holes provide the dominant contribution to the in-plane excitonic g-factors. The holes have been found to dominate the excitonic g-factor changes, and are likely responsible for the sign

change seen in the Eindhoven results.

These results were used in the Eindhoven group's work to conclude that the valence electrons were being measured in their Kerr rotation experiments. The Iowa group's calculations agree with the Eindhoven group's findings of an increasing in-plane g-factor magnitude with increasing magnetic field. Unfortunately, sign information was not able to be obtained to further corroborate the determination that the in-plane hole g-factor was being measured. The hole g-factors found by the Iowa group are still a factor of two too small to agree quantitatively with the Eindhoven group's g-factor magnitude found by Kerr rotation.

Calculations were performed on a number of quantum dots, although only the above two were calculated with electric field. The other dot information was used to provide guidance for further modifications to the quantum dot dimensions. They are reproduced here for reference. The electron (tables 5 and 6 for 1T and 7T, respectively) and hole (tables 7 and 8 for 1T and 7T, respectively) data have been provided in sections III and IV, respectively.

The following tables summarize the exciton g-tensors without applied electric field at magnetic field strengths of both 1T and 7T.

It was unfortunate that the Iowa group was not able to perform calculations over the entire range of electric fields that the Eindhoven group were able to create and use. The dots were able to provide guidance to the Eindhoven group's work, however.

E_g (eV)	$d_{[001]}$	$d_{[110]}$	$d_{[1\bar{1}0]}$	e	$g_{X^0,[001]}$	$g_{X^0,[110]}$	$g_{X^0,[1\bar{1}0]}$
1.252	6.22nm	20.0nm	29.6nm	1.48	.797	.531	.6946
1.256	5.09nm	20.0nm	32.0nm	1.60	.968	.5177	.6586
1.256	5.65nm	20.0nm	32.0nm	1.60	.933	.5367	.6823
1.271	5.65nm	16.8nm	27.2nm	1.62	1.8392	.6408	.8082
1.266	7.35nm	16.8nm	27.2nm	1.62	1.601	.886	1.0858
1.266	5.65nm	18.4nm	28.8nm	1.56	1.526	.5979	.757

Table 9: Summary of neutral exciton g-tensors at 1T without applied electric field;
 $e = \frac{d_{[1\bar{1}0]}}{d_{[110]}}$.

E_g (eV)	$d_{[001]}$	$d_{[110]}$	$d_{[1\bar{1}0]}$	e	$g_{X^0,[001]}$	$g_{X^0,[110]}$	$g_{X^0,[1\bar{1}0]}$
1.252	6.22nm	20.0nm	29.6nm	1.48	.870	.5279	.6918
1.256	5.09nm	20.0nm	32.0nm	1.60	1.035	.5142	.6554
1.271	5.65nm	16.8nm	27.2nm	1.62	1.883	.6336	.8043

Table 10: Summary of neutral exciton g-tensors at 7T without applied electric field;
 $e = \frac{d_{[1\bar{1}0]}}{d_{[110]}}$.

CHAPTER VI

G-TENSOR MODULATION RESONANCE

Resonant manipulation and calculating Rabi times using standard electron spin resonance have been discussed in section I.7.2. A logical continuation of the work in chapters III and IV would be to apply the electric-field-dependent g-tensors to do g-tensor modulation resonance.

VI.1 g-tensor modulation resonance with electron spin

Electron spin resonance would work as a method of spin manipulation, but it has some significant drawbacks. Namely, magnetic fields tend to affect a large area and be slower to respond than an electric field. Therefore, the ability to use a static magnetic field in conjunction with an oscillating *electric* field would be ideal, particularly in quantum dots, where only a few or even a single electron are present. Several methods have been proposed to use a dynamic electric field and static magnetic field, including using fringe magnetic fields[100] or hyperfine gradients[101]. Another promising avenue of research has been using the direction-dependent g-factors of semiconductor nanostructures to control the electron coupling to the magnetic field[102, 85, 103, 104, 105, 106, 107, 96] by coupling to the spin through the spin-orbit interaction. A paper in 2003 by Kato et al.[106] experimentally used a g-factor to perform electron spin resonance with a static magnetic field and an oscillating electric field. This method of manipulating the g-tensor with an electric field to perform ESR is called “g-Tensor Modulation Resonance” or “g-TMR”.

Both g-TMR as well as conventional ESR operate by changing $\vec{\Omega}$. ESR changes

$\vec{\Omega}$ by application of a rotating or oscillating magnetic field transverse to the static magnetic field while possessing a static g-factor; g-TMR changes $\vec{\Omega}$ by keeping the static magnetic field and possessing a changing g-factor. This concept-changing $\vec{\Omega}$ -is illustrated in the figure 49, which is based on a quantum dot to be discussed momentarily.

As defined in equation 90, the axis about which the spin precesses ($\vec{\Omega}$) depends upon both the g-tensor and the static magnetic field:

$$\vec{\Omega} = \begin{pmatrix} g_{[001]} & 0 & 0 \\ 0 & g_{[110]} & 0 \\ 0 & 0 & g_{[1\bar{1}0]} \end{pmatrix} \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} \quad (160)$$

$$(161)$$

As discussed in section III, the electron g-tensor depends sensitively upon the applied electric field. Assuming a linear relation between applied electric field and g-tensor gives

$$\vec{\Omega} = \begin{pmatrix} g_{[001],0} + E(t)m_{[001]} & 0 & 0 \\ 0 & g_{[110],0} + E(t)m_{[110]} & 0 \\ 0 & 0 & g_{[1\bar{1}0],0} + E(t)m_{[1\bar{1}0]} \end{pmatrix} \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} \quad (162)$$

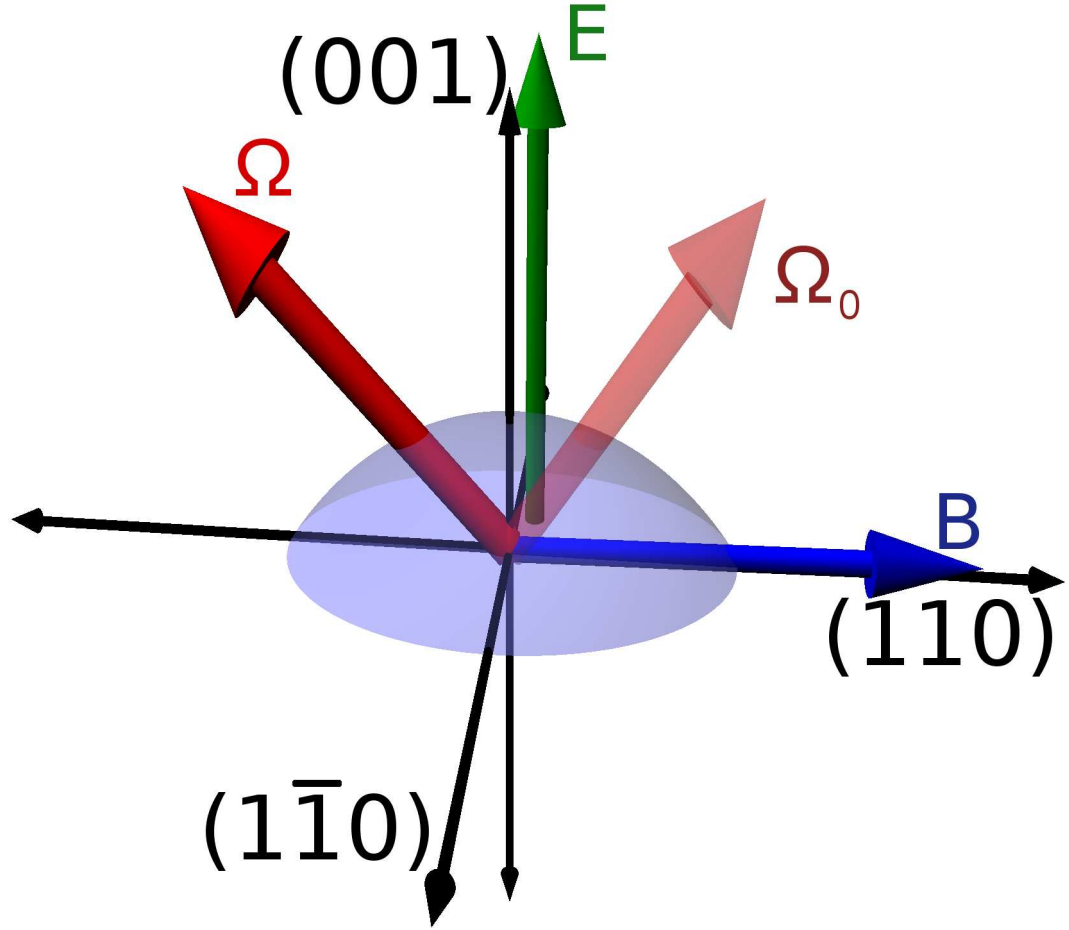


Figure 49: Illustration of g-TMR in a quantum dot being investigated. The blue vector represents the static magnetic field, the green vector represents the electric field, and the red vector represents $\vec{\Omega}$. The faint red vector represents $\vec{\Omega}$ without an electric field applied; the solid red vector is $\vec{\Omega}$ with an electric field applied. The application of the electric field causes $\vec{\Omega}$ to swing 90 degrees.

$$\vec{\Omega} = \left[\begin{pmatrix} g_{[001],0} & 0 & 0 \\ 0 & g_{[110],0} & 0 \\ 0 & 0 & g_{[1\bar{1}0],0} \end{pmatrix} + E(t) \begin{pmatrix} m_{[001]} & 0 & 0 \\ 0 & m_{[110]} & 0 \\ 0 & 0 & m_{[1\bar{1}0]} \end{pmatrix} \right] \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} \quad (163)$$

$$\vec{\Omega} = \begin{pmatrix} g_{[001],0} & 0 & 0 \\ 0 & g_{[110],0} & 0 \\ 0 & 0 & g_{[1\bar{1}0],0} \end{pmatrix} \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} + E(t) \begin{pmatrix} m_{[001]} & 0 & 0 \\ 0 & m_{[110]} & 0 \\ 0 & 0 & m_{[1\bar{1}0]} \end{pmatrix} \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} \quad (164)$$

To achieve an oscillating transverse component of $\vec{\Omega}(t)$, an oscillating electric field need be applied.

$$\begin{pmatrix} g_{[001],0} & 0 & 0 \\ 0 & g_{[110],0} & 0 \\ 0 & 0 & g_{[1\bar{1}0],0} \end{pmatrix} \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} + E \cos(\omega t) \begin{pmatrix} m_{[001]} & 0 & 0 \\ 0 & m_{[110]} & 0 \\ 0 & 0 & m_{[1\bar{1}0]} \end{pmatrix} \cdot \begin{pmatrix} B_{[001]} \\ B_{[110]} \\ B_{[1\bar{1}0]} \end{pmatrix} \quad (165)$$

$$= \vec{\Omega}_0 + \cos(\omega t) \vec{\Omega}_t \quad (166)$$

Because $\vec{\Omega}_t$ need not only oscillate transverse to $\vec{\Omega}_0$, $\vec{\Omega}(t)$ must be decomposed into components parallel and perpendicular to the “z” axis, which lies along $\vec{\Omega}_0$. This is in contrast to the standard ESR method where there is a static and oscillating applied magnetic field, readily decomposing into parallel and perpendicular components.

To start off, the $\hat{z}$ axis is defined using

$$\hat{z} := \frac{\vec{\Omega}_0}{|\vec{\Omega}_0|} \quad (167)$$

Note that if the electric field is chosen to oscillate around a point other than $E(t=0) = 0$, this static offset will be incorporated into $\vec{\Omega}_0$. One may use this to choose the

optimal region for g-TMR, or to point the static component of the spin precession axis in a certain direction.

With this choice of $\hat{z}$, the x-axis may be chosen (or, alternately, the y-axis). A logical choice is the maximal ($\cos(\omega t) = 1$) electric field oscillating component, i.e. $\vec{\Omega}_t$. However, $\vec{\Omega}_t$ may oscillate along $\vec{\Omega}_0$, so this component must be removed.

$$\vec{x} := \vec{\Omega}_t - \frac{\vec{\Omega}_t \cdot \vec{\Omega}_0}{|\vec{\Omega}_0|^2} \vec{\Omega}_0 \quad (168)$$

$$= \vec{\Omega}_t \left(\hat{\Omega}_0 \cdot \hat{\Omega}_0 \right) - \left(\vec{\Omega}_t \cdot \hat{\Omega}_0 \right) \hat{\Omega}_0 \quad (169)$$

$$= \hat{\Omega}_0 \times \left(\vec{\Omega}_t \times \hat{\Omega}_0 \right) \quad (170)$$

The unit vector is then

$$\hat{x} = \frac{\vec{x}}{|\vec{x}|} \quad (171)$$

$$= \frac{\hat{\Omega}_0 \times \left(\vec{\Omega}_t \times \hat{\Omega}_0 \right)}{|\hat{\Omega}_0 \times \left(\vec{\Omega}_t \times \hat{\Omega}_0 \right)|} \quad (172)$$

$$= \frac{\hat{\Omega}_0 \times \left(\vec{\Omega}_t \times \hat{\Omega}_0 \right)}{|\vec{\Omega}_t| |\sin \Theta|} \quad (173)$$

$$= \frac{1}{|\sin(\Theta)|} \left(\hat{\Omega}_0 \times \left(\vec{\Omega}_t \times \hat{\Omega}_0 \right) \right) \quad (174)$$

where Θ is the angle between spin precession axis components.

The $\hat{y}$ axis is then fixed, due to the right-handed coordinate system, to be

$$\hat{y} = \hat{z} \times \hat{x} \quad (175)$$

$$= \frac{1}{|\sin(\Theta)|} \left(\hat{\Omega}_o \times \left(\hat{\Omega}_0 \times \left(\hat{\Omega}_t \times \Omega_0 \right) \right) \right) \quad (176)$$

Because each cross product is perpendicular to the components of the constituent vectors, $\vec{\Omega}_t$ has no component along $\hat{y}$. Therefore, the portion of $\vec{\Omega}_t$ which is perpendicular to $\vec{\Omega}_0$ is *entirely* along $\hat{x}$. Consequently, $\Omega_{t,\perp}$ is simply $\vec{x}$ (which was previously defined as the maximal extent of $\vec{\Omega}_t$ perpendicular to $\hat{\Omega}_0$, or

$$\Omega_{t,\perp} = \hat{\Omega}_0 \times \left(\vec{\Omega}_t \times \hat{\Omega}_0 \right) \quad (177)$$

Then the following results:

$$\hat{x} = \frac{1}{|\sin(\Theta)|} \left(\hat{\Omega}_0 \times \left(\vec{\Omega}_t \times \hat{\Omega}_0 \right) \right) \quad (178)$$

$$\hat{y} = \frac{1}{|\sin(\Theta)|} \left(\hat{\Omega}_o \times \left(\hat{\Omega}_0 \times \left(\hat{\Omega}_t \times \Omega_0 \right) \right) \right) \hat{z} = \frac{\vec{\Omega}_0}{|\vec{\Omega}_0|} \quad (179)$$

$$\Omega_z = \Omega_0 \quad (180)$$

$$\Omega_x = \hat{\Omega}_0 \times \left(\vec{\Omega}_t \times \hat{\Omega}_0 \right) \quad (181)$$

With this in hand, results from the previous section can be used to find the Rabi time or Rabi frequency for using g-TMR given an applied magnetic field,

substituting

$$gB_0 \rightarrow \Omega_z \quad (182)$$

$$gB_t \rightarrow \Omega_x \quad (183)$$

Although the Rabi time or frequency for a single spin in a quantum dot can be found given an applied static magnetic field, the magnetic field has not yet been chosen to be optimal. Because of the complexity in the magnetic field dependence of equations 181, it is most straightforward to maximize the Rabi frequency numerically.

The Rabi time was numerically minimized using Matlab. The g-tensor information was input and an optimal magnetic field direction with a strength of 1T was sought. For the dot in figure 11, an optimal polar angle of $\frac{\pi}{2}$ for a magnetic field in-plane component along $[110]$ and approximately $\frac{\pi}{2} \pm \frac{\pi}{6}$ for a magnetic field in-plane component along $[1\bar{1}0]$ was found. There are a number of similarly optimized magnetic field orientations, as can be seen in figure 50.

VI.2 g-tensor modulation resonance with hole spins

As with the electrons, unoccupied valence electron (i.e. hole) states may also be used for g-tensor modulation resonance and for non-resonant manipulation. Hole spins have recently gained importance as their low nuclear coupling[108] and increased lifetime in low-dimensional nanostructures[97] allow for performance in quantum dots on par with or surpassing that of electrons. Although no g-factor sign

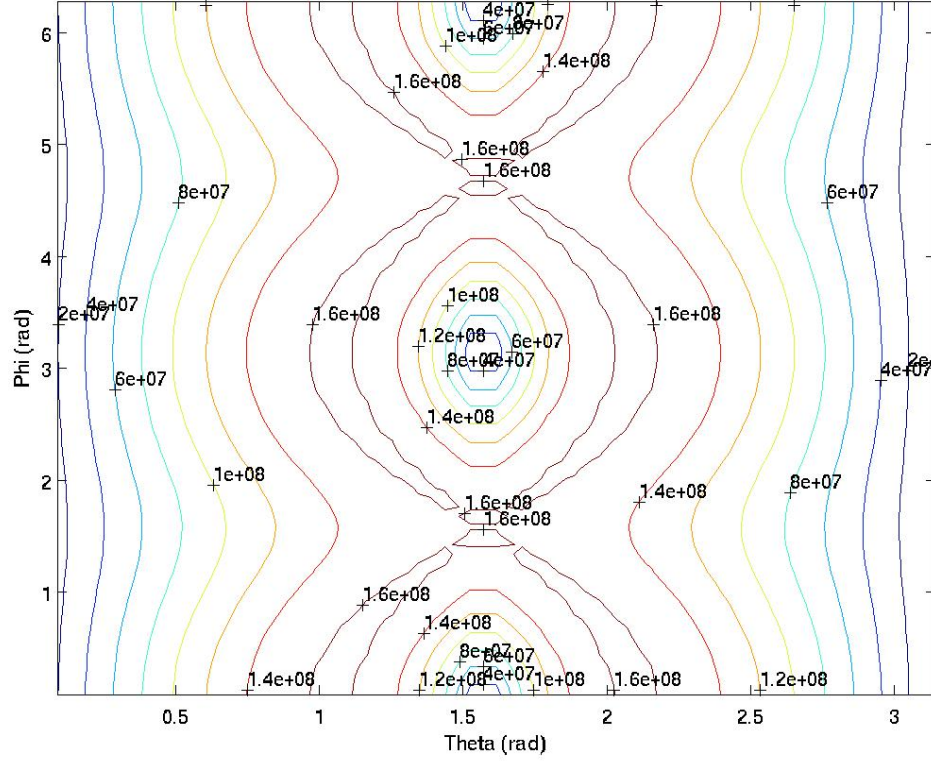


Figure 50: Rabi frequency (contours) as a function of polar (“Theta”) and azimuthal (“Phi”) angle.

change was found in the topmost heavy hole state in InAs/GaAs quantum dots, a sign change was found in the topmost hole state in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dots. The g-tensor of this heavy hole state is plotted in figure 39 as a function of electric field.

Unlike the electron g-factors discussed in a previous section, the hole g-factors have a clear quadratic dependence on the electric field. Regressions were performed on the data in figure 39 using the Gnumeric software package to evaluate the relative importance of the various polynomial components of the g-tensors. For $g_{[001]}$, the

largest component is the quadratic component, as shown in figures 51, 52, and 53. For the other two directions, $g_{[110]}$ (figures 54, 55, and 56) and $g_{[1\bar{1}0]}$ (figures 57, 57, and 57), the linear fit is likely good enough to be usable. The quadratic fit, however, is visually a very close match to the data.

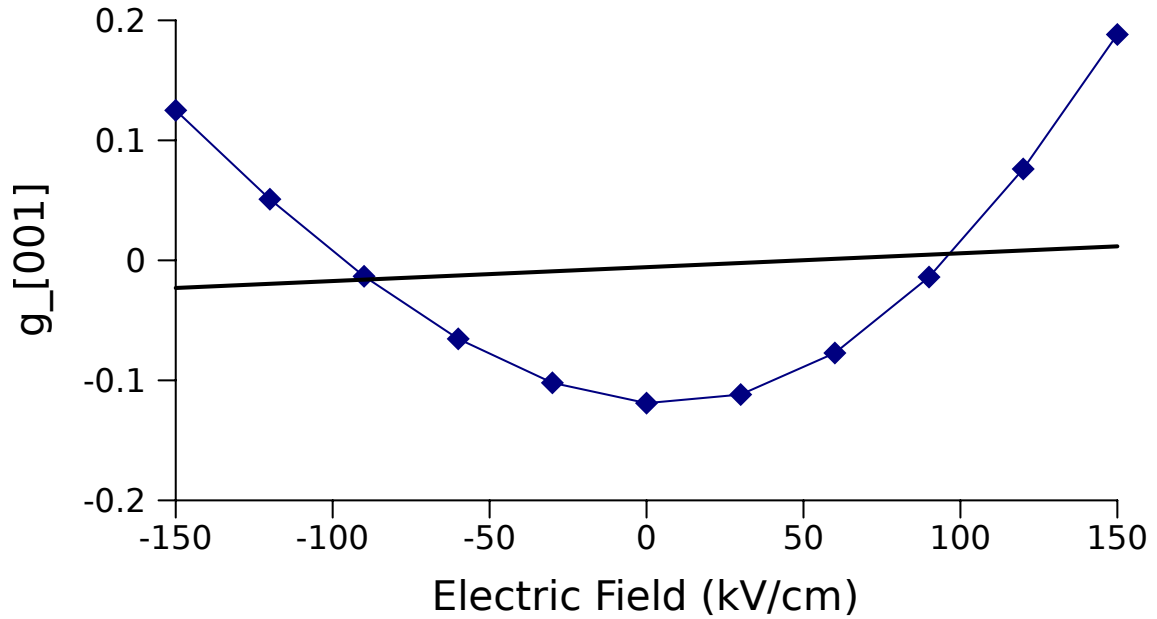


Figure 51: Fit of the hole g -factor along the $[001]$ direction (data from figure 39). A linear regression has been performed and is plotted. Although the cubic regression has the best fit to the eye, the quadratic fit provides a suitable approximation to the data. The linear fit is not suitable.

	$[001]$	$[110]$	$[1\bar{1}0]$
b	-0.0056715458915	0.14805267101402	0.21259916093868
m	0.00011563935419	0.00018828135581	0.00025206398593
$m\epsilon$	0.01734590312889	0.02824220337127	0.03780959789005

Table 11: Polynomial coefficients from a linear regression, and their relative size when accounting for the electric field oscillation magnitudes, using $g_\alpha = m_\alpha E + b_\alpha$; $\epsilon = 150\text{kV/cm}$.

The quadratic model clearly provides a very reasonable match to the data. For that reason, the quadratic model will be used for the rest of this subsection. If

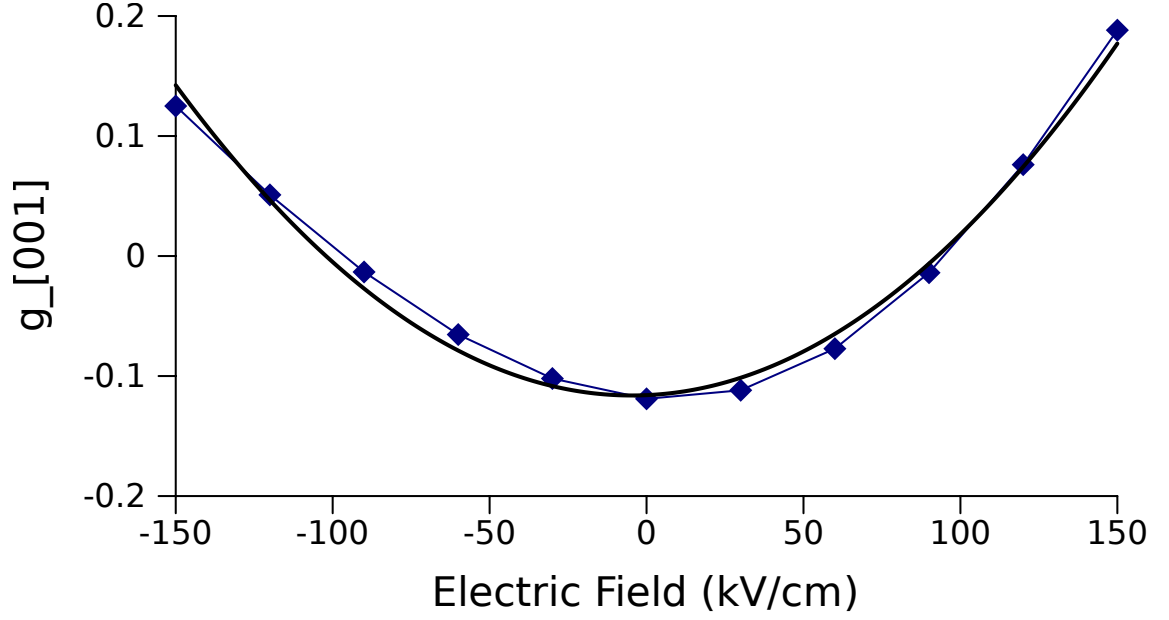


Figure 52: Fit of the hole g-factor along the [001] direction (data from figure 39). A quadratic regression has been performed and is plotted. Although the cubic regression has the best fit to the eye, the quadratic fit provides a suitable approximation to the data. The linear fit is not suitable.

	[001]	[110]	$[\bar{1}\bar{1}0]$
c	-0.1159114101051	0.14337590759878	0.20802740497781
b	0.00011563935419	0.00018828135581	0.00025206398593
a	1.22488738015119E-05	5.19640379470655E-07	5.07972884541353E-07
$b\epsilon$	0.01734590312889	0.02824220337127	0.03780959789005
$a\epsilon^2$	0.27559966053402	0.01169190853809	0.01142938990218

Table 12: Polynomial coefficients from a quadratic regression, and their relative size when accounting for the electric field oscillation magnitudes, using $g_\alpha = a_\alpha E^2 + b_\alpha E + c$; $\epsilon = 150\text{kV/cm}$.

additional precision is desired, however, the discussion below can be generalized to third- and higher-order polynomials. Toward the end of the resonant section extension to higher-order polynomials will be discussed.

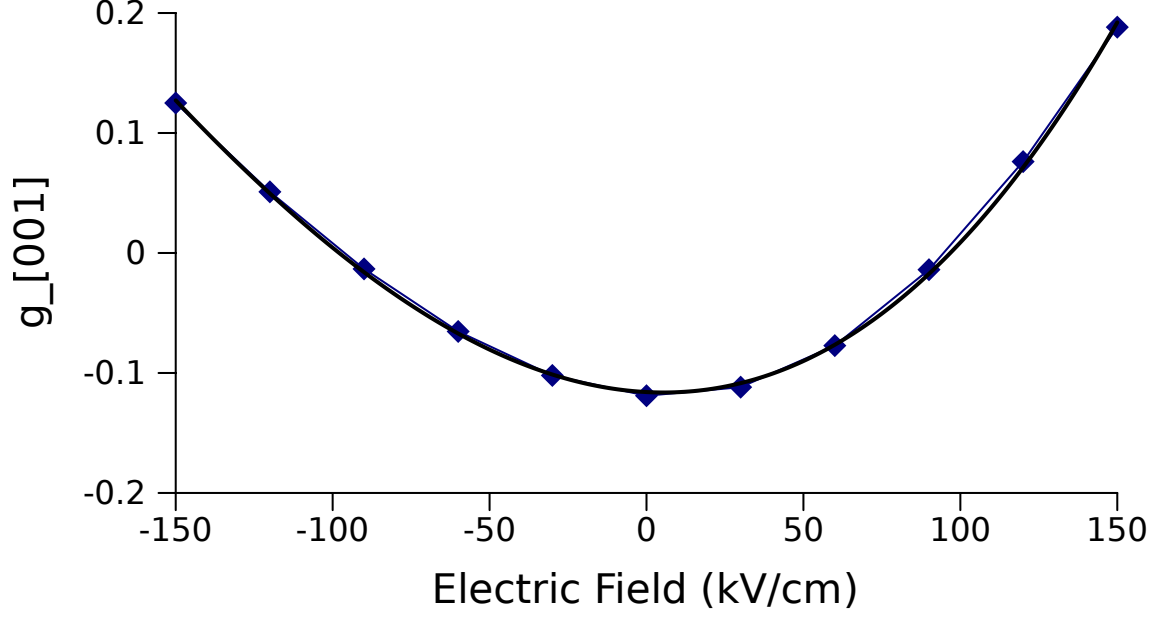


Figure 53: Fit of the hole g-factor along the [001] direction (data from figure 39). A cubic regression has been performed and is plotted. Although the cubic regression has the best fit to the eye, the quadratic fit provides a suitable approximation to the data. The linear fit is not suitable.

	[001]	[110]	$[\bar{1}\bar{1}0]$
d	-0.1159114101051	0.14337590759878	0.20802740497781
c	-0.00013416693211	0.00018267889101	0.00025782413487
b	1.22488738015119E-05	5.19640379470655E-07	5.07972884541353E-07
a	1.55934011424888E-08	3.49716903873096E-10	-3.59559858663176E-10
$c\epsilon$	-0.02012503981651	0.02740183365126	0.03867362023042
$b\epsilon^2$	0.27559966053402	0.01169190853809	0.01142938990218
$a\epsilon^3$	0.0526277288559	0.00118029455057	-0.00121351452299

Table 13: Polynomial coefficients from a cubic regression, and their relative size when accounting for the electric field oscillation magnitudes, using $g_\alpha = a_\alpha E^3 + b_\alpha E^2 + c_\alpha E + d_\alpha$; $\epsilon = 150\text{kV/cm}$.

VI.3 Resonant Hole Spin Manipulation

Because the electron g-tensor depended linearly over the electric field domain which was to be used, the static and transverse oscillating components of the spin precession vector were readily decomposable. A quadratic or higher-order depen-

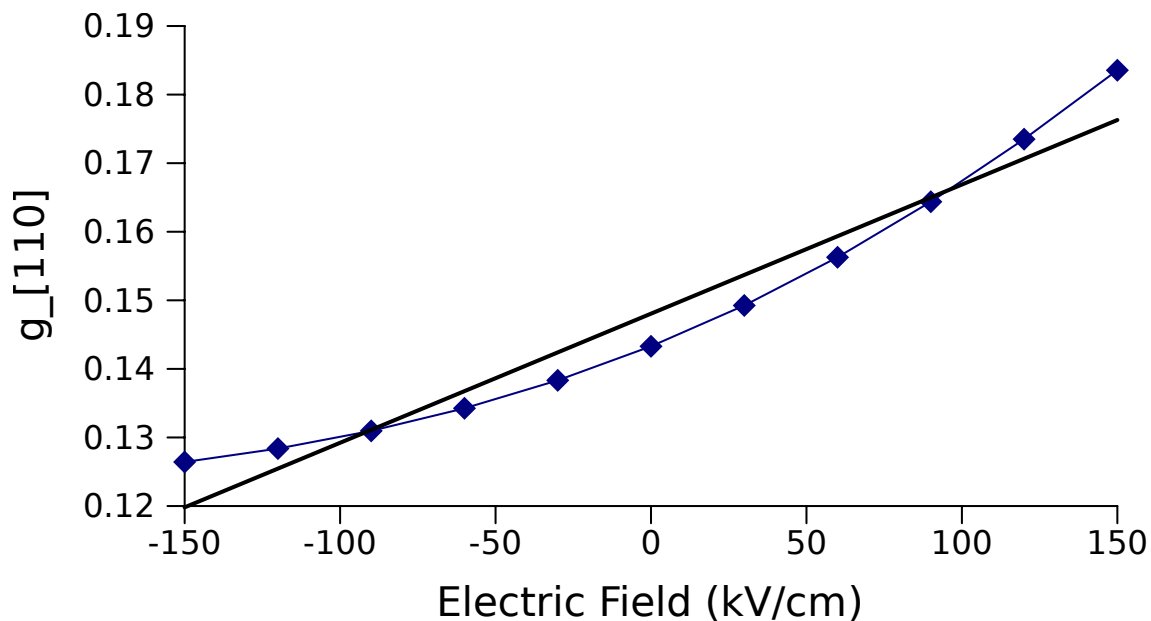


Figure 54: Fit of the hole g-factor along the [110] direction (data from figure 39). A linear regression has been performed and is plotted. The cubic and quadratic fits are visually close to the data. The linear fit is close enough to be usable, however.

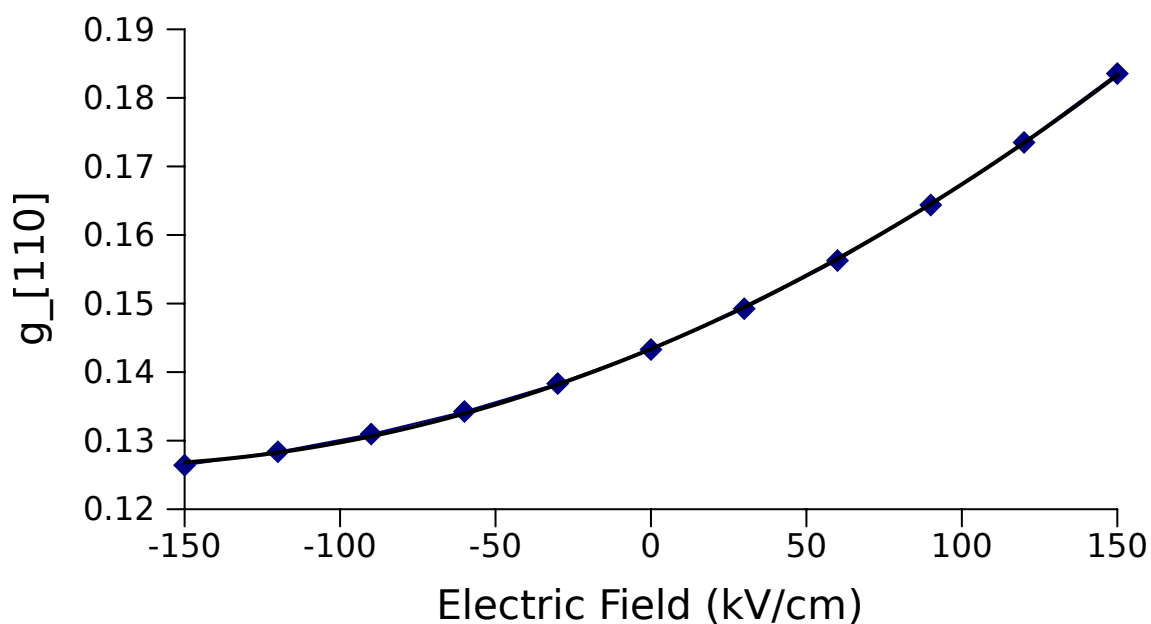


Figure 55: Fit of the hole g-factor along the [110] direction (data from figure 39). A quadratic regression has been performed and is plotted. The cubic and quadratic fits are visually close to the data. The linear fit is close enough to be usable, however.

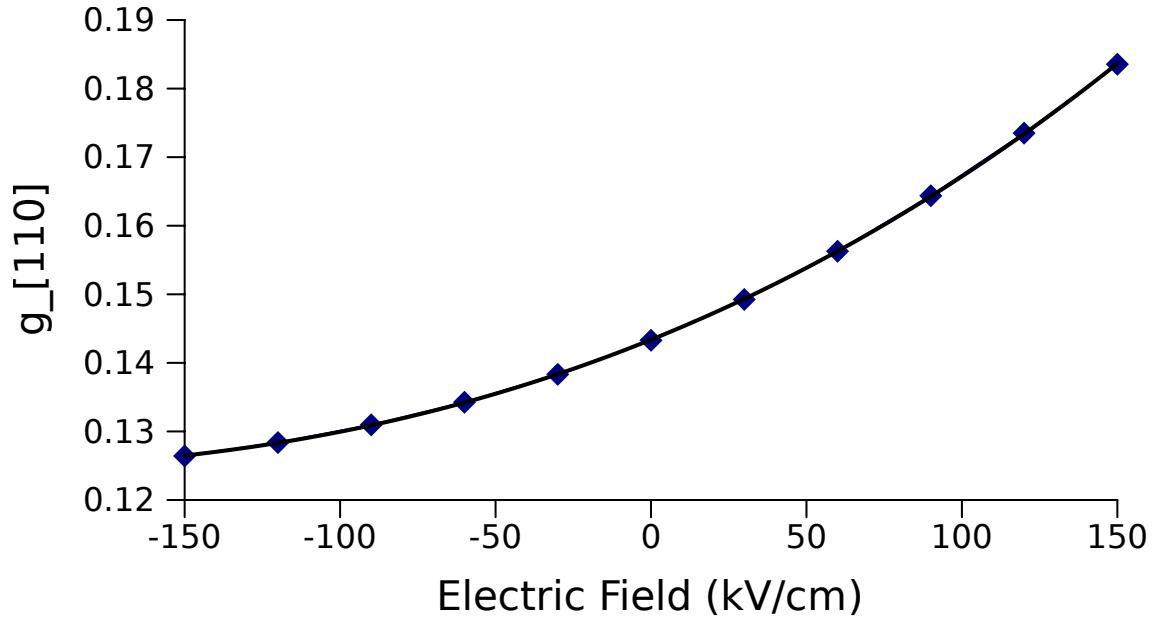


Figure 56: Fit of the hole g-factor along the $[110]$ direction (data from figure 39). A cubic regression has been performed and is plotted. The cubic and quadratic fits are visually close to the data. The linear fit is close enough to be usable, however.

dence of the g-tensor on electric field introduces new complexities. Although the oscillating component of the spin precession vector was along a straight line for the electrons, this is no longer true for the holes, where the electric field dependence is clearly quadratic (or even cubic). A slight quadratic dependence can be thought of as causing the otherwise linear oscillations to start to curve in 3-dimensional space. A strongly quadratic g-tensor would then be strongly curved.

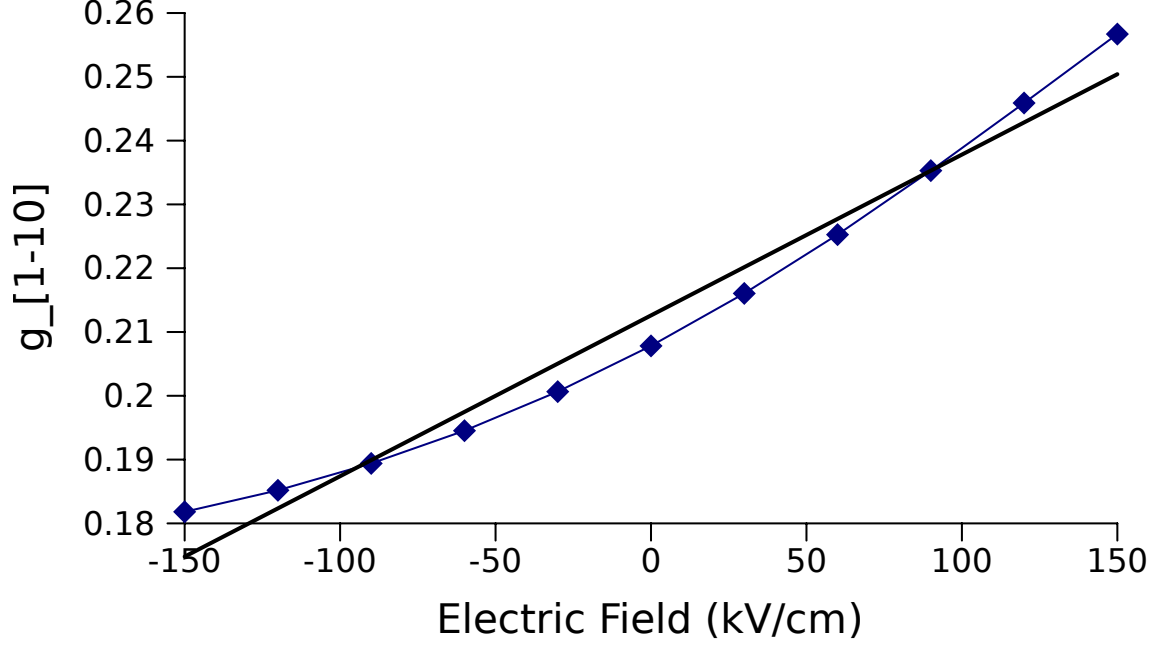


Figure 57: Fit of the hole g-factor along the $[1\bar{1}0]$ direction (data from figure 39). A linear regression has been performed and is plotted. The cubic and quadratic fits are visually close to the data. The linear fit is close enough to be usable, however.

VI.3.1 Resonances from Quadratic Electric Field Dependence

The process of using this g-tensor for resonant spin manipulation begins as it did for electrons. The g-tensor is broken into its polynomial parts:

$$\tilde{g}(E) = \begin{pmatrix} g_x(E) & 0 & 0 \\ 0 & g_y(E) & 0 \\ 0 & 0 & g_z(E) \end{pmatrix} \quad (184)$$

$$= E^2 \begin{pmatrix} a_x & 0 & 0 \\ 0 & a_y & 0 \\ 0 & 0 & a_z \end{pmatrix} + E \begin{pmatrix} b_x & 0 & 0 \\ 0 & b_y & 0 \\ 0 & 0 & b_z \end{pmatrix} + \begin{pmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & c_z \end{pmatrix} \quad (185)$$

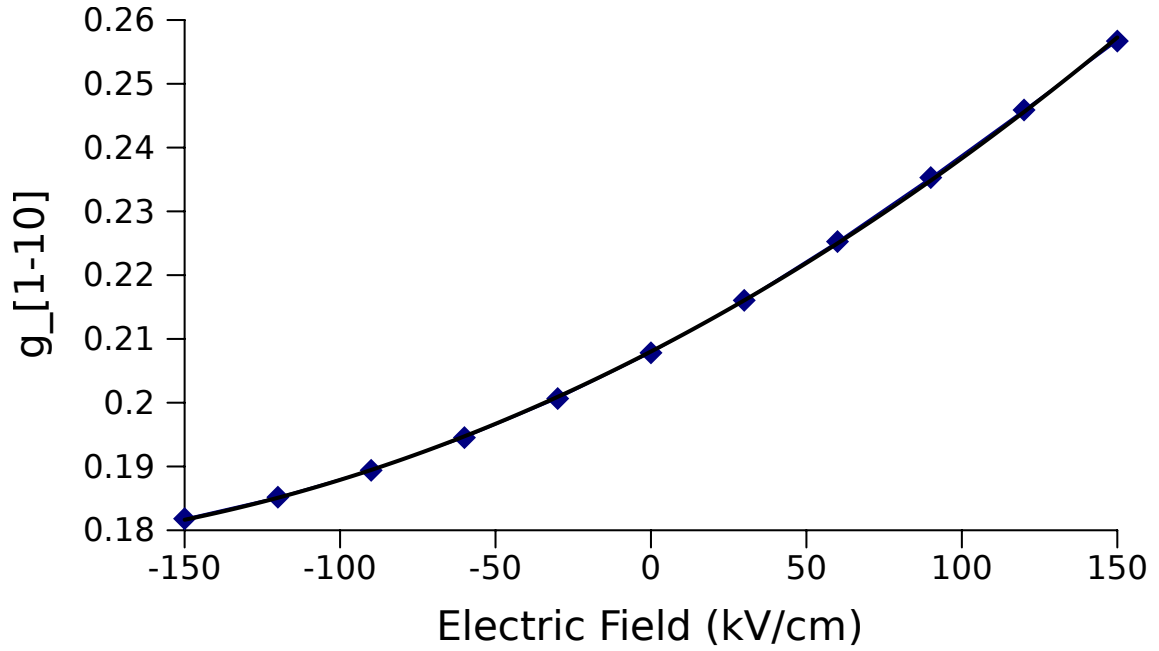


Figure 58: Fit of the hole g-factor along the $[1\bar{1}0]$ direction (data from figure 39). A quadratic regression has been performed and is plotted. The cubic and quadratic fits are visually close to the data. The linear fit is close enough to be usable, however.

Applying an oscillating electric field, $E(t) = E_0 + \epsilon \sin(\omega t)$:

$$\tilde{g}(t) = (E_0 + \epsilon \sin(\omega t))^2 \begin{pmatrix} a_x & 0 & 0 \\ 0 & a_y & 0 \\ 0 & 0 & a_z \end{pmatrix} + (E_0 + \epsilon \sin(\omega t)) \begin{pmatrix} b_x & 0 & 0 \\ 0 & b_y & 0 \\ 0 & 0 & b_z \end{pmatrix} + \begin{pmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & c_z \end{pmatrix} \quad (186)$$

$$(187)$$

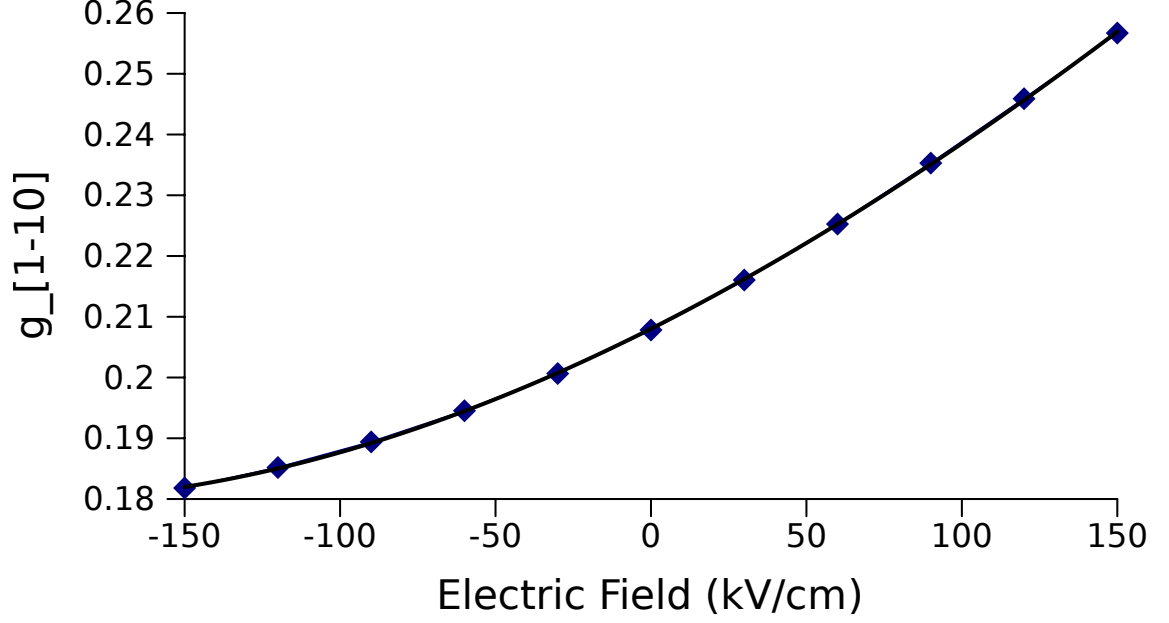


Figure 59: Fit of the hole g-factor along the $[1\bar{1}0]$ direction (data from figure 39). A cubic regression has been performed and is plotted. The cubic and quadratic fits are visually close to the data. The linear fit is close enough to be usable, however.

The quadratic term becomes

$$(E_0 + \epsilon \sin(\omega t))^2 = E_0^2 + \epsilon^2 \sin^2(\omega t) + 2E_0\epsilon \sin(\omega t) \quad (188)$$

$$= E_0^2 + \epsilon^2 \frac{1}{2} (1 - \cos(2\omega t)) + 2E_0\epsilon \sin(\omega t) \quad (189)$$

$$= E_0^2 + \frac{1}{2}\epsilon^2 - \frac{1}{2}\epsilon^2 \cos(2\omega t) + 2E_0\epsilon \sin(\omega t) \quad (190)$$

Collecting constant, $\sin(\omega t)$ and $\cos(2\omega t)$ terms gives

$$\tilde{g}(t) = -\frac{1}{2}\epsilon^2 \cos(2\omega t) \begin{pmatrix} a_x & 0 & 0 \\ 0 & a_y & 0 \\ 0 & 0 & a_z \end{pmatrix}$$

$$\begin{aligned}
& +\epsilon \sin(\omega t) \begin{pmatrix} 2E_0 a_x + b_x & 0 & 0 \\ 0 & 2E_0 a_y + b_y & 0 \\ 0 & 0 & 2E_0 a_z + b_z \end{pmatrix} + \\
& \begin{pmatrix} \left(E_0^2 + \frac{\epsilon^2}{2}\right) a_x + E_0 b_x + c_x & 0 & 0 \\ 0 & \left(E_0^2 + \frac{\epsilon^2}{2}\right) a_y + E_0 b_y + c_y & 0 \\ 0 & 0 & \left(E_0^2 + \frac{\epsilon^2}{2}\right) a_z + E_0 b_z + c_z \end{pmatrix}
\end{aligned}$$

Higher-order polynomials in E can be incorporated similarly to how the quadratic term was integrated above. Geometric identities will cause multiples of $\vec{\omega}$, for instance E^3 :

$$E^3 = (E_0 + \epsilon \sin(\omega t))^3 \quad (191)$$

$$= (E_0 + \epsilon \sin(\omega t)) \left(E_0^2 + \frac{1}{2}\epsilon^2 - \frac{1}{2}\epsilon^2 \cos(2\omega t) + 2E_0\epsilon \sin(\omega t) \right) \quad (192)$$

$$= E_0 \left(E_0^2 + \frac{1}{2}\epsilon^2 - \frac{1}{2}\epsilon^2 \cos(2\omega t) + 2E_0\epsilon \sin(\omega t) \right) \quad (193)$$

$$+ \epsilon \sin(\omega t) \left(E_0^2 + \frac{1}{2}\epsilon^2 - \frac{1}{2}\epsilon^2 \cos(2\omega t) + 2E_0\epsilon \sin(\omega t) \right) \quad (194)$$

$$= E_0^3 + E_0 \frac{1}{2}\epsilon^2 - E_0 \frac{1}{2}\epsilon^2 \cos(2\omega t) + 2E_0^2\epsilon \sin(\omega t) \quad (195)$$

$$+ E_0^2\epsilon \sin(\omega t) + \frac{1}{2}\epsilon^3 \sin(\omega t) - \frac{1}{2}\epsilon^3 \cos(2\omega t) \sin(\omega t) + 2E_0\epsilon^2 \sin^2(\omega t) \quad (196)$$

$$= E_0^3 + \frac{3}{2}E_0\epsilon^2 - \frac{3}{2}E_0\epsilon^2 \cos(2\omega t) + \left(3E_0^2\epsilon + \frac{3}{4}\epsilon^3 \right) \sin(\omega t) - \frac{1}{4}\epsilon^3 \sin(3\omega t) \quad (197)$$

The method then follows like that of the electron g-TMR derivation. The $\hat{z}'$ axis

is defined as the time-independent component of the spin precession vector, namely

$$\vec{\Omega}_0 = \begin{pmatrix} \left(\left(E_0^2 + \frac{\epsilon^2}{2} \right) a_x + E_0 b_x + c_x \right) B_x \\ \left(\left(E_0^2 + \frac{\epsilon^2}{2} \right) a_y + E_0 b_y + c_y \right) B_y \\ \left(\left(E_0^2 + \frac{\epsilon^2}{2} \right) a_z + E_0 b_z + c_z \right) B_z \end{pmatrix} \quad (198)$$

The oscillating component of $\vec{\Omega}(t)$ which lies perpendicular to $\vec{\Omega}_0$ is now sought.

For simplicity of notation, the three components of $\tilde{g}$ ($\vec{\Omega}$ when dotted into the magnetic field) are

$$\vec{\Omega}_0 = \begin{pmatrix} \left(\left(E_0^2 + \frac{\epsilon^2}{2} \right) a_x + E_0 b_x + c_x \right) B_x \\ \left(\left(E_0^2 + \frac{\epsilon^2}{2} \right) a_y + E_0 b_y + c_y \right) B_y \\ \left(\left(E_0^2 + \frac{\epsilon^2}{2} \right) a_z + E_0 b_z + c_z \right) B_z \end{pmatrix} \quad (199)$$

$$\vec{\Omega}_l = \begin{pmatrix} 2E_0 a_x + b_x & 0 & 0 \\ 0 & 2E_0 a_y + b_y & 0 \\ 0 & 0 & 2E_0 a_z + b_z \end{pmatrix} \cdot \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} \quad (200)$$

$$\vec{\Omega}_q = \begin{pmatrix} a_x & 0 & 0 \\ 0 & a_y & 0 \\ 0 & 0 & a_z \end{pmatrix} \cdot \begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} \quad (201)$$

so that now

$$\vec{\Omega}(t) = \vec{\Omega}_0 + \vec{\Omega}_l \epsilon \sin(\omega t) - \vec{\Omega}_q \frac{1}{2} \epsilon^2 \cos(2\omega t) \quad (202)$$

Restricting the problem to two dimensions by only applying the static magnetic field in e.g. the y-z plane restricts the problem to a two-dimensional one and therefore all oscillations must either be parallel to the $\hat{z}'$ axis or the transverse $\hat{y}'$ axis. The largest changes are along the $[001]$ and $[1\bar{1}0]$ axis, so those two are promising candidates for g-TMR. One can identify the $[001]$ axis with $\hat{z}$ and $[1\bar{1}0]$ axis $\hat{x}$ in the preceding equations. Since the $\hat{z}'$ axis has been defined above as $\hat{\Omega}_0$, only the $\hat{y}'$ axis needs defined. This can be done implicitly by finding the transverse time-dependent portion of $\vec{\Omega}(t)$.

$\vec{\Omega}(t)$ can be composed into a static portion ($\vec{\Omega}_0$), a portion oscillating parallel to this static axis ($\vec{\Omega}(t) \cdot \hat{z}'$), and a portion oscillating transverse to the static axis. As with the electron g-TMR, the portion oscillating parallel to the static axis is not important for g-TMR, as it does not cause a spin flip. The portion transverse to the static component, however, is directly responsible for spin manipulation. This component is found like with the electron:

$$\vec{\Omega}_{\perp}(t) = \vec{\Omega}(t) - \left(\vec{\Omega}(t) \cdot \hat{\Omega}_0 \right) \hat{\Omega}_0 \quad (203)$$

This then reduces to

$$\vec{\Omega}_{\perp}(t) = \epsilon \sin(\omega t) \left(\vec{\Omega}_l - \left(\vec{\Omega}_l \cdot \hat{\Omega}_0 \right) \hat{\Omega}_0 \right) - \frac{1}{2} \epsilon^2 \cos(2\omega t) \left(\vec{\Omega}_q - \left(\vec{\Omega}_q \cdot \hat{\Omega}_0 \right) \hat{\Omega}_0 \right) a \quad (204)$$

In the aforementioned two dimensions, subcomponent $\vec{\Omega}_i$ becomes

$$\begin{pmatrix} \Omega_{i,x} \\ \Omega_{i,y} \end{pmatrix} - \begin{pmatrix} \Omega_{i,x} \\ \Omega_{i,y} \end{pmatrix} \cdot \begin{pmatrix} \Omega_{0,x} \\ \Omega_{0,y} \end{pmatrix} \frac{1}{|\vec{\Omega}_0|^2} \begin{pmatrix} \Omega_{0,x} \\ \Omega_{0,y} \end{pmatrix} \quad (205)$$

$$= \frac{1}{|\vec{\Omega}_0|^2} \left[(\Omega_{0,x}^2 + \Omega_{0,y}^2) \begin{pmatrix} \Omega_{i,x} \\ \Omega_{i,y} \end{pmatrix} - (\Omega_{i,x}\Omega_{0,x} + \Omega_{i,y}\Omega_{0,y}) \begin{pmatrix} \Omega_{0,x} \\ \Omega_{0,y} \end{pmatrix} \right] \quad (206)$$

$$= \frac{1}{|\vec{\Omega}_0|^2} \left[\begin{pmatrix} \Omega_{i,x}\Omega_{0,x}^2 + \Omega_{i,x}\Omega_{0,y}^2 \\ \Omega_{i,y}\Omega_{0,x}^2 + \Omega_{i,y}\Omega_{0,y}^2 \end{pmatrix} - \begin{pmatrix} \Omega_{0,x}\Omega_{i,x}\Omega_{0,x} + \Omega_{0,x}\Omega_{i,y}\Omega_{0,y} \\ \Omega_{0,y}\Omega_{i,x}\Omega_{0,x} + \Omega_{0,y}\Omega_{i,y}\Omega_{0,y} \end{pmatrix} \right] \quad (207)$$

$$= \frac{1}{|\vec{\Omega}_0|^2} \begin{pmatrix} \Omega_{i,x}\Omega_{0,x}^2 + \Omega_{i,x}\Omega_{0,y}^2 - \Omega_{0,x}\Omega_{i,x}\Omega_{0,x} - \Omega_{0,x}\Omega_{i,y}\Omega_{0,y} \\ \Omega_{i,y}\Omega_{0,x}^2 + \Omega_{i,y}\Omega_{0,y}^2 - \Omega_{0,y}\Omega_{i,x}\Omega_{0,x} - \Omega_{0,y}\Omega_{i,y}\Omega_{0,y} \end{pmatrix} \quad (208)$$

$$= \frac{1}{|\vec{\Omega}_0|^2} \begin{pmatrix} \Omega_{i,x}\Omega_{0,y}^2 - \Omega_{0,x}\Omega_{i,y}\Omega_{0,y} \\ \Omega_{i,y}\Omega_{0,x}^2 - \Omega_{0,y}\Omega_{i,x}\Omega_{0,x} \end{pmatrix} \quad (209)$$

$$= \frac{1}{|\vec{\Omega}_0|^2} \begin{pmatrix} (\Omega_{i,x}\Omega_{0,y} - \Omega_{0,x}\Omega_{i,y})\Omega_{0,y} \\ -(\Omega_{i,x}\Omega_{0,y} - \Omega_{0,x}\Omega_{i,y})\Omega_{0,x} \end{pmatrix} \quad (210)$$

$$= \frac{(\Omega_{i,x}\Omega_{0,y} - \Omega_{0,x}\Omega_{i,y})}{|\vec{\Omega}_0|^2} \begin{pmatrix} \Omega_{0,y} \\ -\Omega_{0,x} \end{pmatrix} \quad (211)$$

$$= \frac{(\vec{\Omega}_i \times \vec{\Omega}_0)_z}{|\vec{\Omega}_0|^2} \begin{pmatrix} \Omega_{0,y} \\ -\Omega_{0,x} \end{pmatrix} \quad (212)$$

This is similar to the results for the electrons, and is perpendicular to $\hat{\Omega}_0$. However, unlike the electron case, there are two oscillating terms, coming from $\Omega_{\perp l}$ and

$\Omega_{\perp q}$. Continuing with the procedure to find the Rabi frequency, the Hamiltonian is

$$H(t) = -\frac{\mu_B}{\hbar} \vec{S} \cdot \vec{\Omega}_0 - \frac{\mu_B}{\hbar} \vec{S} \cdot \vec{\Omega}_{\perp}(t) \quad (213)$$

$$= -\frac{\mu_B}{\hbar} S_z \Omega_0 - \frac{\mu_B}{\hbar} S_x \Omega_{\perp}(t) \quad (214)$$

$$= -\mu_B \frac{3}{2} \Omega_0 \sigma_z - \mu_B \frac{3}{2} \Omega_{\perp}(t) \sigma_x \quad (215)$$

The oscillatory portion along $\hat{x}$ (with the new unprimed axes being in the newly defined coordinate system and the primed axes being in the original, crystal coordinate system) becomes

$$H_x = -\mu_B \frac{3}{2} \Omega_{\perp}(t) \sigma_x \quad (216)$$

$$= -\mu_B \frac{3}{2} \sigma_x \left[\epsilon \sin(\omega t) \frac{(\vec{\Omega}_l \times \vec{\Omega}_0)_{z'}}{|\vec{\Omega}_0|} + \frac{1}{2} \epsilon^2 \cos(2\omega t) \frac{(\vec{\Omega}_q \times \vec{\Omega}_0)_{z'}}{|\vec{\Omega}_0|} \right] \quad (217)$$

As before, rotating and counter-rotating spin precession vector components are used

$$\hat{x} \sin(\omega t) = \frac{1}{2} [\hat{x} \sin(\omega t) + \hat{y} \cos(\omega t) - \hat{x} \sin(-\omega t) - \hat{y} \cos(-\omega t)] \quad (218)$$

$$\hat{x} \cos(2\omega t) = \frac{1}{2} [\hat{x} \cos(2\omega t) + \hat{y} \sin(2\omega t) + \hat{x} \cos(-2\omega t) + \hat{y} \sin(-2\omega t)] \quad (219)$$

Putting this back into H_x and using the definitions (for notational simplicity)

$$\alpha = \frac{(\vec{\Omega}_l \times \vec{\Omega}_0)_z}{|\vec{\Omega}_0|} \quad (220)$$

$$\beta = \frac{(\vec{\Omega}_q \times \vec{\Omega}_0)_z}{|\vec{\Omega}_0|} \quad (221)$$

gives the intermediate result

$$\begin{aligned} H_x &= -\mu_B \frac{3}{4} \vec{\sigma} \cdot [\hat{x} \epsilon \alpha \sin(\omega t) + \hat{y} \epsilon \alpha \cos(\omega t)] \\ &\quad -\mu_B \frac{3}{4} \vec{\sigma} \cdot [-\hat{x} \epsilon \alpha \sin(-\omega t) - \hat{y} \epsilon \alpha \cos(-\omega t)] \\ &\quad -\mu_B \frac{3}{4} \vec{\sigma} \cdot [-\hat{x} \epsilon^2 \beta \cos(2\omega t) - \hat{y} \epsilon^2 \beta \cos(2\omega t)] \\ &\quad -\mu_B \frac{3}{4} \vec{\sigma} \cdot [-\hat{x} \epsilon^2 \beta \cos(-2\omega t) - \hat{y} \epsilon^2 \beta \cos(-2\omega t)] \end{aligned}$$

Rearranging into forward- and backward rotating groups and substituting exponentials for the geometric functions gives

$$\begin{aligned} H_x &= -\mu_B \frac{3}{8} \vec{\sigma} \cdot [-i \hat{x} \epsilon \alpha (e^{i\omega t} - e^{-i\omega t}) + \hat{y} \epsilon \alpha (e^{i\omega t} + e^{-i\omega t})] \\ &\quad -\mu_B \frac{3}{8} \vec{\sigma} \cdot [-\hat{x} \epsilon^2 \beta (e^{i2\omega t} + e^{-i2\omega t}) + i \hat{y} \epsilon^2 \beta (e^{i2\omega t} - e^{-i2\omega t})] \\ &\quad -\mu_B \frac{3}{8} \vec{\sigma} \cdot [i \hat{x} \epsilon \alpha (e^{-i\omega t} - e^{i\omega t}) - \hat{y} \epsilon \alpha (e^{-i\omega t} + e^{i\omega t})] \\ &\quad -\mu_B \frac{3}{8} \vec{\sigma} \cdot [-\hat{x} \epsilon^2 \beta (e^{-i2\omega t} + e^{i2\omega t}) - \hat{y} \epsilon^2 \beta (e^{-i2\omega t} - e^{i2\omega t})] \end{aligned}$$

Removing the parentheses gives

$$\begin{aligned}
H_x &= -\mu_B \frac{3}{8} \vec{\sigma} \cdot [-\hat{x}\epsilon\alpha e^{i\omega t} + \hat{x}\epsilon\alpha e^{-i\omega t} + \hat{y}\epsilon\alpha e^{i\omega t} + \hat{y}\epsilon\alpha e^{-i\omega t}] \\
&\quad -\mu_B \frac{3}{8} \vec{\sigma} \cdot [-\hat{x}\epsilon^2\beta e^{i2\omega t} - \hat{x}\epsilon^2\beta e^{-i2\omega t} + \hat{y}\epsilon^2\beta e^{i2\omega t} - \hat{y}\epsilon^2\beta e^{-i2\omega t}] \\
&\quad -\mu_B \frac{3}{8} \vec{\sigma} \cdot [\hat{x}\epsilon\alpha e^{-i\omega t} - \hat{x}\epsilon\alpha e^{i\omega t} - \hat{y}\epsilon\alpha e^{-i\omega t} - \hat{y}\epsilon\alpha e^{i\omega t}] \\
&\quad -\mu_B \frac{3}{8} \vec{\sigma} \cdot [-\hat{x}\epsilon^2\beta e^{-i2\omega t} - \hat{x}\epsilon^2\beta e^{i2\omega t} + \hat{y}\epsilon^2\beta e^{-i2\omega t} - \hat{y}\epsilon^2\beta e^{i2\omega t}]
\end{aligned}$$

Bringing $\vec{\sigma}$ into the brackets and dotting into $\hat{x}$ and $\hat{y}$

$$\begin{aligned}
H_x &= -\mu_B \frac{3}{8} [-\epsilon\alpha e^{i\omega t} (\sigma_x + i\sigma_y) + \epsilon\alpha e^{-i\omega t} (\sigma_x - i\sigma_y)] \\
&\quad -\mu_B \frac{3}{8} [\epsilon^2\beta e^{i2\omega t} (-\sigma_x + i\sigma_y) + \epsilon^2\beta e^{-i2\omega t} (-\sigma_x - i\sigma_y)] \\
&\quad -\mu_B \frac{3}{8} [\epsilon\alpha e^{-i\omega t} (\sigma_x + i\sigma_y) - \epsilon\alpha e^{i\omega t} (\sigma_x - i\sigma_y)] \\
&\quad -\mu_B \frac{3}{8} [\epsilon^2\beta e^{-i2\omega t} (-\sigma_x + i\sigma_y) + \epsilon^2\beta e^{i2\omega t} (-\sigma_x - i\sigma_y)]
\end{aligned}$$

Converting the spin matrices to their raising/lowering equivalents ($\sigma_{\pm} = \frac{1}{2}(\sigma_x \pm i\sigma_y)$),

$$\begin{aligned}
H_x &= -\mu_B \frac{3}{4} [-\epsilon\alpha e^{i\omega t} \sigma_+ + \epsilon\alpha e^{-i\omega t} \sigma_-] \\
&\quad -\mu_B \frac{3}{4} [-\epsilon^2\beta e^{i2\omega t} \sigma_- - \epsilon^2\beta e^{-i2\omega t} \sigma_+] \\
&\quad -\mu_B \frac{3}{4} [\epsilon\alpha e^{-i\omega t} \sigma_+ - \epsilon\alpha e^{i\omega t} \sigma_-] \\
&\quad -\mu_B \frac{3}{4} [-\epsilon^2\beta e^{-i2\omega t} \sigma_- - \epsilon^2\beta e^{i2\omega t} \sigma_+]
\end{aligned}$$

Equation 222 may be rephrased in explicit matrix form as

$$H_x = -\mu_B \frac{3}{4} [-\imath \epsilon \alpha e^{\imath \omega t} - \epsilon^2 \beta e^{-\imath 2 \omega t} + \imath \epsilon \alpha e^{-\imath \omega t} - \epsilon^2 \beta e^{\imath 2 \omega t}] \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ -\mu_B \frac{3}{4} [\imath \epsilon \alpha e^{-\imath \omega t} - \epsilon^2 \beta e^{\imath 2 \omega t} - \imath \epsilon \alpha e^{\imath \omega t} - \epsilon^2 \beta e^{-\imath 2 \omega t}] \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Moving this back to the rotating situation from section VI and using the transformations 116 and 117 gives the result

$$H_x = -\mu_B \frac{3}{4} [-\imath \epsilon \alpha e^{\imath(\omega+\gamma)t} - \epsilon^2 \beta e^{-\imath(2\omega-\gamma)t} + \imath \epsilon \alpha e^{-\imath(\omega-\gamma)t} - \epsilon^2 \beta e^{\imath(2\omega+\gamma)t}] \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ -\mu_B \frac{3}{4} [\imath \epsilon \alpha e^{-\imath(\omega+\gamma)t} - \epsilon^2 \beta e^{\imath(2\omega-\gamma)t} - \imath \epsilon \alpha e^{\imath(\omega-\gamma)t} - \epsilon^2 \beta e^{-\imath(2\omega+\gamma)t}] \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

where γ is the frequency of the rotating coordinate system. As has previously been done, the counter-rotating terms, which are therefore rapidly oscillating when away from resonance and therefore negligible, may be removed at this stage. In addition, the new resonance, at 2ω , may be selected instead of ω .

VI.3.2 Rabi Oscillation at the Linear Resonance

Choosing ω gives

$$H_x = -\mu_B \frac{3}{4} \begin{pmatrix} 0 & \imath \epsilon \alpha e^{-\imath(\omega-\gamma)t} \\ -\imath \epsilon \alpha e^{\imath(\omega-\gamma)t} & 0 \end{pmatrix}$$

Choosing the rotating frame to coincide makes simplifies the equation

$$H_x = \mu_B \frac{3\epsilon\alpha}{4} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

or, in a more familiar form,

$$H_x = \mu_B \frac{3\epsilon\alpha}{4} \sigma_y$$

Putting it into the time-dependent, rotating frame Schrödinger equation (125) gives

$$(\omega_0 - \omega) \sigma_z + \omega_t \sigma_y |\chi\rangle = i2 \frac{\partial}{\partial t} |\chi\rangle \quad (222)$$

with the appropriate substitution

$$\omega_t = \frac{\mu_B}{\hbar} \epsilon \frac{3\alpha}{2} \quad (223)$$

Thus, the Rabi frequency equation is identical to the original equation (128), with the above definition of ω_t and using σ_y instead of σ_x .

The quadratic dependence on electric field has two effects. First, the linear and quadratic terms cause the angle or magnitude (or both) of the static spin precession vector component $\vec{\Omega}_0$ to change from the 0-electric-field value. This comes from the definition of $\vec{\Omega}_0$,

$$\Omega_i = \left[\left(E_0^2 + \frac{\epsilon^2}{2} \right) \alpha_i + E_0 b_x + c_x \right] B_i \quad (224)$$

Because the DC electric field and magnetic field are identical for all components of $\vec{\Omega}_0$, the magnitude of $\vec{\Omega}_0$, which determines the Larmor frequency, will change as the $a_i B_i$ and $b_i B_i$ change identically for all i . Changing *individual* components a_i and b_i will cause $\vec{\Omega}_0$ to swing. Although this has no effect on the Larmor frequency, it will cause other changes which will be discussed below. For the moment, only changes which modify the magnitude of $\vec{\Omega}_0$ will be discussed.

A change in the Larmor frequency will cause a shift in the resonant frequency. Both the linear and quadratic terms in $\vec{\Omega}_0$ cause a shift which is dependent upon the DC electric field. Unlike the linear ($a_i = 0$) shift, which is solely DC-based, the oscillating electric field amplitude also causes a shift from the quadratic term.

In addition to a shift in the Larmor frequency, the Rabi frequency itself is shifted by the quadratic term. The quadratic coefficient of the electric field dependence causes a DC-field shift in the angle and the magnitude of $\vec{\Omega}_l$, as can be seen in the definition of the i -th component of $\vec{\Omega}_l$,

$$\Omega_{l,i} = (a_i E_0 + b_i) B_i \quad (225)$$

The shift in magnitude of $\vec{\Omega}_l$ causes a shift in the Rabi frequency, which is dependent upon the sign of the DC electric field and the g-tensor coefficient a_i . This can be seen more clearly by rewriting equation 220:

$$\alpha = \left| \vec{\Omega}_l \right| \left(\hat{\Omega}_l \times \hat{\Omega}_0 \right)_z \quad (226)$$

The angle between $\vec{\Omega}_l$ and $\vec{\Omega}_0$ influences the magnitude of α and therefore ω_t in addition to the the aforementioned magnitude effects. Because the changes in $\vec{\Omega}_l$ and $\vec{\Omega}_0$ are not straightforward when the cross-product is taken, it is difficult to generalize what the relative effects of changing a_i and b_i are. Nevertheless, it is possible to numerically maximize the magnitude of $\vec{\Omega}_l$ and angle of the two vectors and thereby change the Rabi frequency by changing the DC and AC electric field.

VI.3.3 Rabi Oscillation at the Quadratic Resonance

Choosing to focus on the 2ω terms in equation 222 gives

$$H_x = -\mu_B \frac{\hbar}{4} \begin{pmatrix} 0 & i\epsilon\alpha e^{-i(2\omega-\gamma)t} \\ -i\epsilon\alpha e^{i(2\omega-\gamma)t} & 0 \end{pmatrix}$$

Choosing the rotating frame frequency γ to be 2ω makes simplifies the equation again to

$$H_x = \mu_B \frac{3\epsilon\alpha}{4} \sigma_y$$

Putting it into the time-dependent, rotating frame Schrödinger equation (125) and noting that $\gamma = 2\omega$ (or, more usefully, $\frac{\gamma}{2} = \omega$) gives

$$\left(\omega_0 - \frac{\omega}{2}\right) \sigma_z + \omega_t \sigma_y |\chi\rangle = i2 \frac{\partial}{\partial t} |\chi\rangle \quad (227)$$

with the appropriate substitution

$$\omega_t = \frac{\mu_B}{\hbar} \epsilon \frac{3\beta}{2} \quad (228)$$

Thus, the Rabi frequency for the quadratic term is modified from the original equation (128) and becomes

$$|\vec{\Omega}| = \sqrt{\left(\frac{\omega_0 - \frac{\omega}{2}}{2}\right)^2 + \left(\frac{\omega_t}{2}\right)^2} \quad (229)$$

Because of the modification from $\omega_0 - \omega$ in the Rabi frequency equation above to $\omega_0 - \frac{\omega}{2}$, the resonance occurs at *double* the Larmor frequency. It is reasonable to assume that higher-order terms (e.g. cubic) will see similar effects, e.g. the cubic Rabi resonance will be at triple the Larmor frequency for similar reasons. The same shift in the Larmor frequency as was discussed in the linear Rabi resonance term is to be expected in the quadratic Rabi term.

Rephrasing the definition of β as was done for α gives a similar result,

$$\beta = \left| \vec{\Omega}_q \right| \left(\hat{\Omega}_q \times \hat{\Omega}_0 \right)_z \quad (230)$$

No shift in the Rabi frequency will occur without the inclusion of higher-order terms in electric field, because to second order in electric field, the quadratic Rabi term is

dependent upon $\vec{\Omega}_q$, which is dependent solely upon a_i :

$$\Omega_{q,i} = a_i B_i \quad (231)$$

It is still possible, however, to tune the angle between $\vec{\Omega}_q$ and $\vec{\Omega}_0$ by tuning the DC and AC electric field components. Unlike with the linear effects, however, the electric field tuning only modifies $\vec{\Omega}_0$.

VI.3.4 Rabi Oscillations in the $In_{0.5}Ga_{0.5}As/GaAs$ Hole Systems

In order to finish developing the above theory into a more practical application, and to give an idea of the actual performance of such a device, numerical calculations of the Rabi frequencies were performed. The highest Rabi frequency (lowest manipulation time) was sought. Because of the complexity of the problem, Matlab was used to find the highest Rabi frequency numerically. To get a feel for the problem, several graphs were produced to see how the Rabi frequency behaved as the input parameters, ϵ and ϕ , were changed. ϕ is the angle of the magnetic field from the $[1\bar{1}0]$ axis in the $[001]$ - $[1\bar{1}0]$ plane. A DC electric field of 0 was assumed, to allow the largest AC electric field amplitude. The results of these calculations are plotted in figures 60, 61 for electric field amplitudes up to 150kV/cm and 62, 63 for electric field amplitudes up to 300kV/cm.

Because the data was limited to 150kV/cm, the final calculations were limited to that range as well. Figure 62 shows that higher electric fields will lead to higher Rabi frequencies at the linear resonance. The maximum Rabi frequency (lowest Rabi

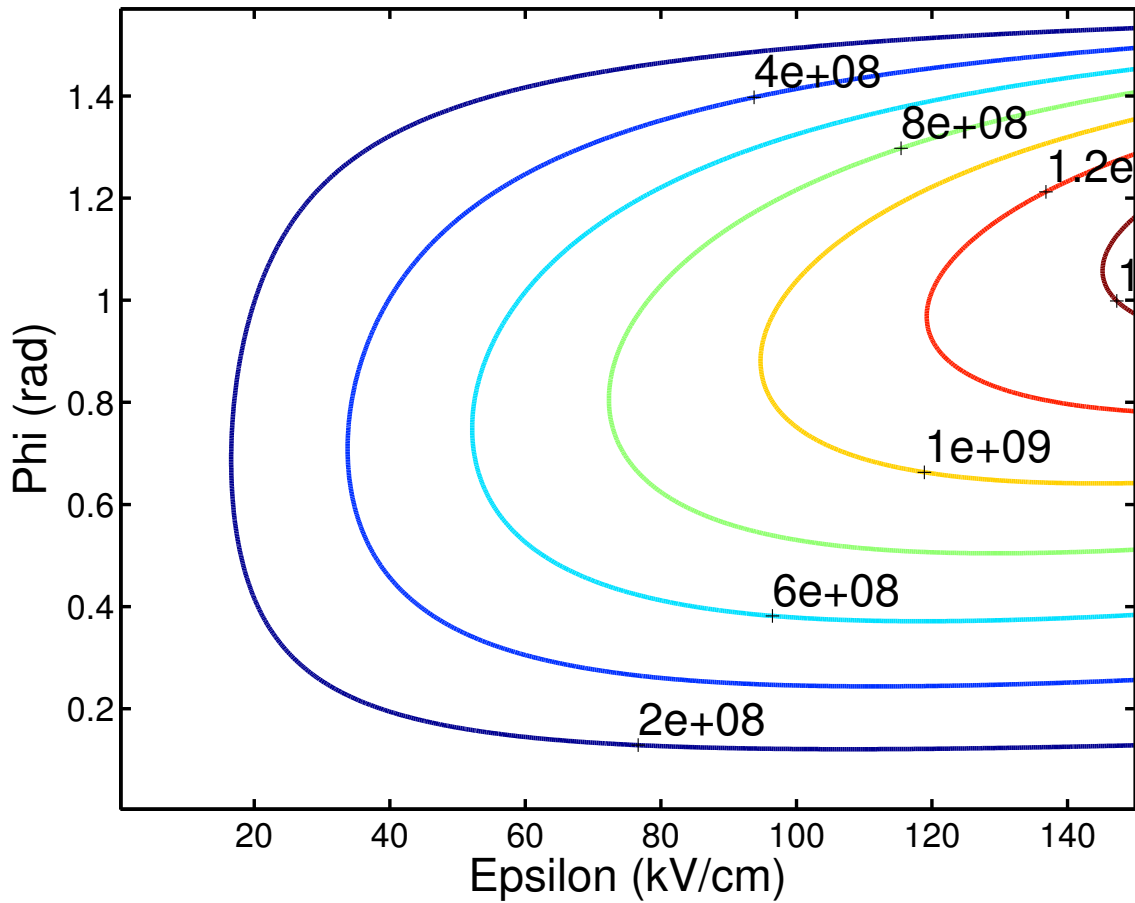


Figure 60: Contour plot of Rabi frequency at linear resonance as a function of AC electric field amplitude ϵ (horizontal axis) up to 150kV/cm and magnetic field angle ϕ (vertical axis). Frequency is in Hz. A trend of higher electric field amplitude and higher magnetic field angle is visible.

time) was found numerically for an electric field amplitude of 150kV/cm. The magnetic field angle was found to be 1.0741 radians, with a Rabi frequency of 1.4GHz. This gives the time required to make a full rotation of 0.696ns. At 300kV/cm, an extremely high electric field, the Rabi frequency becomes even higher. For an electric field amplitude of 300kV/cm, the maximum frequency of 2.74GHz (0.364ns) is found at a magnetic field angle of 1.3928 radians. Figure 62 shows that indeed, higher electric field amplitudes have higher frequencies, and at higher magnetic field

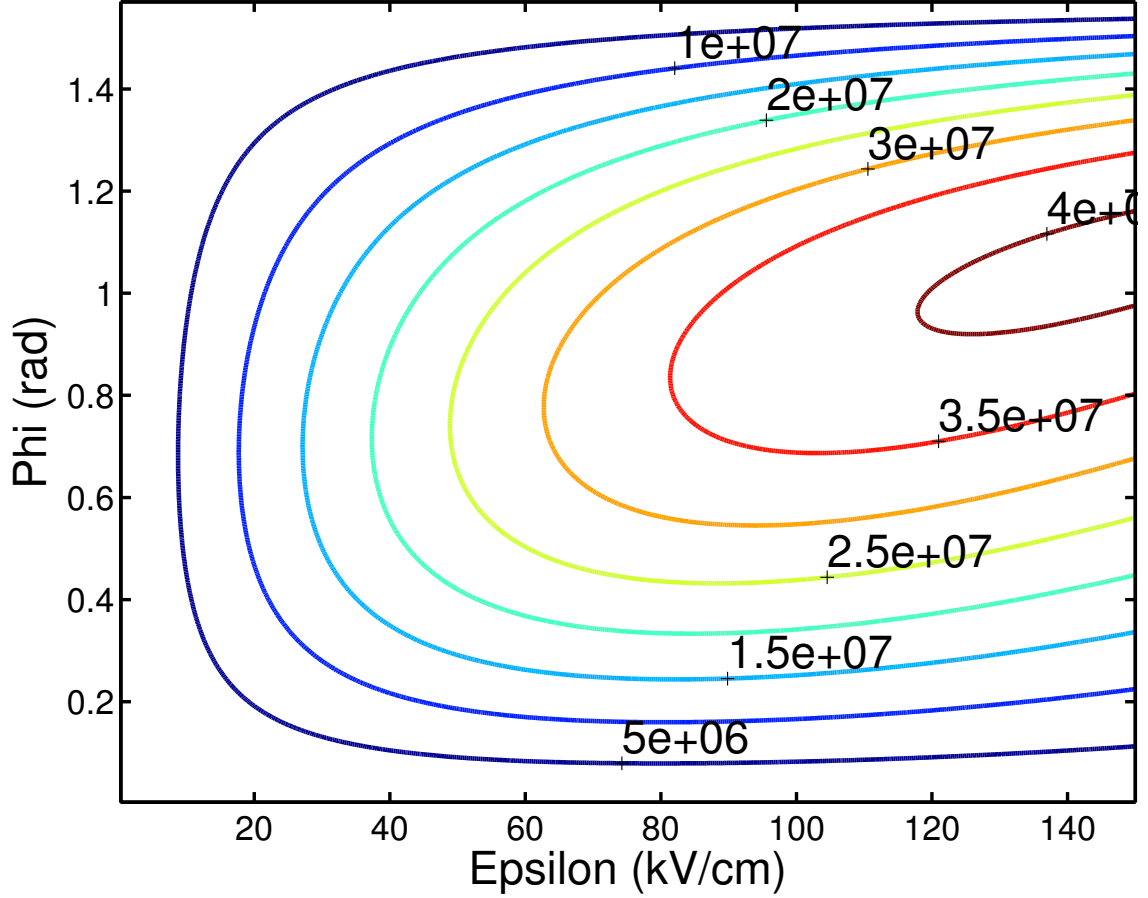


Figure 61: Contour plot of Rabi frequency at quadratic resonance as a function of AC electric field amplitude ϵ (horizontal axis) up to 150kV/cm and magnetic field angle ϕ (vertical axis). Frequency is in Hz. A trend of higher electric field amplitude and higher magnetic field angle is visible.

angles, i.e. with the magnetic field pointing more along the $[001]$ axis, due to the strong quadratic coefficient of $g_{[001]}$ and linearity of $g_{[1\bar{1}0]}$.

An electric field amplitude was sought to minimize the Rabi time at the quadratic resonance as well. Unlike the linear resonances (figure 62), which shows constantly higher Rabi frequencies as electric field increases, the quadratic resonance (figure 63) shows that there is a local maximum frequency of 41.0MHz (24.4ns) at 150kV/cm, magnetic field angle of approximately 1.07 radians. Another minimum is present

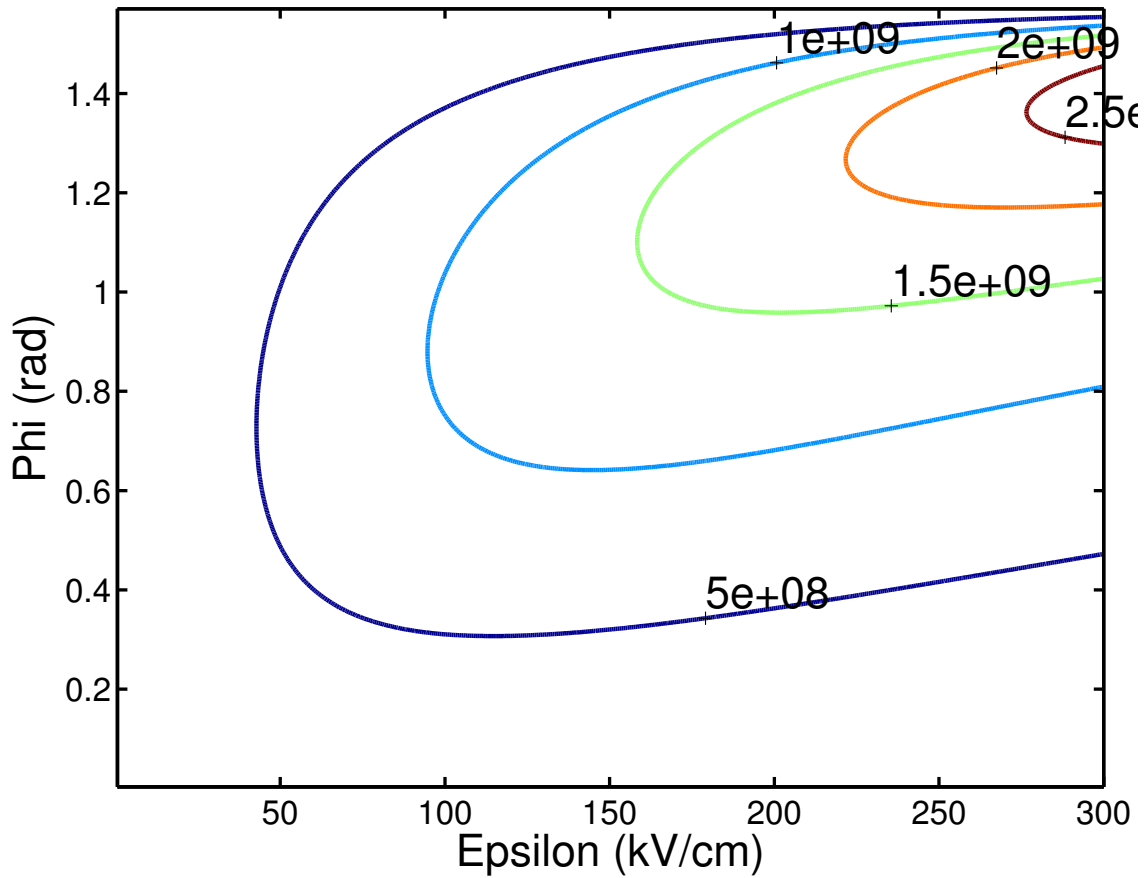


Figure 62: Contour plot of Rabi frequency at linear resonance as a function of AC electric field amplitude ϵ (horizontal axis) up to 300kV/cm and magnetic field angle ϕ (vertical axis). Frequency is in Hz. A trend of higher electric field amplitude and higher magnetic field angle is visible. The quadratic resonance has a maximum Rabi frequency near an AC electric field amplitude of 150kV/cm and magnetic field angle of 1.0733. The fastest Rabi frequency is 1.4GHz (0.696ns) at this linear resonance for electric field amplitudes up to 150kV/cm. Faster Rabi frequencies are possible at this resonance with higher electric field amplitudes.

at unrealistically high electric fields, e.g. 600kV/cm, far past the breakdown voltage. The same general trend of higher electric field amplitude giving higher Rabi frequency with a maximum at higher magnetic field angle as is seen at the linear resonance. Due to the much faster speed, the clear gains as electric field amplitude is increased, and the lower resonant frequency, the linear resonance is likely the most

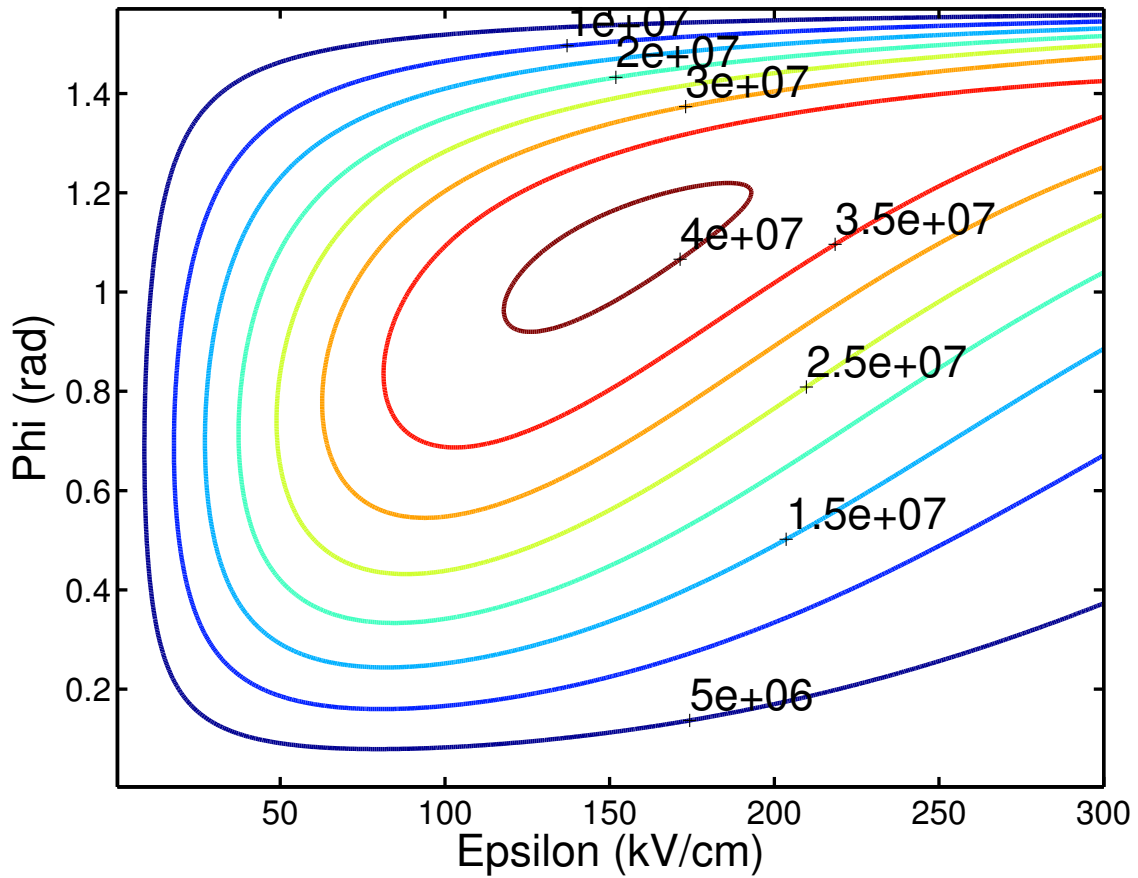


Figure 63: Contour plot of Rabi frequency at quadratic resonance as a function of AC electric field amplitude ϵ (horizontal axis) up to 300kV/cm and magnetic field angle ϕ (vertical axis). Frequency is in Hz. A trend of higher electric field amplitude and higher magnetic field angle is visible. The n quadratic resonance has a maximum Rabi frequency near an AC electric field amplitude of 150kV/cm and magnetic field angle of 1.0733. The fastest Rabi frequency 41MHz (24ns) at this quadratic resonance for electric field amplitudes up to 150kV/cm.

advantageous of the two resonances for spin manipulation.

CHAPTER VII

NONRESONANT SPIN CONTROL OVER THE ENTIRE BLOCH SPHERE

Although one may use g-tensor modulation resonance to coherently manipulate a single spin in a quantum dot, another method is also possible. To understand this second method, the g-tensor will be discussed from a different perspective, following up with the proposed device in abstract. The results of section III will be used to illustrate and predict the performance of such a device.

Resonant methods are ideal when a single dot in an ensemble of dots is to be addressed. Not every dot will be resonant at exactly the same frequency (due to variation of the g-factor from structural or from minor applied field (electric or magnetic) fluctuations. Because of this, one may use a resonant method to only address a specific dot within the ensemble, as shown in figure 8. If, however, a single dot can be addressed directly, non-resonant manipulation may be used if certain conditions are met. These conditions will be discussed shortly.

VII.1 Overview of the Method

Full Bloch-sphere control of a single spin requires the ability to switch between two orthogonal spin precession axes. This is illustrated in figure 70. Flipping a spin then requires positioning the spin at 45° from the spin precession axis (or, equivalently, positioning the spin precession axis at 45° from the spin. The spin then precesses about the spin precession axis until it is 45° from the spin precession axis on the other side. The spin precession axis is then switched off, and the orthogonal spin precession axis switched on. From there, the spin precesses about the new axis

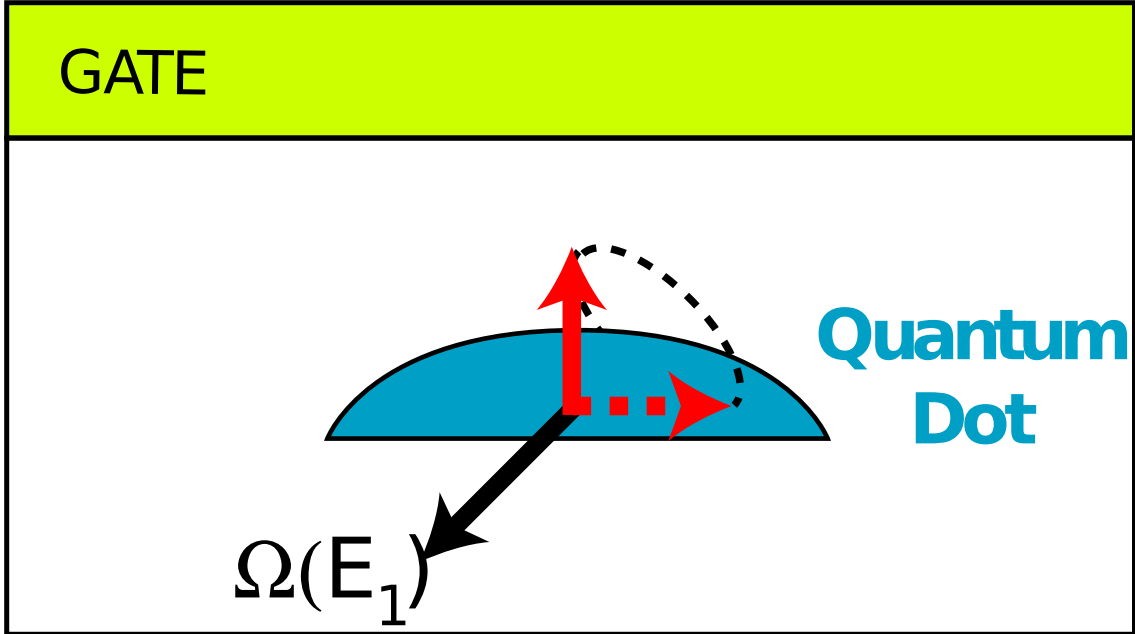


Figure 64: Abstract schematic of the proposed device during spin manipulation. The spin precesses about the spin precession axis, going from “up” to an intermediate state.

(starting out 45° from that axis) until it is 45° from the axis on the other side. The spin is now “down”. This is illustrated diagrammatically in figures 64 (precession about the first axis) and 65 (precession about the second axis).

VII.2 Orthogonality of the single spin precession axis

Conventional ESR, described in section I.7.2, is conventionally performed using two applied magnetic fields: one static magnetic field and one transverse “rotating” (oscillating) magnetic field. Using standard spin manipulation techniques, to create two orthogonal spin precession axes, one would similarly apply two perpendicular magnetic fields. However, it will become apparent that, under certain circumstances, full Bloch-sphere control of the carrier spin may be achieved through a single electrical gate.

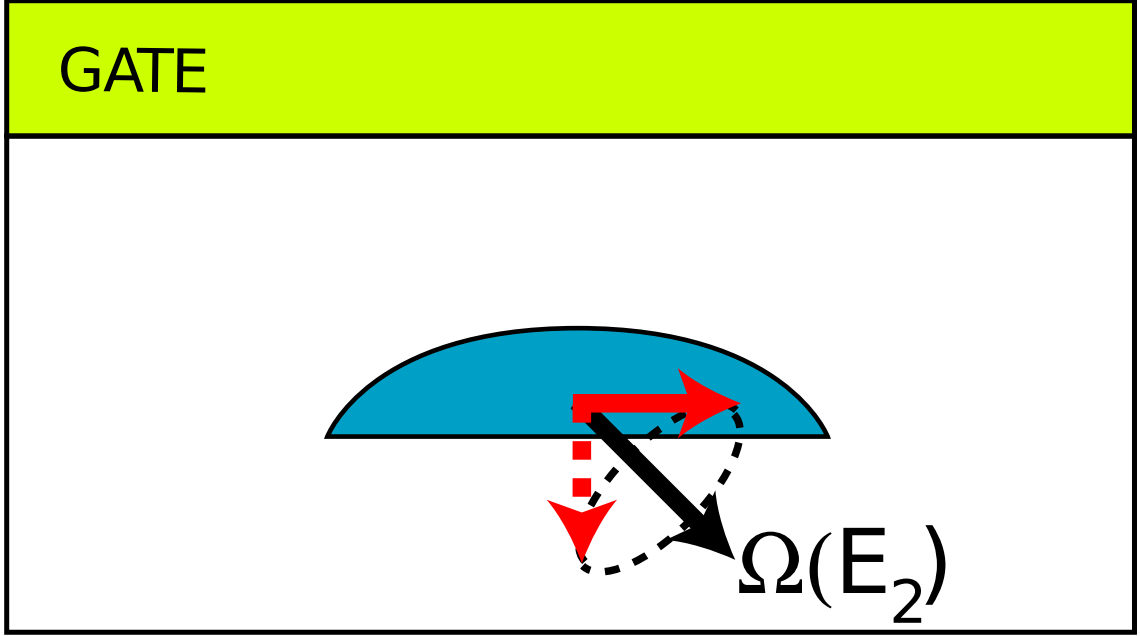


Figure 65: Abstract schematic of the proposed device during spin manipulation. The spin completes precession to “down” by precessing around the second axis, which is orthogonal to the spin precession axis in 64.

By using the electron g-tensor and modifying it through application of an electric field from a gate deposited above the dot, it is possible to achieve full Bloch sphere control of the electron spin. The spin precession axis is dependent upon the g-tensor and magnetic field:

$$\vec{\Omega} = \begin{pmatrix} g_{[001]}(E)B_{[001]} \\ g_{[110]}(E)B_{[110]} \\ g_{[1\bar{1}0]}(E)B_{[1\bar{1}0]} \end{pmatrix} \quad (232)$$

the dependence upon the electric field strength (the electric field being oriented along the growth direction) being made explicit.

Then, having two orthogonal spin precession axes requires the following equation

to be satisfied:

$$0 = \vec{\Omega}(E_1) \cdot \vec{\Omega}(E_2) \quad (233)$$

$$= \begin{pmatrix} g_{[001]}(E_1)B_{[001]} \\ g_{[110]}(E_1)B_{[110]} \\ g_{[1\bar{1}0]}(E_1)B_{[1\bar{1}0]} \end{pmatrix} \cdot \begin{pmatrix} g_{[001]}(E_2)B_{[001]} \\ g_{[110]}(E_2)B_{[110]} \\ g_{[1\bar{1}0]}(E_2)B_{[1\bar{1}0]} \end{pmatrix} \quad (234)$$

$$= g_{[001]}(E_1)g_{[001]}(E_2)B_{[001]}^2 + g_{[110]}(E_1)g_{[110]}(E_2)B_{[110]}^2 + g_{[1\bar{1}0]}(E_1)g_{[1\bar{1}0]}(E_2)B_{[1\bar{1}0]}^2 \quad (235)$$

In two dimensions, this simplifies to

$$0 = g_\alpha(E_1)g_\alpha(E_2)B_\alpha^2 + g_\beta(E_1)g_\beta(E_2)B_\beta^2 \quad (236)$$

$$g_\alpha(E_1)g_\alpha(E_2)B_\alpha^2 = -g_\beta(E_1)g_\beta(E_2)B_\beta^2 \quad (237)$$

$$\frac{g_\alpha(E_1)g_\alpha(E_2)}{g_\beta(E_1)g_\beta(E_2)B_\beta^2} = -\frac{B_\beta^2}{B_\alpha^2} \quad (238)$$

$$\frac{g_\alpha(E_1)g_\alpha(E_2)}{g_\beta(E_1)g_\beta(E_2)B_\beta^2} = -\tan^2(\theta) \quad (239)$$

where θ is the angle of the magnetic field relative to the β axis. In the two-dimensional case, then, there can be a solution only if there is a sign change in the g-factor along one and *only* one axis. In three dimensions, one or two sign changes are required, three will not work. This then fixes one component of the magnetic field (one angle if using a spherical coordinate system). In a 3-D system, the other two parameters (second angle, magnetic field magnitude) are still free to

be chosen as needed (i.e. optimized).

Restricting the problem to two dimensions for simplicity, it remains to be shown that a quantum dot which has an electron g-factor that changes sign as a function of electric field is possible. By reviewing figure 11, it is apparent that found just such a dot has been found.

In figure 11, there is a clear sign change along only the $[110]$ direction. Simplifying the situation by limiting ourselves to two directions, we can choose either the $[001]$ or the $[1\bar{1}0]$ direction for our second direction. Noting that at an electric field strength of roughly 50kV/cm , $g_{[110]}$ becomes 0, and that this therefore implies that $\vec{\Omega}(E = 50\text{kV/cm}) = g_{\alpha}(E = 50\text{kV/cm})B_{\alpha}\hat{\alpha}$ i.e. that the spin precession axis becomes oriented along the $\hat{\alpha}$ axis. It has been shown[54] that orienting the spin along the $[001]$ axis is advantageous for optical interactions. Therefore, $\hat{\alpha} = \hat{e}_{[001]}$ is selected.

VII.3 Calculating device performance with electron spins

The first order of business is to find the spin precession time around one axis $\vec{\Omega}_1 = \vec{\Omega}(E_1), t_1$. This is done by constructing the spin precession vector at “45 degrees” on each side of the spin precession axis. Without loss of generality, the $\hat{z}$ axis can be defined to lie along $\vec{\Omega}_1$. In the z-basis, the $|+z\rangle$ state is $|\uparrow\rangle$, and the $|+x\rangle$ state is $\frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle)$. The first “45 degrees” state will thus be the state $k \left[|\uparrow\rangle + \frac{1}{\sqrt{2}}(|\uparrow\rangle + |\downarrow\rangle) \right]$ (with normalization constant k). The $|-x\rangle$ state is $\frac{1}{\sqrt{2}}(|\uparrow\rangle - |\downarrow\rangle)$, which gives a “45 degrees” state 180 degrees about $\hat{z}$, opposite from

the first of $k' \left[|\uparrow\rangle + \frac{1}{\sqrt{2}} (|\uparrow\rangle - |\downarrow\rangle) \right]$. The normalization constants are found to be

$$k = k' = \frac{1}{\sqrt{4 + 2\sqrt{2}}} \quad (240)$$

and the states then are

$$|\psi_0\rangle = \frac{1}{\sqrt{4 + 2\sqrt{2}}} \left[(\sqrt{2} + 1) |\uparrow\rangle + |\downarrow\rangle \right] \quad (241)$$

$$|\psi_0\rangle = \frac{1}{\sqrt{4 + 2\sqrt{2}}} \left[(\sqrt{2} + 1) |\uparrow\rangle - |\downarrow\rangle \right] \quad (242)$$

The initial state $|\psi_0\rangle$ is then precessed until it overlaps completely with the final state, $|\psi_1\rangle$. That is,

$$\left| \langle \psi_1 | e^{-i\frac{\Omega t}{2\hbar} \sigma_z} | \psi_0 \rangle \right|^2 = 1 \quad (243)$$

is sought. This then becomes

$$\begin{aligned}
1 &= \left| \langle \psi_1 | \cos\left(\frac{\Omega t}{2\hbar}\right) - i\sigma_z \sin\left(\frac{\Omega t}{2\hbar}\right) | \psi_0 \rangle \right|^2 \\
1 &= \left| \frac{(\sqrt{2}+1) \langle \uparrow | + \langle \downarrow |}{\sqrt{4+2\sqrt{2}}} \left[\cos\left(\frac{\Omega t}{2\hbar}\right) - i\sigma_z \sin\left(\frac{\Omega t}{2\hbar}\right) \right] \frac{(\sqrt{2}+1) |\uparrow\rangle - |\downarrow\rangle}{\sqrt{4+2\sqrt{2}}} \right|^2 \\
1 &= \frac{1}{(4+2\sqrt{2})^2} \left| \left[(\sqrt{2}+1)^2 - 1 \right] \cos\left(\frac{\Omega t}{2\hbar}\right) - i \sin\left(\frac{\Omega t}{2\hbar}\right) \left[(\sqrt{2}+1)^2 + 1 \right] \right|^2 \\
1 &= \frac{1}{(4+2\sqrt{2})^2} \left| \left[2+2\sqrt{2} \right] \cos\left(\frac{\Omega t}{2\hbar}\right) - i \sin\left(\frac{\Omega t}{2\hbar}\right) \left[4+2\sqrt{2} \right] \right|^2 \\
1 &= \left(\frac{2+2\sqrt{2}}{4+2\sqrt{2}} \right)^2 \cos^2\left(\frac{\Omega t}{2\hbar}\right) + \left(\frac{4+2\sqrt{2}}{4+2\sqrt{2}} \right)^2 \sin^2\left(\frac{\Omega t}{2\hbar}\right) \\
1 &= \left(\frac{2+2\sqrt{2}}{4+2\sqrt{2}} \right)^2 \cos^2\left(\frac{\Omega t}{2\hbar}\right) + \sin^2\left(\frac{\Omega t}{2\hbar}\right) \\
1 &= \left(\frac{2+2\sqrt{2}}{4+2\sqrt{2}} \right)^2 \cos^2\left(\frac{\Omega t}{2\hbar}\right) + 1 - \cos^2\left(\frac{\Omega t}{2\hbar}\right)
\end{aligned}$$

giving the final equation

$$0 = \frac{(2+2\sqrt{2})^2 - (4+2\sqrt{2})^2}{(4+2\sqrt{2})^2} \cos^2\left(\frac{\Omega t}{2\hbar}\right) \quad (244)$$

This equation is solved when

$$\frac{\Omega t}{2\hbar} = \frac{\pi}{2} \quad (245)$$

and therefore the time required to flip a spin 180 degrees about the $\hat{z}$ axis is

$$t = \frac{\pi\hbar}{\Omega} \quad (246)$$

It then remains to be shown how this is to be achieved using a single electrical gate and a static magnetic field.

VII.4 Review of device operation

The design, then, includes for parts: initialization, manipulation, manipulation along an orthogonal axis, and then readout. This is illustrated in figure 70. In figure 66, the electric field is turned off, which orients $\vec{\Omega}(E = 0)$ along $\hat{e}_{[001]}$. An electron is injected optically and sits in the dot. The electric field is turned on ($E = E_1$) (figure

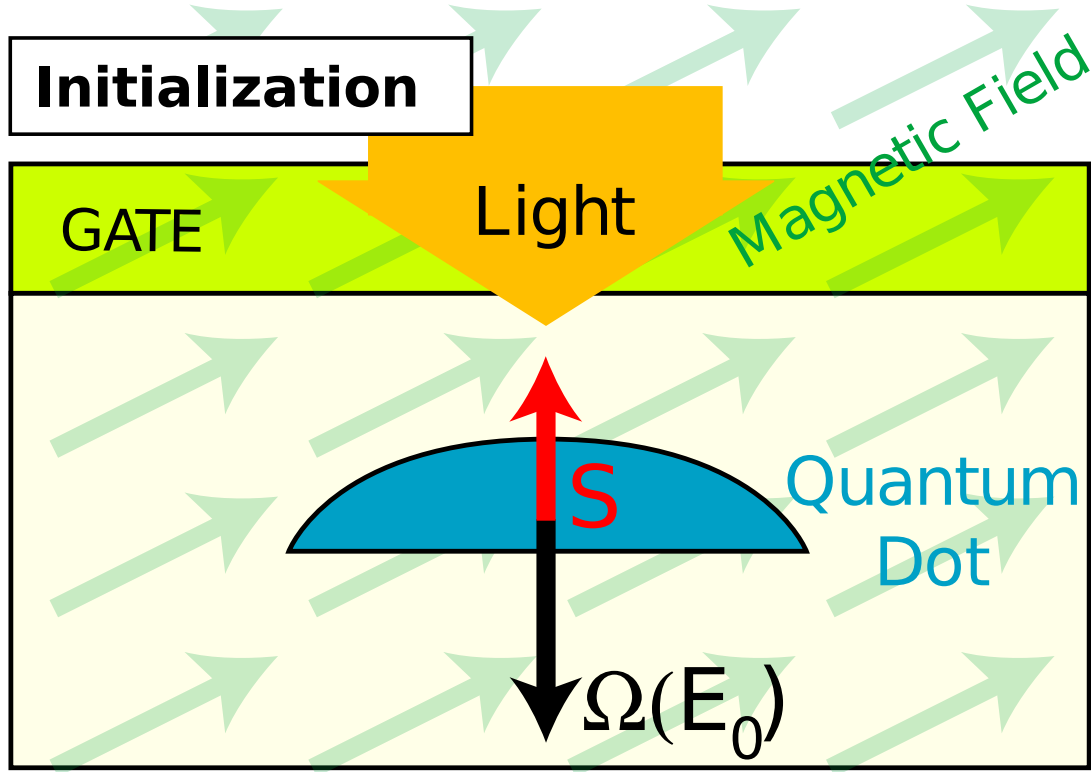


Figure 66: The spin in the quantum dot is initialized by turning off the electric field and optically injecting an electron.

67), which orients $\vec{\Omega}(E_1)$ at an angle of 45 degrees from $\hat{e}_{[001]}$. The spin begins to

precess around $\vec{\Omega}(E_1)$. At a time τ_1 (figure 68), the spin has precessed 180 degrees

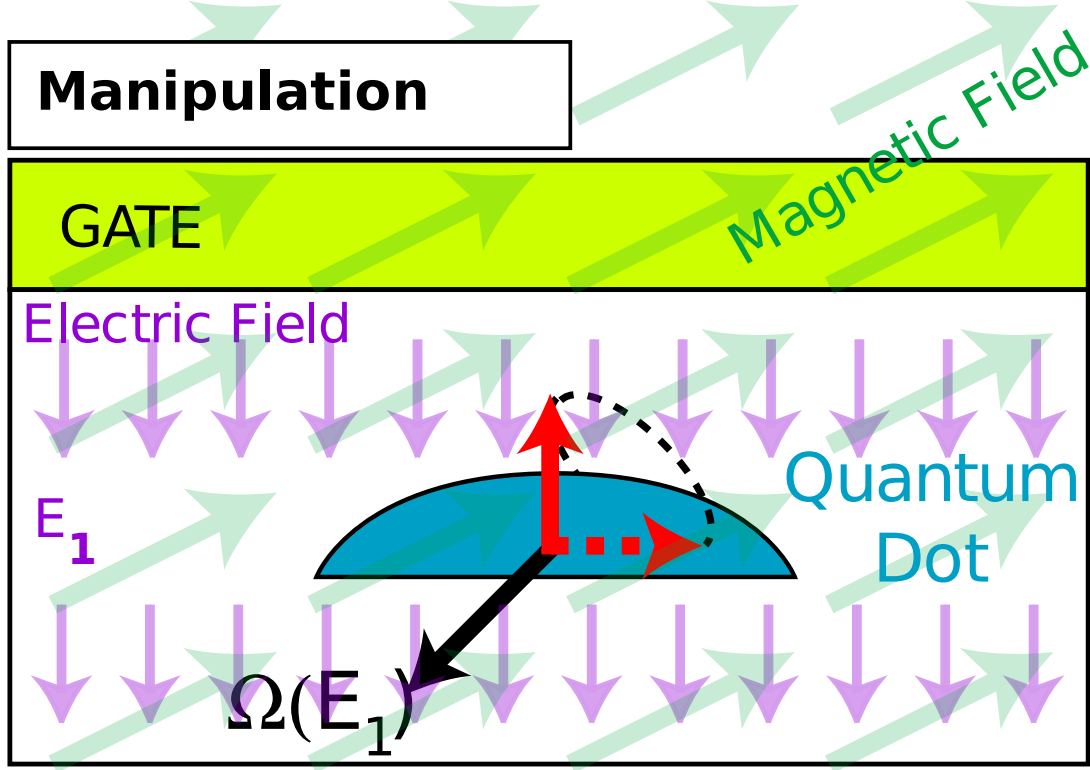


Figure 67: The electric field is turned on, and the spin begins to precess.

around $\vec{\Omega}(E_1)$. The electric field is now switched to E_2 and the spin precession axis switches to $\vec{\Omega}(E_2)$, which has been set to be fully orthogonal to $\vec{\Omega}(E_1)$. The spin begins to precess about this axis, and at some time τ_2 (figure 69) the spin is then oriented along $-\hat{e}_{[001]}$. The electric field is switched off, causing the spin precession axis to be aligned along $-\hat{e}_{[001]}$. As the spin and the spin precession axis are parallel, the spin no longer precesses. The spin may now be read out at some later time by Kerr or Faraday rotation.

To help visualize the process, figure 70 summarizes figures 66, 67, 68, and 69.

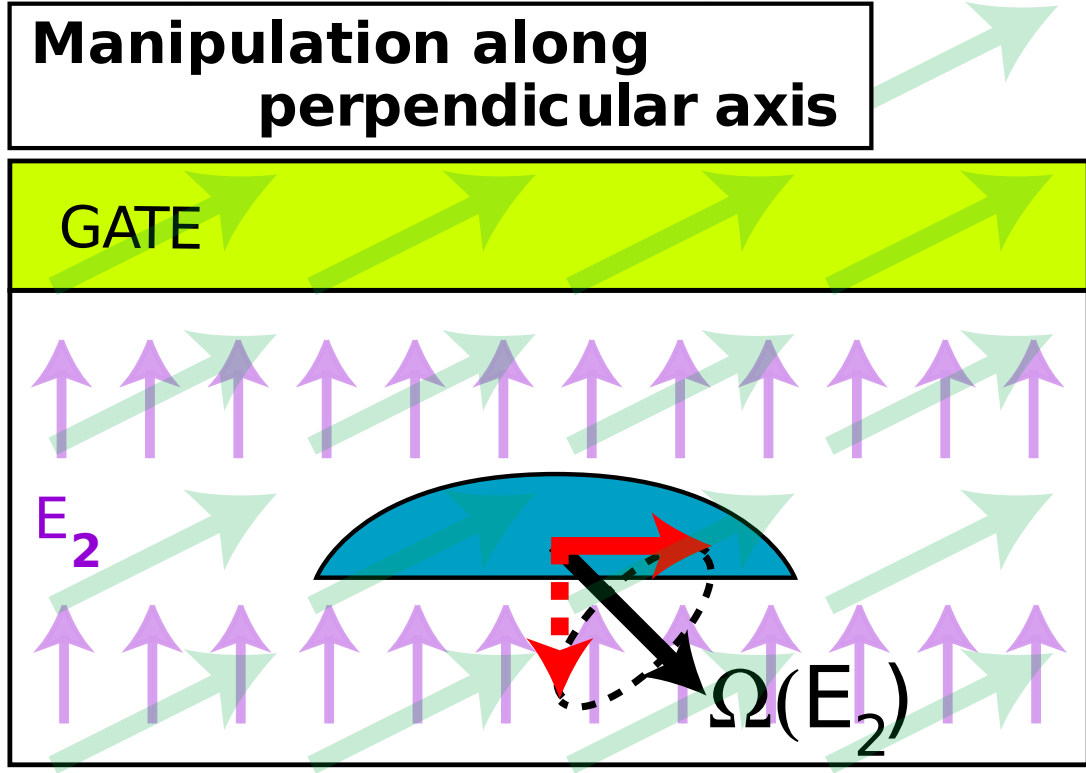


Figure 68: Once the spin has precessed 180 degrees about $\vec{\Omega}(E_1)$, the electric field is changed to E_2 and the spin begins to precess about $\vec{\Omega}(E_2)$.

VII.5 Generality of the Result

Given the above discussion, it might be construed that only a very few dots can be used to construct the outlined device. On the contrary, there is a *range* of dots which have a sign change along one (or more) axes, as shown in figures 13 and 14. The lines show where the g-factor along the given axis crosses 0. Clearly, the choice of dot above was not unique.

One may have noticed a disconnect between figure 11 and the g-tensor being used in figure 70. Indeed, the g-tensor in figure 11 would have resulted in a slightly

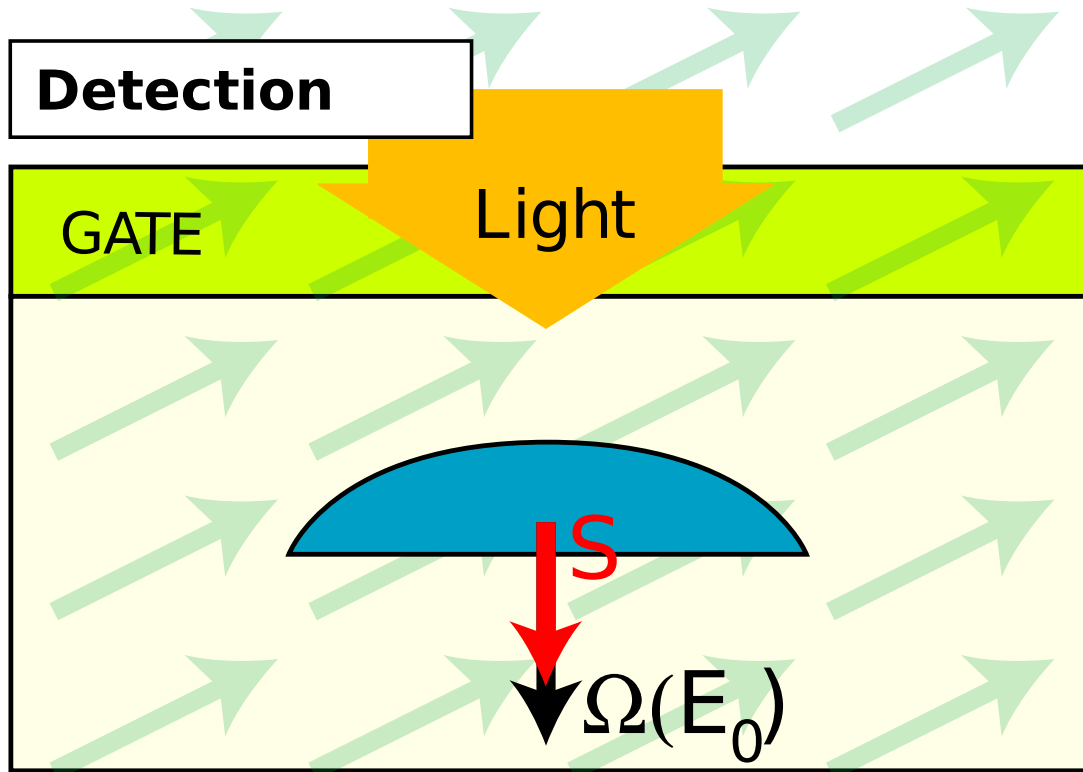


Figure 69: Once the spin has precessed 180 degrees about $\vec{\Omega}(E_2)$, the electric field is turned off and the spin ceases precession. The spin is then read out optically by Kerr or Faraday rotation.

different set of electrical fields (figure 71).

The g-tensor used to determine the electric fields in figure 70 was taken from interpolating between two points which straddled a sign change along one axis, and finding a dot which should have a zero g-factor at zero electric field (nicknamed “zero-zero”).

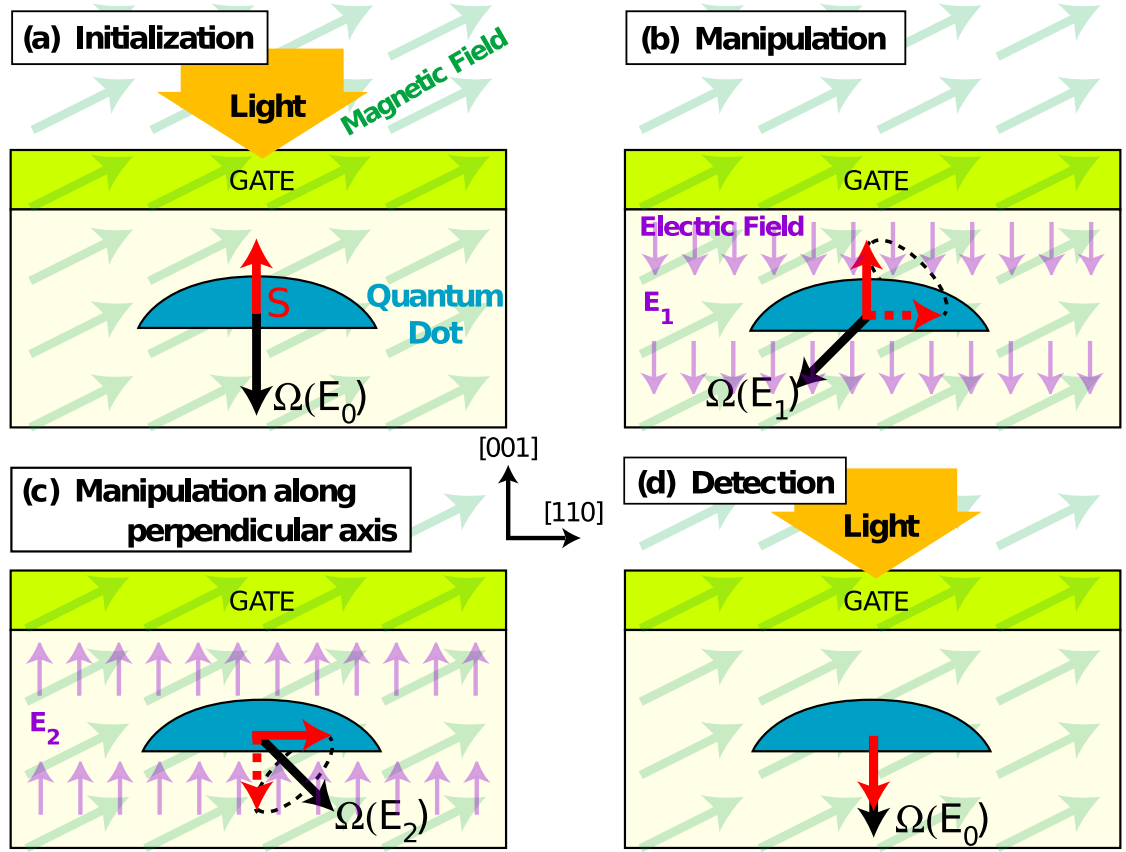


Figure 70: Schematic illustrating the operation of the device. a) The spin in the quantum dot is initialized by turning off the electric field and optically injecting an electron. b) The electric field is turned on, and the spin begins to precess. c) Once the spin has precessed 180 degrees about $\vec{\Omega}(E_1)$, the electric field is changed to E_2 and the spin begins to precess about $\vec{\Omega}(E_2)$. d) Once the spin has precessed 180 degrees about $\vec{\Omega}(E_2)$, the electric field is turned off and the spin ceases precession. The spin is then read out optically by Kerr or Faraday rotation.

VII.6 Non-resonant spin manipulation with hole spins

As with the non-resonant spin manipulation method for electrons, the non-resonant method can be adapted for use in holes. Because the spin operator for holes was defined as three times the spin operator for electrons, the solution to

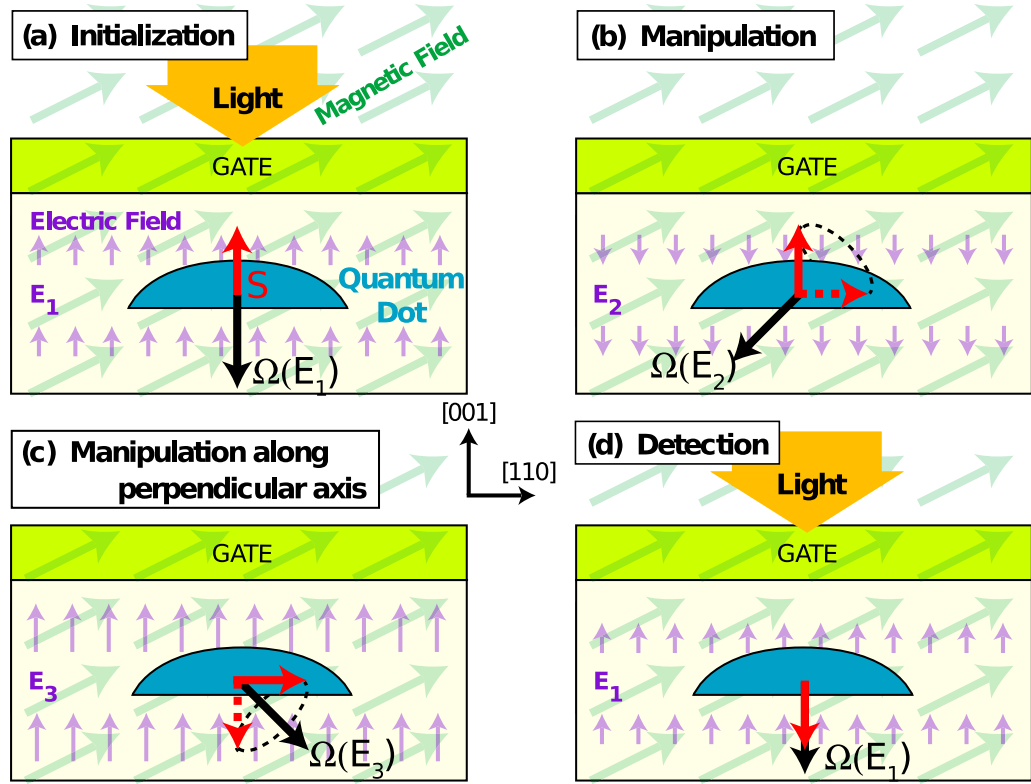


Figure 71: Illustration of using the device with the electron g-tensor from figure 11.

equation 244 becomes

$$\frac{3\Omega t}{2\hbar} = \frac{\pi}{2} \quad (247)$$

and therefore the time required to flip a spin 180 degrees about the $\hat{z}$ axis for holes is

$$t = \frac{\pi\hbar}{3\Omega} \quad (248)$$

The electron g-tensor in section VII had an approximately constant (with respect to electric field) g-factor along the [001] direction. Because of this, it was relatively

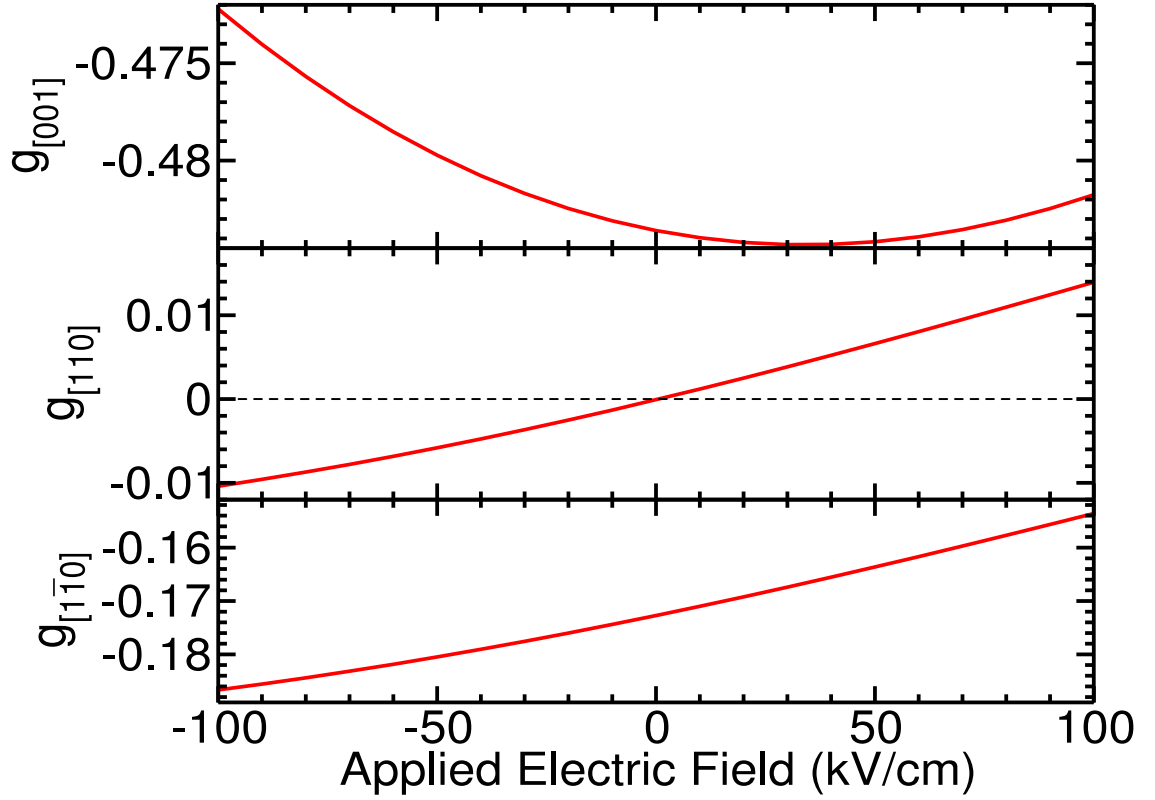


Figure 72: $\mathbf{g}$ tensor components of an InAs/GaAs quantum dot as a function of an applied electric field along the $[001]$ direction. The lens-shaped quantum dot has a height of 6.2 nm, base diameter in the $[110]$ direction of 10.7 nm and base diameter in the $[1\bar{1}0]$ direction of 6.2 nm. The dashed black line is $g_{[110]} = 0$, showing $g_{[110]}$ crossing from negative to positive near 0 kV/cm.

straightforward to create two orthogonal spin precession axes using the constant $g_{[001]}$ and linear in-plane g-factor with sign change. The hole case is more complicated. None of the g-factors are approximately constant across a range, and one of them ($g_{[001]}$) even has a strong quadratic dependence on electric field. Because of this complexity, it was necessary to numerically find the optimum electric field values.

As with the electrons, two orthogonal spin precession axes are required to per-

form the spin manipulation. Equation 239 shows that of the three independent parameters which may be changed— E_1 , E_2 , and ϕ —one of these free parameters is fixed from the orthogonality condition. ϕ (θ in equation 239) is arguably the simplest of the three to determine from the orthogonality condition. In addition, E_1 and E_2 are not *completely* independent, as there must be precisely one sign change in order for the orthogonality condition to be solvable. The sign change is present at high electric fields, visible in the top graph ($g_{[001]}$) in figure 39. Choosing the $[001]$ and $[1\bar{1}0]$ axes due to their large g-factor changes leaves only assigning E_1 and E_2 . E_1 is arbitrarily chosen as being in less than the lower solution of $g_{[001]} = 0$ (called E_l) but greater than -150kV/cm, and E_2 as constrained to be within the two solutions of $g_{[001]} = 0$, E_u and E_l . From the quadratic fit above, the two solutions of $g_{[001]} = 0$ are

$$E_u = 92.6721\text{kV/cm} \quad (249)$$

$$E_l = -102.1130\text{kV/cm} \quad (250)$$

and the constraints on the electric field are, restated,

$$-150\text{kV/cm} \leq E_1 \leq E_l = -102.1130\text{kV/cm} \quad (251)$$

$$-102.1130\text{kV/cm} = E_l \leq E_2 \leq E_u = 92.6721\text{kV/cm} \quad (252)$$

For each combination of E_1 and E_2 , ϕ is determined from the orthogonality

condition. The ϕ values are presented as a contour plot in figure 73.

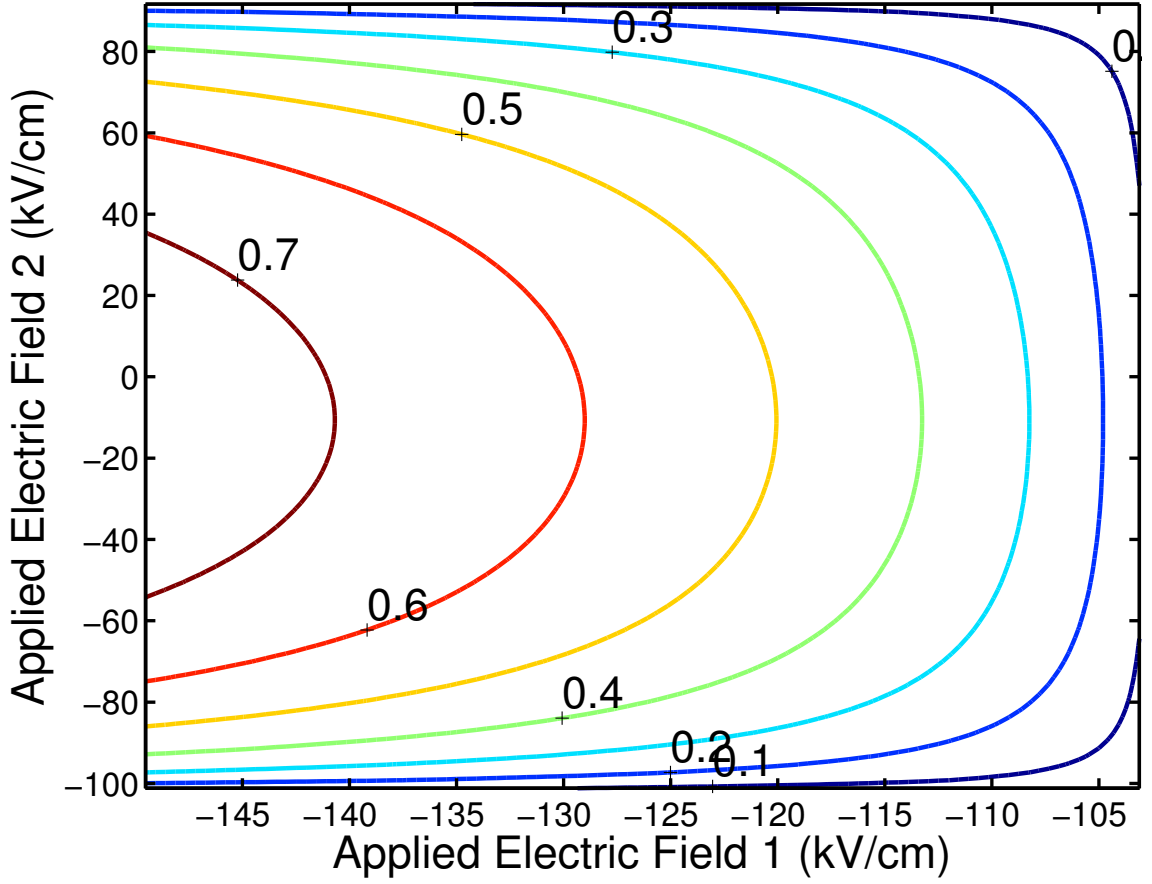


Figure 73: Magnetic field angle ϕ (from the $[1\bar{1}0]$ axis) in the $[001]$ - $[1\bar{1}0]$ plane. For each combination of E_1 (horizontal axis) and E_2 (vertical axis), the value of ϕ which is required to have two fully orthogonal spin precession axes is plotted. Angles are in radians.

Assigning ϕ based upon the orthogonality condition is sufficient to make the two spin precession axes $\vec{\Omega}_1$ and $\vec{\Omega}_2$ fully orthogonal and enable full Bloch-sphere control of the spin. This does not, however, optimize the speed. For instance, $\vec{\Omega}_1$ may be very small and $\vec{\Omega}_2$ may be very large, which may not be optimal. Because of the complicated dependence of $\vec{\Omega}_1$ and $\vec{\Omega}_2$ on electric field, a numerical approach to optimizing the total spin manipulation time was taken. As was done previously,

Matlab was used to numerically find the optimal value of E_2 which minimizes the total spin manipulation time. A plot of the total spin manipulation time (time required to take a spin oriented along $[001]$ and flip it to the $[00\bar{1}]$ direction) is given in figure 74

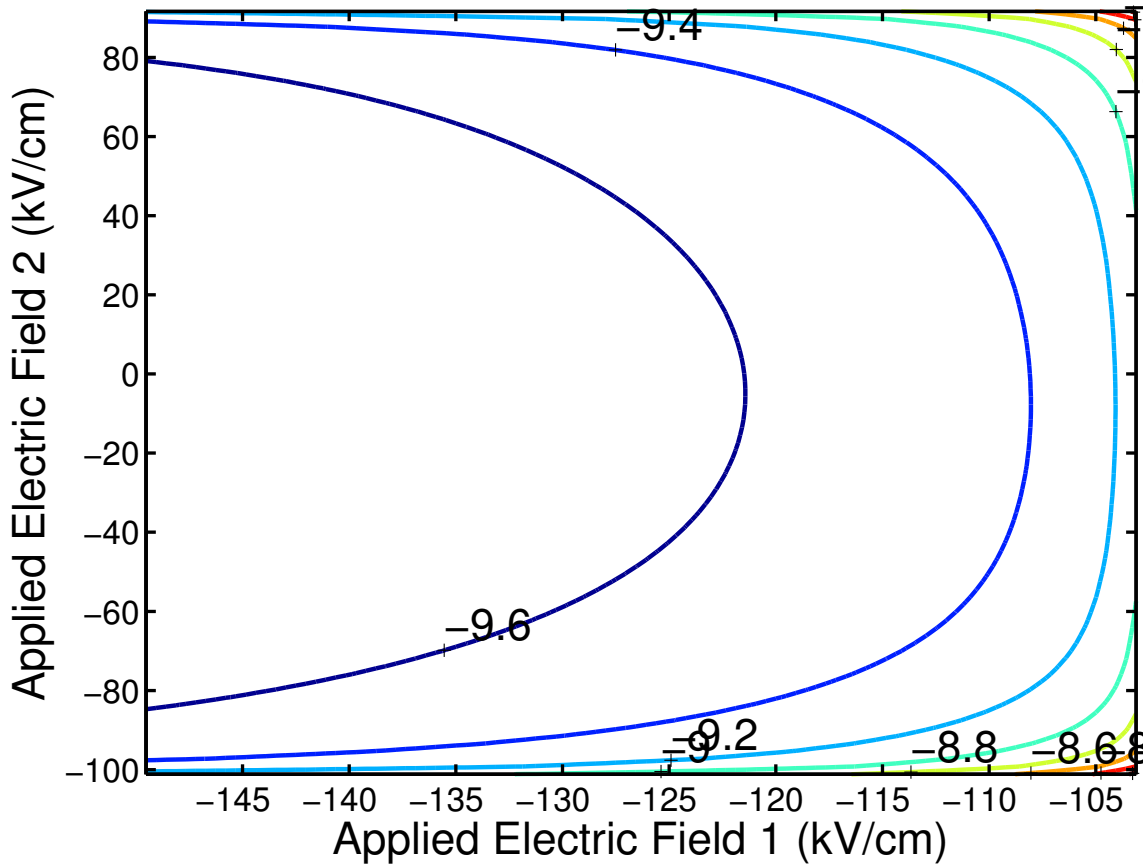


Figure 74: Total spin manipulation time for all valid electric field combinations E_1 and E_2 . For each combination of E_1 (horizontal axis) and E_2 (vertical axis), the time for full spin manipulation (time required to take a spin oriented along $[001]$ and flip it to the $[00\bar{1}]$) is plotted. In this figure, plots the log (base 10) of the time required to flip a spin, to better clarify the structure.

Figure 74 indicates that a maximum field E_1 and setting E_2 to close to 0 will be close to optimal. Using Matlab to numerically solve the maximum value of E_2 for $E_1 = -150\text{kV/cm}$ results in a value for E_2 of 3.1kV/cm . This combination gives a

magnetic field angle of $\phi = 0.759$ radians, which results in a total spin manipulation time of 0.180 ns (88 ps about $\vec{\Omega}_1$ and 92 ps about $\vec{\Omega}_2$). As with the Rabi calculations, the electric field values were restricted to the range -150 kV/cm to 150 kV/cm. Figure 74 that additional speed may be gained by increasing the magnitude of E_1 , if possible.

CHAPTER VIII CONCLUSIONS

In this dissertation, the background has been laid for a fairly thorough understanding of g-tensor modulation resonance (g-TMR) and non-resonant spin manipulation. A quick introduction to $k \cdot p$ theory was provided in the first few chapters, with an eye to including electric and magnetic fields and the spin-orbit interaction.

The $k \cdot p$ method was used to calculate the g-tensors of electrons and holes in a variety of structures. The g-tensor behavior of these dots as a function of electric field and quantum dot shape was then used to perform g-TMR as well as, in the presence of a sign change, perform non-resonant manipulation. In addition, a device was proposed which uses the electric field dependence of the g-tensor to perform non-resonant manipulation. A resonant spin manipulation time of 6.7ns and a non-resonant spin manipulation time of 3.9ns were found for electrons in InAs/GaAs quantum dots.

A resonant spin manipulation time of 0.64ns and a non-resonant manipulation time of 180ps for holes in $\text{In}_{0.5}\text{Ga}_{0.5}\text{As}/\text{GaAs}$ quantum dots was found in section VII. A second Rabi resonance at double the Larmor frequency was found in hole systems due to the strong quadratic dependence of the hole g-factor along [001]. Unfortunately, no sign change was able to be identified in InAs/GaAs quantum dots.

APPENDIX A CRAIG PRYOR'S DOTCODE

A.1 From Geometry to Realspace Grid

To begin with, the a grid is created that will describe the quantum dot and surrounding material. Each grid site is assigned one material type or another (e.g. GaAs, InAs, or vacuum) but not more than one. Material assignments are done based upon a text file which describes the geometry.

```
material GaAs

material InAs(0.50)GaAs(0.50)

#   principle radii   center

inside ellipsoid    0 0 -2.7    1 1 0 9.01    -1 1 0 13.31    0 0 1 5.4

z> plane 0 0 1    0 0 -2.71
```

In this example (actually used for some of the subsequent calculations), the default (here, barrier) material is specified as GaAs. If no other material is specified, it will be assigned to the grid site. Special regions follow, each started with a “material” line. In this case, a quantum dot of $In_{0.5}Ga_{0.5}As$ is specified (`InAs(0.50)GaAs(0.50)`). This InGaAs is to be assigned everywhere within the AND of the subsequent conditions. Here, the InGaAs is to be assigned inside of an ellipsoid centered at 0, 0, -2.7 (x, y, z coordinates), with axes of 9.01nm along [1, 1, 0]; 13.31nm along [-1, 1, 0]; and 5.4nm along [0, 0, 1]. In addition, $In_{0.5}Ga_{0.5}As$ is to be used for grid sites with the z coordinate above a plane perpendicular to the [0, 0, 1] axis located at 0, 0, -2.71nm. An extra 0.01 has been added

to ensure that the grid sites which are exactly on the line will be included in the quantum dot.

The center of the grid is placed at the origin of the coordinate system used in the geometry file. This allows structures to be placed around the resulting grid as desired, e.g. placing quantum dots on or embedded in an interface. To get a maximally predictable structure, care should be taken to position the nanostructure geometry consistently within the final grid structure, e.g. having the border consistently at the center of a grid site. This is difficult to achieve for a general structure, as the structure geometry may be of an intermediate size. Careful choice of grid density can be taken to assist in creating a structure.

If investigating the effects of a change (e.g. changing a quantum dot height), the change should be large enough to create as nearly identical border conditions between structures as possible. That is, the continuous geometric borders should fall within roughly the same location inside a grid site, and the change should be an integer number of grid sites. This is simplified by using a script which automatically takes this into consideration when taking user input, creating the geometry file, and making the grid. Such a script has been created by Craig Pryor, `makeSphericalCap.pl`. However, discretization errors and slight discrepancies between the continuous and discrete representations of a structure will continue to exist, although they are minimized.

A.2 Electric Field Difficulties

The electric field is applied as a linear potential along the $\hat{z}$ axis. The electron wavefunction leaks out of the quantum dot to some extent, as one would expect when the barrier height is lowered (see figure A.1). Because of the small Zeeman splitting (it is on the order of a few tens of μeV at 1T), it is necessary to use a larger number of grid sites than is common for other applications. The electric field is applied as a linear potential along $\hat{z}$, and thus will be negative at some places along the $\hat{z}$ axis and will pull the “barrier potential” down with it. Therefore, for any non-zero electric field, there will be some distance from the dot where the electric potential will negate the barrier potential and then cause it to go below the bound state energy. The larger the electric field applied, the closer this crossing point gets to the quantum dot. This can cause the electron to leak completely out of the dot, and become localized at the bottom of the calculating box, leading to non-physical g-factors. To solve this problem, the electric potential is set to zero outside of certain distance from the center of the dot, as illustrated in figure A.1, creating an electric field “box” around the quantum dot. After performing tests to make sure the electric field box size used did not significantly affect calculations, this box was used to prevent electron leakage during the calculations.

A.3 Additional Magnetic Field Considerations

As mentioned in chapter I.4, the differentiation operator is represented by a finite difference of values on the realspace grid:

$$\frac{d\psi}{dx} \approx \frac{\psi(x+d) - \psi(x)}{d} \quad (253)$$

In the presence of a magnetic field, however, the finite differencing operator must be changed to preserve the Aharonov-Bohm phase around a plaquette through which the magnetic field points:

$$\frac{d\psi}{dx} \approx \frac{\psi(x+d)U_d(x) - \psi(x)U_d(x)}{d} \quad (254)$$

where, for example,

$$U_x(\vec{r})U_y(\vec{r} + \epsilon\hat{x})U_x^\dagger(\vec{r} + \epsilon\hat{y})U_y^\dagger(\vec{r}) = e^{i\frac{e}{\hbar}B_\perp\epsilon^2} \quad (255)$$

To prevent a class of numerical artifacts from occurring, the flux through each of these plaquettes is kept integer by taking the excess magnetic field and placing it on the exterior few lattice sites where the wavefunction is approximately zero.[109][88]

A.4 Calculating Spin States in a Strained Quantum Dot

The $\vec{k} \cdot \vec{p}$ method has been outlined in section I.4. Strain was introduced to the model using Bahder's method[1], discussed in section I.5. Adding the magnetic field was introduced in section I.6. The pieces are now brought together into a whole

picture.

Each portion of the overall Hamiltonian—strain-dependent 8-band $\vec{k} \cdot \vec{p}$, magnetic field interaction, and electric field—are assembled into a final Hamiltonian. The eigenvalues and eigenstates of the Hamiltonian are found by means of the Lanczos algorithm.[110] The energies in which the eigenenergies are expected to fall are bracketed above and below, and must be estimated based upon experience or test calculation.

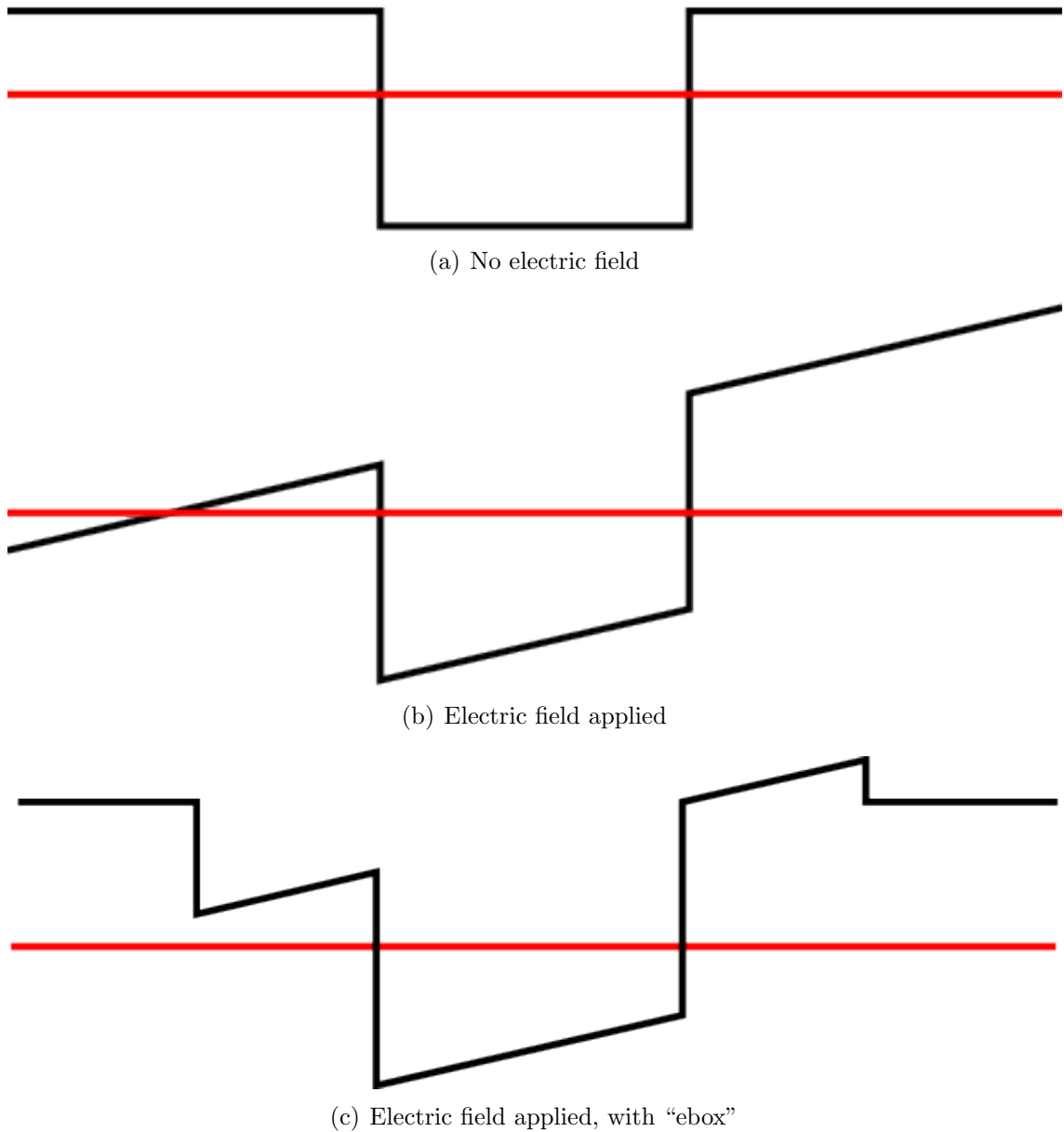


Figure A.1: Cartoon of the zone-center band structure of a quantum dot along the z -axis. 1(a) Quantum dot without applied electric field. The ground state energy is represented by the red line. 1(b) Same quantum dot with applied constant electric field (linear potential). Note the point on the left at which the potential crosses the first ground state energy and the electron is no longer confined to the dot. 1(c) The same quantum dot, with a constant electric field (linear electric potential) applied. Outside of a specified range from the dot center (the “ebox”, the electric potential is set to 0, regaining containment.

APPENDIX B

MAKESPHERICALCAP.PL

```
#!/usr/bin/perl

use strict;
use Getopt::Std;

getopts('d');

use vars '$opt_d';

# args:  <height>           height of the dot in grid
#        units (an integer)
#        <avg radius>       in the plane
#        <x>
#        <y>                x,y specify the direction
#        of elongation
#        <aspect ratio>     (diameter along x,y) /
#        (diameter along perpendicular direction)
#        <grid spacing>[nm]
#        <well material>
#        <barrier material>
#        <matdb filename>

if( $#ARGV != 12 ){ print STDERR " args:  <output
filename>\n"; print STDERR "           <height>
height of the dot in grid units (an integer)\n";
print STDERR "           <r_xy>           radius
in the x-y plane in grid units (an integer)\n"; print
STDERR "           <x>\n"; print STDERR "           <y>
x,y specify the direction of elongation\n"; print
STDERR "           <aspect ratio>         (diameter along
x,y) / (diameter along perpendicular direction)\n";
print STDERR "           <grid spacing>[nm]\n"; print
STDERR "           <well material>\n"; print STDERR "
<barrier material>\n"; print STDERR "           <matdb
filename>\n"; print STDERR "           <x box size>\n";
print STDERR "           <y box size>\n"; print STDERR
"           <z box size>\n"; exit(1);
}
```

```

# generate a grid file for a spherical-cap shaped dot
# with possible elongation

# and make a geom file with this:
#
# inside ellipsoid  0 0 <c>  <dir0> <r>  <dir1> <r>
# 0 0 1 <r>
# z> plane 0 0 1  0 0 0
#
# h = height of dome, rxy = radius in xy plane, cp=z coord
#   of the plane defining the bottom of the dome (<0)
#   h =.; rxy =.;
#ans = Solve[ (rxy + 1/2)^2 + (c + cp)^2 == (-c + h -
# 1/2)^2, c ];
#Simplify[c /. ans[[1]]]

my ($outfile,      # filename of output (a *.displ file)
    $n,            # height in grid units (integer) $rxy,
    # mean radius in x-y plane $x,                # $y,
    # x,y specify direction of elongation $aspectRatio, #
    (diameter along x,y) / (diameter along perpendicular
    direction) $a,                # grid spacing in
    nanometers $wellMaterial, # material comprising the
    dot $barrMaterial, # material comprising the barrier
    $matdb,          # name of file to be used for the
    material database $xb,          # $yb,          #
    $zb              # x,y,z dimensions of box (in number
    of grid sites) ) = @ARGV;

my $cp = -0.0001; # z-coord of the plane.  Make the plane
    barely below a grid site plane.
# z-coord of the center of the sphere:
my $c = ( $cp*$cp + $n - $n*$n + $rxy + $rxy*$rxy )/(
    1.0 - 2.0*$cp - 2.0*$n);
my $r = $n - $c - 0.5; # radius of the sphere

$c -= int( $n / 2.0 ); # shift the dome downward by
    1/2 its height, so it is centered at the
$cp -= int( $n / 2.0 ); # origin of the calculation box.

# scale to physical grid spacing
$c *= $a;

```

```

$cp *= $a;
$r  *= $a;

my $dir0 = " $x $y 0";
my $dir1 = "-$y $x 0";

my $r1 = $r * sqrt( $aspectRatio );
my $r2 = $r / sqrt( $aspectRatio );
my $r3 = $r;

my $hh = sprintf("%3.2f",$a * $n);

open FOUT, "+>tmpgeom";

print FOUT "# makeSphericalCap.pl ", "@ARGV", "\n";
print FOUT "material GaAs\n";
print FOUT "material InAs\n";
print FOUT "# principle radii center\n";
print FOUT "inside ellipsoid 0 0 $c $dir0 $r1 $dir1
    $r2 0 0 1 $r3\n";
print FOUT "z> plane 0 0 1 0 0 $cp\n";
close FOUT;

my $cmd1 = "makeGrid tmpgeom $matdb $outfile.displ
    $xb $yb $zb $a > tmpmakeGrid.log"; # rm
    *.$barrMaterial.scatter";
sleep 10;
my $cmd2 = "mv tmpmakeGrid.log
    $outfile.displ/.makeGrid.log; rm tmpgeom";

if( $opt_d ){ print "$cmd1\n$cmd2\n";
}else{ system( $cmd1 ); system( $cmd2 );

    my $c1 = "grep InAs $outfile.displ/.makeGrid.log |
grep -v x" ;
    my $out = ` $c1 `;
    chomp $out;
    $out =~ s/^\s+//;
    $out =~ s/\s+$//;
    my @tem = split "\n", $out;
    $out = $tem[$#tem];
    my $volume = (split /\s+/, $out)[2]; # dot volume in
grid sites

```



```

#my $aa = "tail -1 $outfile.displ/.makeGrid.log";
#print STDERR "aa=$aa\n";
#$aa = '$aa';
#print STDERR "aa=$aa\n";
#$aa = (split /\s+/, $aa)[2];
#my @aa = split 'x' , $aa;
# 1      InAs      6.36396 x 6.36396 x 3
#my @aa = (split /\s+/, $aa);
#$aa[0] = $aa[2];
#$aa[1] = $aa[4];
#$aa[2] = $aa[6];

    #printf( "d($dir0)=%3i (%5.4f nm)\n", $aa[0], $aa[0]
* $a);
    #printf( "d($dir1)=%3i (%5.4f nm)\n", $aa[1], $aa[1]
* $a);
    #printf( "d(0 0 1)=%3i (%5.4f nm)\n", $aa[2], $aa[2]
* $a);
    #printf( "d($dir0)/d($dir1)=%5.4f\n", $aa[0]/$aa[1]);
    #printf( "d($dir0)/d(0 0 1)=%5.4f\n", $aa[0]/$aa[2]);
    #printf( "d($dir1)/d(0 0 1)=%5.4f\n", $aa[1]/$aa[2]);
    #print "@aa","\n";
    #print "volume = $volume grid sites\n";
    #print "volume = ", $volume * $a * $a * $a , "
nm^3\n"; }

```

This has been modified slightly from Craig Pryor's version. The temporary geometry file has been kept in place. A 10-second sleep has been added. The \$aa and @aa variable and array have been commented out. For the dots in sections VI and VII, the materials were manually set to InAs/GaAs.

APPENDIX C ELECTRON CODE

The following Matlab code was used to calculate and optimize the Rabi times and the nonresonant spin manipulation times for the electrons in InAs/GaAs.

```
function [g0, gt, E] = get_gs
%%calculate the Rabi time
%our z g-factor
gz = -5.046969E-01;
gz = gz + -5.050866E-01;
gz = gz + -5.053451E-01;
gz = gz + -5.054759E-01;
gz = gz + -5.054805E-01;
gz = gz + -5.053633E-01;
gz = gz + -5.051262E-01;
gz = gz + -5.047700E-01;
gz = gz + -5.042985E-01;
gz = gz + -5.037138E-01;
gz = gz + -5.030170E-01;
%average the values
gz = gz/11;
%Our g-factors min/max along 110
gmin_110 = -6.180173E-03;
gmax_110 = 8.386213E-03;
%Our g-factors min/max along 1-10
gmin_1m10 = -1.768331E-01;
gmax_1m10 = -1.571847E-01;
%electric field min/max
Emin = 0;
Emax = 100;
%The center about which we'll oscillate
E0 = 50;
%Amplitude of the oscillations (minus sign because it's
    a neg slope)
E=50;

m_110 = (gmax_110 - gmin_110)/(Emax - Emin);
m_1m10 = (gmax_1m10 - gmin_1m10)/(Emax - Emin);
g0_110 = gmin_110;
g0_1m10 = gmin_1m10;
check = m_110*Emax + g0_110;
check = m_1m10*Emax + g0_1m10;
```

```

g0_110 = g0_110 + m_110*E0;
g0_1m10 = g0_1m10 + m_1m10*E0;

g0 = [ g0_1m10 0 0; 0 2*g0_110 0; 0 0 gz ];
gt = [ m_1m10 0 0; 0 2*m_110 0; 0 0 0 ];
%Gets the direction of the magnetic field for setting up
    two orthogonal,
% equal magnitude precession axes.

[g0, gt, E] = get_gs_zero_zero;

%Omega 1 is electric field -E
%Omega 2 is electric field +E
gx1=g0(1, 1) - E*gt(1, 1);
gy1=g0(2, 2) - E*gt(2, 2);
gz1=g0(3, 3) - E*gt(3, 3);

gx2=g0(1, 1) + E*gt(1, 1);
gy2=g0(2, 2) + E*gt(2, 2);
gz2=g0(3, 3) + E*gt(3, 3);

gamma=(gz1*gz1 - gz2*gz2)/(gz1*gz2);
phi = atan(sqrt((gx1*gx1 - gx2*gx2) -
    gamma*gx1*gx2)/(gamma*gy1*gy2 - (gy1*gy1 - gy2*gy2)));
theta = atan(sqrt(-gz1*gz2/(gx1*gx2*cos(phi)*cos(phi) +
    gy1*gy2*sin(phi)*sin(phi))));

phiopi = phi/pi
thetaopi = theta/pi
[phin, phid] = rat(phiopi, 1e-4);
[thetan, thetad] = rat(thetaopi, 1e-4);
fprintf(1, 'B-field direction: phi=%d/%d pi theta=%d/%d
    pi\n', phin, phid, thetan, thetad)

%(magnitude of) electron charge (c)
qe=1.602176487e-19;
%c (m/s)
c=299792458;
%Electron mass, in eV/c
me=8.18710438e-14;
%me, in eV

```

```

me=8.18710438e-14*c^2;
%http://physics.nist.gov/cgi-bin/cuu/Value?mubev|
%   search_for=bohr+magneton
uB = 5.7883817555e-5;
%hbar, in eV*s
hbar=6.58211899e-16;

%Find the time to rotate from up to down (2*precession
    around one omega at
% 45 degree angle.)
B = vb(1, phi, theta)
Bx = B(1)
By = B(2)
Bz = B(3)
%bsquare = dot(B, B)
%gminus = (g0 - E*gt)
%gplus = (g0 + E*gt)
omega1 = (g0 - E*gt)*B
omega2 = (g0 + E*gt)*B
%check1 = omega1(1) - (g0(1, 1) - E*gt(1,1))*B(1)
%check2 = omega1(2) - (g0(2, 2) - E*gt(2,2))*B(2)
%check3 = omega1(3) - (g0(3, 3) - E*gt(3,3))*B(3)
%check4 = omega1(1) - gx1*Bx
%check5 = omega1(2) - gy1*By
%check6 = omega1(3) - gz1*Bz
%check7 = omega2(1) - gx2*Bx
%check8 = omega2(2) - gy2*By
%check9 = omega2(3) - gz2*Bz
omega1_mag=sqrt(dot(omega1, omega1))
omega2_mag=sqrt(dot(omega2, omega2))
omega_check=dot(omega1, omega2)
time=2*pi*hbar/(uB*omega2_mag)
time=2*pi*hbar/(uB*omega1_mag)%Gets the direction of the
    magnetic field for setting up two orthogonal,
%   equal magnitude precession axes.

%g_001 at E=0kV/cm
g_001_E0=-0.483594814816858;
%g_001 at E=-100kV/cm
g_001_E1=-0.472218681926037;
%g_001 at E=+100kV/cm
g_001_E2=-0.481756667219331;

%g_110 at E=0kV/cm

```

```

g_110_E0=-8.560250827868e-05
%g_110 at E=-100kV/cm
g_110_E1=-0.0104034257947905
%g_110 at E=+100kV/cm
g_110_E2=0.0139389389803865

%(magnitude of) electron charge (c)
qe=1.602176487e-19;
%c (m/s)
c=299792458;
%Electron mass, in eV/c
me=8.18710438e-14;
%me, in eV
me=8.18710438e-14*c^2;
%http://physics.nist.gov/cgi-bin/cuu/Value?mubev|
%   search_for=bohr+magneton
uB = 5.7883817555e-5;
%hbar, in eV*s
hbar=6.58211899e-16;

theta =
    atan(sqrt(-(g_110_E1*g_110_E2)/(g_001_E1*g_001_E2)))
B_001=sin(theta);
B_110=cos(theta);
B=[B_001; B_110];

omega1=[g_001_E1 0; 0 g_110_E1]*B
omega2=[g_001_E2 0; 0 g_110_E2]*B

omega1_mag=sqrt(dot(omega1, omega1))
omega2_mag=sqrt(dot(omega2, omega2))
omega_check=dot(omega1, omega2)
time2=pi*hbar*2/(uB*omega2_mag)
time1=pi*hbar*2/(uB*omega1_mag)
freq1=1/time1
freq2=1/time2
%total_time=time2+time1
average_time=(time2+time1)/2
freq=1/total_timefunction [g0, gt, E] = get_gs
%%calculate the Rabi time
%our z g-factor
gz = -5.046969E-01;
gz = gz + -5.050866E-01;
gz = gz + -5.053451E-01;

```

```

gz = gz + -5.054759E-01;
gz = gz + -5.054805E-01;
gz = gz + -5.053633E-01;
gz = gz + -5.051262E-01;
gz = gz + -5.047700E-01;
gz = gz + -5.042985E-01;
gz = gz + -5.037138E-01;
gz = gz + -5.030170E-01;
%average the values
gz = gz/11;
%Our g-factors min/max along 110
gmin_110 = -6.180173E-03;
gmax_110 = 8.386213E-03;
%Our g-factors min/max along 1-10
gmin_1m10 = -1.768331E-01;
gmax_1m10 = -1.571847E-01;
%electric field min/max
Emin = 0;
Emax = 100;
%The center about which we'll oscillate
E0=50;
%Amplitude of the oscillations (minus sign because it's
    a neg slope)
E=50;

m_110 = (gmax_110 - gmin_110)/(Emax - Emin);
m_1m10 = (gmax_1m10 - gmin_1m10)/(Emax - Emin);
g0_110 = gmin_110 + m_110*E0;
g0_1m10 = gmin_1m10 + m_1m10*E0;
%check = m_110*E + g0_110 - gmax_110;
%check = m_1m10*E + g0_1m10 - gmax_1m10;
%check = m_110*(-E) + g0_110 - gmin_110;
%check = m_1m10*(-E) + g0_1m10 - gmin_1m10;

%gmin = [ gmin_1m10 0 0; 0 gmin_110 0; 0 0 gz ];
%gmax = [ gmax_1m10 0 0; 0 gmax_110 0; 0 0 gz ];

g0 = [ g0_1m10 0 0; 0 g0_110 0; 0 0 gz ];
gt = [ m_1m10 0 0; 0 m_110 0; 0 0 0 ];

%gmin = (g0 - E*gt);
%gmax = (g0 + E*gt);function result=gimme_rabi(angles)
%RETURNS RABI_FREQUENCY
[g0, gt, E] = get_gs;

```

```

%1T
B=1;
result=rabi_frequency(B, angles(1), angles(2), g0, gt, E);
function result = maximize_me(angles)
result=-gimme_rabi(angles);
function result = minimize_me(angles)
result=gimme_rabi(angles);
function freq = rabi_frequency(B, phi, theta, g0, gt, E)
vB = vb(B, phi, theta);
Omega0 = g0*vB(B, phi, theta);
wt=gt*vB;
%(magnitude of) electron charge (c)
qe=1.602176487e-19;
%c (m/s)
c=299792458;
%Electron mass, in eV/c
me=8.18710438e-14;
%me, in eV
me=8.18710438e-14*c^2;
%http://physics.nist.gov/cgi-bin/cuu/Value?mubev|
%   search_for=bohr+magneton
uB = 5.7883817555e-5;
%hbar, in eV*s
hbar=6.58211899e-16;

%Assemble magnitudes
mag2_Omega0=dot(Omega0, Omega0);
mag_Omega0=sqrt(mag2_Omega0);
mag2_wt=dot(wt, wt);
mag_wt = sqrt(mag2_wt);

%E is in kV/cm
freq = uB/(hbar*4)*E*mag_wt*sqrt(1-dot(wt/mag_wt,
    Omega0/mag_Omega0)^2);

[E, g0, gt] = get_gs;

%Start at random points to find the global highest rabi
    freq.
nsamples=250
phi=randn(nsamples, 1)*2*pi;
theta=randn(nsamples, 1)*pi;
%our min rabi time and index

```

```

index=1;
pt_pair = fminsearch(@maximize_me, [phi(index),
    theta(index)]);
freq=gimme_rabi(pt_pair)
for i=2:nsamples %Search with our random starting point.
    pt = fminsearch(@maximize_me, [phi(i), theta(i)]);
    tfreq=gimme_rabi(pt); %searching for global maximum here.
    Is this better than previous? if(tfreq > freq)
        freq=tfreq
        index=i
        pt_pair=pt;
        pt_pair/pi
        [phin,phid]= rat(pt_pair(1)/pi, 1e-2);
        [thetan,thetad]= rat(pt_pair(2)/pi, 1e-2);
        fprintf(1, 'NEW pt pair: phi=%d/%d pi theta=%d/%d
            pi\n', phin, phid, thetan, thetad) end
end

i

pt_pair
pt_pair/pi
[phin,phid]= rat(pt_pair(1)/pi, 1e-2);
[thetan,thetad]= rat(pt_pair(2)/pi, 1e-2);
fprintf(1, 'NEW pt pair: phi=%d/%d pi theta=%d/%d pi\n',
    phin, phid, thetan, thetad)

%Plug in the optimal angle to find the shortest Rabi time.
freq=gimme_rabi(pt_pair)
time=1/freq

nphi=500
ntheta=nphi/2
phis=(1:nphi)/nphi*2*pi;
thetas=(1:ntheta)/ntheta*pi;

freqs=zeros(nphi, ntheta);
times=zeros(nphi, ntheta);
for p=1:nphi for t=1:ntheta
    freqs(p, t) = gimme_rabi([phis(p); thetas(t)]);
    %times(p, t) = 1/freqs(p, t); end
end
size(freqs)
size(times)

```



```

size(phis)
size(thetas)

%surf(thetas, phis, freqs, 'XLabel', 'Theta (rad)',
      'YLabel', 'Phi (rad)', 'ZLabel', 'Rabi Freq (Hz)')
[C, h] = contour(thetas, phis, freqs);
xlabel('Theta (rad)');
ylabel('Phi (rad)');
zlabel('Rabi Freq. (Hz)');
clabel(C);
%surf(thetas, phis, times)
%xlabel('Theta (rad)');
%ylabel('Phi (rad)');
%zlabel('Rabi Time (s)');
function Bvec = vb (B, phi, theta)
Bvec = B*[cos(phi)*sin(theta); sin(phi)*sin(theta);
          cos(theta)];

```

APPENDIX D

HOLE RABI MATLAB CODE

The following Matlab code was used to calculate and optimize the Rabi times of holes in InGaAs/GaAs quantum dots.

```
function hbar = get_hbar()
%Returns hbar in eV*s
hbar = 6.58211899e-16;function mu_b = get_mu_b()
%Bohr magneton in eV/T
mu_b = 5.7883817555e-5;function [omega_0_x, omega_0_y] =
    get_omega_0(a_x, b_x, c_x, a_y, b_y, c_y, epsilon, E0,
    B, phi)

omega_0_x = (a_x*(E0*E0 - epsilon*epsilon/2) + E0*b_x +
    c_x)*B*cos(phi);
omega_0_y = (a_y*(E0*E0 - epsilon*epsilon/2) + E0*b_y
    + c_y)*B*sin(phi);function [omega_l_x, omega_l_y] =
    get_omega_l(a_x, b_x, c_x, a_y, b_y, c_y, epsilon, E0,
    B, phi)

omega_l_x = (2*a_x*E0+b_x)*B*cos(phi);
omega_l_y = (2*a_y*E0+b_y)*B*sin(phi);

function [omega_q_x, omega_q_y] = get_omega_q(a_x, b_x,
    c_x, a_y, b_y, c_y, epsilon, E0, B, phi)

omega_q_x = a_x*B*cos(phi);
omega_q_y = a_y*B*sin(phi);
function omega_tl = get_omega_tl(a_x, b_x, c_x, a_y, b_y,
    c_y, epsilon, E0, B, phi)
%Gets the Rabi frequency

[omega_0_x, omega_0_y] = get_omega_0(a_x, b_x, c_x, a_y,
    b_y, c_y, epsilon, E0, B, phi);
[omega_l_x, omega_l_y] = get_omega_l(a_x, b_x, c_x, a_y,
    b_y, c_y, epsilon, E0, B, phi);

alpha = 3*(omega_l_x*omega_0_y -
    omega_l_y*omega_0_x)/(omega_0_x*omega_0_x +
    omega_0_y*omega_0_y);

omega_tl = get_mu_b()*epsilon*alpha/4/get_hbar()/pi;
```

```

function omega_tq = get_omega_tq(a_x, b_x, c_x, a_y, b_y,
    c_y, epsilon, E0, B, phi)
%Gets the Rabi frequency for the quadratic resonance

[omega_0_x, omega_0_y] = get_omega_0(a_x, b_x, c_x, a_y,
    b_y, c_y, epsilon, E0, B, phi);
[omega_q_x, omega_q_y] = get_omega_q(a_x, b_x, c_x, a_y,
    b_y, c_y, epsilon, E0, B, phi);

beta = 3*(omega_q_x*omega_0_y -
    omega_q_y*omega_0_x)/(omega_0_x*omega_0_x +
    omega_0_y*omega_0_y);

omega_tq = get_mu_b()*epsilon*beta/4/get_hbar()/pi;
function [ax, bx, cx, ay, by, cy] =
    get_quadratic_fit_params()
%Gets the fit parameters for our dot.

%x is 001, y is 1-10
%Quadratic coefficients
ax = 1.22488738015119e-5;
ay = 5.19640379470655e-7;

%Linear coefficientst
bx = 0.00011563935419;
by = 0.00018828135581;

%Constant
cx = -0.1159114101051;
cy = 0.14337590759878;
function inv_omega_tl = minimize_linear(args)

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

phi=args(1);
epsilon=args(2);
inv_omega_tl = 2/get_omega_tl(ax, bx, cx, ay, by,
    cy, epsilon, 0, 1, phi);function inv_omega_tl =
    minimize_linear_phi(phi)

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

epsilon=300;

```

```

inv_omega_tl = 2/get_omega_tl(ax, bx, cx, ay, by, cy,
    epsilon, 0, 1, phi);
function inv_omega_tl = minimize_linear_phi(phi)

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

epsilon=150;
inv_omega_tl = 2/get_omega_tl(ax, bx, cx, ay, by, cy,
    epsilon, 0, 1, phi);
function inv_omega_tq = minimize_quadratic(args)

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

phi=args(1);
epsilon=args(2);
inv_omega_tq = 2/get_omega_tq(ax, bx, cx, ay, by, cy,
    epsilon, 0, 1, phi);
function inv_omega_tq = minimize_quadratic_phi(phi)

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

epsilon=150;
inv_omega_tq = 2/get_omega_tq(ax, bx, cx, ay, by, cy,
    epsilon, 0, 1, phi);
%* Prints a 2D graph of the Rabi frequencies for epsilon,
    phi
%* Finds the fastest Rabi time possible.

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

[test_Omega_0_x, test_Omega_0_y] = get_omega_0(ax, bx,
    cx, ay, by, cy, 150, 0, 1, pi/4)
[test_Omega_1_x, test_Omega_1_y] = get_omega_1(ax, bx,
    cx, ay, by, cy, 150, 0, 1, pi/4)
omega_tl_test = get_omega_tl(ax, bx, cx, ay, by, cy, 150,
    0, 1, pi/4)
rabi_freq_test=omega_tl_test/2
rabi_time_test=1/rabi_freq_test

%graphing
nphi=500;
nepsilon=nphi;
phis=(1:nphi)/nphi*pi/2;

```

```

epsilons=(1:nepsilon)/nepsilon*150;
freqs=zeros(nphi, nepsilon);
times=zeros(nphi, nepsilon);
for p=1:nphi for t=1:nepsilon
    freqs(p, t) = 1/minimize_linear([phis(p);
    epsilons(t)]); end
end
[C, h] = contour(epsilons, phis, freqs,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Epsilon (kV/cm)', 'FontSize',20);
ylabel('Phi (rad)', 'FontSize',20);
zlabel('Rabi Freq. @ Linear Resonance (Hz)',
'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'rabi_linear_150.pdf'

```

```

epsilons=(1:nepsilon)/nepsilon*200;
freqs=zeros(nphi, nepsilon);
times=zeros(nphi, nepsilon);
for p=1:nphi for t=1:nepsilon
    freqs(p, t) = 1/minimize_linear([phis(p);
    epsilons(t)]); end
end
[C, h] = contour(epsilons, phis, freqs,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Epsilon (kV/cm)', 'FontSize',20);
ylabel('Phi (rad)', 'FontSize',20);
zlabel('Rabi Freq. @ Linear Resonance (Hz)',
'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'rabi_linear_200.pdf'

```

```

epsilons=(1:nepsilon)/nepsilon*300;
freqs=zeros(nphi, nepsilon);
times=zeros(nphi, nepsilon);
for p=1:nphi for t=1:nepsilon
    freqs(p, t) = 1/minimize_linear([phis(p);
    epsilons(t)]); end
end
[C, h] = contour(epsilons, phis, freqs,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);

```

```

xlabel('Epsilon (kV/cm)', 'FontSize',20);
ylabel('Phi (rad)', 'FontSize',20);
zlabel('Rabi Freq. @ Linear Resonance (Hz)',
    'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'rabi_linear_300.pdf'

epsilons=(1:nepsilon)/nepsilon*600;
freqs=zeros(nphi, nepsilon);
times=zeros(nphi, nepsilon);
for p=1:nphi for t=1:nepsilon
    freqs(p, t) = 1/minimize_linear([phis(p);
    epsilons(t)]); end
end
[C, h] = contour(epsilons, phis, freqs,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Epsilon (kV/cm)', 'FontSize',20);
ylabel('Phi (rad)', 'FontSize',20);
zlabel('Rabi Freq. @ Linear Resonance (Hz)',
    'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'rabi_linear_600.pdf'

phi_min = fminbnd(@minimize_linear_phi, 0, pi/2)
fastest_rabi_time=minimize_linear_phi(phi_min)
fastest_rabi_freq=1/fastest_rabi_time

phi_min_300 = fminbnd(@minimize_linear_phi2, 0, pi/2)
fastest_rabi_time_300=minimize_linear_phi2(phi_min_300)
fastest_rabi_freq_300=1/fastest_rabi_time_300
%* Prints a 2D graph of the Rabi frequencies for epsilon,
    phi
%* Finds the fastest Rabi time possible.

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

[test_Omega_0_x, test_Omega_0_y] = get_omega_0(ax, bx,
    cx, ay, by, cy, 150, 0, 1, pi/4)
[test_Omega_q_x, test_Omega_q_y] = get_omega_q(ax, bx,
    cx, ay, by, cy, 150, 0, 1, pi/4)
omega_tq_test = get_omega_tq(ax, bx, cx, ay, by, cy, 150,
    0, 1, pi/4)
rabi_freq_test=omega_tq_test/2

```

```

rabi_time_test=1/rabi_freq_test

%graphing
nphi=500;
nepsilon=nphi;
phis=(1:nphi)/nphi*pi/2;

epsilons=(1:nepsilon)/nepsilon*150;
freqs=zeros(nphi, nepsilon);
times=zeros(nphi, nepsilon);
for p=1:nphi for t=1:nepsilon
    freqs(p, t) = 1/minimize_quadratic([phis(p);
    epsilons(t)]); end
end
[C, h] = contour(epsilons, phis, freqs,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Epsilon (kV/cm)', 'FontSize',20);
ylabel('Phi (rad)', 'FontSize',20);
zlabel('Rabi Freq. @ Quadratic Resonance (Hz)',
    'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'rabi_quadratic_150.pdf'

epsilons=(1:nepsilon)/nepsilon*200;
freqs=zeros(nphi, nepsilon);
times=zeros(nphi, nepsilon);
for p=1:nphi for t=1:nepsilon
    freqs(p, t) = 1/minimize_quadratic([phis(p);
    epsilons(t)]); end
end
[C, h] = contour(epsilons, phis, freqs,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Epsilon (kV/cm)', 'FontSize',20);
ylabel('Phi (rad)', 'FontSize',20);
zlabel('Rabi Freq. @ Quadratic Resonance (Hz)',
    'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'rabi_quadratic_200.pdf'

epsilons=(1:nepsilon)/nepsilon*300;
freqs=zeros(nphi, nepsilon);
times=zeros(nphi, nepsilon);

```

```

for p=1:nphi for t=1:nepsilon
    freqs(p, t) = 1/minimize_quadratic([phis(p);
    epsilons(t)]); end
end
[C, h] = contour(epsilons, phis, freqs,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Epsilon (kV/cm)', 'FontSize',20);
ylabel('Phi (rad)', 'FontSize',20);
zlabel('Rabi Freq. @ Quadratic Resonance (Hz)',
    'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'rabi_quadratic_300.pdf'

epsilons=(1:nepsilon)/nepsilon*600;
freqs=zeros(nphi, nepsilon);
times=zeros(nphi, nepsilon);
for p=1:nphi for t=1:nepsilon
    freqs(p, t) = 1/minimize_quadratic([phis(p);
    epsilons(t)]); end
end
[C, h] = contour(epsilons, phis, freqs,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Epsilon (kV/cm)', 'FontSize',20);
ylabel('Phi (rad)', 'FontSize',20);
zlabel('Rabi Freq. @ Quadratic Resonance (Hz)',
    'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'rabi_quadratic_600.pdf'

phi_almost_min = fminbnd(@minimize_quadratic_phi, 0, pi/2)
almost_fastest_rabi_time=minimize_quadratic_phi(phi_almost_min)
almost_fastest_rabi_freq=1/almost_fastest_rabi_time

min = fminsearch(@minimize_quadratic, [1.1, 150])
fastest_rabi_time=minimize_quadratic([min(1), min(2)])
fastest_rabi_freq=1/fastest_rabi_time

```


APPENDIX E

HOLE NONRESONANT MATLAB CODE

The following Matlab code was used to calculate and optimize the Rabi times of holes in InGaAs/GaAs quantum dots.

```
function [gx, gy] = get_gs(ax, bx, cx, ay, by, cy, E)
%Gets [gx, gy] as a function of the a, b, and c's

gx=ax*E*E + bx*E + cx;
gy=ay*E*E + by*E + cy;function hbar = get_hbar()
%Returns hbar in eV*s
hbar = 6.58211899e-16;function mu_b = get_mu_b()
%Bohr magneton in eV/T
mu_b = 5.7883817555e-5;function [omega_x, omega_y] =
    get_omega(a_x, b_x, c_x, a_y, b_y, c_y, E, B, phi)

omega_x = (a_x*E*E + b_x*E + c_x)*B*cos(phi);
omega_y = (a_y*E*E + b_y*E + c_y)*B*sin(phi);
function phi = get_phi(gx1, gy1, gx2, gy2)

phi = atan(sqrt((-gx1*gx2)/(gy1*gy2)));function t =
    get_pi_time(omega_x, omega_y)

t = pi*get_hbar()/get_mu_b()/3/sqrt(omega_x*omega_x +
    omega_y*omega_y);function [ax, bx, cx, ay, by, cy]
    = get_quadratic_fit_params()
%Gets the fit parameters for our dot.

%x is 001, y is 1-10
%Quadratic coefficients
ax = 1.22488738015119e-5;
ay = 5.19640379470655e-7;

%Linear coefficientst
bx = 0.00011563935419;
by = 0.00018828135581;

%Constant
cx = -0.1159114101051;
cy = 0.14337590759878;
function t = get_total_time(args)
%Returns to total time for [phi, epsilon1, epsilon2].
```

```

% Useful for numerical minimization.

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

phi=args(1);
epsilon1=args(2);
epsilon2=args(3);

[omega_x, omega_y] = get_omega(ax, bx, cx, ay, by, cy,
    epsilon1, 1, phi);
time = get_pi_time(omega_x, omega_y);
[omega_x, omega_y] = get_omega(ax, bx, cx, ay, by, cy,
    epsilon2, 1, phi);
t = time + get_pi_time(omega_x, omega_y);
function time = minimize_nonresonant(e2)

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

%Best time is at e1=-150kV/cm
e1=-150;

[g1x, g1y] = get_gs(ax, bx, cx, ay, by, cy, e1);
[g2x, g2y] = get_gs(ax, bx, cx, ay, by, cy, e2);
phi = get_phi(g1x, g1y, g2x, g2y);
[omega1_x, omega1_y] = get_omega(ax, bx, cx, ay, by, cy,
    e1, 1, phi);
[omega2_x, omega2_y] = get_omega(ax, bx, cx, ay, by, cy,
    e2, 1, phi);
time = get_pi_time(omega1_x, omega1_y) +
    get_pi_time(omega2_x, omega2_y);
%* Prints a 2D graph of the Rabi frequencies for epsilon,
    phi
%* Finds the fastest Rabi time possible.

[ax, bx, cx, ay, by, cy] = get_quadratic_fit_params();

E0m = (-bx - sqrt(bx*bx-4*ax*cx))/(2*ax)
E0p = (-bx + sqrt(bx*bx-4*ax*cx))/(2*ax)

[test_omega_x, test_omega_y] = get_omega(ax, bx, cx, ay,
    by, cy, 150, 1, pi/4)
t_test = get_pi_time(test_omega_x, test_omega_y)
[g1x, g1y] = get_gs(ax, bx, cx, ay, by, cy, -150)
[g2x, g2y] = get_gs(ax, bx, cx, ay, by, cy, 0)

```

```

phi = get_phi(g1x, g1y, g2x, g2y)
[test1_omega_x, test1_omega_y] = get_omega(ax, bx, cx,
    ay, by, cy, -150, 1, phi)
[test2_omega_x, test2_omega_y] = get_omega(ax, bx, cx,
    ay, by, cy, 0, 1, phi)
i_should_be_zero = test1_omega_x*test2_omega_x +
    test1_omega_y*test2_omega_y
t1 = get_pi_time(test1_omega_x, test1_omega_y)
t2 = get_pi_time(test2_omega_x, test2_omega_y)

%graphing
nepsilon=100;
epsilon1s=(1:nepsilon)/nepsilon*(E0m-1+150)-ones(1,nepsilon)*150;
epsilon2s=(1:nepsilon)/nepsilon*((E0p-1)-(E0m-1))+ones(1,nepsilon)
    *(E0m-1);

times=zeros(nepsilon, nepsilon);
ltimes=zeros(nepsilon, nepsilon);
phis=zeros(nepsilon, nepsilon);
for p=1:nepsilon %fprintf(1, 'p=%d\n', p); for t=1:nepsilon
    [g1x, g1y] = get_gs(ax, bx, cx, ay, by, cy,
        epsilon1s(t));
    [g2x, g2y] = get_gs(ax, bx, cx, ay, by, cy,
        epsilon2s(p));
    phis(p, t) = get_phi(g1x, g1y, g2x, g2y);
    [omega1_x, omega1_y] = get_omega(ax, bx, cx, ay,
        by, cy, epsilon1s(t), 1, phis(p, t));
    [omega2_x, omega2_y] = get_omega(ax, bx, cx, ay,
        by, cy, epsilon2s(p), 1, phis(p,t));
    times(p, t) = get_pi_time(omega1_x, omega1_y) +
        get_pi_time(omega2_x, omega2_y);
    ltimes(p, t) = log10(get_pi_time(omega1_x,
        omega1_y) + get_pi_time(omega2_x, omega2_y));
    %fprintf(1, 't=%g\t', times(p,t)); end
end
[C, h] = contour(epsilon1s, epsilon2s,
    times,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Applied Electric Field 1 (kV/cm)', 'FontSize',20);
ylabel('Applied Electric Field 2 (kV/cm)', 'FontSize',20);
zlabel('Time to Flip Spin using Mechanism (s)',
    'FontSize',20);
title('Time to Flip Spin using Mechanism (s)',

```

```

        'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'pi_time.pdf'
[C, h] = contour(epsilon1s, epsilon2s, phis,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Applied Electric Field 1 (kV/cm)', 'FontSize',20);
ylabel('Applied Electric Field 2 (kV/cm)', 'FontSize',20);
zlabel('Magnetic Field Angle for Orthogonal Axes (rad)',
        'FontSize',20);
title('Magnetic Field Angle for Orthogonal Axes (rad)',
        'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'pi_phi.pdf'
[C, h] = contour(epsilon1s, epsilon2s,
        ltimes,'LineWidth',2);
set(gca,'FontSize',15);
set(gca,'LineWidth',1.5);
xlabel('Applied Electric Field 1 (kV/cm)', 'FontSize',20);
ylabel('Applied Electric Field 2 (kV/cm)', 'FontSize',20);
zlabel('Log_{10} of Time to Flip Spin using Mechanism (s)',
        'FontSize',20);
title('Log_{10} of Time to Flip Spin using Mechanism (s)',
        'FontSize',20);
clabel(C,'FontSize',20);
print -dpdf 'pi_ltime.pdf'

%Clearly optimum is max applied E-field
e1_min=-150
e2_min = fminbnd(@minimize_nonresonant, E0m, E0p);
fastest_time = minimize_nonresonant(e2_min);
[g1x, g1y] = get_gs(ax, bx, cx, ay, by, cy, e1_min);
[g2x, g2y] = get_gs(ax, bx, cx, ay, by, cy, e2_min);
phi = get_phi(g1x, g1y, g2x, g2y);
[omega1_x, omega1_y] = get_omega(ax, bx, cx, ay, by, cy,
        e1_min, 1, phi);
[omega2_x, omega2_y] = get_omega(ax, bx, cx, ay, by, cy,
        e2_min, 1, phi);
i_should_be_zero = test1_omega_x*test2_omega_x +
        test1_omega_y*test2_omega_y;
t1 = get_pi_time(test1_omega_x, test1_omega_y);
t2 = get_pi_time(test2_omega_x, test2_omega_y);
fprintf(1, 'Best time was %gs (t1=%gs t2=%gs); e1=%gkV/cm,

```

```
e2=%gkV/cm, phi=%grad (orthogonality verification:  
%g (should be 0))\n', fastest_time, t1, t2, e1_min,  
e2_min, phi, i_should_be_zero);
```

APPENDIX F SCRIPTS

Two main scripts were used in calculating g-tensors when varying a parameter. The first script run is a structure generator script. The example one below is called “e:1.667_r:v_h:10_gs”, meaning that it generates dots with an ellipticity of 1.667 and a height of 10 gridsites, varying the dot footprint radius based upon the job id given it.

```
#!/usr/bin/perl -w
#THIS IS A STRUCTURE GENERATION SCRIPT; THIS DOES NOT
  COMPUTE ANYTHING; IT
# MERELY GENERATES THE STRUCTURES FOR FURTHER CALCULATIONS
#This script conforms to the nexstruct naming system
#
#This script will generate a series of different-radius
  dots.
use strict;
use File::Basename;
use File::Path;

#Get out job number:
my $job=$ARGV[0];
if(!defined($job)){die("Unable to find job number!\n");}
#this is needed for job control
my $slkdp_is_gonna_die=0;
my $get_the_heck_out=0;
#SEt driver name for later use.
my $driver=basename($0);

my $project_base="$ENV{HOME}/dots";
#Radius change
my $radmin = 6;
my $radmax = 11;
my $radstep = 1;
#Structure creation information
my $structure_cmd = "$ENV{HOME}/bin/makeSphericalCap.pl";
my $structure_base_path = "$project_base/structures";
#Height varies here
my $hmin = 5;
my $hmax = 11;
my $hstep = 1;
#Set our height.
```

```

#my $structure_height = $hmin + $hstep*$job;
my $structure_height =10;
my $structure_ellipticity = 1.667;
my $structure_size = "100 100 100";
my $matdb = "/usr/local/share/dotcode/matdb.txt";
my $structure_xy_elong_dir = "1 1";
my $structure_cellsize = 0.56532;
my $structure_materials = "InAs GaAs";

#Derived quantities, for use in just a bit.
#my $radsteps = ($radmax - $radmin)/$radstep + 1;

#Derive axis length
my $axis_length_id = $job;
my $axis_length = $radmin + $radstep*$axis_length_id;

#Finally establish the final structure dir and the
  working dir.
my $fs_base =
  $structure_base_path."/e:". $structure_ellipticity."_r:".
  $axis_length."_h:". $structure_height."/src_structure";
my $final_structure = $fs_base.".displ";
my $working_dir = $fs_base;
#Temp structure
my $working_structure = $working_dir."/struct";

#Create the directory and start outputting information
  to it.
eval {mkpath($working_dir) };
if($?) { die("UNABLE TO CREATE $working_dir\n");
}
my $output_file="$working_dir/vc8-output.txt";
#Now, let's reopen our output to the appropriate files
  before we exec it.
open (STDOUT, ">>$output_file") or die("Unable to append
  to program output file \"$output_file\"!\n");
open (STDERR, ">>$output_file") or die("Unable to append
  to program output file \"$output_file\"!\n");
print "Beginning recorded script output for driver $driver;
  job $job\n";

chdir($working_dir) or die("UNABLE TO CD TO
  $working_dir\n");
#Create the unstrained struct

```

```

#Assemble the struct creation command
my $struct_creation_command =
    $structure_cmd." ".$working_structure."
    ".$structure_height." ".$axis_length."
    ".$structure_xy_elong_dir."
    ".$structure_ellipticity." ".$structure_cellsize."
    ".$structure_materials." ".$matdb." ".$structure_size;
#And run it
print STDOUT "\n\n--Running big/unstrained structure
    creation command \"$struct_creation_command\"\n\n\n";
system($struct_creation_command);
#When run, the struct_creation_command tacks on a .displ
    (displacement) suffix.
my $big_structure = $working_structure.".displ";
if (! -d $big_structure ) { die("ERROR: unstrained
    structure ($big_structure) not created (command is
    \"$struct_creation_command\")!");
}
#Strain it
my $strain_cmd = "strain $big_structure";
print STDOUT "\n\n--Running strain command
    \"$strain_cmd\"\n\n\n";
system($strain_cmd);
#We no longer truncate it; individual scripts take care
    of that.
print "STRUCTURE GENERATION COMPLETE.\n";
#Now move the final struct to its final name. We don't
    have to worry about fs boundaries
rename($big_structure, $final_structure) or die("ERROR:
    unable to rename finished temp structure ($big_structure)
    to final structure ($final_structure) (error $!)\n");
print "RENAMING SUCCEEDED.\n";

```

After the structure generator script created the files necessary to calculate on a structure, another set of scripts performed the calculations. The following script is an example, and is named “e:1.667-r:v-h:10-B:1-1”. This means that the script calculates the properties of dots with a height of 10 gridsites, ellipticity of 1.667, applied magnetic field of 1T, varying the height based upon the job id given it. The electric field is implicitly set at +100kV/cm, −100kV/cm, and 0, depending on job ID number.

```

#!/usr/bin/perl -w
use strict;
use File::Basename;
use File::Path;

```



```

#Get out job number:
my $job=$ARGV[0];
if(!defined($job)){die("Unable to find job number!\n");}
#this is needed for job control
my $slkdp_is_gonna_die=0;
my $get_the_heck_out=0;
#SEt driver name for later use.
my $driver=basename($0);

#Standard args for vc8
#Number of states to require to converge
my $nstates=2;
my $nvectors=$nstates + 1;
my $program="vc8";
my $args="repeps=1.0e-8 nreq=$nstates evecs=$nvectors
    Hamiltonian=G6+G7+G8";
#A vc8 run:
#vc8 trun.displ Bx=0.0707106781 By=-0.0707106781
    repeps=1.0e-8 eVmin=0 eVmax=1 nreq=4
    Hamiltonian=G6+G7+G8 Ez_mV_per_nm=9 >& vc8-output.txt &
#Electric field step information
#B-field magnitude
my $B=1;
my $project_base="$ENV{HOME}/dots";
#Source structure is determined based on our job id
#my
    $source_structure="$project_base/structures/$driver.displ";
my $too_small=1e-6;
#Structure creation information
my $structure_base_path = "$project_base/structures";
my $matdb = "$ENV{HOME}/dist/share/dotcode/matdb";
#Height is constant.
my $height=10;
#Radius change
my $radmin = 8;
my $radmax = 11;
my $radstep = 1;
my $radsteps = ($radmax - $radmin)/$radstep + 1;
#Modified since we now know our Nx.
my $Nx = 25;
my $Ny = $Nx;
my $Nabove = $Nx;
my $Nbelow = $Nx;

```

```

#EBox is now constant.
my $EBox = 17;
#E-field changes
my $efield_min = -10;
my $efield_max = 10;
my $efield_step = 10;
my $efield_steps = ($efield_max - $efield_min)/$efield_step
    + 1;

#Get E, B, and radius here.
my $remainder = $job;
my $chunksize = $efield_steps*$radsteps;
#Bfield is highest-"digit"
my $Bfield_id = ($remainder -
    $remainder%$chunksize)/$chunksize;
$remainder = $remainder - $Bfield_id*$chunksize;
#Efield is next
$chunksize = $radsteps;
my $Efield_id = ($remainder -
    $remainder%$chunksize)/$chunksize;
$remainder = $remainder - $Efield_id*$chunksize;
#Remainder is now axis length id (lowest-"digit")
my $axis_length_id = $remainder;

#Now create the actual B and E field info from these
    numbers.
#Max e-field (maximal push of the electron out of the dot)
my $Bx;
my $By;
my $Bz;
if($Bfield_id == 0){ $Bx = 0; $By = 0; $Bz = 1;
}elseif($Bfield_id == 1) { $Bx = 1; $By = 1; $Bz = 0;
}elseif ($Bfield_id == 2) { $Bx = 1; $By = -1; $Bz = 0;
}else{ die("NO SUCH B-FIELD ID ($Bfield_id)\n");
}
my $Ez = $Efield_id*$efield_step + $efield_min;
my $estring;
if(abs($Ez) >= $too_small) { $estring = "Ez_mV_per_nm=$Ez";
}else{ $estring = "";
}

#Now for the axis length
#Smallest axis length

```

```

my $axis_length = $radmin + $radstep*$axis_length_id;
#This was a pretty good value through all of the points.
my $evmin=0;
#But the CB varies quite a bit.
#Pairs are min,max and their respective radii
my $cb_minrad_evmax=0.56;
my $cb_minrad=8;
my $cb_maxrad_evmax=0.53;
my $cb_maxrad=11;
#Determine cb_scaling factor
my $cbscaling=($cb_maxrad_evmax -
    $cb_minrad_evmax)/($cb_maxrad - $cb_minrad);
#Now we know where to put evmax
#Based on y=mx+b with y0,x0 and y1,x1 known
my $evmax = $cbscaling*$axis_length + ($cb_minrad_evmax -
    $cbscaling*$cb_minrad);

#Output dir based on B-field direction
my $output_directory =
    $project_base."/".$driver."/r:$axis_length/B:$Bx$By$Bz/E:$Ez";
#Now that we have the B-field info, we need to make the
    actual B params.
my $bdirmag=sqrt($Bx*$Bx + $By*$By + $Bz*$Bz);
$Bx = $B*$Bx/$bdirmag;
$By = $B*$By/$bdirmag;
$Bz = $B*$Bz/$bdirmag;
my $bxstring;
my $bystring;
my $bzstring;
if(abs($Bx) >= $too_small) { $bxstring = "Bx=$Bx";
}else{ $bxstring .= "";
}
if(abs($By) >= $too_small) { $bystring = "By=$By";
}else{ $bystring .= "";
}
if(abs($Bz) >= $too_small) { $bzstring = "Bz=$Bz";
}else{ $bzstring .= "";
}
#Check to make sure we *have* a B-field
if(abs($B < $too_small)) { $bzstring = "";
}
#Set the EBox string
my $eboxstring="Erad=$EBox:1";

```

```

#Truncation info
#Determine the final struct info
#The base of the working and final struct
my $final_struct_subbase =
    $structure_base_path."/e:1.667_r:${axis_length}_h:$height";
my $final_struct_base = $final_struct_subbase."/Nxyz:$Nx";
#The source (100x100x100) structure
my $src_struct =
    $final_struct_subbase."/src_structure.displ";
#The final, truncated structure
my $final_struct = $final_struct_base.".displ";
#The working directory
my $working_dir=$final_struct_base;
#The struct that is output from manip
my $working_struct=$working_dir."/trun.displ";
#Truncation output file
my $manip_outfile=$working_dir."/manip-output.txt";

#Create the directory and start outputting information
to it.
eval {mkpath($output_directory) };
if($?) { die("UNABLE TO CREATE $output_directory\n");
}
my $output_file="$output_directory/vc8-output.txt";
#Now, let's reopen our output to the appropriate files
before we exec it.
open (STDOUT, ">>$output_file") or die("Unable to append
to program output file \"$output_file\"!\n");
open (STDERR, ">>$output_file") or die("Unable to append
to program output file \"$output_file\"!\n");
print "Beginning recorded script output for driver $driver;
job $job\n";
#####
#END OF DETERMINING (simple) VARIABLES

#First off, has the final struct already been created?
print "Do we have the final struct? ";
if(!( -d $final_struct)) { print "Nope.\nIs someone
making it? "; #Nope. Is someone already making it?
#NOTE: manip_outfile is a *much* more reliable judge
of whether the # construction has started or not.
if(!( -e $manip_outfile)) {
    print "Nope.\nAre we sure they didn't just finish? ";
    #One last check. Maybe they finished between checks.

```

```

if(!( -d $final_struct)) { #Nope. Gotta make it
    ourselves. eval {mkpath($working_dir) }; if($?) {
die("UNABLE TO CREATE $working_dir\n"); } #And
    cd into it chdir($working_dir) or die("UNABLE
    TO CD TO $working_dir\n"); #Now create the new
    truncated file my $struct_command="manip $src_struct
    interactive=0 Nxside=$Nx Nyside=$Ny Nabove=$Nabove
    Nbelow=$Nbelow"; print "Yep. Starting the truncation
    process (command: $struct_command). Output will
    be redirected to $manip_outfile\n"; open (STDOUT,
    ">>$manip_outfile") or die("Unable to append to
    program output file \"$manip_outfile\"!\n"); open
    (STDERR, ">>$manip_outfile") or die("Unable to append
    to program output file \"$manip_outfile\"!\n");
    system($struct_command); print STDERR "struct
    generation command ($struct_command) gave return
    value of $?\n"; #Source structure is now created.
    Tell program how to find it. #We need to use
    the relative path, since we have an = sign in the
    path # (since it thinks the = signifies a key)
    #$source_structure="trun.displ"; #We have to do our
    error checking down here; manip doesn't seem to give
    reliable # return values! if ( ! -d $working_struct
    ) {
die("COULDN'T CREATE $working_struct (command
    is [$struct_command])\n"); } #Finally, move the
    struct to the final name. rename($working_struct,
    $final_struct) or die("ERROR: unable to move
    truncated structure ($working_struct) to final name
    ($final_struct) (error: $!)\n");
} else{ print "Nope\nThey just got done.\n";
} } else{
print "Yep.\n";
#Someone's working on it. Just sleep until the file
    is ready.
print "Waiting for final structure ($final_struct)
    to be created";
while(!( -d $final_struct)) { printf("."); sleep 5;
}
print "\n"; } } else{ print "Yep.\n"; }

#Now, let's re-reopen our output to the appropriate files
    before we exec it.
open (STDOUT, ">>$output_file") or die("Unable to append

```

```

    to program output file \"$output_file\"!\n");
open (STDERR, ">>$output_file") or die("Unable to append
    to program output file \"$output_file\"!\n");
print "Beginning actual program run.\n";

#####
#PREPARE FOR RUNNING VC8!!!

#Now assemble the full command and go to the directory
we'll be
# working in.
my $command="$program $final_struct $args eVmin=$evmin
    eVmax=$evmax $estring $bxstring $bystring $bzstring
    $eboxstring";
chdir($output_directory) or die("UNABLE TO CD TO
    $output_directory\n");

#Now exec vc8!
#(this code inherited from slkdp drivers)
##Before we fork, we need to set up a signal handler to
listen for sigchild.
##Otherwise, slkdp can die, and we just hang here, waiting
for it to write files it'll never write.
##And, to top it all off, slkdp's still hanging around,
so nodespam thinks we're still busy.
sub clean_slkdp { if(!$slkdp_is_gonna_die){ print STDERR
    "ERROR: the program has died (signal $_[0])!
    Cleaning up... "; } #Call wait to clean up the zombies.
    wait;
    if(!$slkdp_is_gonna_die){ print STDERR "alerting wait
        loop..."; } $get_the_heck_out = 1;
    if(!$slkdp_is_gonna_die){ print STDERR "done\n"; }
}
#Set the signal handler
$SIG{CHLD}=\&clean_slkdp;

#Fork here.
print STDERR "going to fork!\n";
my $pid = fork();
if(!defined($pid)){ die("ERROR! Fork was unsuccessful!
    ERROR!");
}
if($pid == 0){ #OK. Almost done. Let's make our slkdp

```

```

command, and then exec it.  print STDERR "\n\n--exec'ing
command ($command)\n\n\n"; print STDERR "---OUTPUT OF
COMMAND---\n"; #And, finally, exec!  exec($command);
}else{ print STDERR "'$command', job $job is PID $pid\n";
  do { #Did slkdp die?  if($get_the_heck_out){ print
    STDERR "Wait loop: noticed that
      program died.  Exiting.\n"; exit 100; } #Now, sleep
    for a second, so we don't spam the cpu.  #print STDERR
    "beep\n"; sleep(10); #A quick explanation on the last
    kill bit: #Kill with a 0 signal will not actually send
    a signal; it'll just
      check to make sure that the pid is #  a valid process
    to send the signal to (i.e. process is running
      and is killable by us) #For more info, see 'perldoc
    -f kill' }until (!kill(0, $pid));
  #Alright!  The files we need are in place!  #Sleep for
    2 seconds to
  let things get written, then go on.  }

```

A second script, “gather-data-and-vars” gathered the data that was output from all the runs along with the parameters used. This was then put into one big table for further postprocessing.

```

#!/usr/bin/perl -w
use strict;
use Getopt::Std;
use File::Basename;
#This program has been created to gather
  whitespace-delimited table files
#  containing data, with additional parameters specified
  in the path.
my $program=basename($0);
#A list of entries (in order of appearance) with keys that
  have been set.
#  (list of hashes)  We store the whole table in memory
  for assembly.
my @TheBigTable=();
#A hash to store all known vars
my %known_vars=();
my %ops;
#the final options hash, as described before its
  translation function.
my %options;
getopt('hp:v:d:V:f:o:a:A:x:', \%ops);
store_options();

```

```

my $file;
my %params;
my $key;
my @line;
my $item = -1;
my $datum;
my $datumnum;
my $xcmd;
for $file (@{$options{files}}) { #print STDERR $file.":\n";
    #Get the params for this file #print STDERR "Getting
    varhash\n"; %params = %{get_varhash(get_varlist($file))};
    #Now process the file. Each line a new item!
    open(INFILE, "<$file") or die("ERROR: Unable to open
    file for input! ($file) (error: $!)"); while (<INFILE>) {
        #print STDERR " $_";
        chomp;
        #Ignore comments
        s/\#.*$//;
        #Removing leading whitespace
        s/^\s+//;
        if(/^$/){ #Blank line. Next! #print STDERR "
            [blank]\n"; next;
        }
        #print STDERR " $_\n";
        #New item!
        $item++;
        #Save off the filename to be safe
        $TheBigTable[$item]{filename} = $file;
        #Put the variables in the table
        foreach $key (sort keys %params) {
            $TheBigTable[$item]{vars}{$key} = $params{$key};
            #Set this key to be known. $known_vars{$key}=1;
        }
        #Break the line up
        #print STDERR "$options{delimiters}\n";
        #print STDERR " $_\n";
        @line = split(/[$options{delimiters}]+/);
        #print STDERR " " . join(' ', @line) . "\n";
        #Datum counter
        $datumnum = 0;
        #Save these values.
        foreach $datum (@line) {
            $TheBigTable[$item]{data}[$datumnum] = $datum;
            $datumnum++;
        }
    }
}

```



```

    }
    #How much data is in there?
    $TheBigTable[$item]{datumnum} = $datumnum;
    #Now, if needed, exec our driver
    if(defined($options{execdriver})) { #Assemble the
        command $xcmd=$options{execdriver}." '$file'";
        foreach $key (sort keys %params) {
$xcmd .= " '$params{$key}'"; } foreach $key
        (@{$TheBigTable[$item]{data}}) {
$xcmd .= " '$key'"; } #Now run the driver.
        #print STDERR "xcmd=$xcmd"; my $result='$xcmd';
        die_if_nonzero_retval($?, 1, $xcmd,
            $TheBigTable[$item]{vars}{xdriver}); my $r; my
            $num=0; my $name; for $r (split(/\s+/, $result)) {
$name = "xdriver_". $num;
$TheBigTable[$item]{vars}{$name}=$r;
#Set this key to be known.
$known_vars{$name}=1;
$num++; }
        } } close(INFILE);
    }
    my $maxitem = $item;
    #Table assembled.
    #Open the output file
    open(OUTFILE, ">$options{outfile}") or die("ERROR: Unable
        to open output file ($options{outfile}) for output
        (error: $!)!");
    my $var;
    #First off, output table titles.
    print OUTFILE "filename\t";
    foreach $var (sort keys %{$options{paramlist}}) { print
        OUTFILE $var."\t";
    }
    #output the named vars from the first line.
    foreach $var (sort keys %known_vars) { print OUTFILE
        $var."\t";
    }
    print OUTFILE "\n";
    for($item = 0; $item <= $#TheBigTable; $item++) { #print
        STDERR "item $item\n"; my $string; #First output the
        filename $string .= $TheBigTable[$item]{filename}."\t";
        #First output the user-defined vars. foreach $var
        (sort keys %{$options{paramlist}}) {
            $string .= ${options{paramlist}}{$var}."\t"; } #Now

```

```

output all of the known vars.  foreach $var (sort keys
%known_vars) {
    if(defined($TheBigTable[$item]{vars}{$var})) {
        #Output the key value.  $string .=
        $TheBigTable[$item]{vars}{$var}; #print STDERR "$var:
        $TheBigTable[$item]{vars}{$var}";
    }
    $string .= "\t";
    #print STDERR "\t"; } #Now output the values for
($datum = 0; $datum < $TheBigTable[$item]{datumnum};
$datum++){
    $string .= $TheBigTable[$item]{data}[$datum]."\t";
    #print STDERR $TheBigTable[$item]{data}[$datum]."\t";
} chomp $string; $string .= "\n"; #print STDERR "\n";
print OUTFILE $string;
}
close (OUTFILE);

#The options available:
#CONSTANT:
# defdelimits  default delimiters
# defvarseps   default varseps
# defvarparts  default varparts
# defanonvarsep default anonymous variable separator
# defanonvarname default anonymous variable name
#USER-SPECIFIED:
# paramlist  an array of parameters given explicitly
on invocation
# fileseek  an array of files to be sought in
directory structures
# outfile  file to output the final table to
# execdriver  program (and args) to run when a file
is found.
#DERIVED:
# files  an array of files from which data is
gathered
# delimiters  additional characters separating columns
in the files
# varseps  characters used to separate variables
from each other
# varparts  characters used to separate variable
names from values
# anonvarsep  anonymous variable separator
# anonvarname  anonymous variable collection name.

```

```

sub store_options { $Options{defdelimits}="\s";
  $Options{defvarseps}="_,\\";
  $Options{defvarparts}=":"; $Options{defanonvarsep}="";
  $Options{defanonvarname}="anonymous_vars"; #Delimiters
  and things are needed later on in the args, so we need
  them first #Delimiters if(defined($ops{d})){
    $Options{delimiters} = $Options{defdelimits} . $ops{d};
  }else{
    $Options{delimiters} = $Options{defdelimits}; }
  #Variable separators if(defined($ops{v})){
    $Options{varseps} = $Options{defvarseps} . $ops{v};
  }else{
    $Options{varseps} = $Options{defvarseps}; } #variable
  delimiters if(defined($ops{V})){
    $Options{varparts} = $Options{defvarparts} . $ops{V};
  }else{
    $Options{varparts} = $Options{defvarparts}; }
  if(defined($ops{A})){
    $Options{anonvarsep}=$ops{A}; }else{
    $Options{anonvarsep}=$Options{defanonvarsep}; }
  if(defined($ops{a})) {
    $Options{anonvarname}=$ops{a}; }else{
    $Options{anonvarname}=$Options{defanonvarname}; }
  #Additional parameters if(defined($ops{p})){
    #We simply call the get_varlist function. We use the
    same parsing rules
    # as later on. Consistency!
    #NOTE: get_varhash now returns a *reference* to
    the hash!
    print STDERR "Getting varhash (paramlist)\n";
    $Options{paramlist} =
    get_varhash(get_varlist($ops{p})); } #Files to be
    sought in a dir structure. Separated by params.
    if(defined($ops{f})){
      my @list = split(/ /, $ops{f});
      $Options{fileseek} = \@list; } #Need the output file
    if(defined($ops{o})){
      $Options{outfile} = $ops{o}; }else{
      print STDERR "ERROR: no output file specified!\n";
      $ops{h}=1; } #The exec driver if(defined($ops{x})) {
      $Options{execdriver} = $ops{x}; } #Gather files
      and dirs specified on the command line into one list
      of files. #This is done via a recursive function.
      #THIS MUST BE THE LAST THING DONE, SINCE WE NEED

```

```

VARIABLES TO BE # APPROPRIATELY SET!  my @files =
get_file_list(@ARGV); $options{files} = \@files;
    if(defined($ops{h})){
        do_help();
        exit; }
}
#Gets help.
sub do_help { print "---\n"; print "Help for $program:\n";
    print "Gathers whitespace-delimited data from a set
of files\n"; print "USAGE:\n"; print "  $program
[options] [dir/file [dir/file [...]]]\n"; print
"REQUIRED OPTIONS:\n"; print "  -o filename  file
to which the completed table is output\n"; print
"OPTIONAL OPTIONS:\n"; print "  -d chars  Adds
chars to the list of column delimiters\n"; print "
[default delimiters are $options{defdelimits}]\n"; print
"  -v chars      Adds chars to the list of variable
separators\n"; print "      [default
delimiters are $options{defvarseps}]\n"; print "  -V
chars          Adds chars to the list of variable-value
part separators\n"; print "      [default
separators are $options{defvarparts}]\n"; print
"  -f filelist  Filenames sought when examining a
directory structure\n"; print "      Files
are separated by a space\n"; print "  -p paramlist
Additional variable-value pairs to be specified
in the file\n"; print "      Pairs
are separated by variable separators\n"; print "
Variable-value parts separated by variable-value
separators\n"; print "  -x xdriver  Execute a program
for each line in the target files (-f).  This\n"; print "
can perform useful functions such as finding the sign of
g.\n"; print "      SEE NOTES BELOW\n"; print
"  -h          Prints this friendly help\n"; print
"NOTES:\n"; print "* xdriver output will be directly
piped to the field.\n"; print "* xdriver MAY return
multiple fields.  This is done by placing a tab\n"; print
"      character between each successive field.  The fields
will be labeled\n"; print "  \"xdriver_<fieldnum>\"
(<fieldnum> is the field number, starting at 0\n";
print "* xdriver MUST NOT return a newline character.
This would begin the next line, and input\n"; print "
will almost certainly still follow.\n";
}

```

```

#Gets a list of variable=value pairs separated by varseps
#ARGS:
#  a list of strings containing variable=value pairs
#    (separated by varparts),
#    each sepaprated by varseps
#RETURNS
#  a list of variable=value pairs
sub get_varlist { my $varpairs; my @varlist; my
  @skippedlist; foreach $varpairs (@_) {
    #Just blindly push; policy will be dealt with down
    the stack where it's appropriate.
    push(@varlist, split(/\[Options{varseps}\] +/,
      $varpairs)); } return @varlist;
}
#Gets a hash consisting of variable -> value pairs.
#ARGS:
#  a list of variable=value pairs (separated by varparts)
#RETURNS:
#  a *reference* to a hash containing the variable-value
  pairs.
sub get_varhash { my $varpair; my @var; my %varhash;
  foreach $varpair (@_) {
    #print "varpair: $varpair\n";
    @var = split(/\[Options{varparts}\] +/, $varpair);
    #print "  parts: ".join(', ', @var)."\n";
    if($#var != 1) { #This is an anonymous var.
      if(!defined($varhash{Options{anonvarname}})) {
#anonymous variable name doesn't exist yet.
$varhash{Options{anonvarname}}=$varpair; }else{
$varhash{Options{anonvarname}} .=
    Options{anonvarsep}.$varpair; }
    }else{ #Is good!  Add it.  $varhash{$var[0]} = $var[1];
    } } return \%varhash;
}

#Gets the files we are going to examine.
#If a file is passed to this puppy, it just naiively
  adds it.
#  Behaviour is different if you pass it a dir, and it
    recurses to the _r
#  variant (just after this one)
sub get_file_list { my $piece; my @dirfiles; my $direntry;

```

```

my @files; for $piece (@_) {
    #print $piece."\n";
    if ( -d $piece ) { #recurse!  push(@files,
        get_file_list_r($piece));
    }else{ #Must be a file.  Just add it.  push(@files,
        $piece);
    } } return @files;
}

#Same as before, but made for recursion.
sub get_file_list_r { my $piece; my @dirfiles; my
    $dirent; my @files; for $piece (@_) {
        #print $piece."\n";
        if ( -d $piece ) { #print "  is dir\n"; #Dir.
            if(!opendir(DIR, $piece)){
print STDERR "WARNING: Unable to open directory
    ($piece) for reading!  Skipping!\n";
next; } @dirfiles = readdir(DIR); closedir DIR;
            #print " contains: "; for $dirent (@dirfiles) {
#print $dirent." ";
#Files to always be skipped
if (($dirent eq ".") || ($dirent eq "..")) {
    next;
}
#recurse!
push(@files, get_file_list_r("$piece/$dirent"));
        } #print "\n";
        }else{ #Is this a file we want?  if
            (is_wanted_file($piece)) {
#print "  is wanted\n";
push(@files, $piece); #}else{
#print "  is NOT wanted\n"; } #Otherwise,
            we're done.
        } } return @files;
    }
}

#Determines if the file passed is wanted or not.
sub is_wanted_file { my $file;
    #Obviously not; we don't want it!
    if(!defined($options{fileseek})) {
        return 0; } for $file (@{$options{fileseek}}) {
        if (basename($file) eq basename($_[0])) { return 1;
        } } #If we've not found it by now, it's not wanted!
    return 0;
}

```

```

#Adds new vars to the list and returns the new list.
#ARGS:
# A *reference* to a key-value hash
# A *reference* to the list
#RETURNS:
# Any new keys from the hash and the old key list
sub add_new_vars { my $key; my $item; my %hash = %{$_[0]};
  my @list = @{$_[1]}; my @newitems; keyloop: foreach $key
    (sort keys %hash) {
      #print STDERR "key $key: ";
      itemloop: foreach $item (@list) { #print STDERR "$item
        ("; if ($key eq $item) {
#print STDERR "Y)\n";
next keyloop; #}else{
#print STDERR "N)"; } #print STDERR "\n";
      }
      #If we've made it here, this is a new key!
      push(@newitems, $key); } push(@newitems, @list);
    return @newitems;
  }

#Die with error foo if there was a nonzero retval.
#ARGS
# the return value ($?)
# the value to return when we exit
# string containing the command that was run
# string containing the text (if any) output by the
  command to STDOUT.
# @oklist containing all errors which should be ignored.
#RETURNS
# nothing. NOTE: THIS MAY WELL EXIT THE PROGRAM.
#NOTES:
# code ripped shamelessly from 'perldoc -f system'
sub die_if_nonzero_retval { my $err=shift(); my
  $reterr=shift(); my $cmd=shift(); my $cmdtext=shift();
  my @oklist = @_; #print STDERR "oklist=".join(', ',
  @oklist)."\n"; my $i; if($err == 0) {
    #Command executed OK. Just return.
    return; } if($err == -1) {
    #Error forking.
    print "ERROR: couldn't execute command ($cmd): $!\n";
    exit $reterr; } if($err & 127) {
    printf "ERROR: xdriver ($cmd) exited with signal %d,
    %s coredump. Command output: %s\n", ($err & 127),

```

```

($err & 128) ? 'with' : 'without', $cmdtext;
    exit $reterr; } $err = $err >> 8; foreach $i
(@oklist) {
    if($err == $i) { #We have been told to ignore this
        error. #print STDERR "err=$err == i=$i\n"; return;
    }
    #print STDERR "err=$err != i=$i\n"; } printf
"ERROR: xdriver ($cmd) exited with non-zero status %s.
Command output: %s\n", $err, $cmdtext; exit $reterr;
}

```

A sub-script (xdriver) was called by the gather-data-and-vars script, to find the g-factor or just its magnitude, depending on the name that was called.

```

#!/usr/bin/perl -w
use strict;
use File::Basename;

use vars qw/ $max_deltaE $max_deltaE_factor
    $propconst_electron $propconst_hole
    $same_energy_threshold /;

my $prog = basename($0);

#Alternate program names do alternate things.
my $gmagname = "xdriver-findgmag";
my $varbname = "xdriver-findg_varb";

#exit -1;

BEGIN { #Initializer: #Maximum difference in
    energies to be considered a spin splitting (eV);
    $max_deltaE=2e-3; $max_deltaE_factor=1; #Default
    magnetic field magnitude (T) #Proportionality
    constant of spin-splitting and mag field: (E1-E0) =
    g*B*propconst $propconst_electron=5.788381749E-05;
    $propconst_hole=$propconst_electron*3.0;
    $same_energy_threshold=1e-8;
}
#print STDERR "ARGS=@ARGV\n";
#Number of args which this program is passed directly
#ARGS TO THIS SCRIPT:
# 0) index of the first energy level
# 1) index of the b direction
my $nargs;

```



```

if ($prog eq $gmagname) { $nargs=1;
} elsif ($prog eq $varbname) { $nargs=3;
} else { $nargs=2;
}
#Which arg has B direction? Step over our private args
to the
# args passed by gather-data-and-vars
if (scalar(@ARGV) < $nargs) { print STDERR "ERROR:
insufficient number of arguments!\n"; print STDERR
"USAGE: $prog <first energy index>"; if ($prog ne
$gmagname) {
    print STDERR " <B-field direction index>"; } if ($prog
eq $varbname) {
    print STDERR " <B-field magnitude index>"; } print
STDERR "\n"; exit 1;
}
my $firstEnergy=$ARGV[0]+$nargs;
my $Bdir_arg;
my $Bdir;
if ($prog eq $gmagname) { $Bdir_arg=undef; $Bdir=undef;
}else{ $Bdir_arg=$ARGV[1]+$nargs; $Bdir=$ARGV[$Bdir_arg];
}
#What is the target path (directory containing the target
file)
#eigenvalues.dat file with ocmplete path is first argument
past the privat ones.
my $path=dirname($ARGV[0+$nargs]);
my $target_path=dirname($ARGV[0+$nargs]);
#print STDERR "path=$path\n";
#exit 5;
my @raw_states=@ARGV[$firstEnergy.. $#ARGV];
my $Bx=undef;
my $By=undef;
my $Bz=undef;
my $bmag=undef;
my $x;
my $y;
my $z;
#Direction of the magnetic field
if (defined($Bdir)) { #print STDERR "Bdir=$Bdir\n"; if
($Bdir eq '1-10') {
    $Bx = 1;
    $By = -1;
    $Bz = 0; } elsif ($Bdir == 1) {

```

```

    $Bx = 0;
    $By = 0;
    $Bz = 1; } elsif ($Bdir == 110) {
    $Bx = 1;
    $By = 1;
    $Bz = 0; } else {
    die("ERROR: unknown magnetic field direction:
    $Bdir\n"); } $x=$Bx/sqrt($Bx*$Bx + $By*$By +
    $Bz*$Bz); $y=$By/sqrt($Bx*$Bx + $By*$By + $Bz*$Bz);
    $z=$Bz/sqrt($Bx*$Bx + $By*$By + $Bz*$Bz); $bmag =
    sqrt($x*$x + $y*$y + $z*$z);
} else { #Default B magnitude is 1 $bmag = 1;
}
#Change the B-field magnitude
if ($prog eq $varbname) { $bmag *= $ARGV[$ARGV[2]+$nargs];
    #print STDERR "VARB: bmag=$bmag Bx=$Bx By=$By Bz=$Bz\n";
}

#print STDERR unify(sort {$a <=> $b} @raw_states)."\n";
my @spinstates = group_spinsplits(uniquify(sort {$a <=>
    $b} @raw_states));

#OK. Now we have a list of spin-splittings.
#Find the g-factor!
my @gfactors = find_gfactors($bmag, $Bx, $By, $Bz, $path,
    @spinstates);

#Now print out the g-factors.
my $spinpair;
for ($spinpair=0; $spinpair < scalar(@spinstates);
    $spinpair++) { #Print a connecting \t between subsequent
    sets of data. if ($spinpair != 0) {
        print "\t"; } map {
        print "$_\t"; } @{$spinstates[$spinpair]};
    #Maintain backwards compitibility wrt output ordering.
    if (defined($gfactors[$spinpair]{sign})) {
        print
        "$gfactors[$spinpair]{sign}\t$gfactors[$spinpair]{mg}\t
        $gfactors[$spinpair]{g}";
    } else {
        print "$gfactors[$spinpair]{mg}"; }
}

#Gets the g-factors for a (list of) spin states.

```

```

#ARGS:
#  bmag magnetic field magnitude
#  Bx   x-direction of mag field
#  By
#  Bz
#  (list of) refs to lists of spin-split energy levels.
#RETURNS:
#  (list of) refs to hashes consisting of {sign} and {mg}
#      and {g}
#      (g-factor sign, magnitude, and g-factor,
#      respectively)
#NOTE:
#  If used in scalar context, will return results from
#  FIRST list (so you can
#  use it for a single or list of spin-splittings.
sub find_gfactors { my $bmag = shift(); my $bx=shift();
  my $by=shift(); my $bz=shift(); my $path=shift(); my
  $splitting; my $hash; my @hlist = (); #print STDERR
  "find_gfactors: bmag=$bmag bx=$bx by=$by bz=$bz\n";
  foreach $splitting (@_) {
    $hash = {};
    ${$hash}{mg} = get_gfactor_mag($bmag, @{$splitting});
    if (defined($bx)) { ${$hash}{sign} =
      get_gfactor_sign($bx, $by, $bz, $path,
        @{$splitting}); ${$hash}{g} = ${$hash}{mg} *
        ${$hash}{sign};
    }
    push(@hlist, $hash); } if (wantarray) {
  return @hlist; } else {
  return $hlist[0]; }
}

#Gets the magnitude of the g-factor for a list of
#  spin-split energies
#ARGS:
#  bmag  magnitude of magnetic field
#  (sorted) list of energies
#RETURNS:
#  g-factor magnitude
sub get_gfactor_mag { my $bmag = shift(); my $propconst;
  if (scalar(@_) != 2) {
    die("ERROR: get_gfactor_mag: more than 2 spin-split
    energy levels? This isn't spin-1/2!\n"); } if ($_[0]
  < 0) {

```

```

    #Use hole propconst;
    $propconst = $propconst_hole; } else {
    $propconst = $propconst_electron; } #print STDERR
"\tget_gfactor_mag: e1=$_[1] e2=$_[0] delta=".(($_[1] -
$_[0])). " propconst=$propconst bmag=$bmag"; #print STDERR
" g-factor=" . ((($_[1] - $_[0]))/($propconst*$bmag))
. "\n"; return ((($_[1] - $_[0]))/($propconst*$bmag));
}

#Gets the g-factor sign from a (sorted) list of g-factor
energies
#ARGS:
# bx x-direction of magnetic field
# by
# bz
# dir directory holding the energy eigenstates
# (sorted) list of energies
#RETURNS:
# +1 if g-factor is positive
# -1 if g-factor is negative
sub get_gfactor_sign { my $bx = shift(); my $by = shift();
my $bz = shift(); my $dir = shift(); #Take the lowest
energy-level. my $energy = shift(); my $state =
""; $state=find_state_for_energy($energy, $dir);
if(!defined($state)) {
    die("ERROR: unable to find state $state for energy
$energy in dir $dir!\n"); } my $cmd = "(cd $dir &&
angularMomentum -S -d $bx $by $bz $state)"; my $result =
'$cmd'; if ($? != 0) {
    die("ERROR: got non-zero error value from
angularMomentum command (cmd=$cmd): $?!\n"); } #print
STDERR "\tresult=$result\n"; #Cut off leading/trailing
whitespace. $result =~ s/^\s*//; #print STDERR
"\tresult=$result\n"; $result =~ s/\s*$//; my
$rstring=$result; #print STDERR "\tresult=$result\n";
$result = (split(/\s+/, $result))[0]; #print STDERR
"\tresult=$result\n"; $result = (split(/\s*\+I*\s*/,
$result))[0]; #print STDERR "\tresult=$result\n"; #First,
assume electron. my $sign=undef; if ($result < 0) {
    $sign=1; } elsif ($result > 0) {
    $sign=-1; } else {
    print STDERR "WARNING: get_gfactor_sign:
pseudospin angular momentum ($result)(first column
is path=".$target_path.")(string was '$rstring')

```

```

is zero!\n";
    $sign=0; } #print STDERR "\tsign=$sign "; if ($energy <
0) {
    #Flip sign; it's a hole.
    $sign *= -1; } #print STDERR "\tsign=$sign\n";
return $sign;
}

#Break the energies into pairs.
#ARGS:
# list of energies
#RETURNS:
# list of refs to lists of (ordered) spin-splittings.
sub group_spinsplits { my @epairs = (); #print STDERR
    "states=".join(', ', @_)." \n"; if ((!defined($_[0])) ||
    (!defined($_[1]))) {
        die("group_spinsplits: insufficient number of states
        (must be at least two; we got $#_ (script args were
        @ARGV)\n"); } my $state; #Energies are sorted at this
        point, so if the next one isn't # the right energy, no
        others are gonna be either. #Take off the last energy.
        my $laststate = shift(); #Initialize the spin-split list
        ref. my $spinlist = []; push(@{$spinlist}, $laststate);
        foreach $state (@_) {
            #print STDERR "state=$state\n";
            #Is this is a spin splitting?
            if ($state - $laststate < 0) { die("group_spinsplits:
            energy difference between two states is less than
            zero!! (E1=$laststate, E2=$state)\n");
            }
            #Meaning of all this:
            # 1) States must always be less than max_deltaE.
            # 2) If there is already a spin-split pair, the next
            state MUST be within
            # max_deltaE_factor times the last spin-split
            pair (to avoid bringing
            # in neighboring spin splittings.
            if ((scalar(@{$spinlist}) < 2) ||
                ((${$spinlist}[$#{ $spinlist}] -
                ${ $spinlist}[$#{ $spinlist}-1])*$max_deltaE_factor >
                ($state - $laststate))) && ($state - $laststate
                < $max_deltaE)) { #if ($state - $laststate <
                $max_deltaE) { #print STDERR "A\n"; #Yes. Record the
                previous state. push(@{$spinlist}, $state); #print

```

```

STDERR "spinlist=".join(', ', @{$spinlist})."\n";
#_this_ spin-split state will be recorded on the
next iteration.
} else { #print STDERR "B ".scalar(@{$spinlist})."
" . (${$spinlist}[$#{ $spinlist}] -
${$spinlist}[$#{ $spinlist}-1])." " . ($state -
$laststate)."\n"; #No. If the list isn't empty,
save the previous one off, then # save off the
list itself, and then create a new list ref.
if (scalar(@{$spinlist}) > 1) {
push(@epairs, $spinlist);
$spinlist=[];
#The current state now becomes the first of a
new pair.
push(@{$spinlist}, $laststate);
} elsif (scalar(@{$spinlist}) == 0) {
#Warn the user if we have only one in the current
spin pairing.
print STDERR "WARNING: the following state was
unpaired: @{$spinlist}\n"; } #If
spinlist is empty; we've nothing to do.
}
#Make sure we update the previous energy
$laststate = $state; } #Finish off the last state if
we need to. if (scalar(@{$spinlist}) > 1) {
#print STDERR "saved in-prep spinlist
(@{$spinlist})\n";
push(@epairs, $spinlist); } elsif (scalar(@{$spinlist})
== 0) {
#Warn the user if we have only one in the current
spin pairing.
print STDERR "WARNING: the following state was
unpaired: @{$spinlist}\n"; } #print STDERR "Done:
epairs=@epairs\n"; return @epairs;
}

#Returns a list of *unique* items passed
#ARGS:
# @list
#RETURNS
# @uniquified_list
sub uniquify { my @results=(); my $a; my $b; #print STDERR
"uniquify: args=@_\n"; #Try to place each arg....
outloop: foreach $a (@_) { #See if it's already in

```

```

@results: foreach $b (@results) {
    #If they're the same, go to the next arg ($a)
    if ($a eq $b) { next outloop;
    } } #If we've gone through the results and found no
    matches, this is unique push(@results, $a);
}
#print STDERR "uniquify: returning @results\n";
return @results; }

#finds the eigenstate for a specified energy.
#ARGS:
# target energy
# directory holding the states.
#RETURNS:
# Name of the eigenstate for this energy or undef if
  there was no such state.
sub find_state_for_energy{ my $energy=shift();
  my $dir=shift(); opendir(STATESDIR, $dir) or
  die("Unable to open directory ($dir) to get the
  eigentstate for energy $energy!\n"); my @states =
  grep(/(0-)?[0-9]*.[0-9]+eV\.psi8/, readdir(STATESDIR));
  closedir(STATESDIR); my $state; my $state_e; my
  $final_state=undef; foreach $state (@states) {
    $state_e=$state;
    $state_e =~ s/eV\.psi8$//;
    $state_e =~ s/^0-/-/;
    if(abs($energy-$state_e) <
      abs($same_energy_threshold*$energy)) {
      if(defined($final_state)) {
die("ERROR: found TWO states (there may be more!
  I'm quitting now!) for $energy: $final_state,
  $state\n"); }else{
#print STDERR "FOUND state for energy: $energy ->
  $state (energy in filename was $state_e)\n";
$final_state=$state; }
    }else{ #print STDERR "REJECTED state: $energy -!->
      $state (energy in filename was $state_e)\n";
    } } #if final_state is undefined, there was no similar
    state. return $final_state;
}

```

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